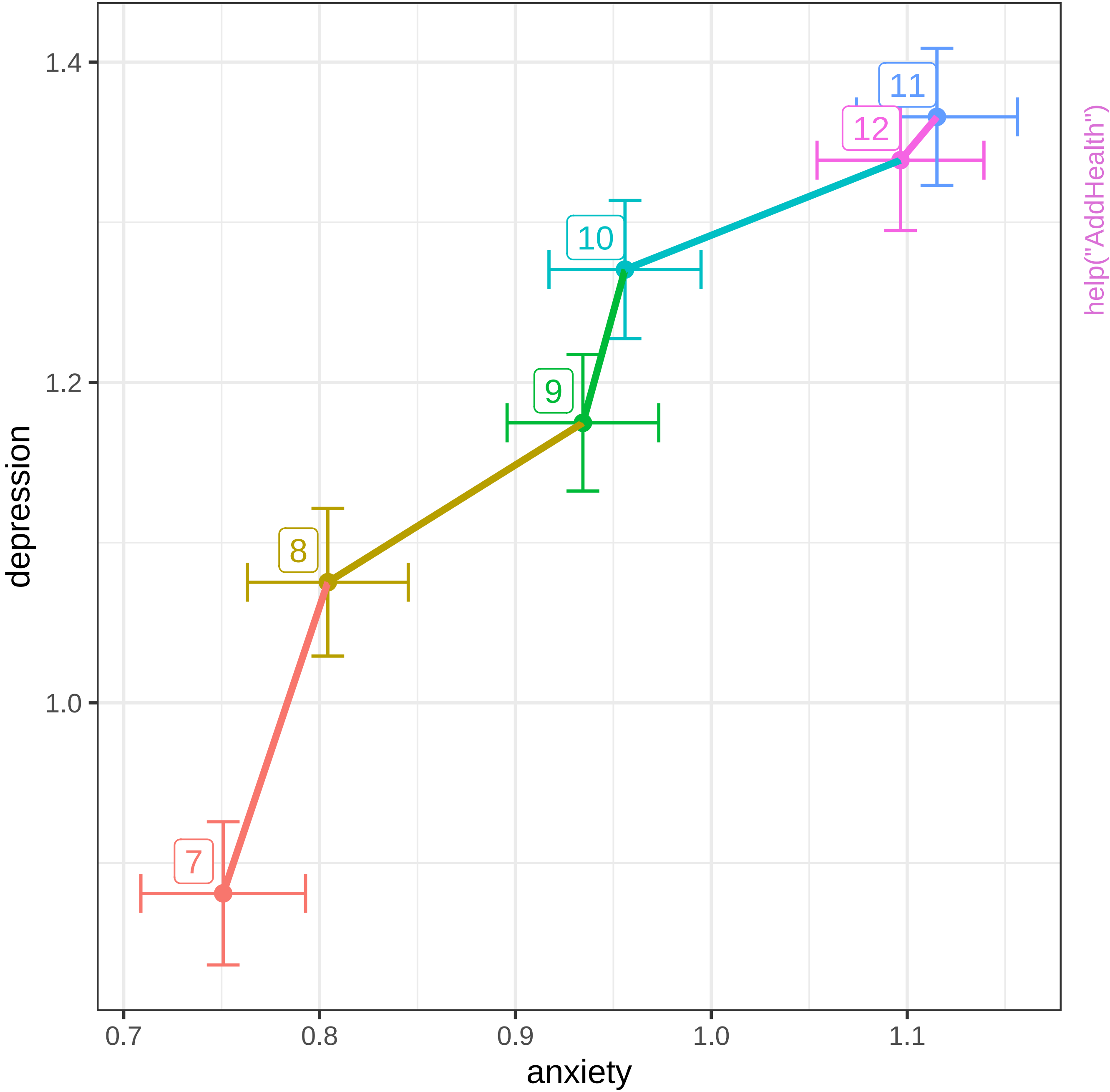
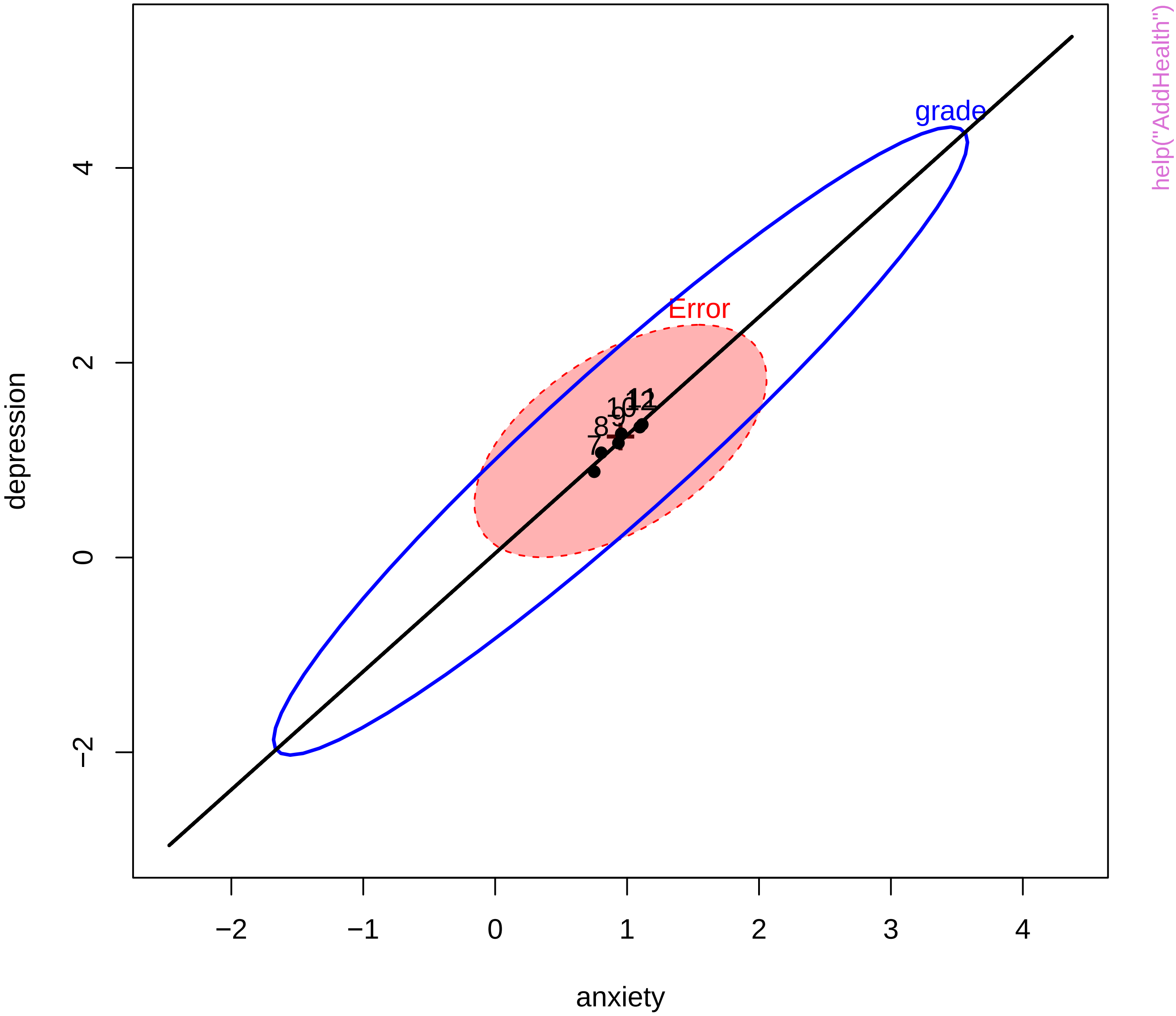
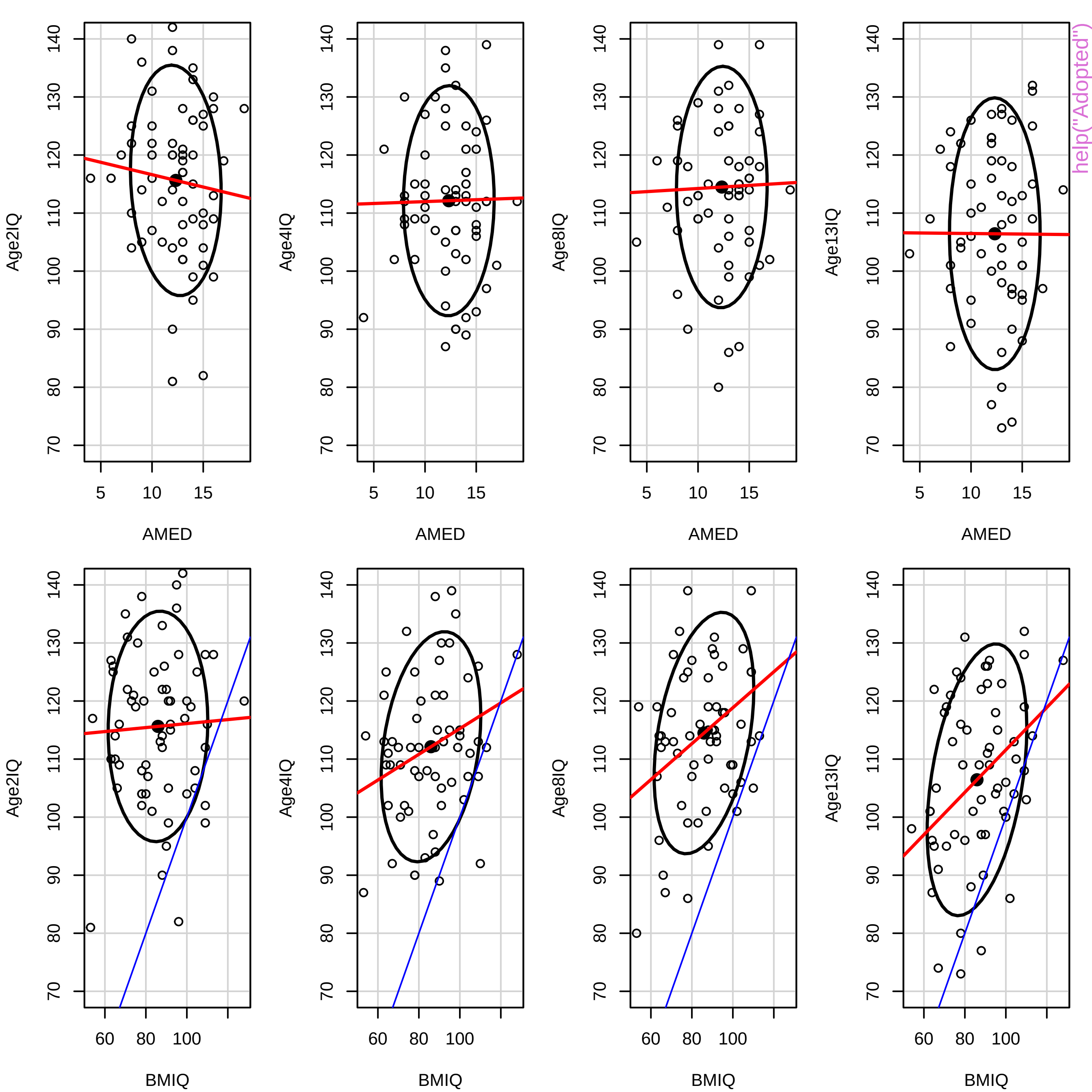
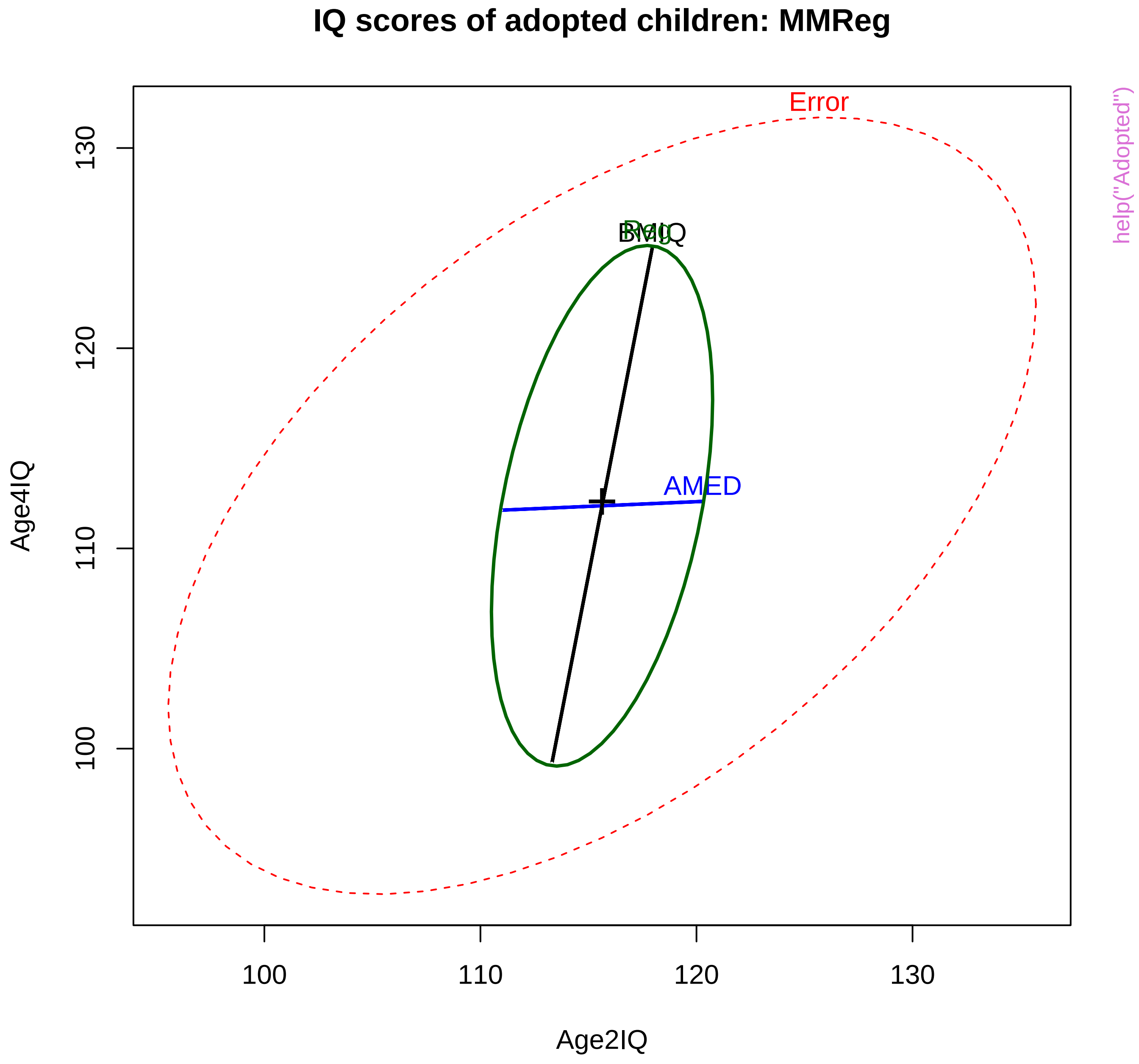


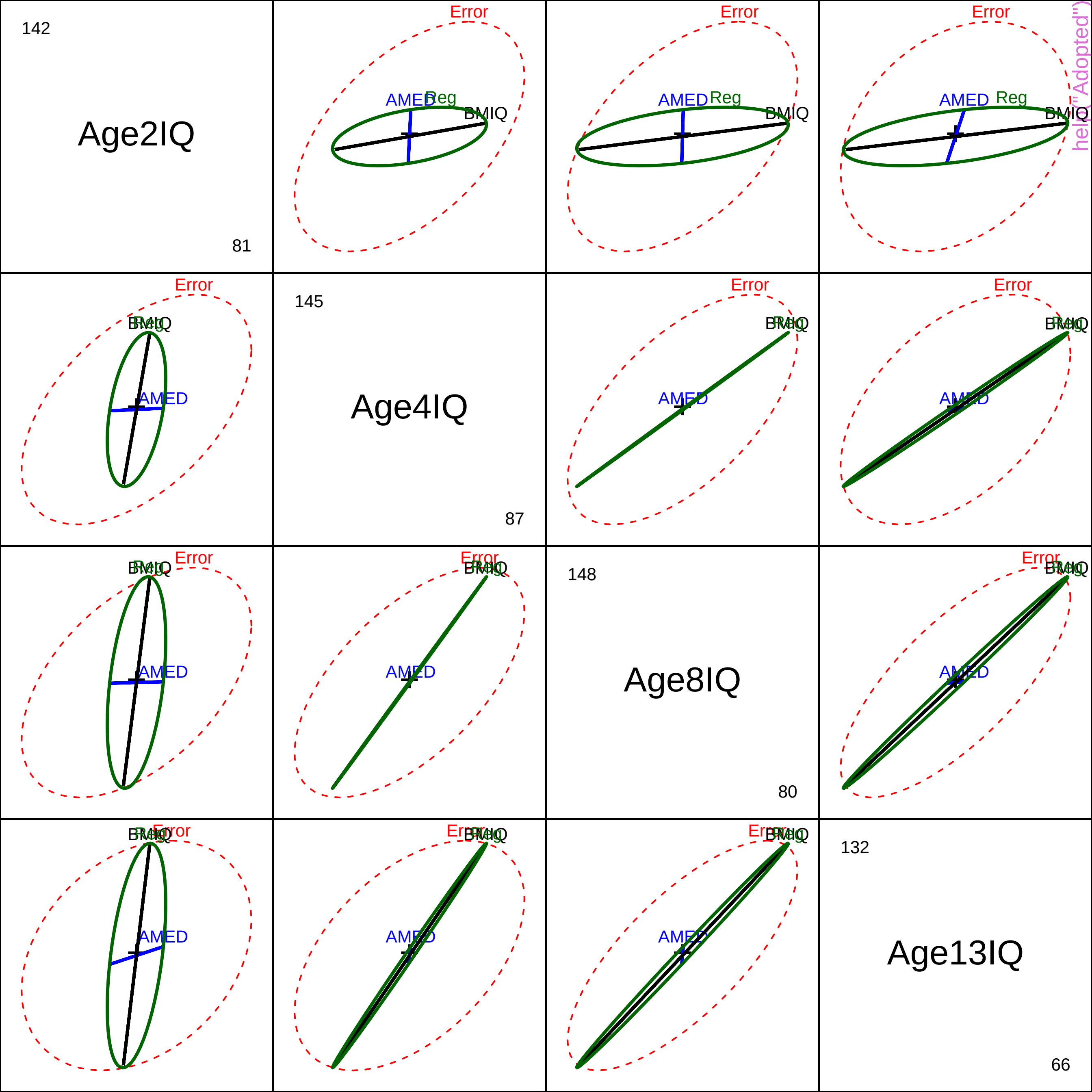
1.4
1.2
1.0
depression
0.7
0.8
0.9
1.0
1.1
anxiety
7
8
9
10
11
12
help("AddHealth")



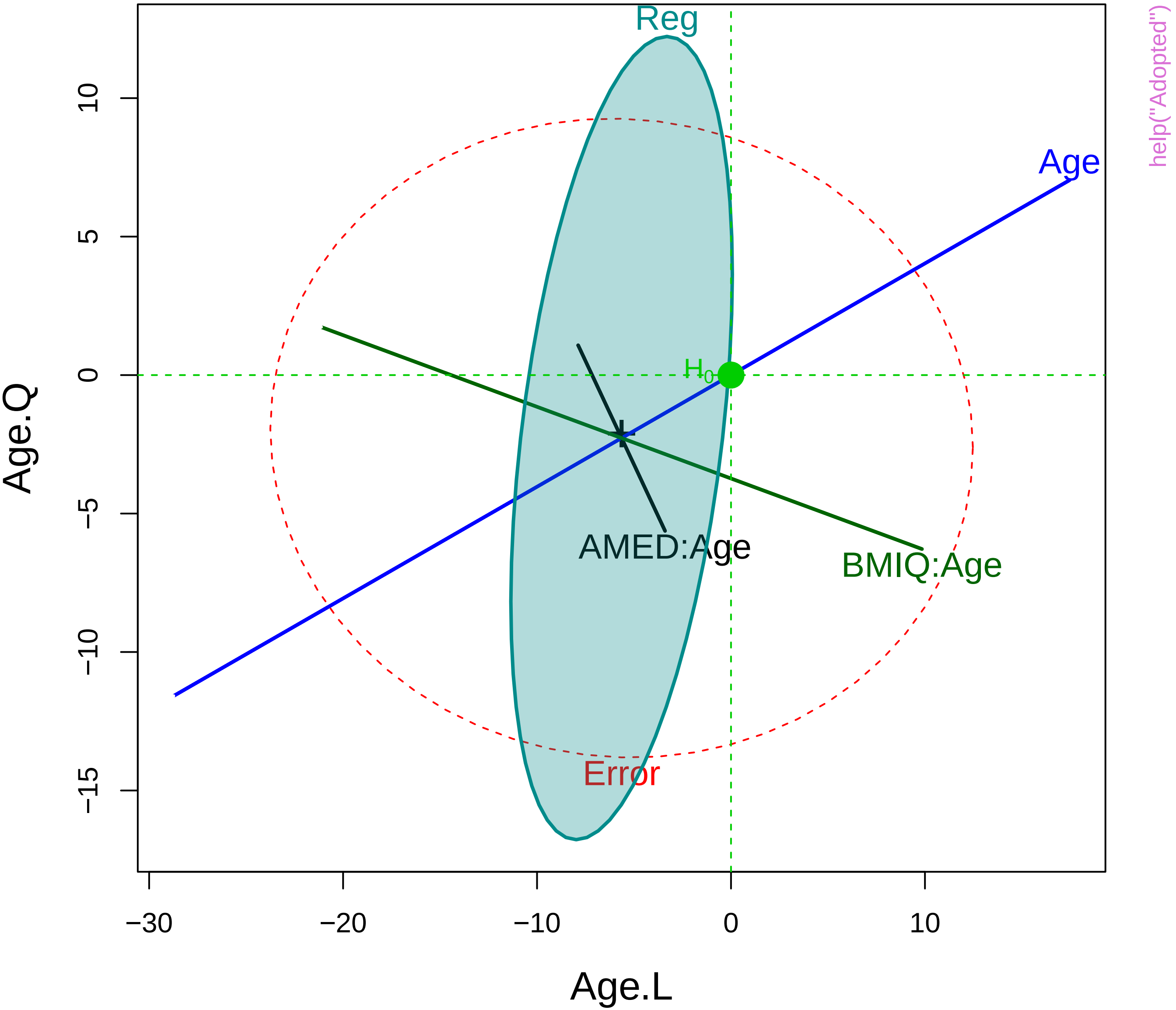
grade
Error
7
8
9
10
11
12
depression
anxiety
−2
−1
0
1
2
3
4
−2
0
2
4
help("AddHealth")


help("Adopted")
Age2IQ
Age4IQ
Age8IQ
Age13IQ
AMED
BMIQ

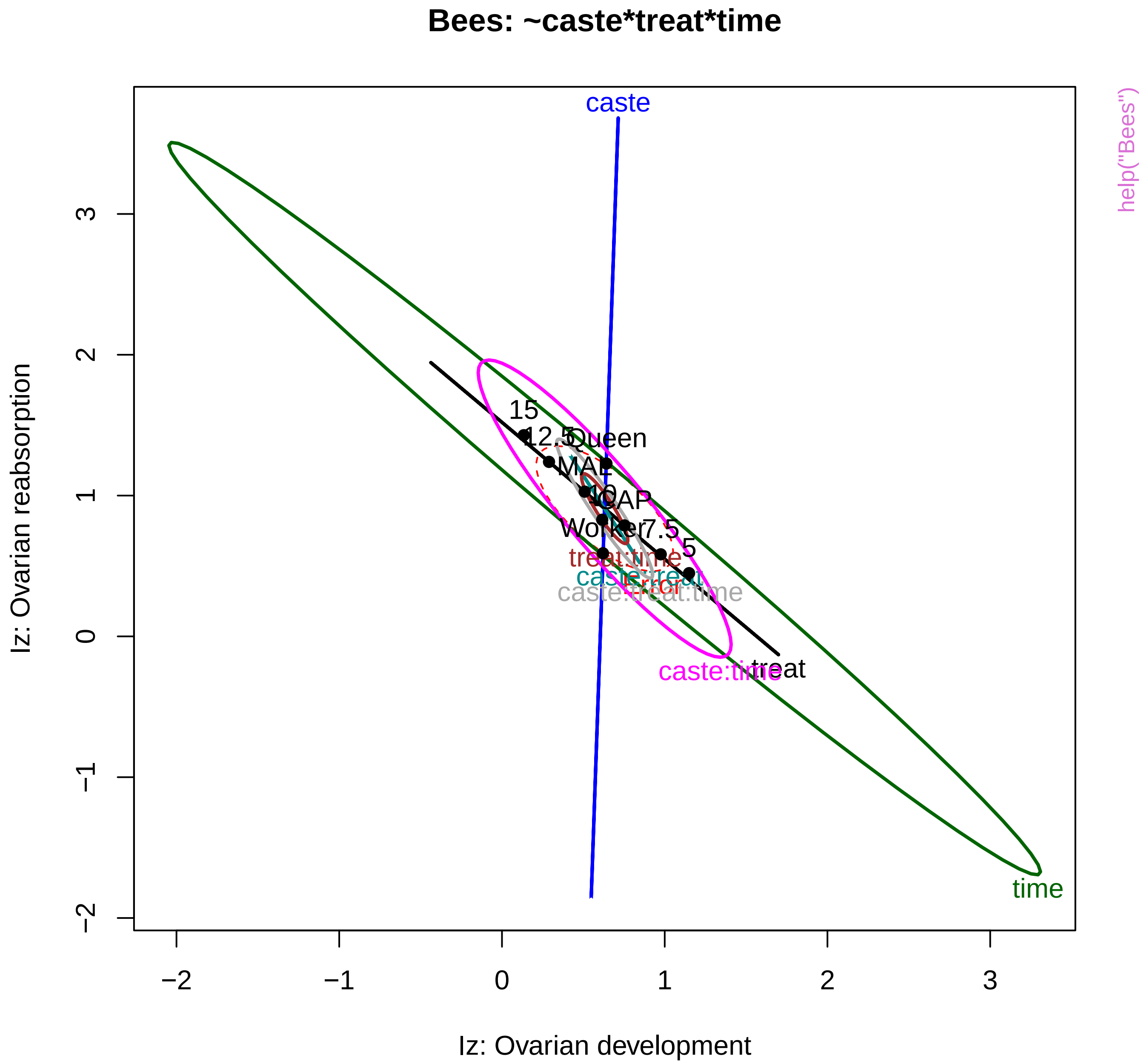

IQ scores of adopted children: MMReg
Error
Reg
BMIQ
AMED
130
120
110
100
Age4IQ
100
110
120
130
Age2IQ
help("Adopted")

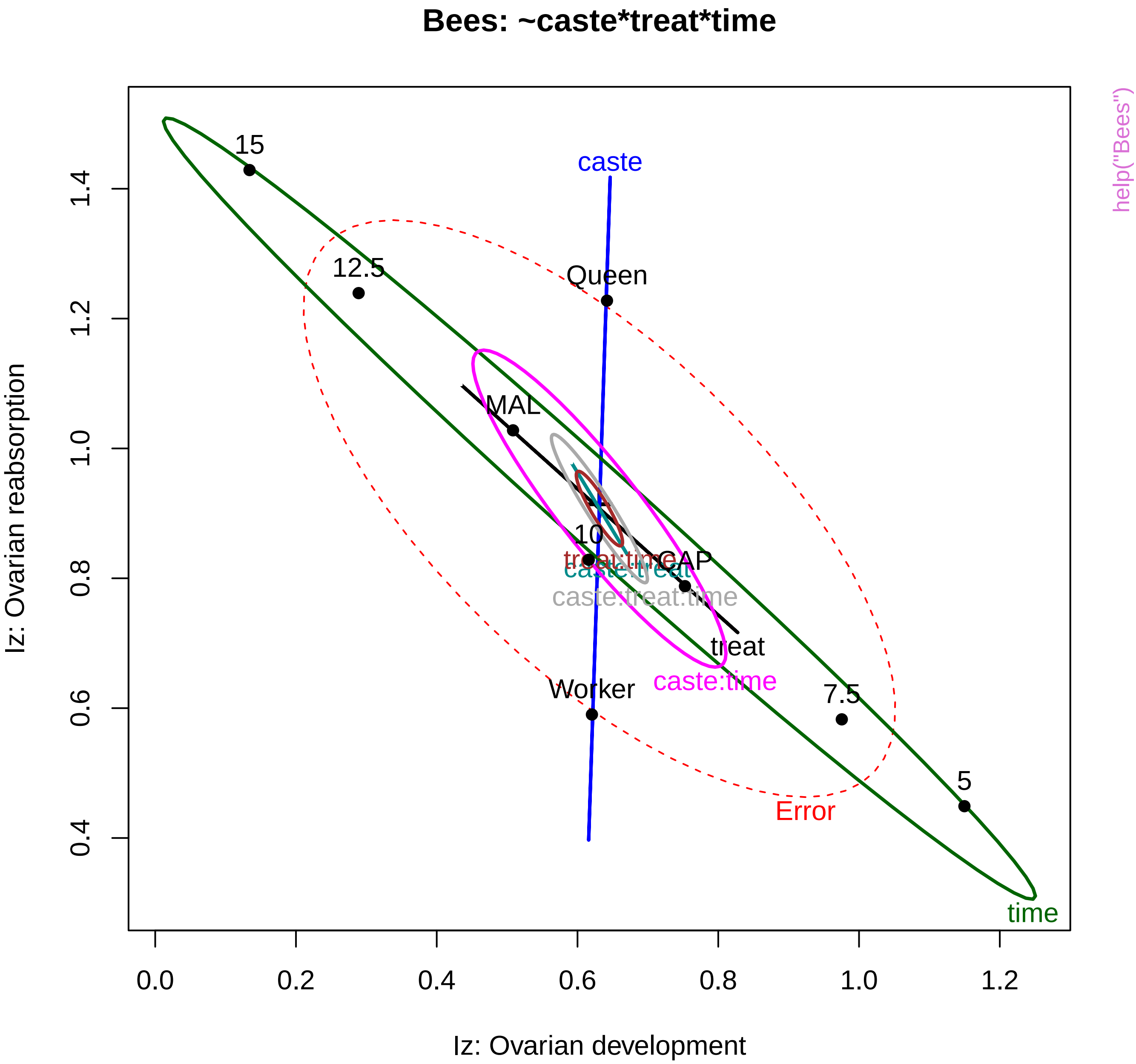


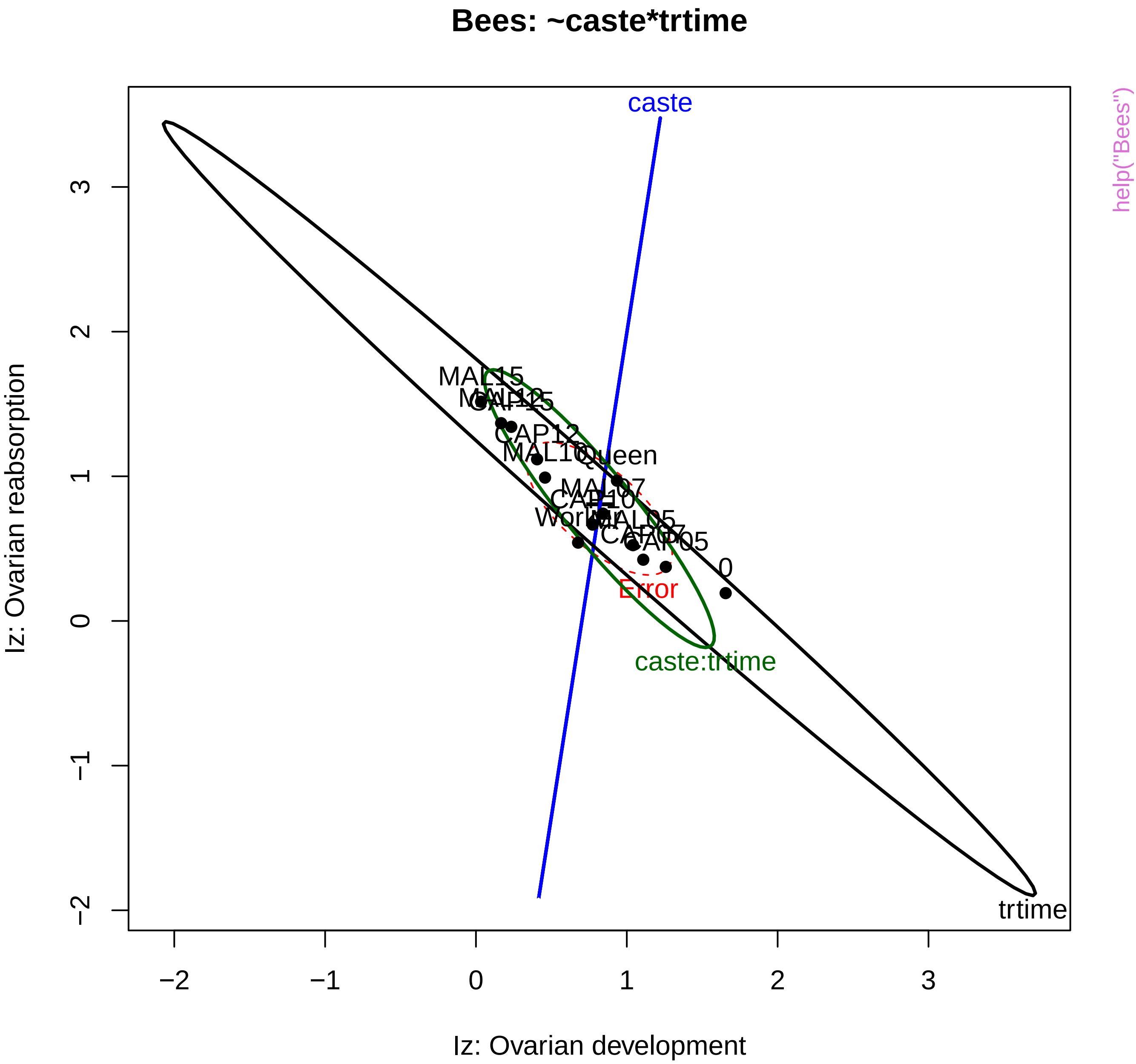
142
Age2IQ
81
Error
AMED
Reg
BMIQ
145
Age4IQ
87
148
Age8IQ
80
132
Age13IQ
66
help("Adopted")

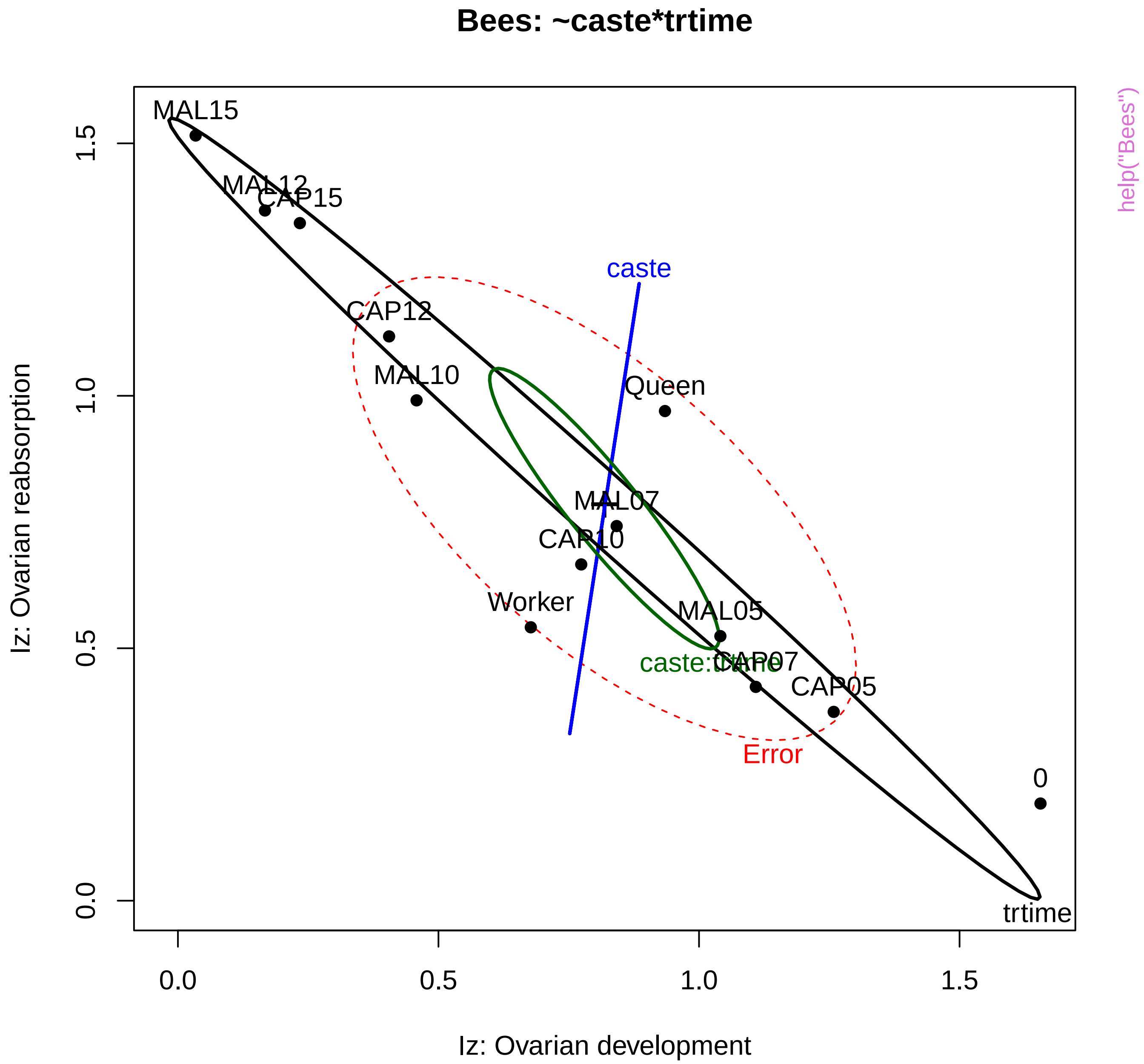


Reg
Age
help("Adopted")
H0
AMED:Age
BMIQ:Age
Error
Age.Q
Age.L
10
5
0
−5
−10
−15
−30
−20
−10
0
10

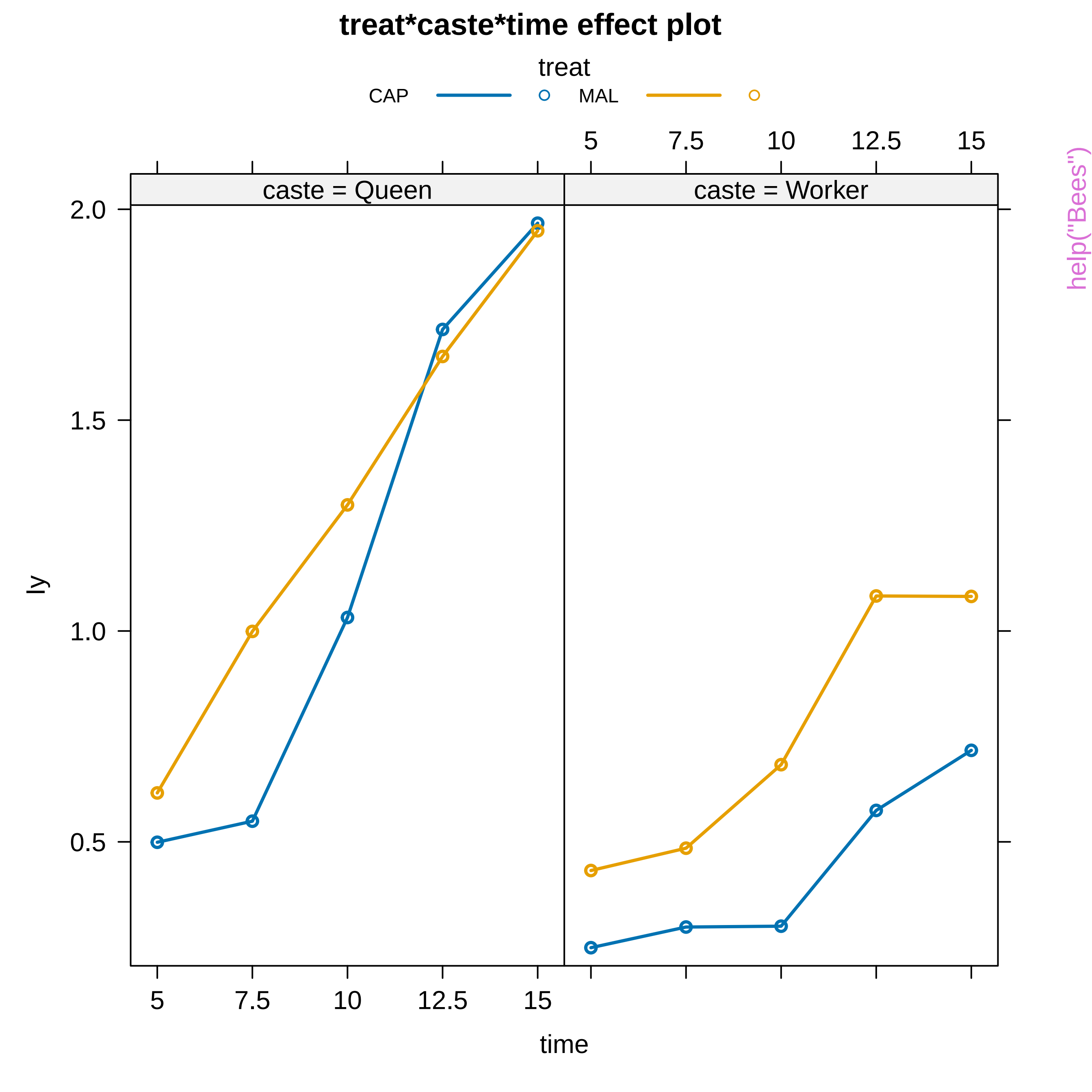

Bees: ~caste*treat*time
caste
15
12.5
Queen
MAL
10
CAP
Worker
7.5
5
treat:time
caste:treat
Error
caste:treat:time
caste:time
treat
time
help("Bees")
Iz: Ovarian reabsorption
Iz: Ovarian development
3
2
1
0
−1
−2
−2
−1
0
1
2
3

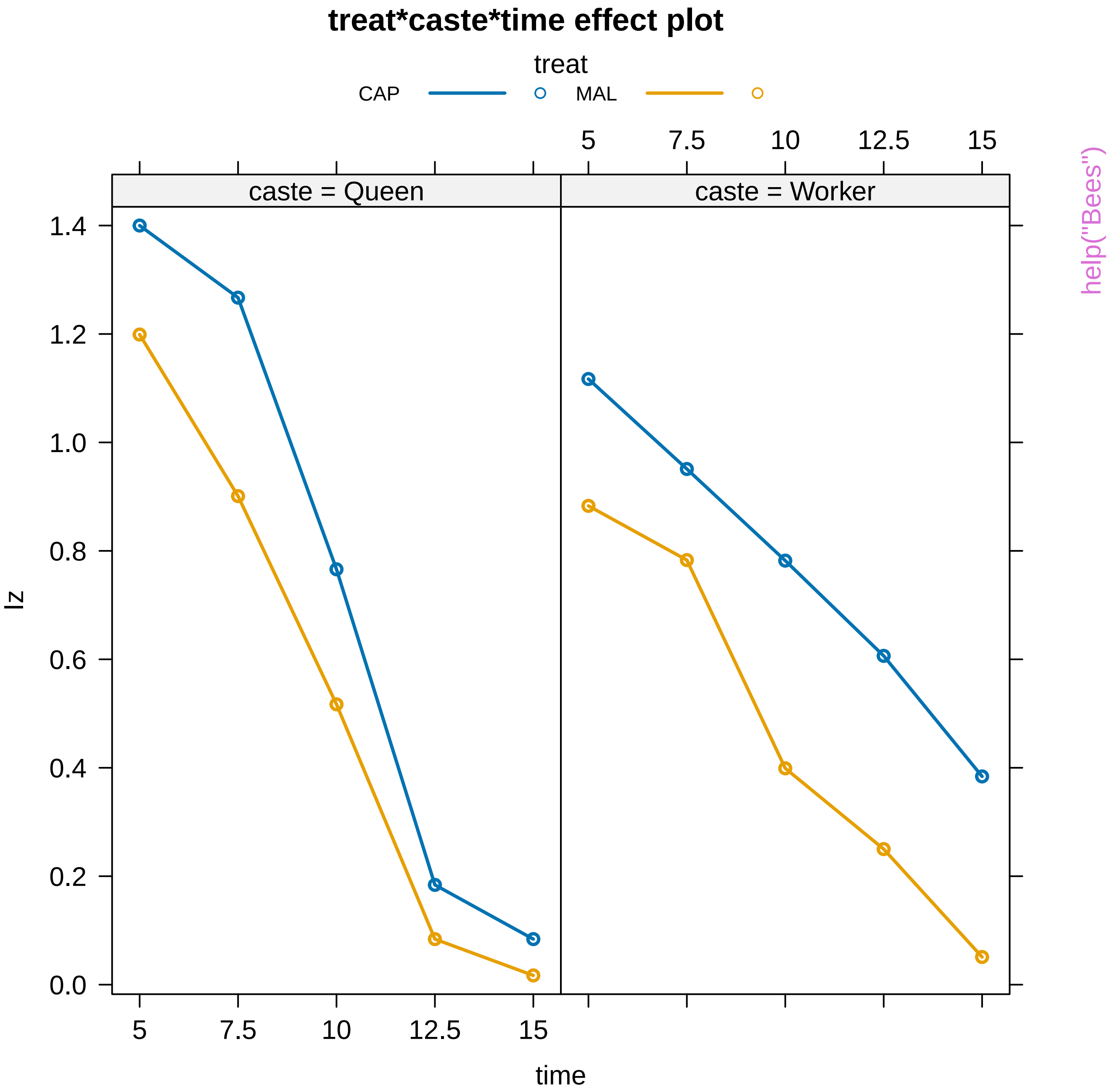

Bees: ~caste*treat*time
help("Bees")
15
12.5
caste
Queen
MAL
10
treat:time
caste:treat
CAP
caste:treat:time
treat
caste:time
Worker
7.5
5
Error
time
1.4
1.2
1.0
0.8
0.6
0.4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Iz: Ovarian reabsorption
Iz: Ovarian development

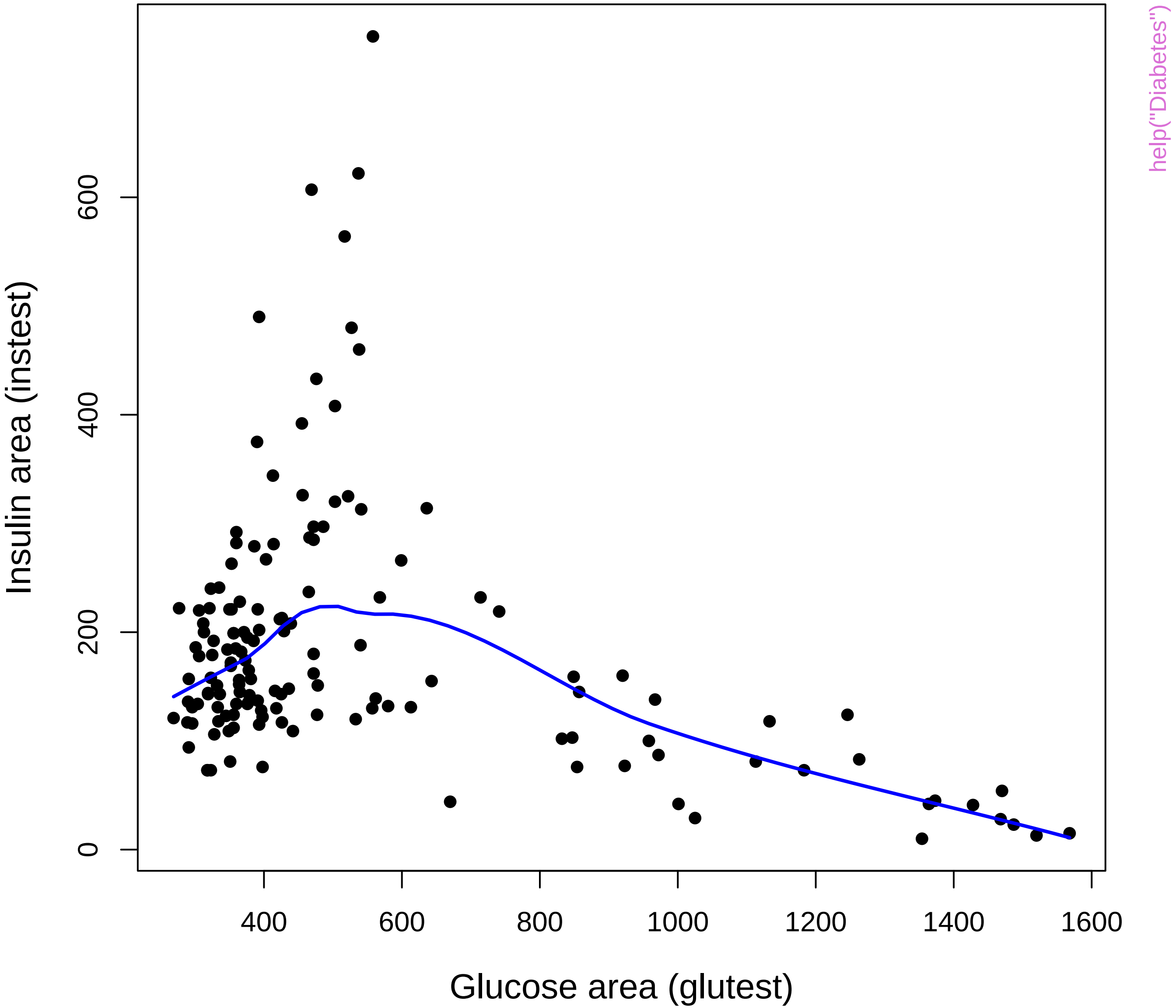

Bees: ~caste*trtime
caste
help("Bees")
MAL15
CAP12
MAL10
Queen
MAL07
MAL05
CAP05
0
Error
caste:trtime
trtime
Iz: Ovarian reabsorption
Iz: Ovarian development
−2
−1
0
1
2
3



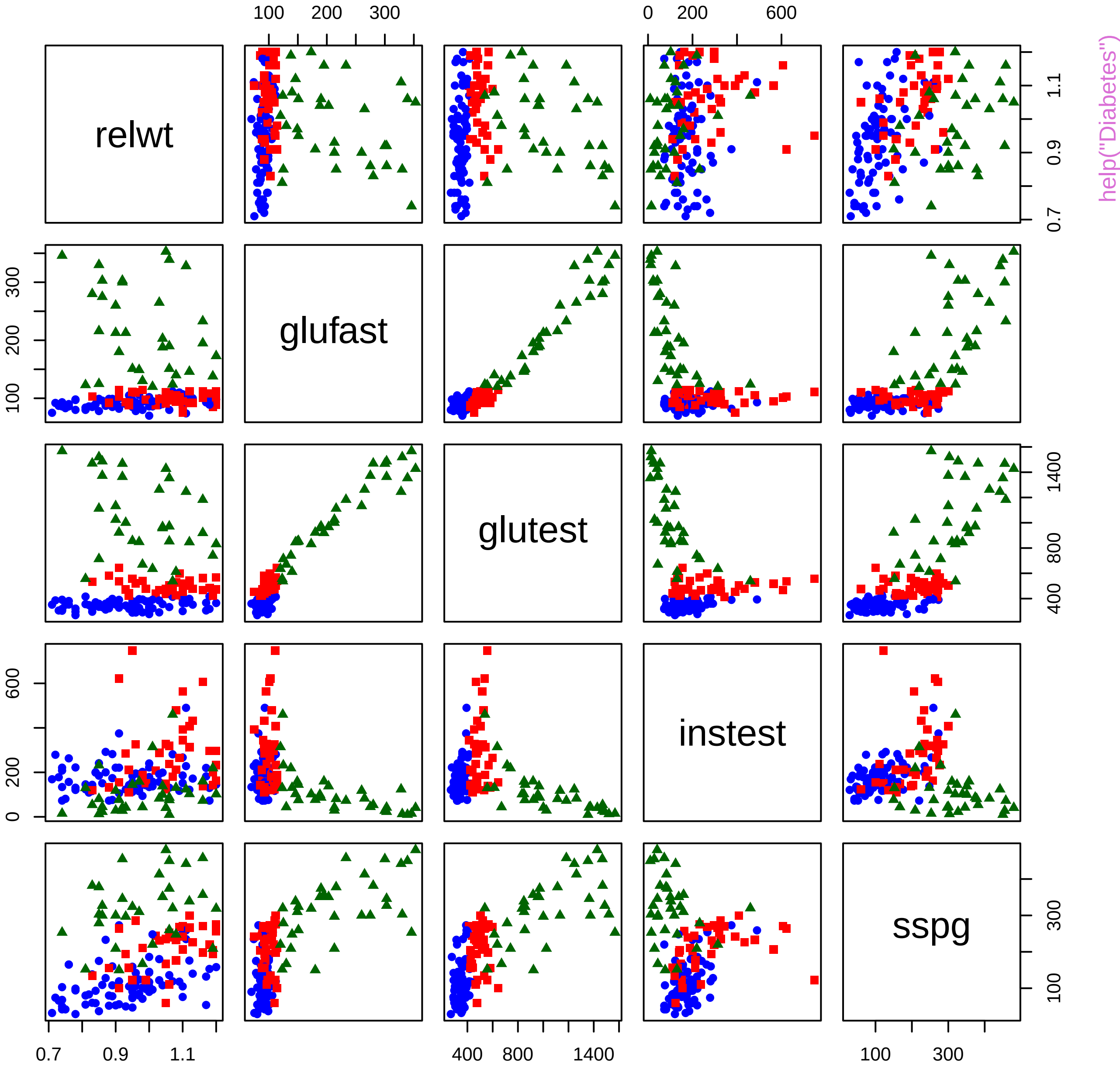
Bees: ~caste*trtime
help("Bees")
MAL15
MAL12
CAP15
caste
CAP12
MAL10
Queen
MAL07
CAP10
Worker
MAL05
caste:trtime
CAP07
CAP05
Error
0
trtime
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
Iz: Ovarian reabsorption
Iz: Ovarian development


treat*caste*time effect plot
treat
CAP
MAL
caste = Queen
caste = Worker
ly
2.0
1.5
1.0
0.5
5
7.5
10
12.5
15
time
help("Bees")

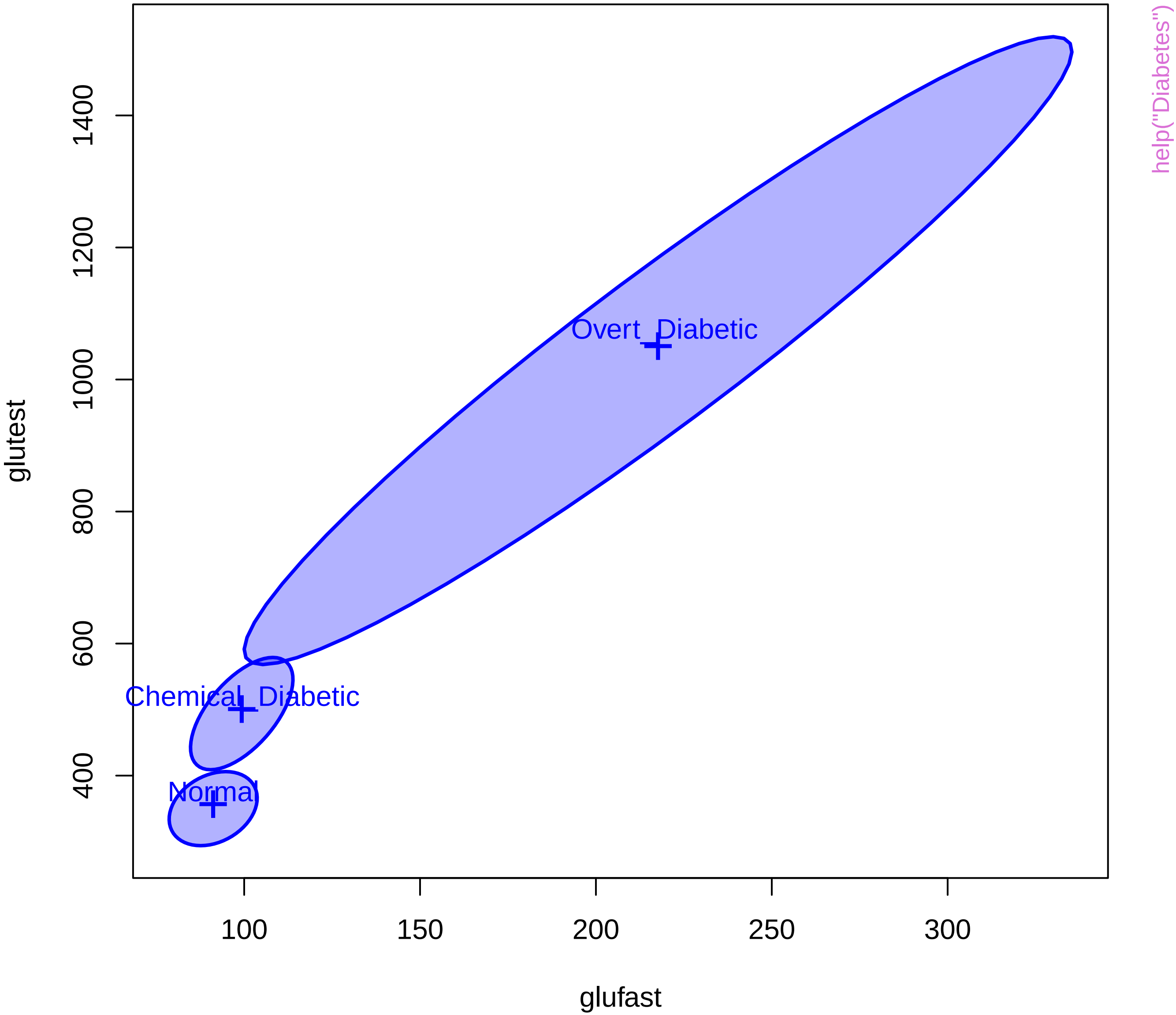

treat*caste*time effect plot
treat
CAP
MAL
caste = Queen
caste = Worker
lz
time
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
5
7.5
10
12.5
15
help("Bees")



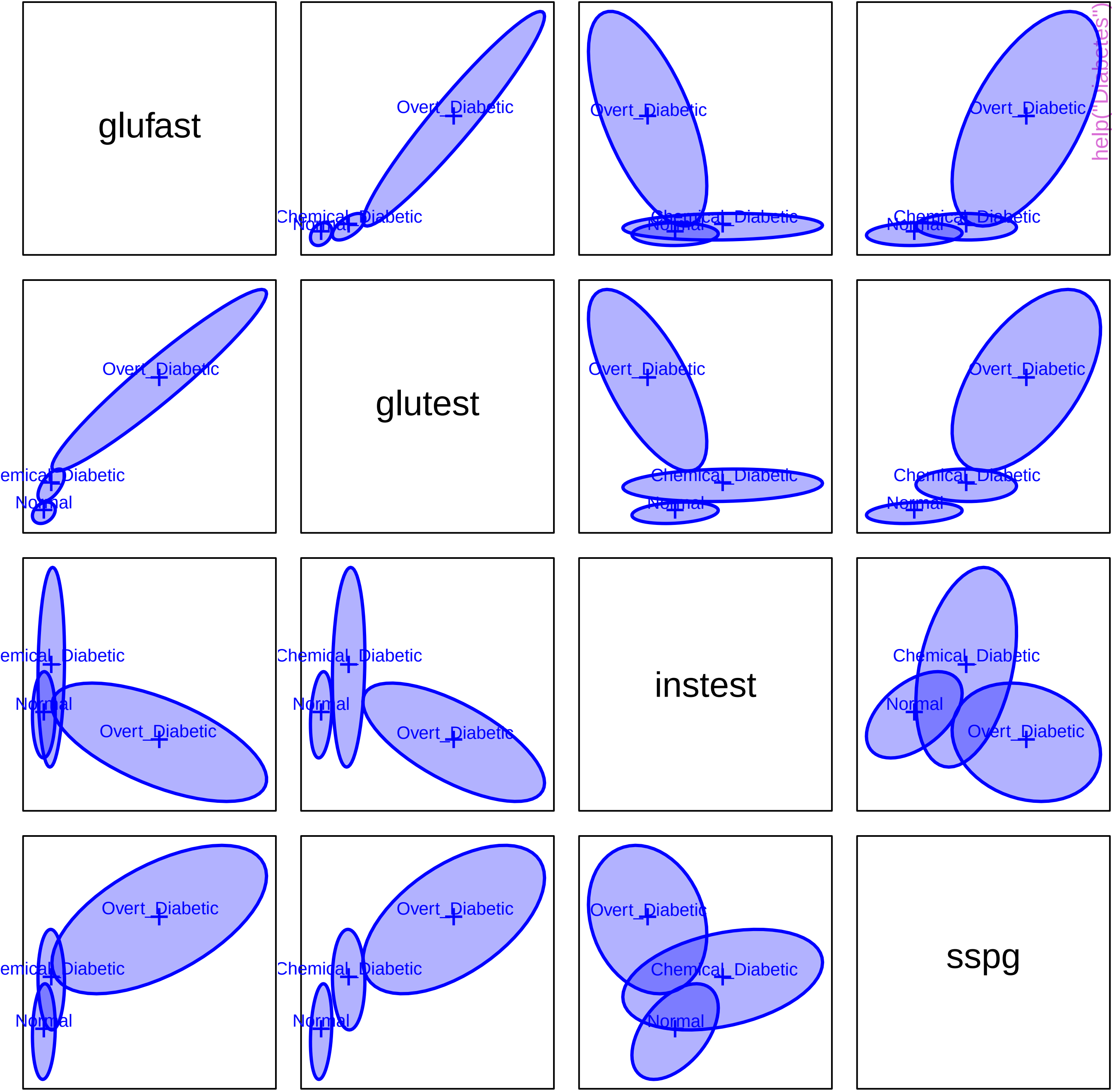
Insulin area (instest)
0
200
400
600
400
600
800
1000
1200
1400
1600
Glucose area (glutest)
help("Diabetes")

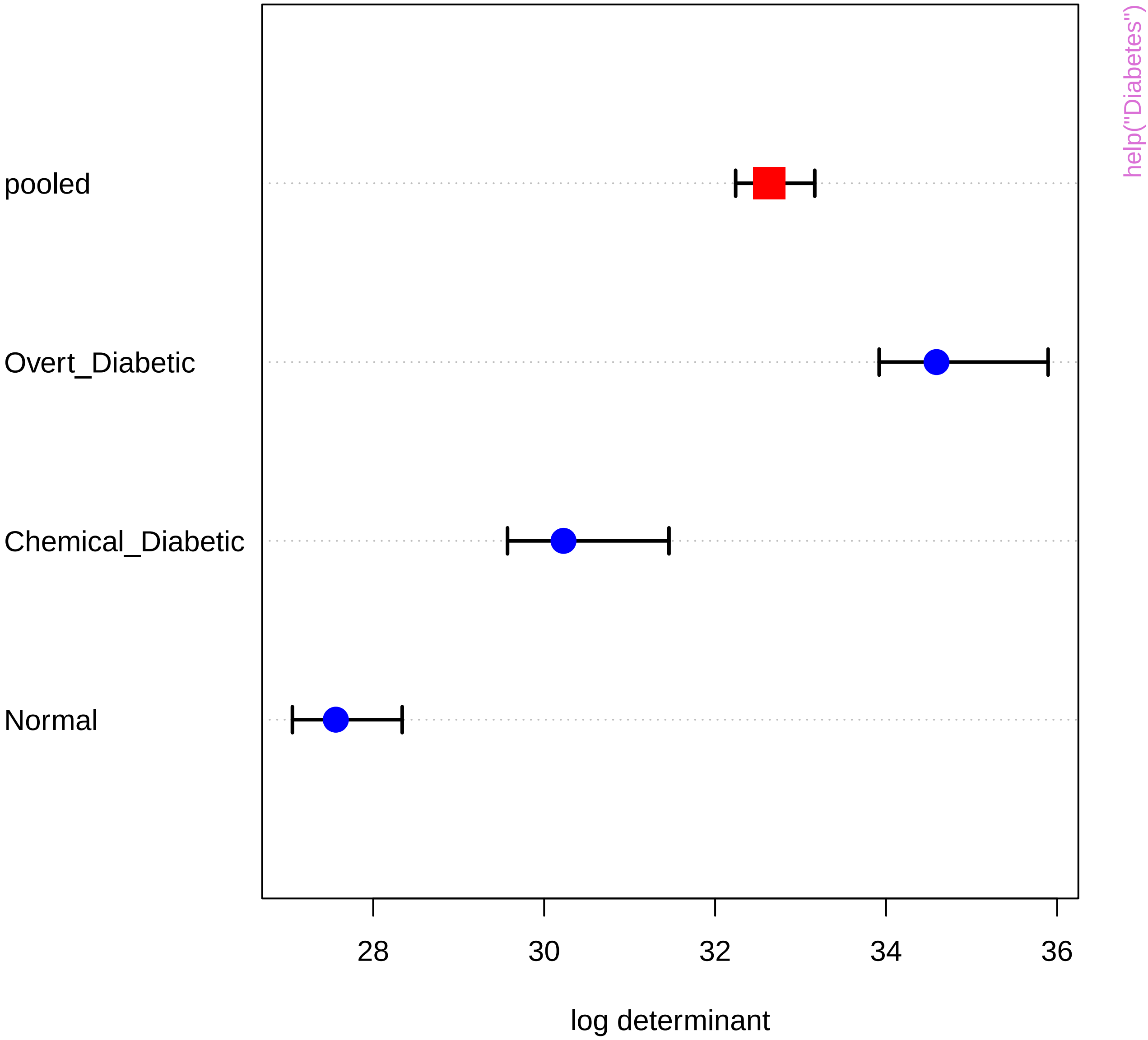


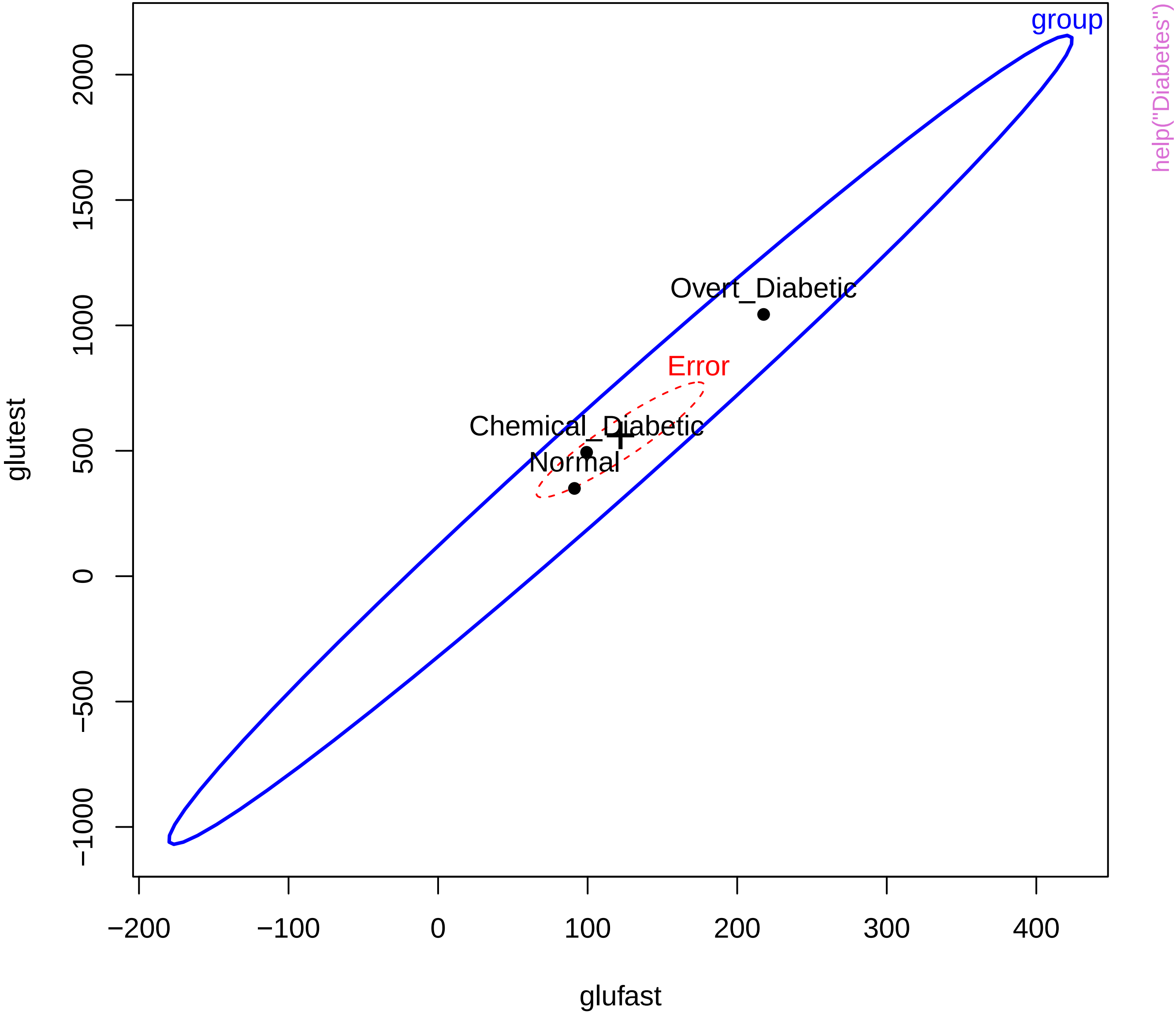
relwt
glufast
glutest
instest
sspg
help("Diabetes")
100
200
300
0
200
600
0.7
0.9
1.1
400
800
1400
100
300

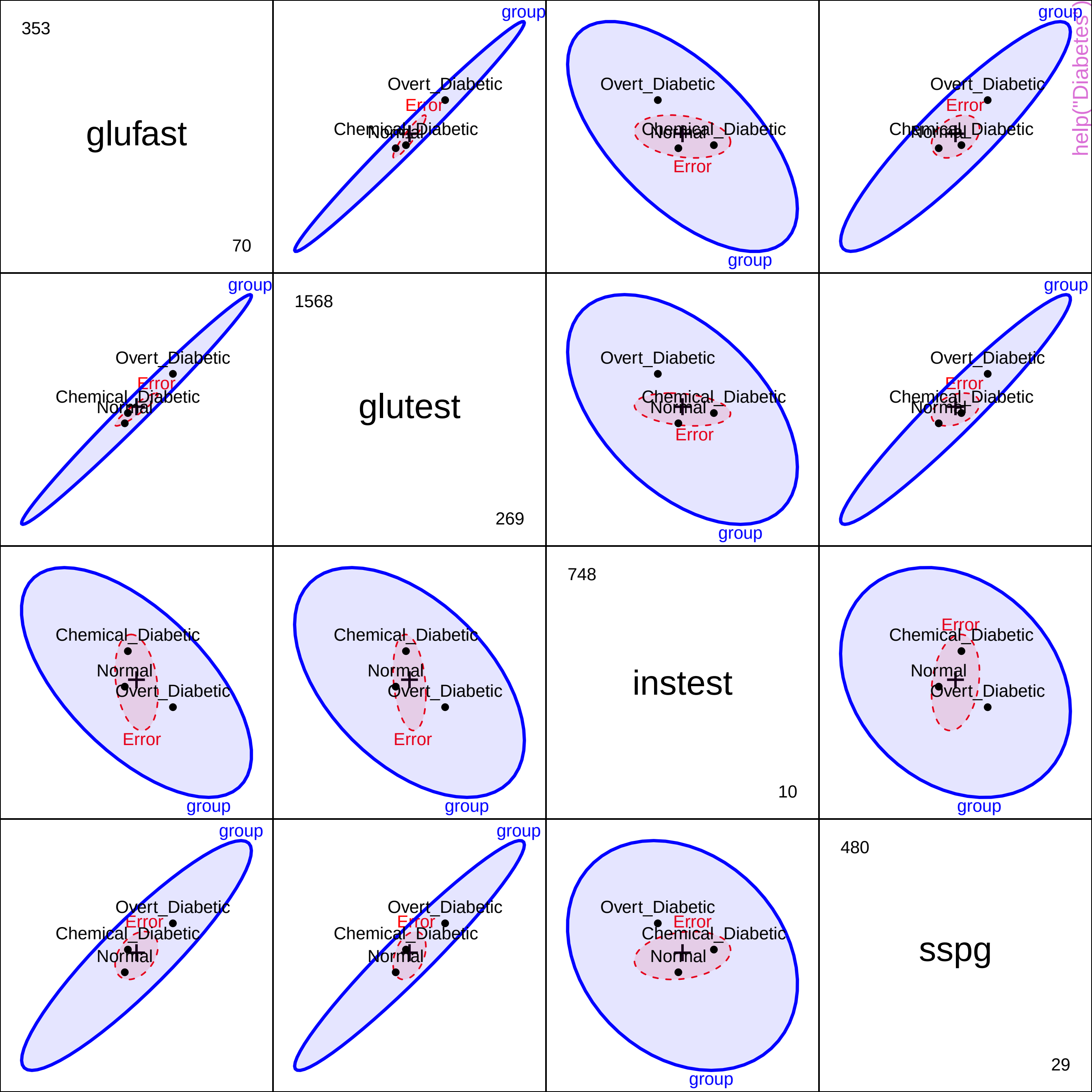


Overt_Diabetic
Chemical_Diabetic
Normal
glutest
400
600
800
1000
1200
1400
100
150
200
250
300
glufast
help("Diabetes")

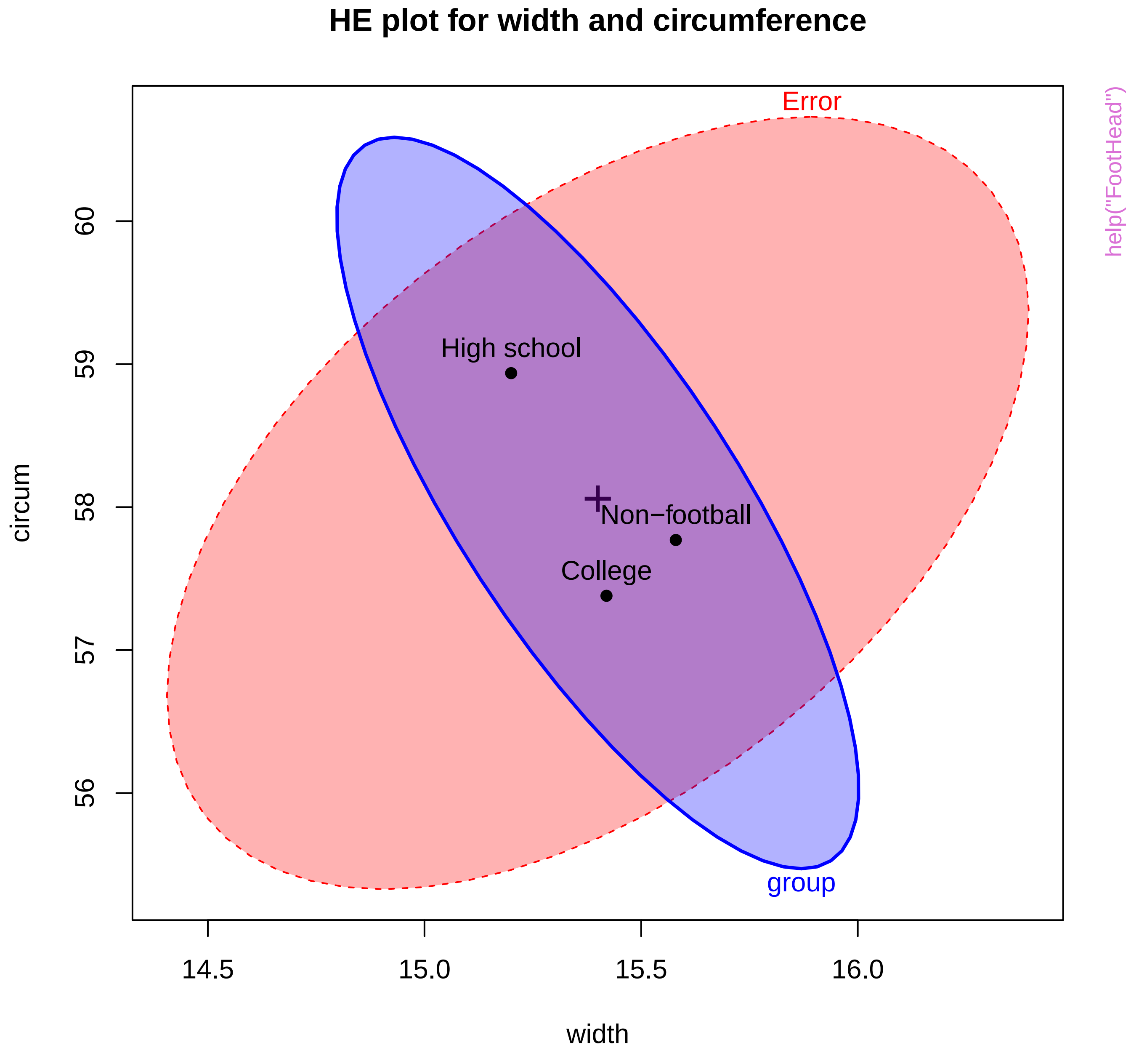

glufast
glutest
instest
sspg
Overt_Diabetic
Chemical_Diabetic
Normal
help("Diabetes")


pooled
Overt_Diabetic
Chemical_Diabetic
Normal
28
30
32
34
36
log determinant
help("Diabetes")

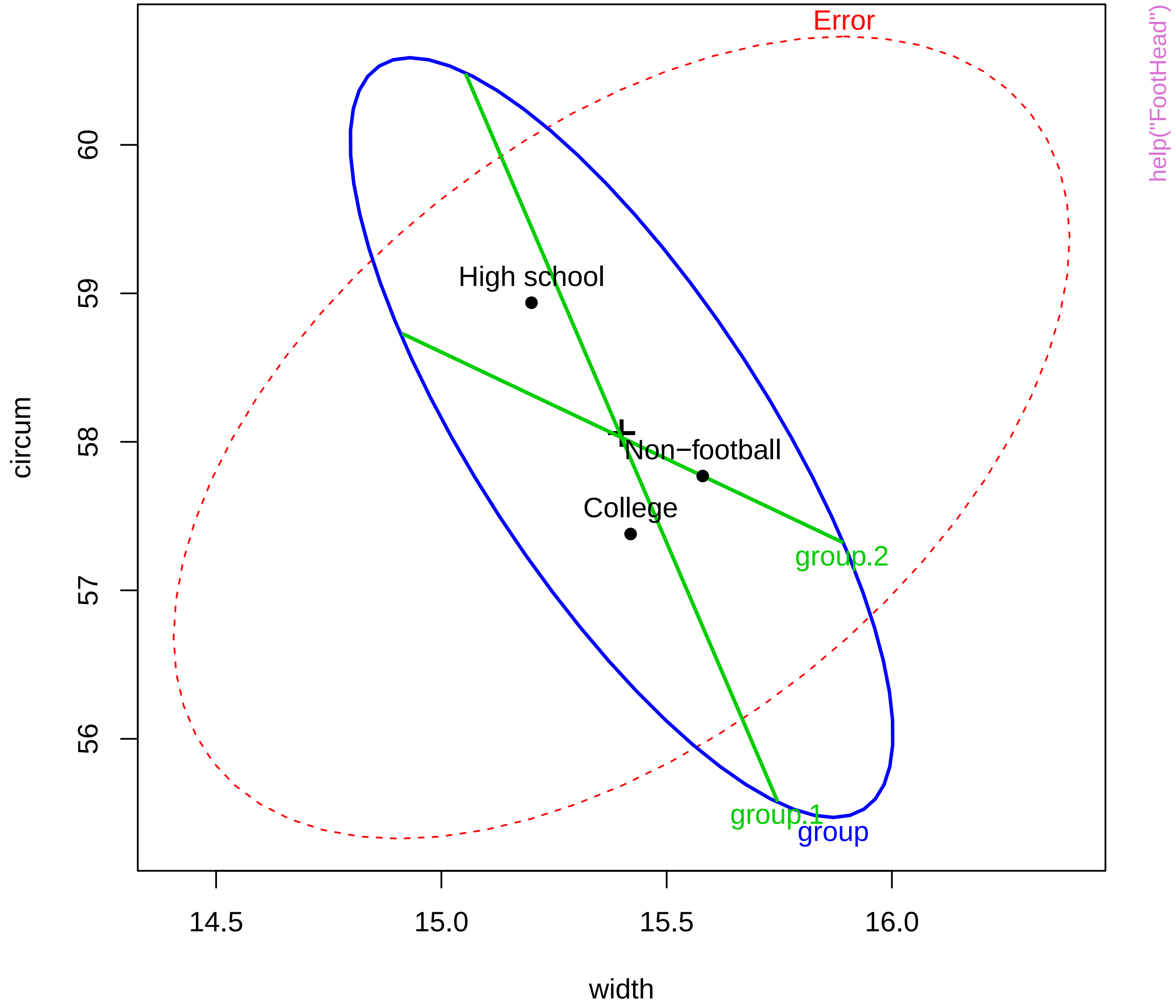

group
Overt_Diabetic
Error
Chemical_Diabetic
Normal
2000
1500
1000
500
0
−500
−1000
−200
−100
0
100
200
300
400
glutest
glufast
help("Diabetes")

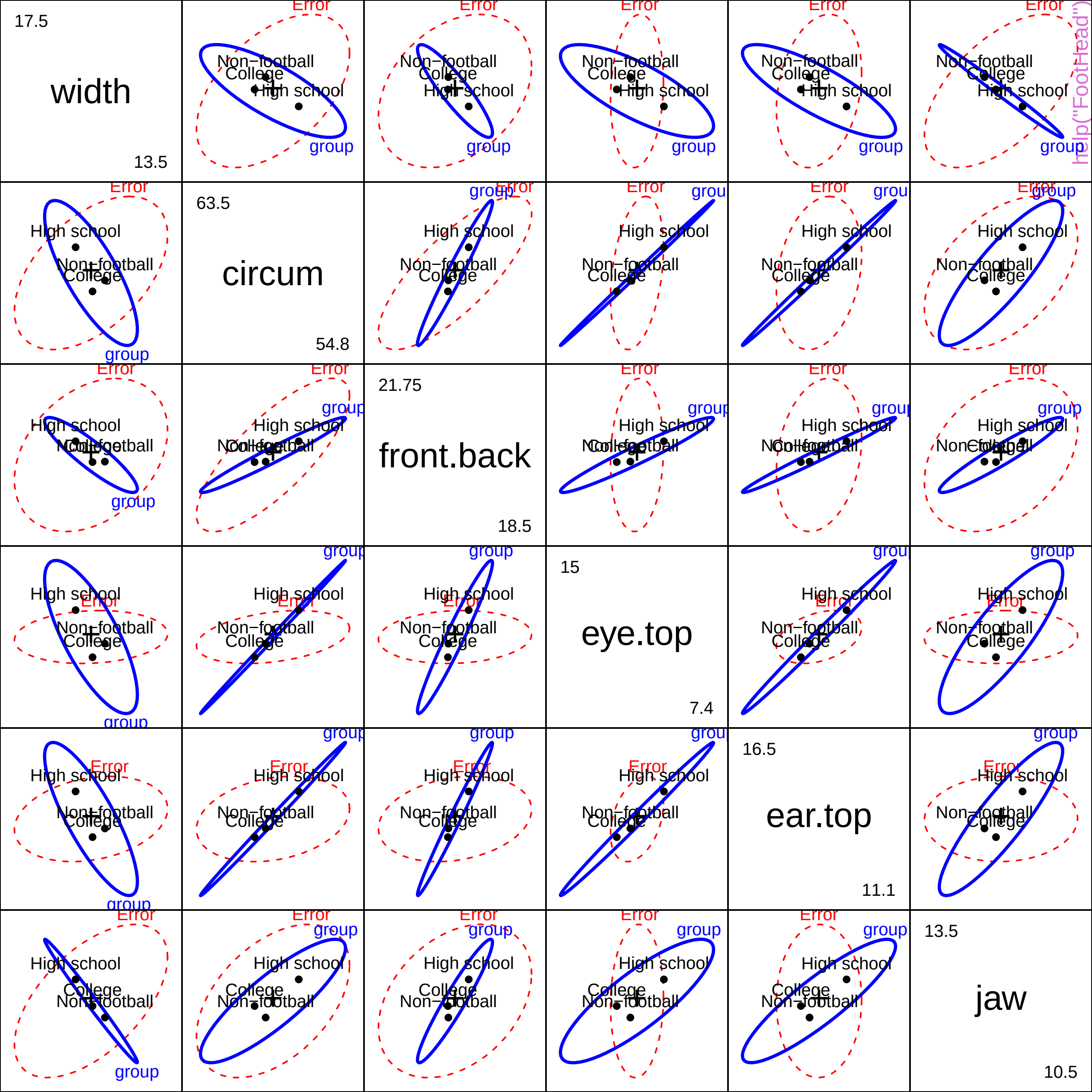


353
glufast
70
1568
glutest
269
748
instest
10
480
sspg
29
group
Overt_Diabetic
Error
Chemical_Diabetic
Normal
help("Diabetes")

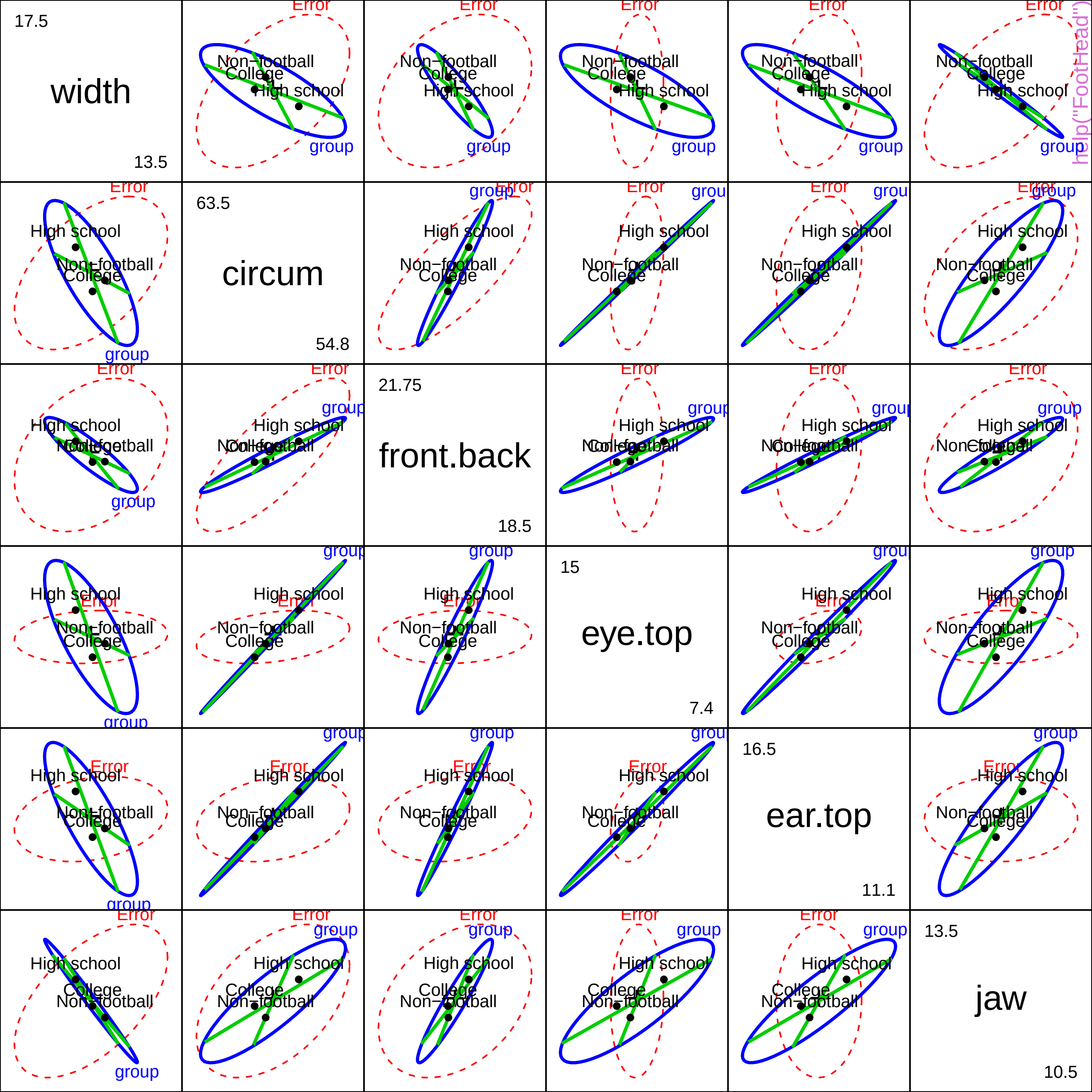

HE plot for width and circumference
Error
group
High school
Non−football
College
60
59
58
57
56
circum
14.5
15.0
15.5
16.0
width
help("FootHead")

HE plot with orthogonal Helmert contrasts

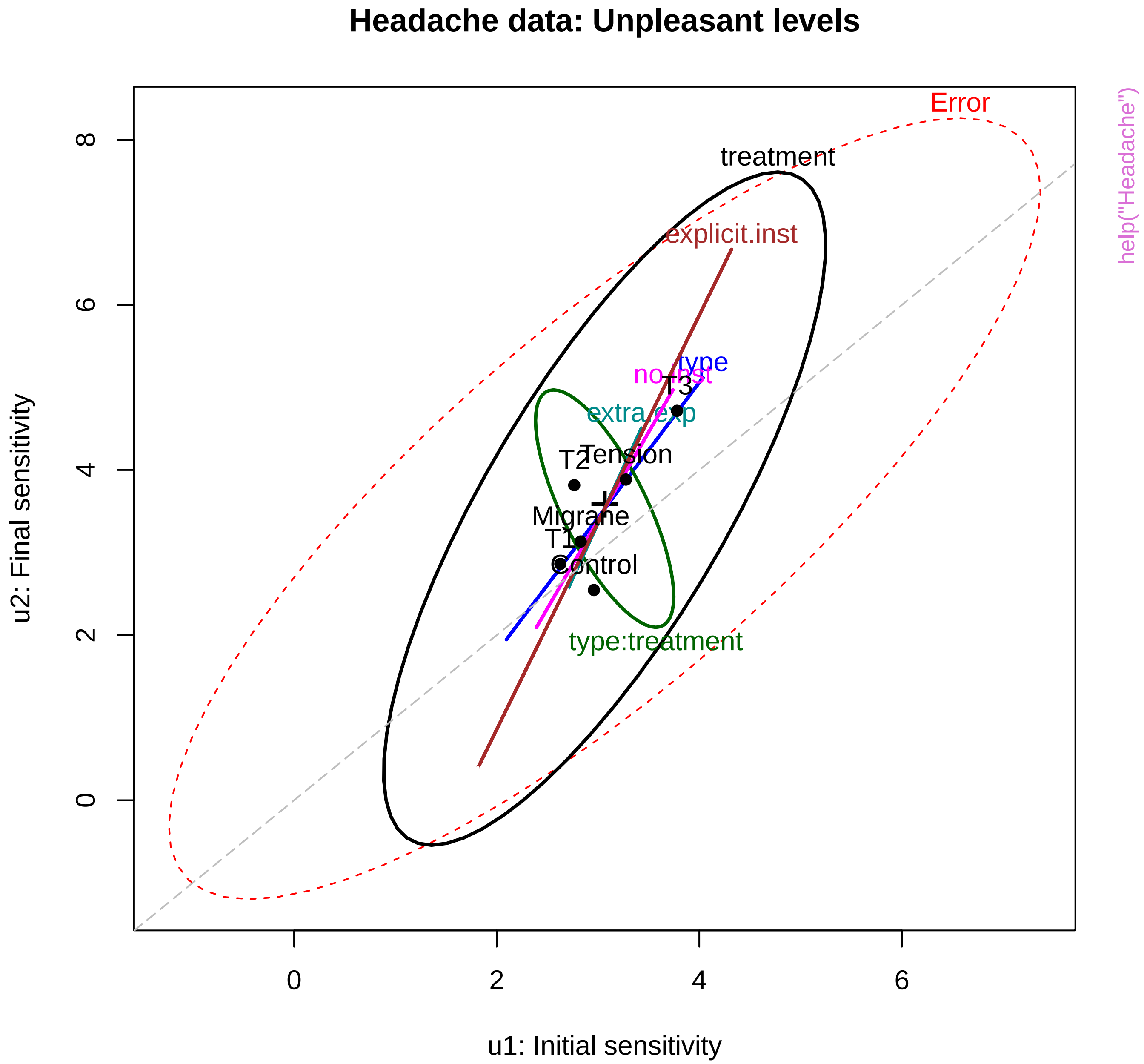


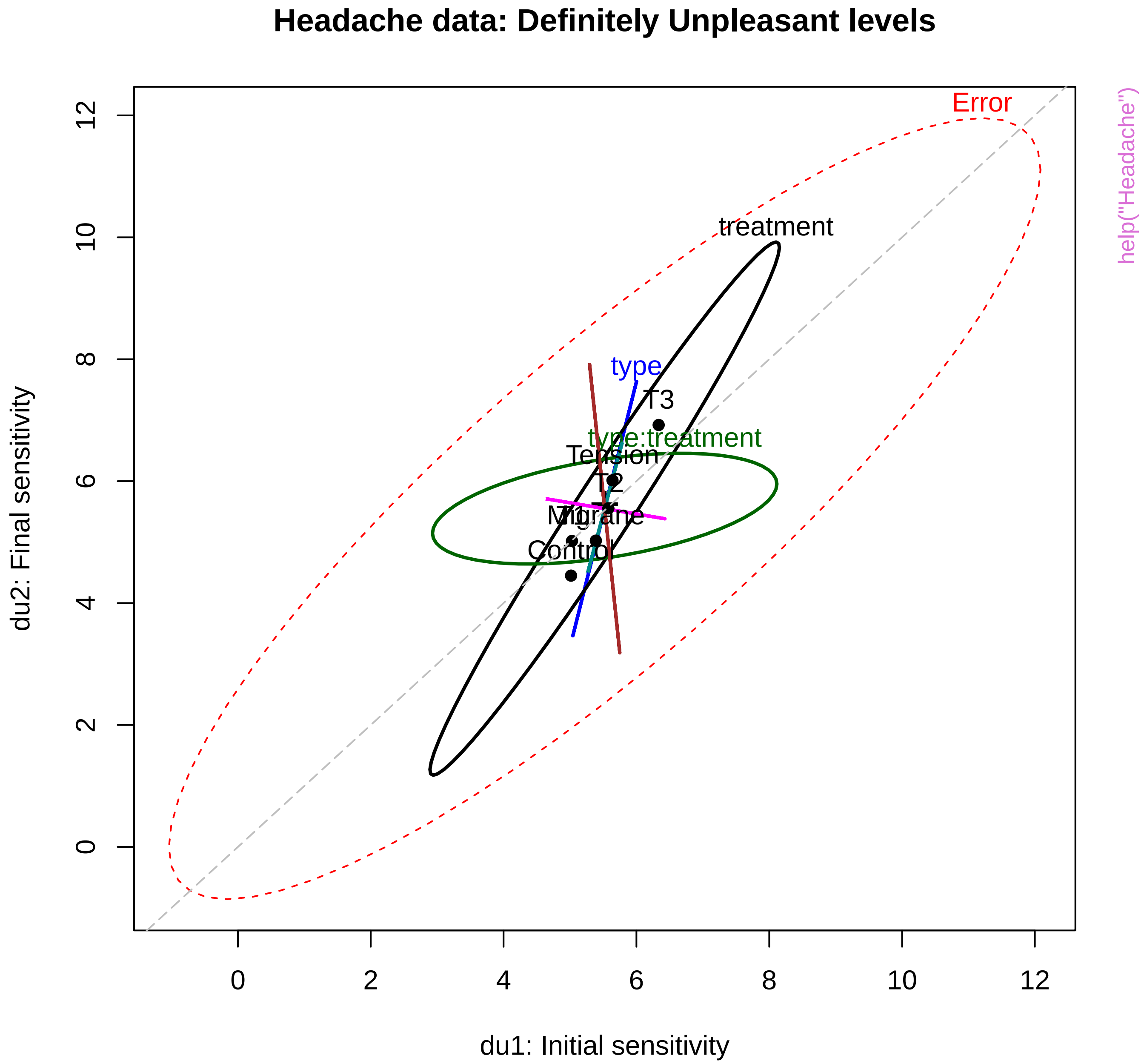


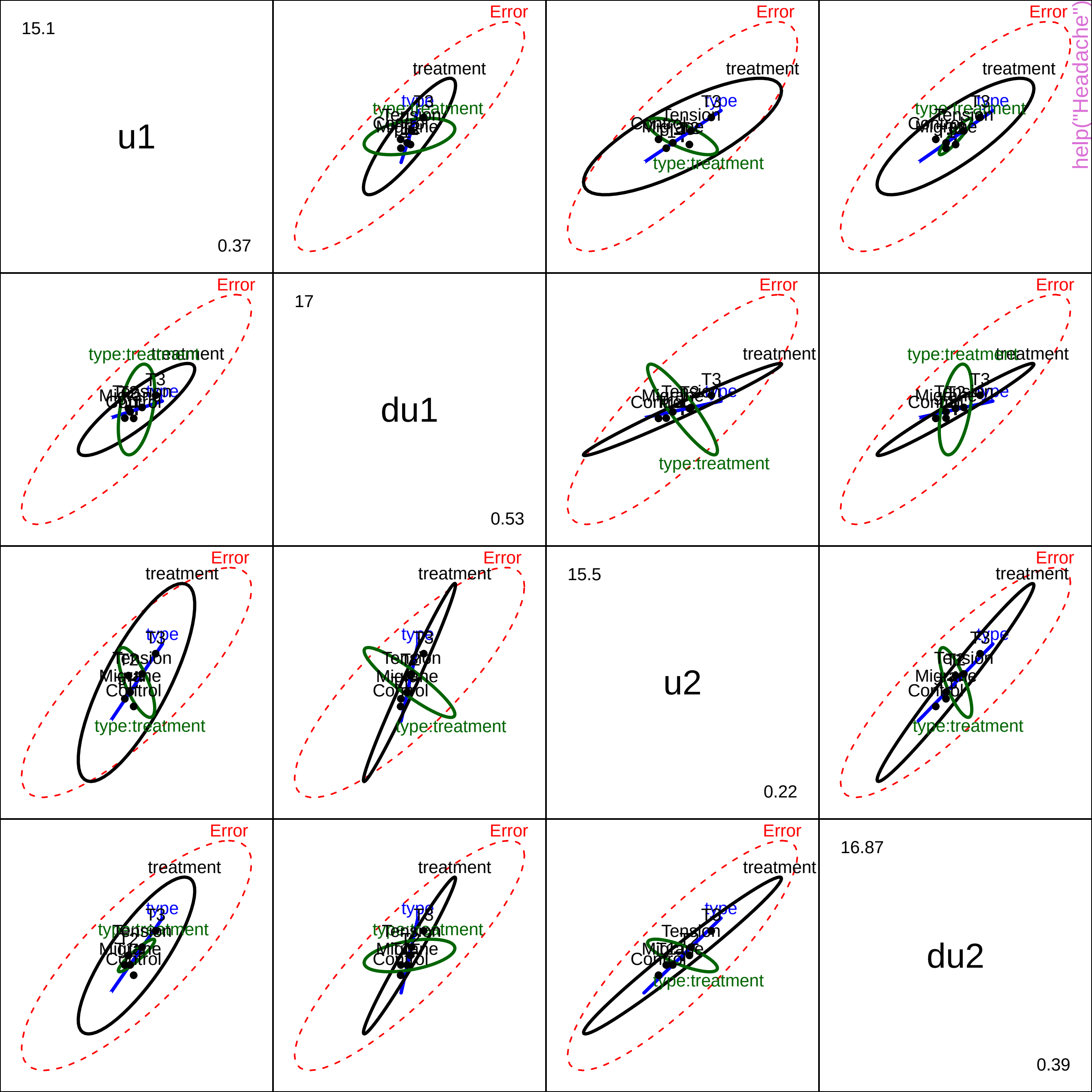
width
17.5
13.5
circum
63.5
54.8
front.back
21.75
18.5
eye.top
15
7.4
ear.top
16.5
11.1
jaw
13.5
10.5
Error
group
High school
Non−football
College
help("FootHead")



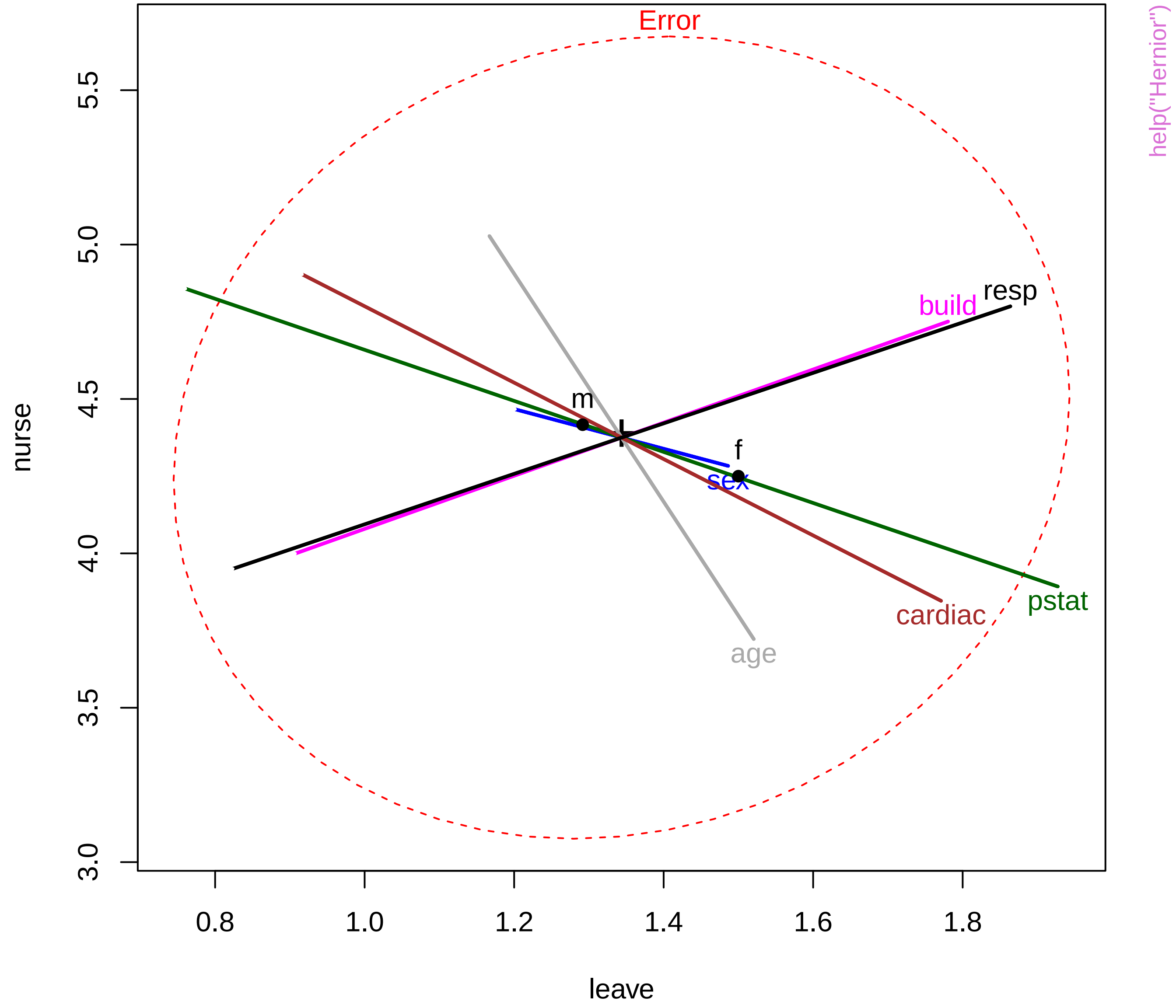
17.5
width
13.5
63.5
circum
54.8
21.75
front.back
18.5
15
eye.top
7.4
16.5
ear.top
11.1
13.5
jaw
10.5
Error
group
Non−football
College
High school
help("FootHead")

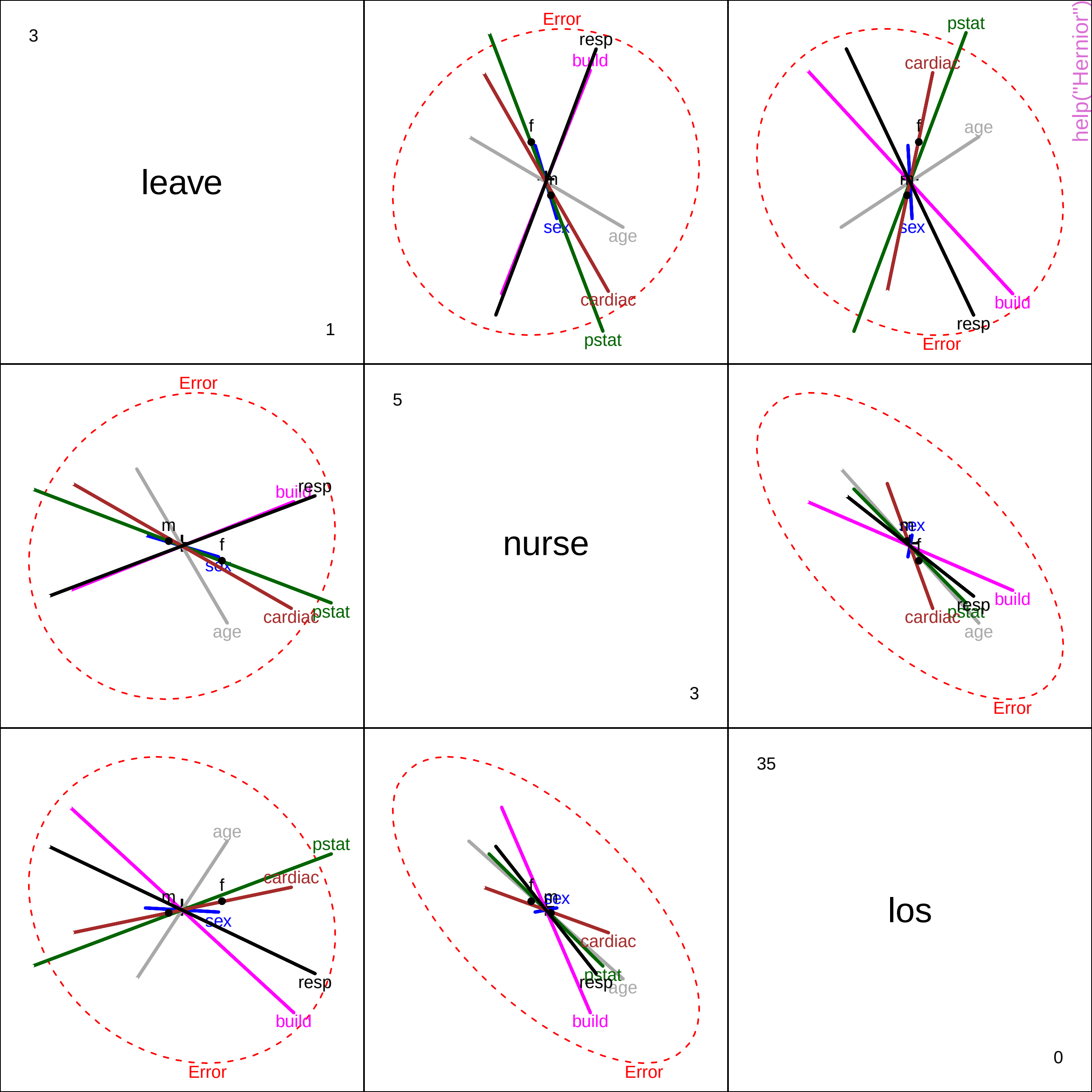

Headache data: Unpleasant levels
Error
treatment
explicit.inst
type
no.inst
T3
extra.exp
T2
Tension
Migrane
T1
Control
type:treatment
u1: Initial sensitivity
u2: Final sensitivity
0
2
4
6
8
help("Headache")


Headache data: Definitely Unpleasant levels
Error
treatment
type
T3
type:treatment
Tension
T2
Migrane
T1
Control
help("Headache")
du2: Final sensitivity
du1: Initial sensitivity
0
2
4
6
8
10
12

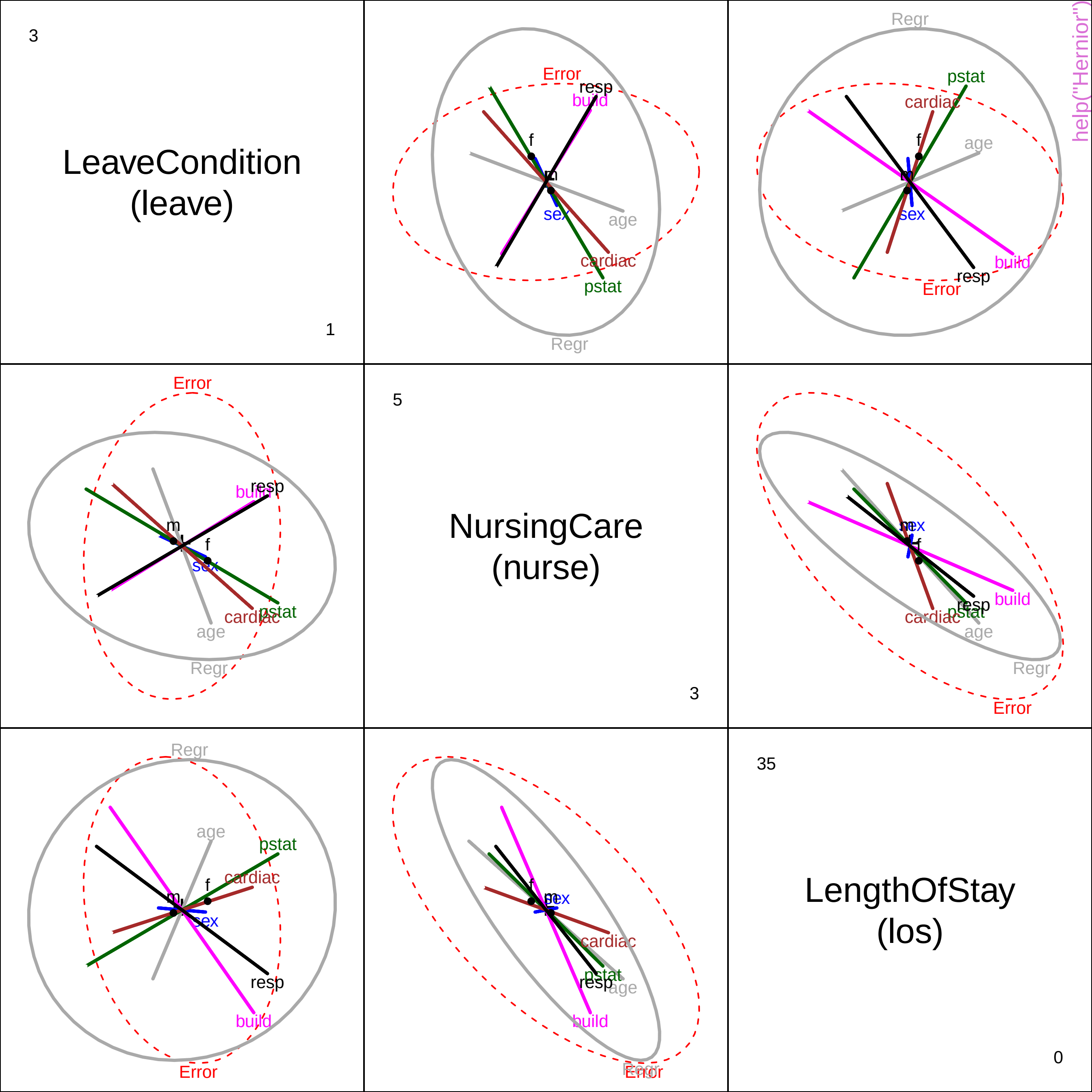


u1
15.1
0.37
du1
17
0.53
u2
15.5
0.22
du2
16.87
0.39
Error
treatment
type
type:treatment
Tension
Migraine
Control
T3
help("Headache")

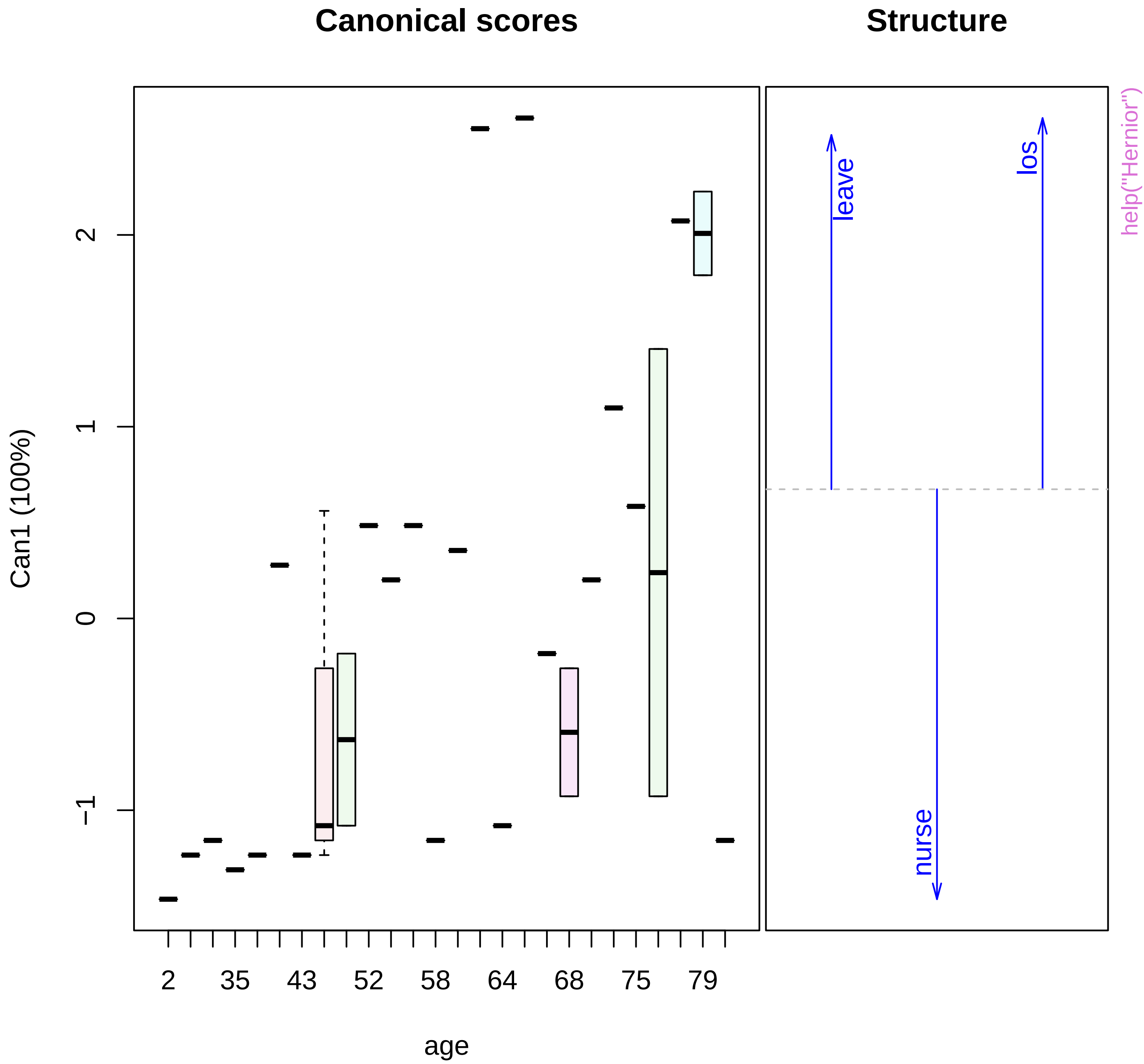

Error
resp
build
m
f
sex
pstat
cardiac
age
5.5
5.0
4.5
4.0
3.5
3.0
nurse
0.8
1.0
1.2
1.4
1.6
1.8
leave
help("Hernior")



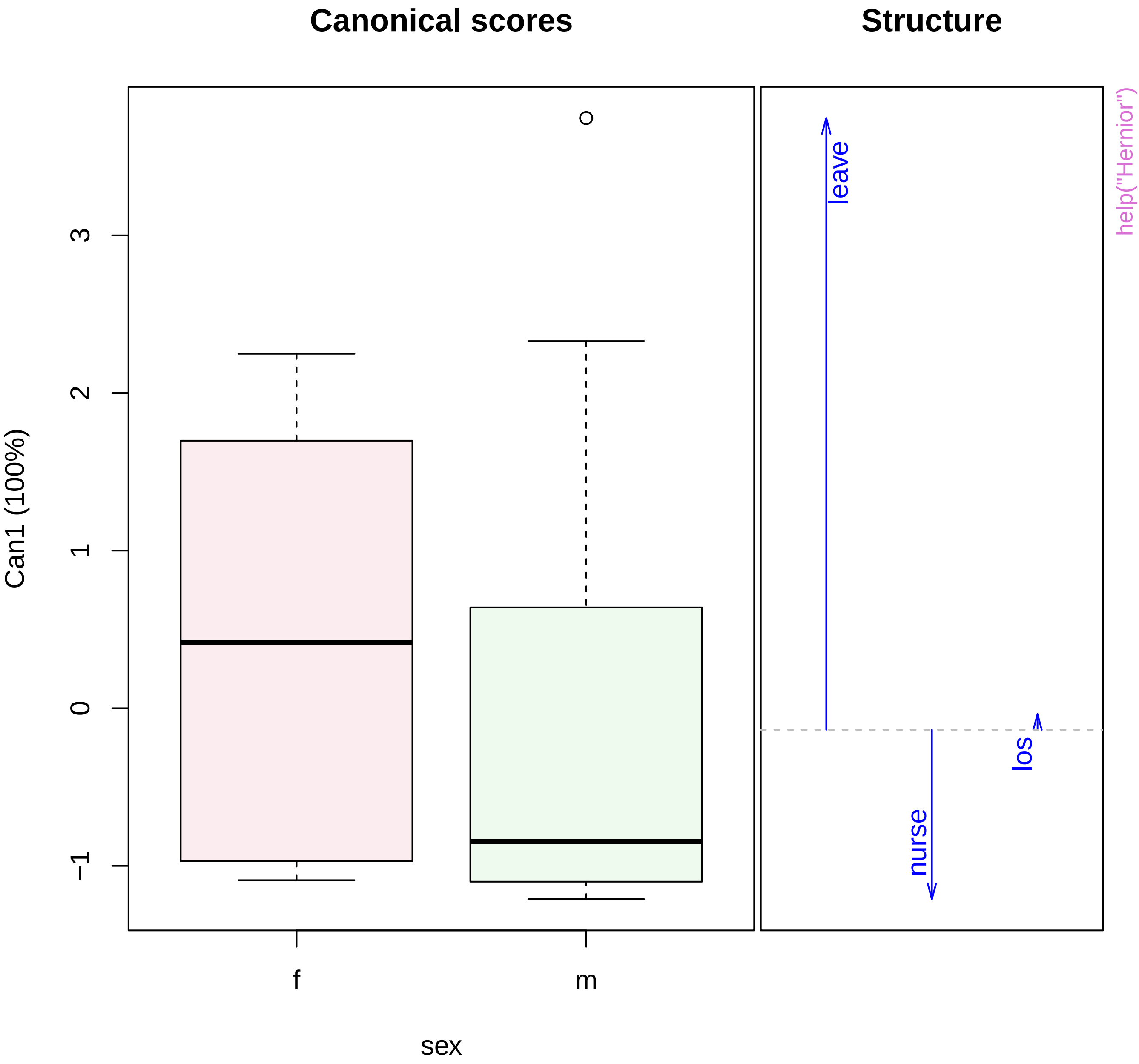
3
leave
1
Error
resp
build
f
m
sex
age
cardiac
pstat
help("Hernior")
5
nurse
3
35
los
0



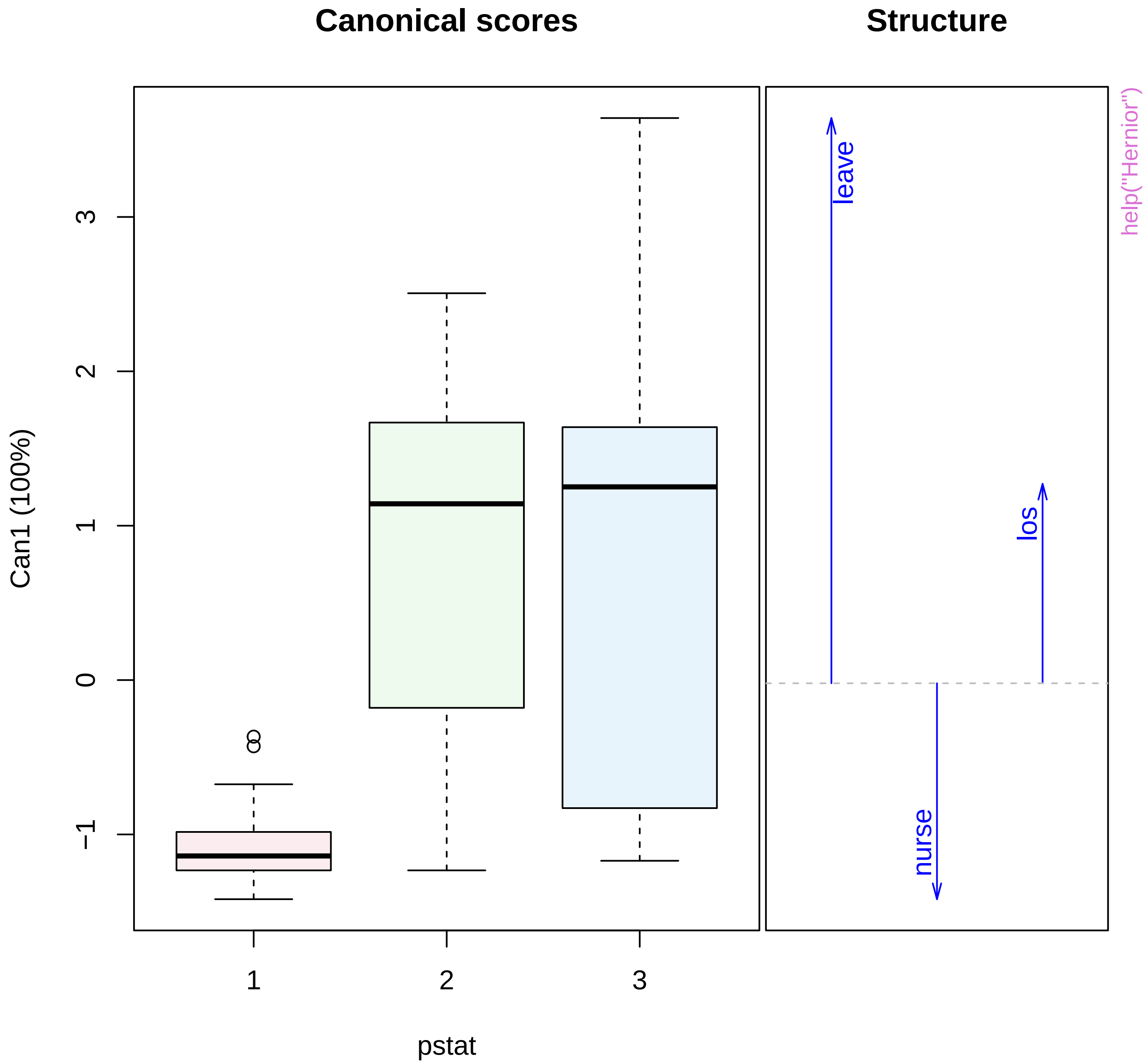
3
LeaveCondition
(leave)
1
Error
resp
build
f
m
sex
age
cardiac
pstat
Regr
help("Hernior")
5
NursingCare
(nurse)
3
35
LengthOfStay
(los)
0

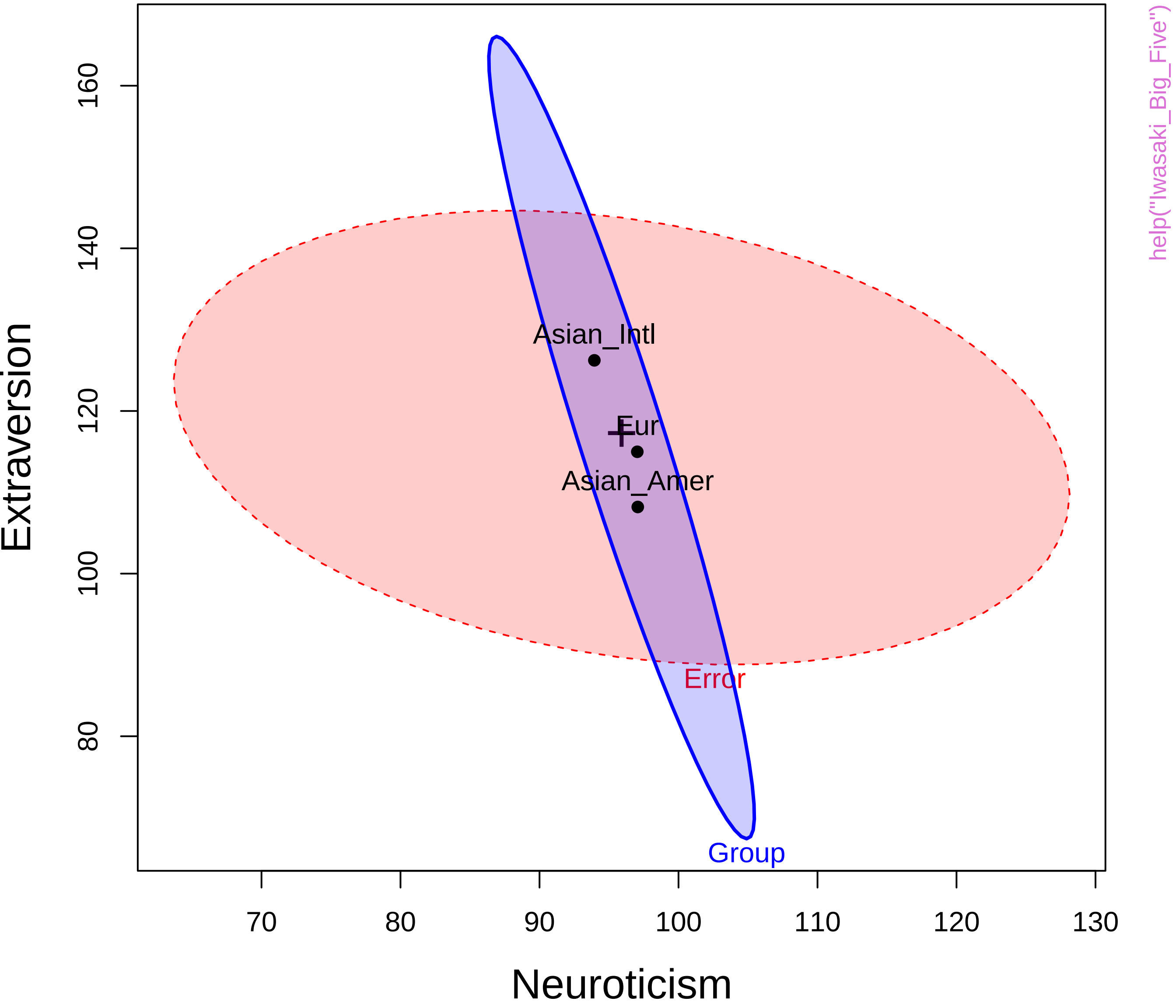


Canonical scores
Structure
Can1 (100%)
2
1
0
−1
2
35
43
52
58
64
68
75
79
age
leave
los
nurse
help("Hernior")

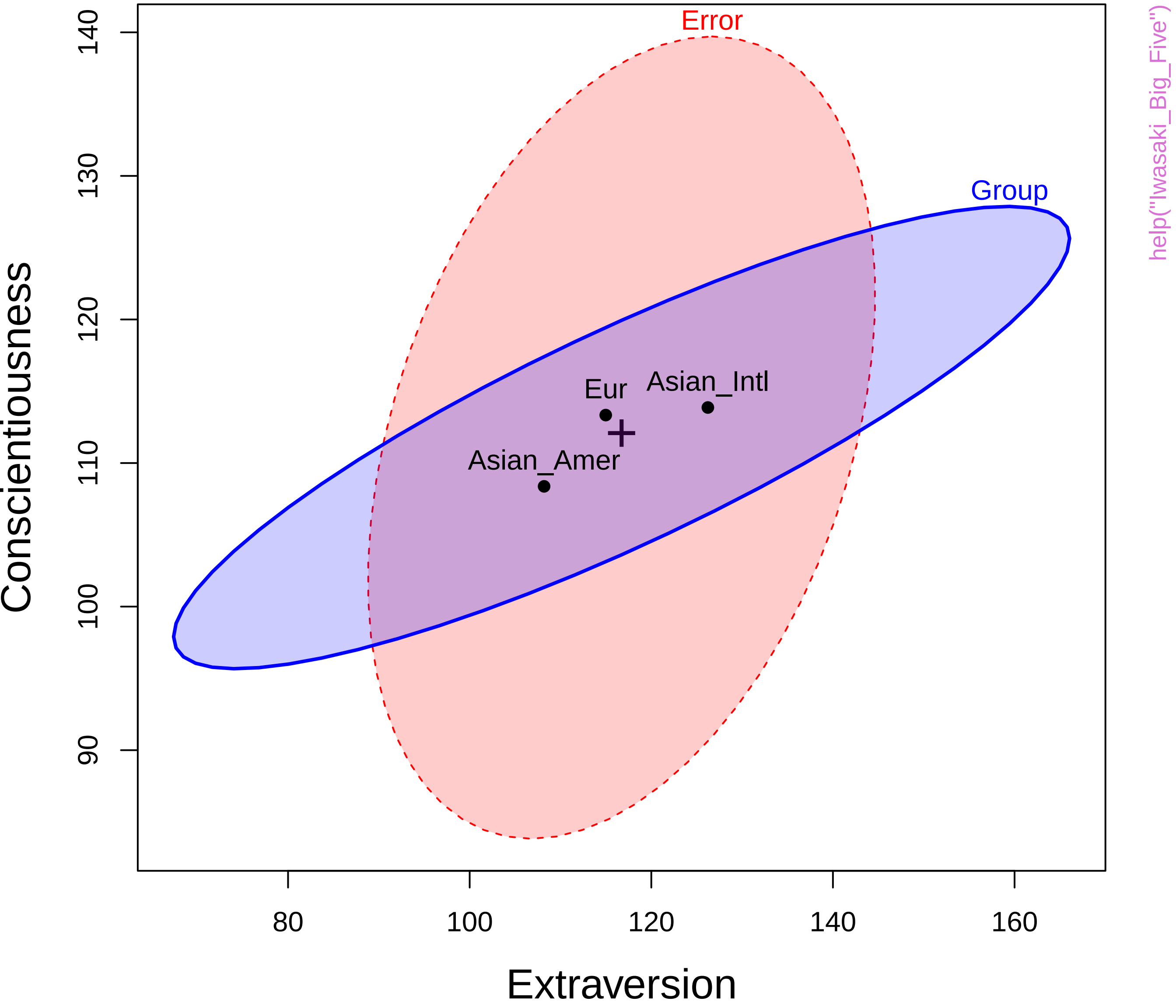


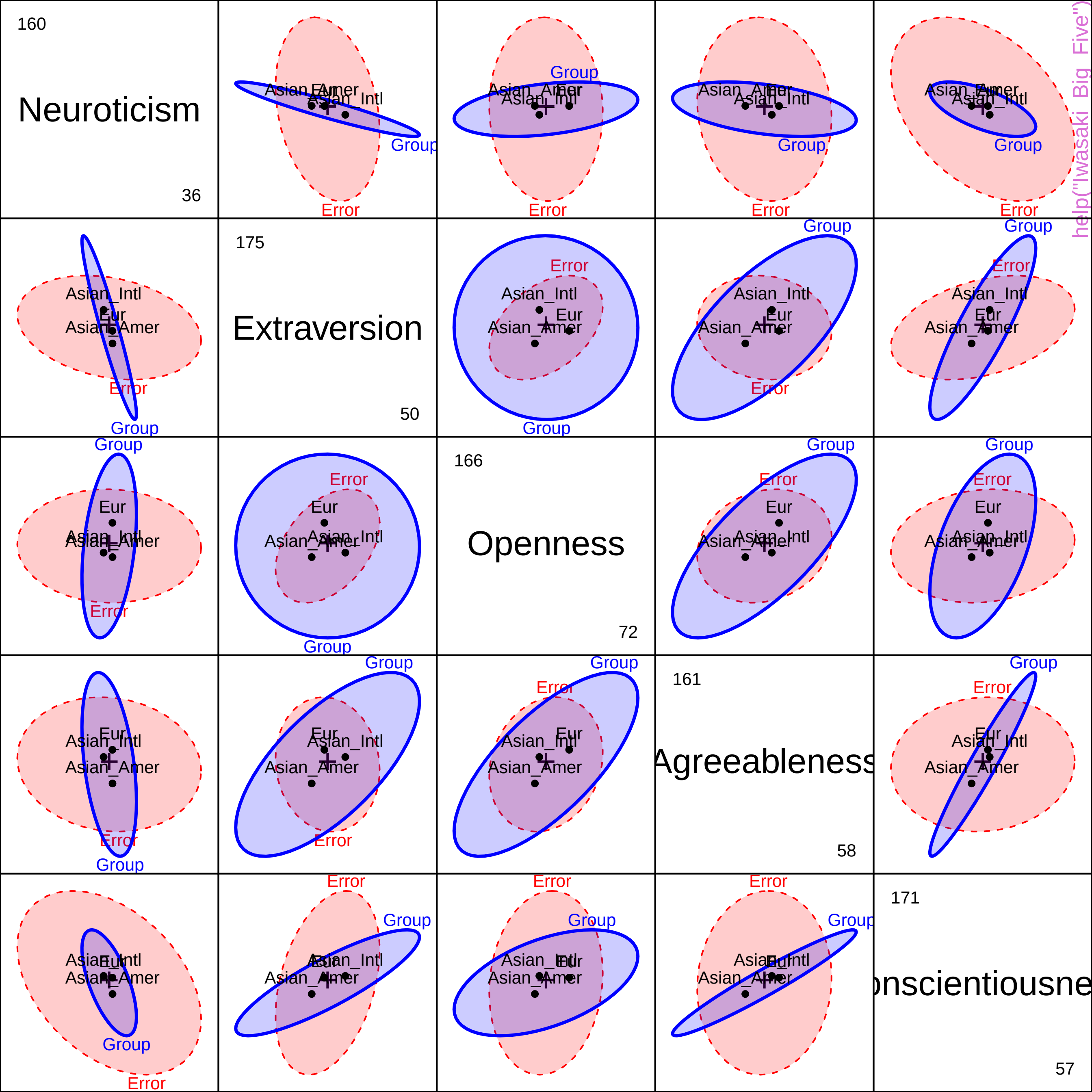
Canonical scores
Structure
Can1 (100%)
3
2
1
0
−1
f
m
sex
leave
nurse
los
help("Hernior")

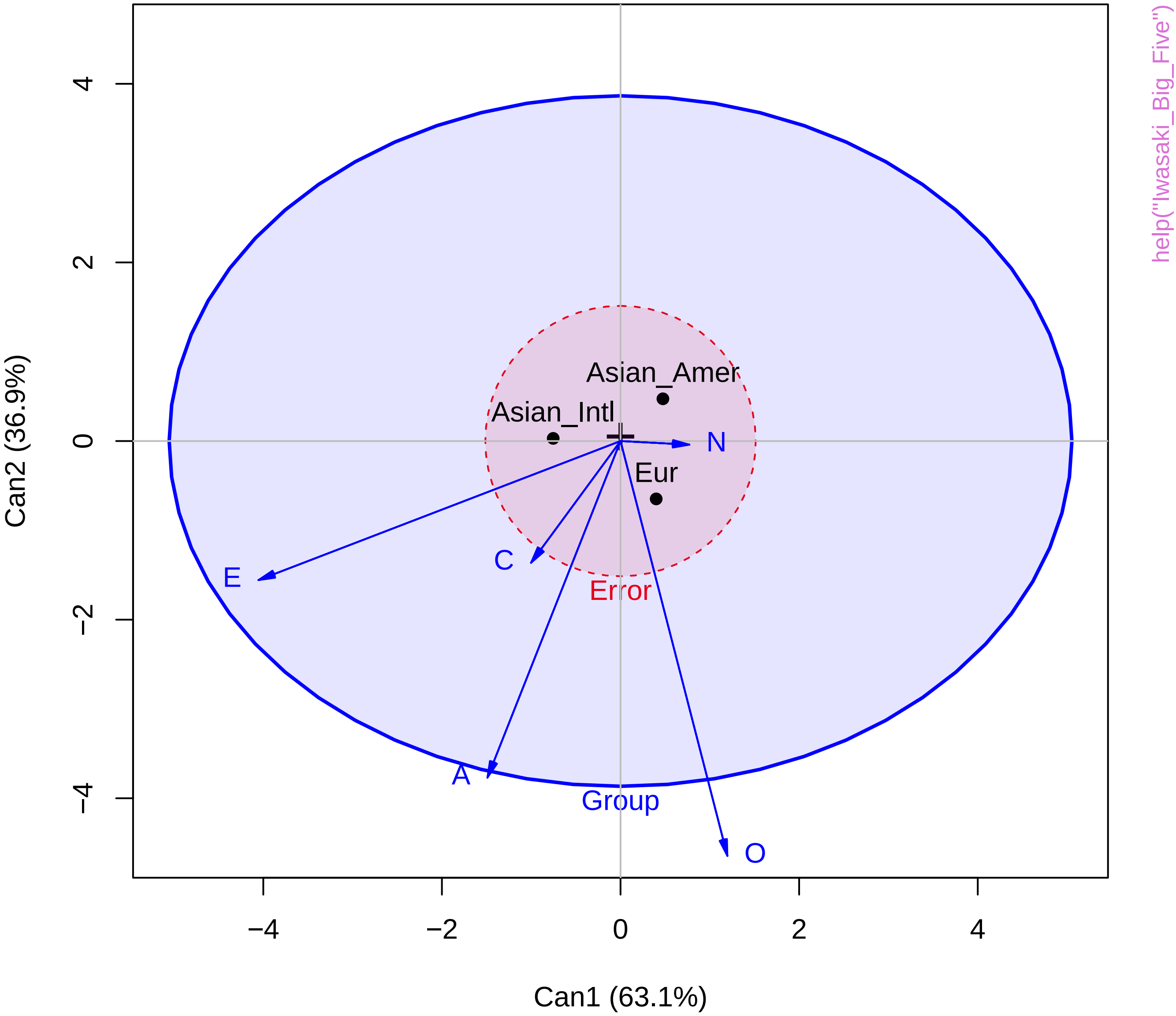

Canonical scores
Structure
Can1 (100%)
-1
0
1
2
3
1
2
3
pstat
leave
los
nurse
help("Hernior")

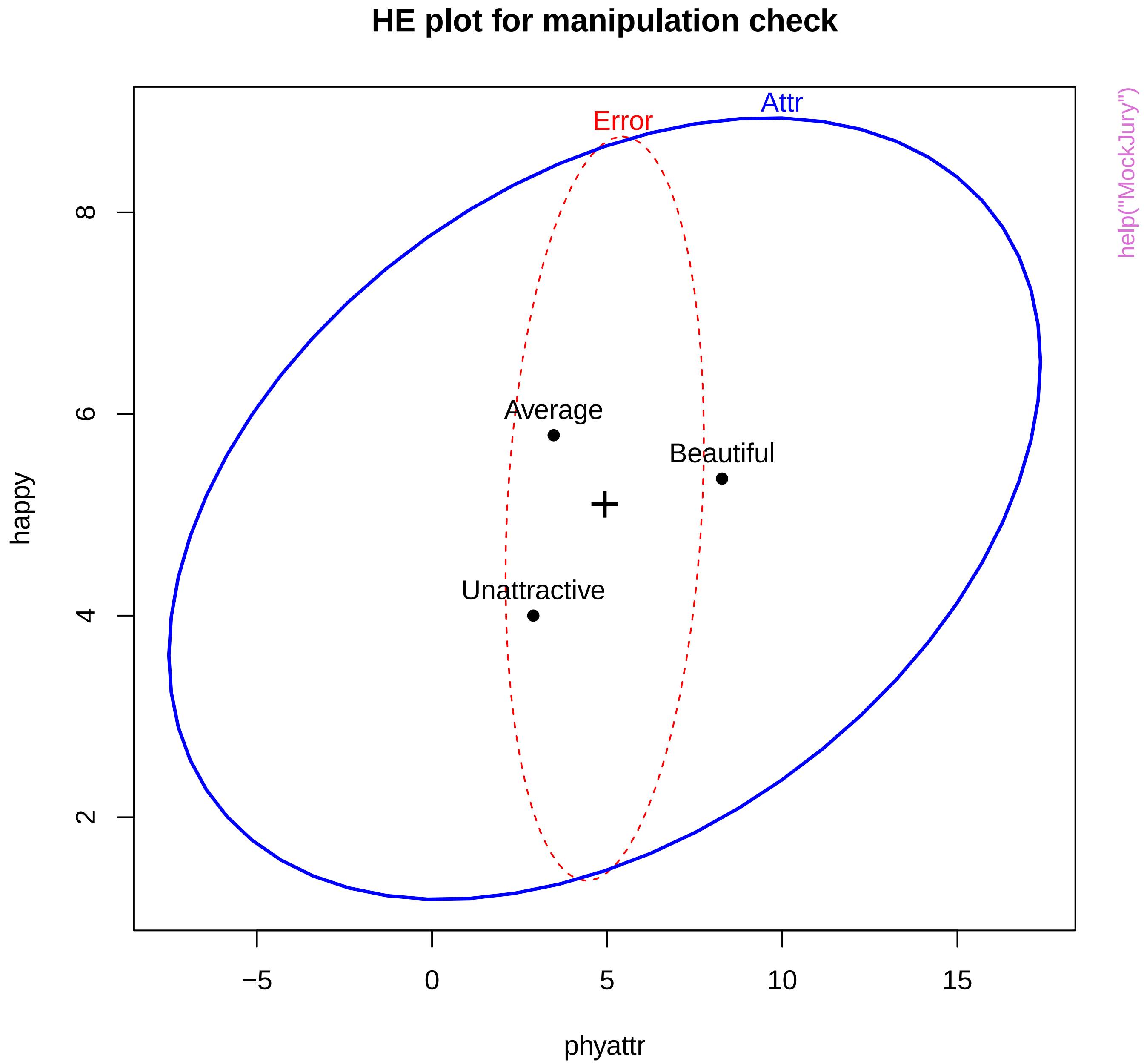


Extraversion
Neuroticism
160
140
120
100
80
70
80
90
100
110
120
130
Asian_Intl
Eur
Asian_Amer
Error
Group
help("Iwasaki_Big_Five")

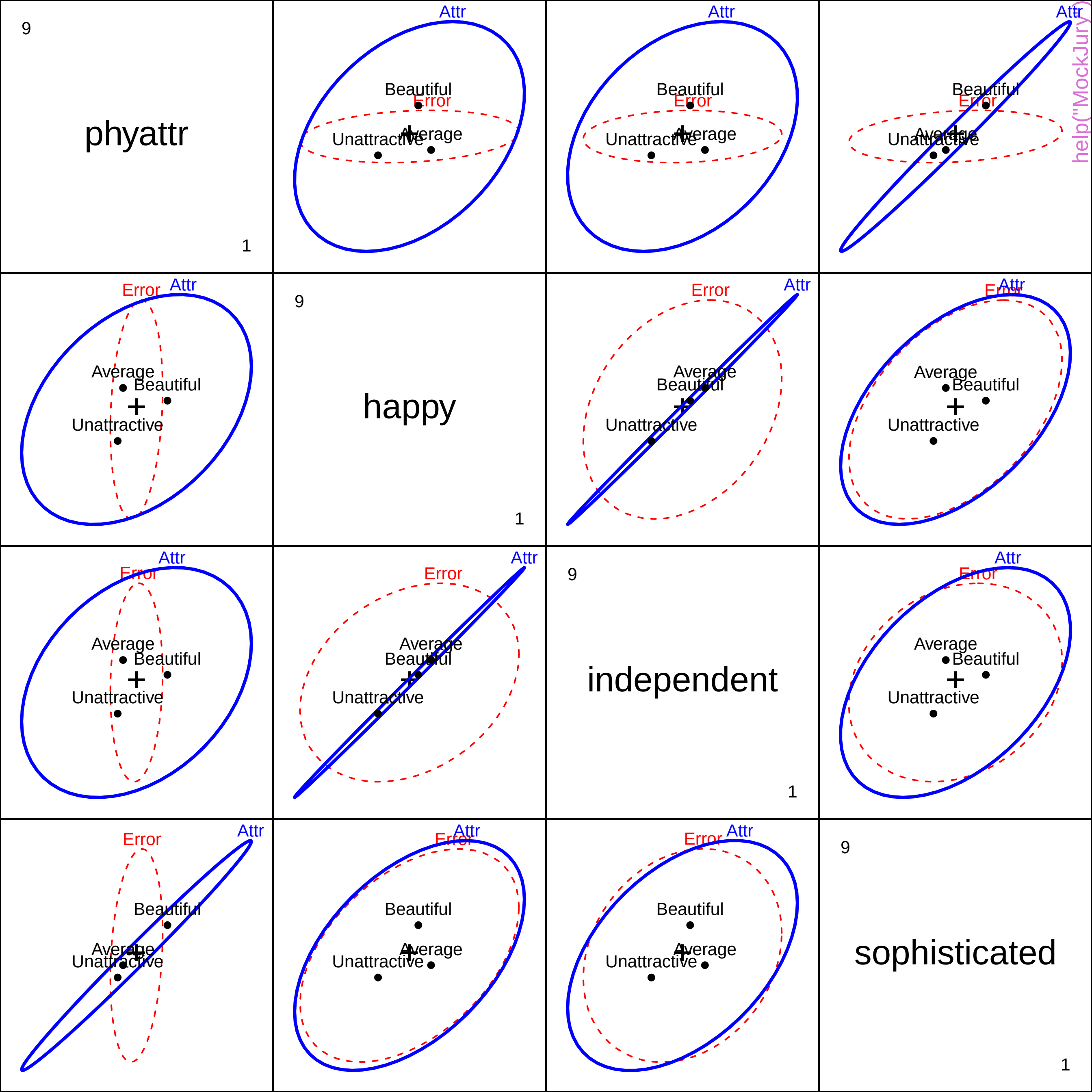

Error
Group
Eur
Asian_Intl
Asian_Amer
140
130
120
110
100
90
Conscientiousness
80
100
120
140
160
Extraversion
help("Iwasaki_Big_Five")


Neuroticism
Extraversion
Openness
Agreeableness
onscientiousne
160
36
175
50
166
72
161
58
171
57
Group
Error
Asian_Amer
Asian_Intl
Eur
help("Iwasaki_Big_Five")

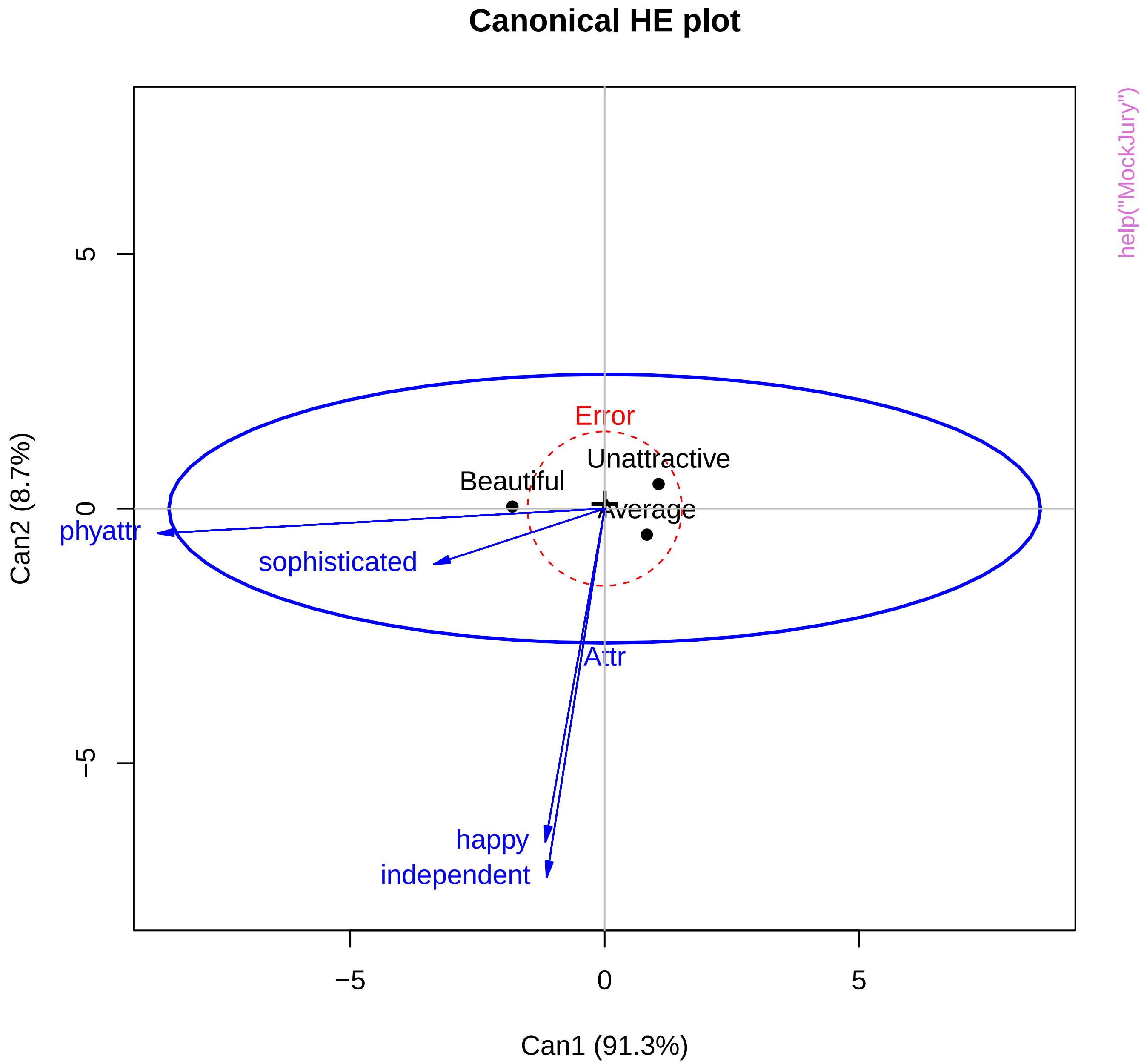

Asian_Amer
Asian_Intl
N
Eur
C
E
Error
A
Group
O
Can2 (36.9%)
Can1 (63.1%)
−4
−2
0
2
4
help("Iwasaki_Big_Five")

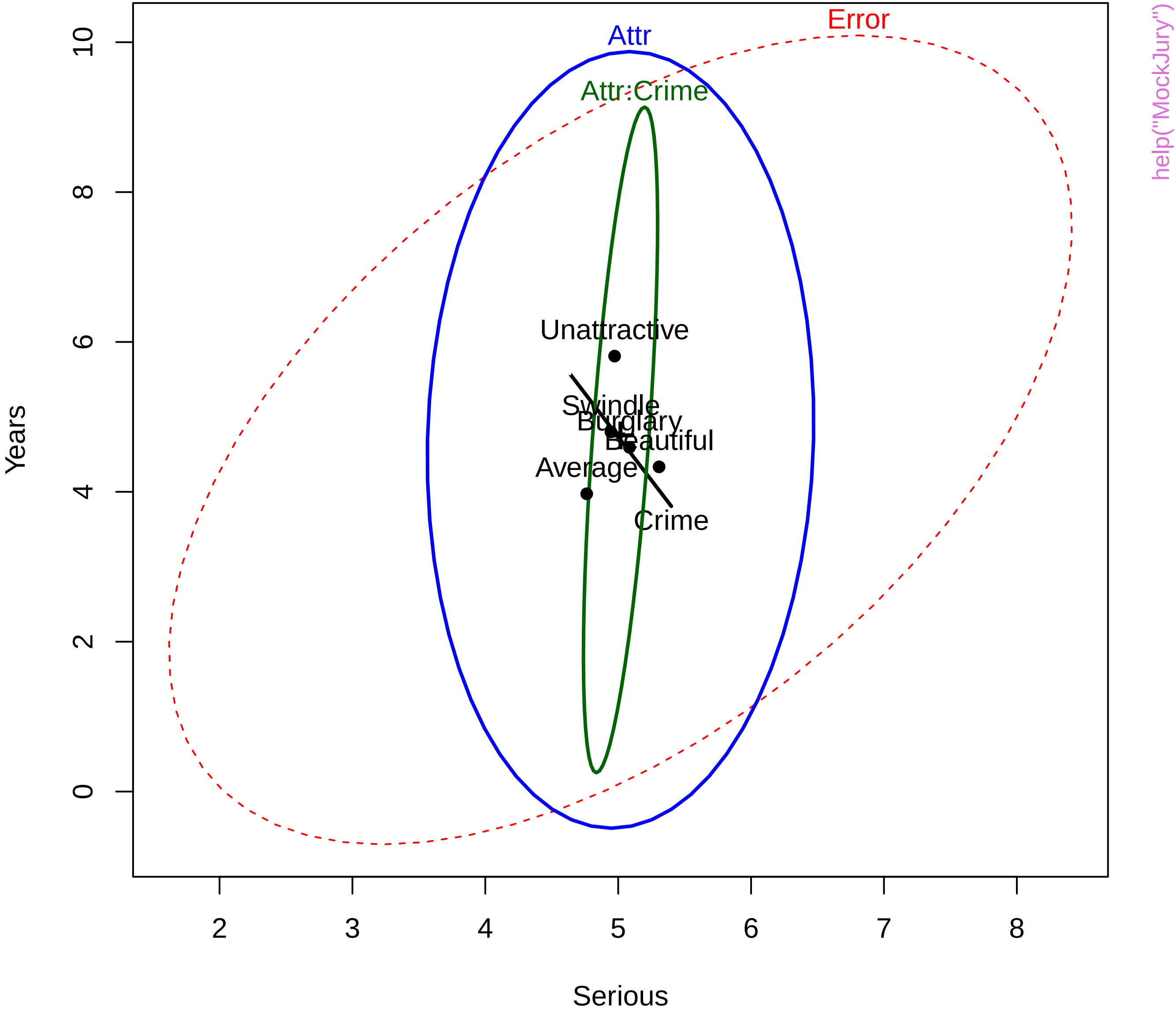


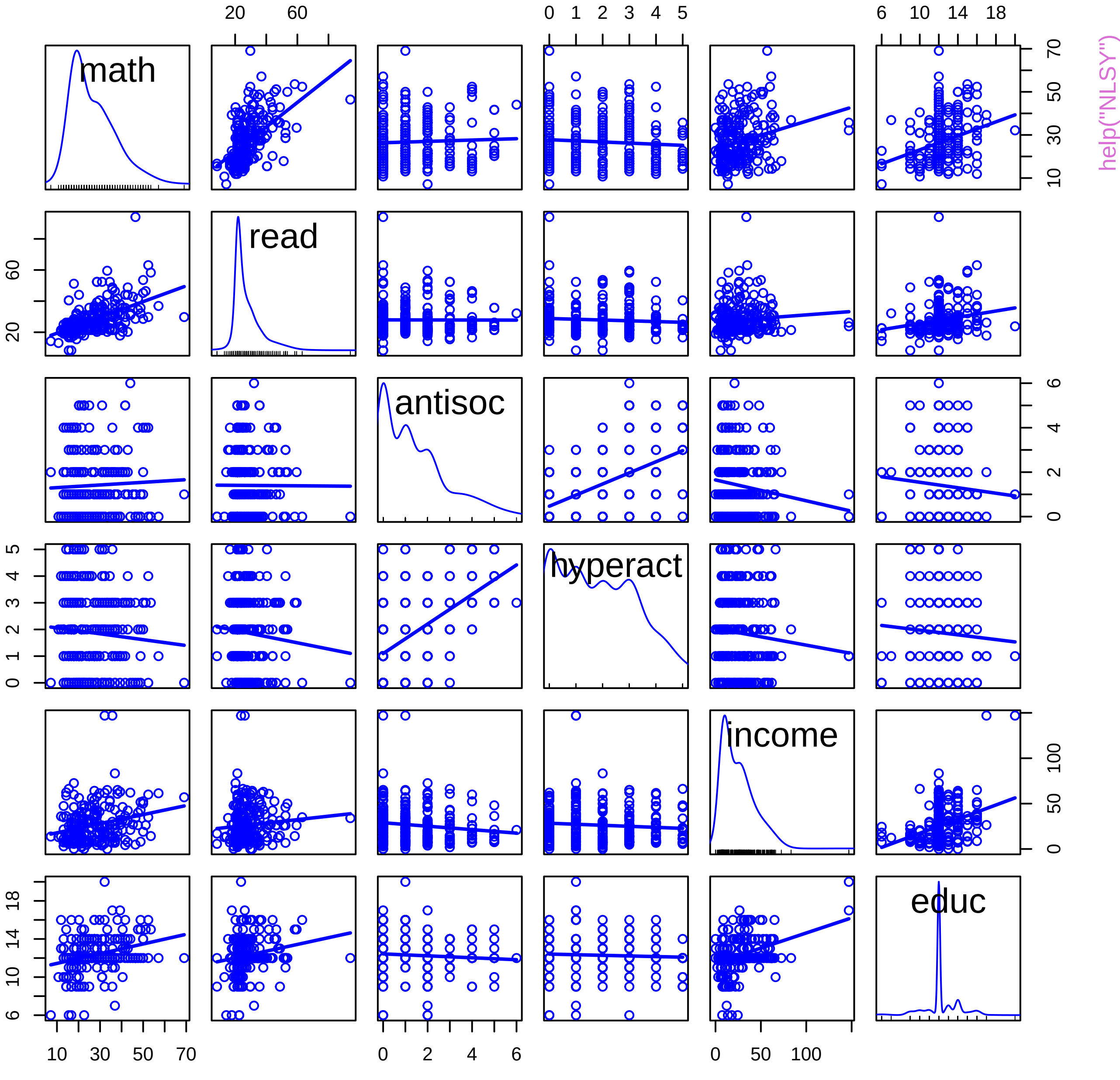
HE plot for manipulation check
Error
Attr
Average
Beautiful
Unattractive
help("MockJury")
happy
phyattr
8
6
4
2
−5
0
5
10
15

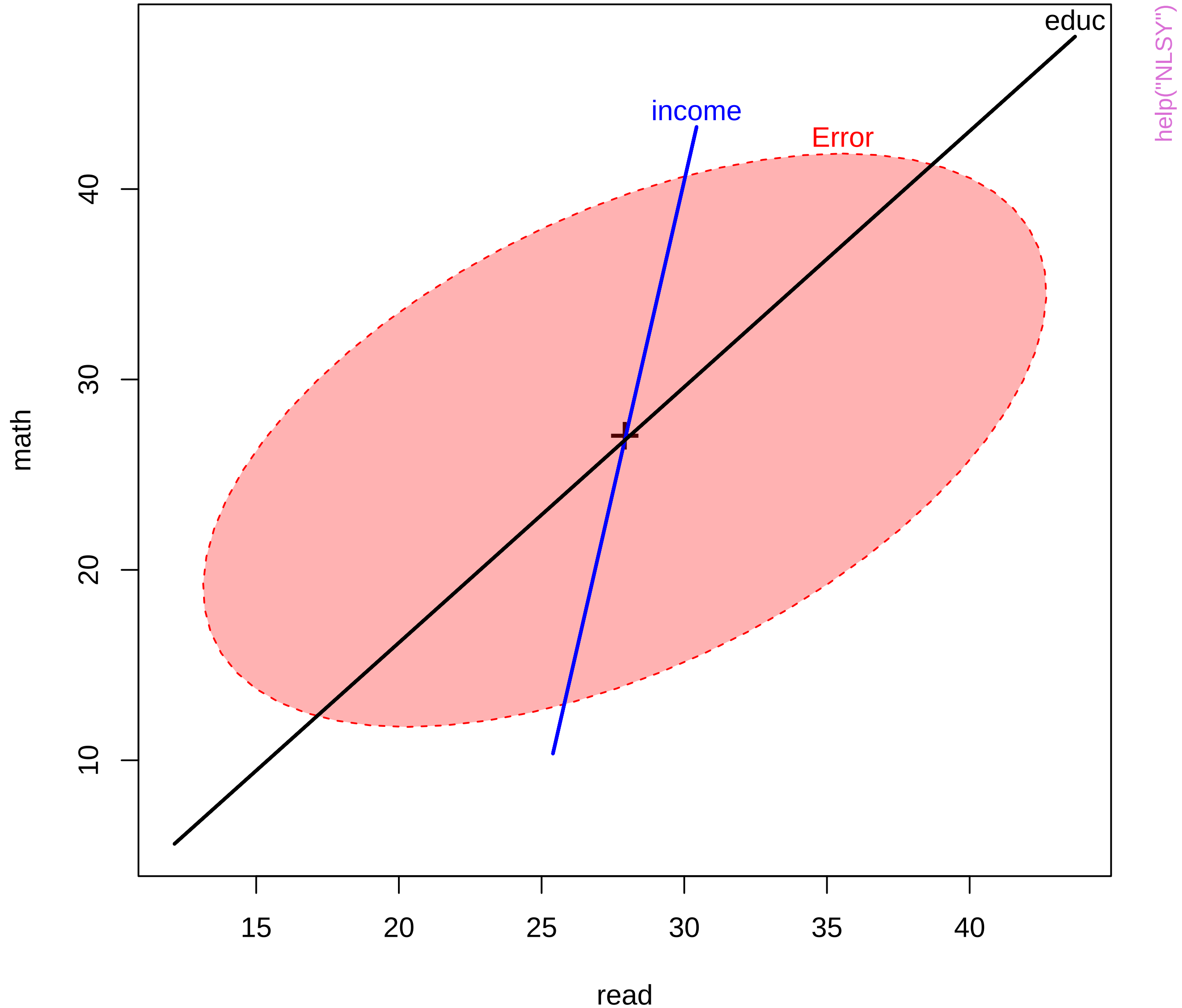


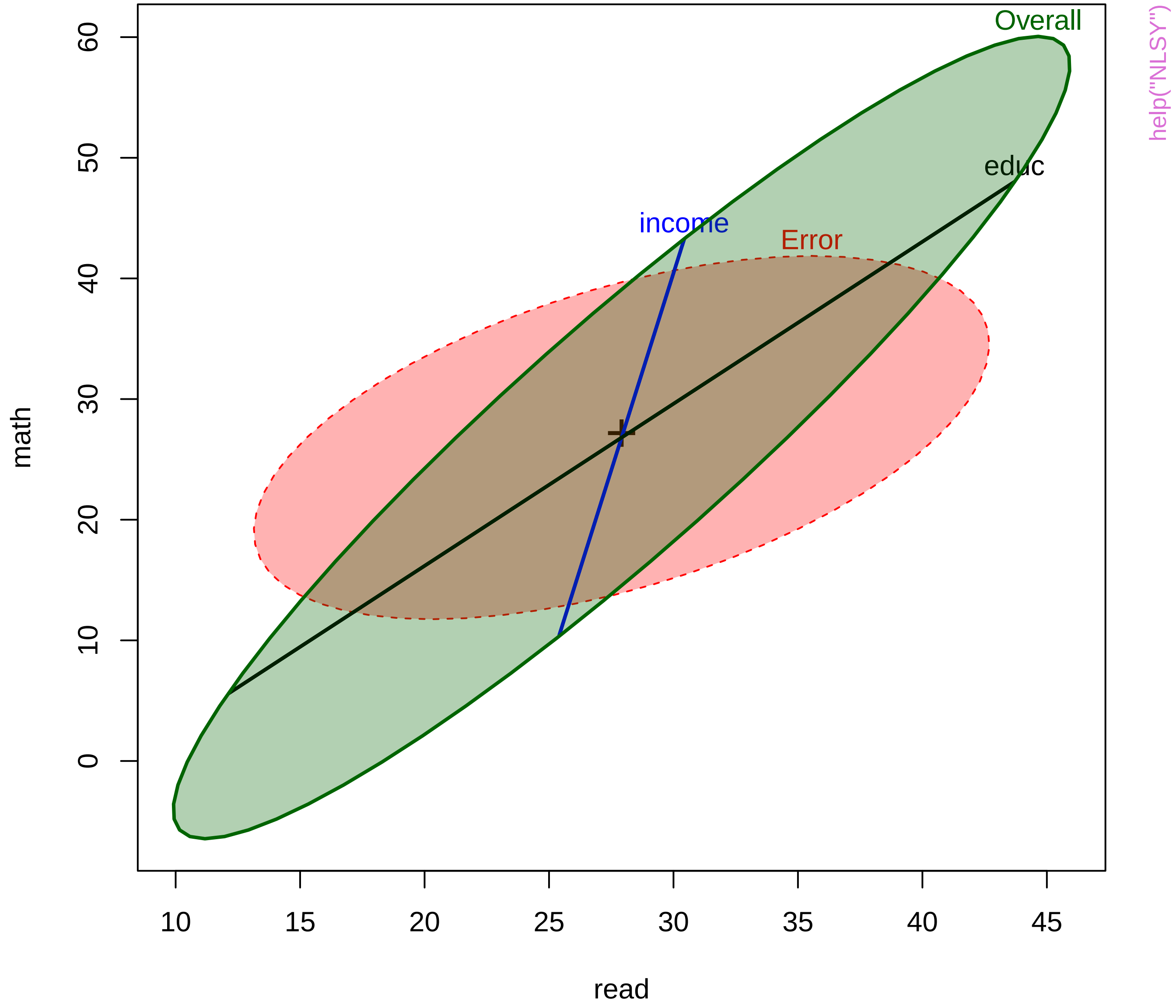
phyattr
happy
independent
sophisticated
9
1
Attr
Error
Beautiful
Average
Unattractive
help("MockJury")

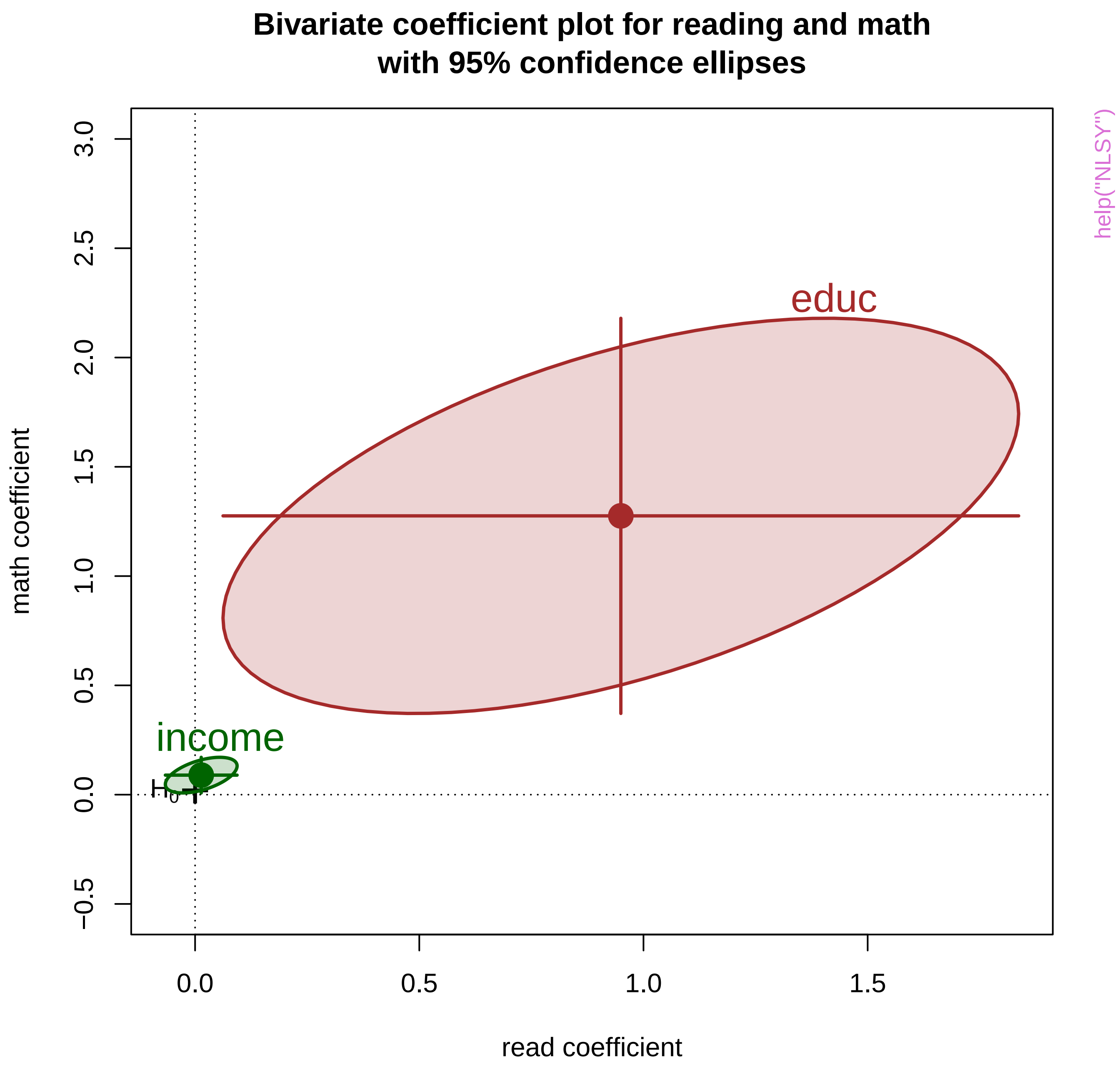

Canonical HE plot
Error
Unattractive
Beautiful
Average
phyattr
sophisticated
Attr
happy
independent
Can2 (8.7%)
Can1 (91.3%)
5
0
−5
−5
0
5
help("MockJury")


Error
Attr
Attr:Crime
Unattractive
Swindle
Burglary
Beautiful
Average
Crime
Years
Serious
0
2
4
6
8
10
2
3
4
5
6
7
8
help("MockJury")

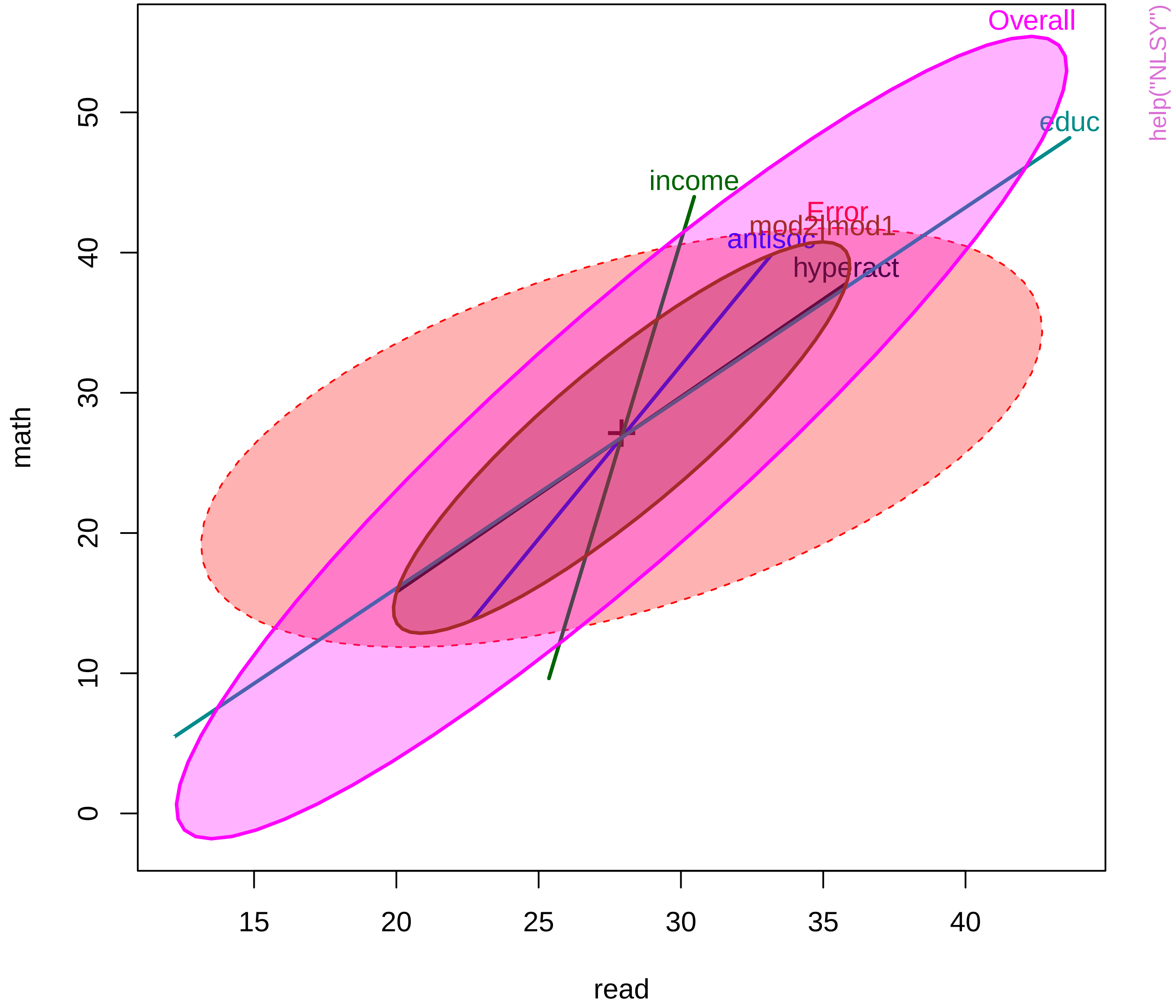

math
read
antisoc
hyperact
income
educ
help("NLSY")


educ
income
Error
help("NLSY")
math
read
10
20
30
40
15
20
25
30
35
40

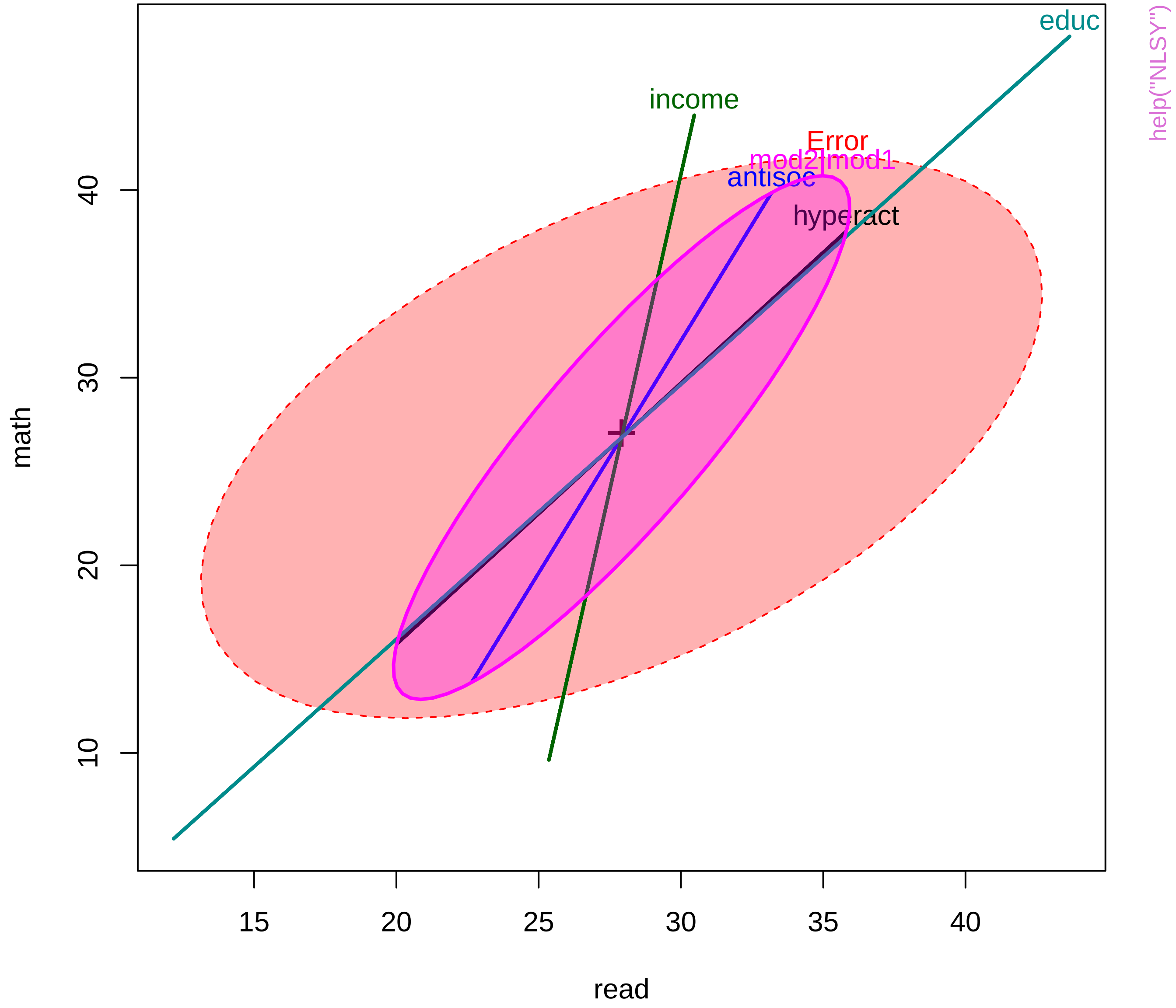

Overall
educ
income
Error
help("NLSY")
math
read
0
10
20
30
40
50
60
10
15
20
25
30
35
40
45



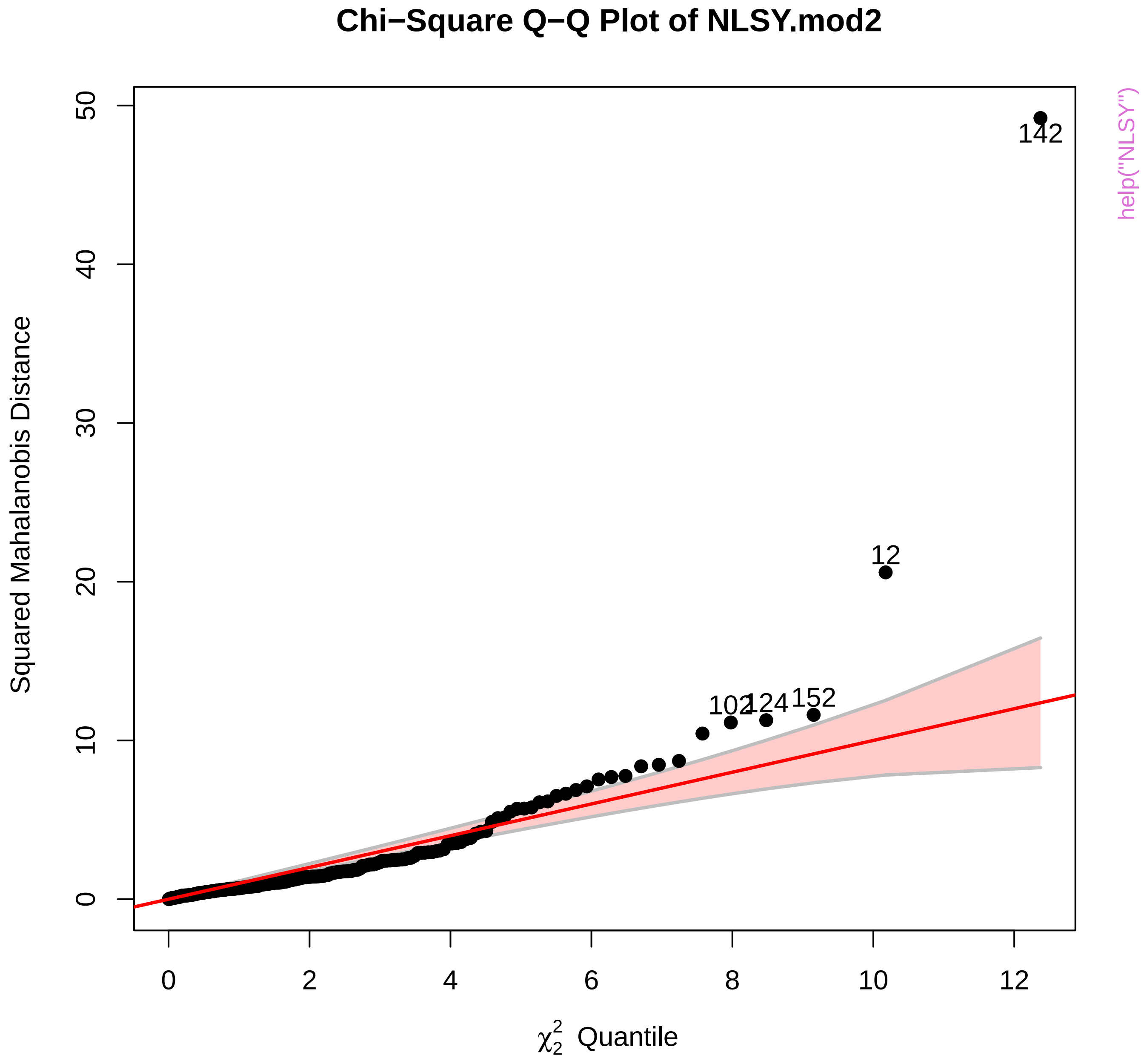
Bivariate coefficient plot for reading and math
with 95% confidence ellipses
educ
income
H0
math coefficient
read coefficient
help("NLSY")
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0.0
0.5
1.0
1.5



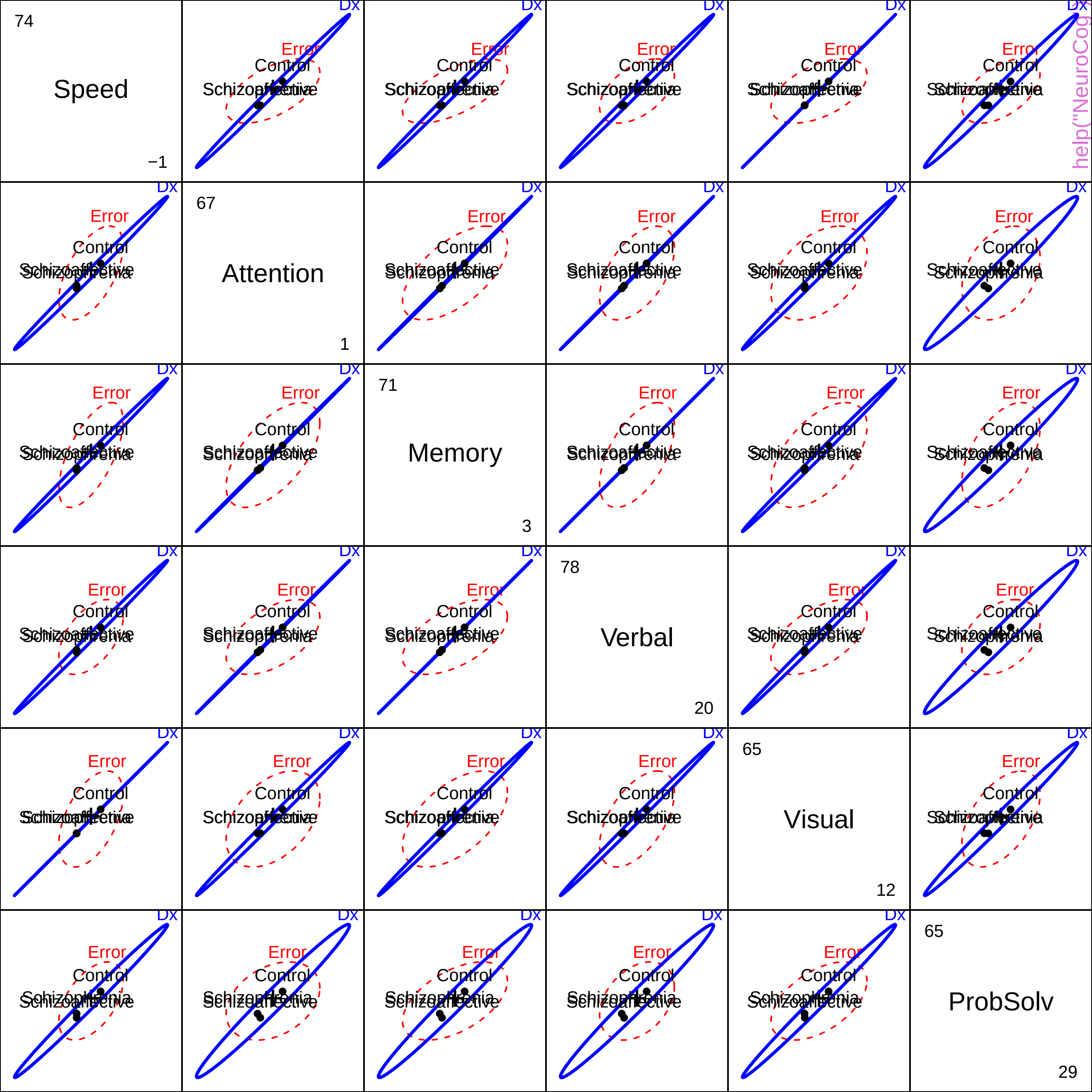
Overall
educ
income
Error
mod2|mod1
antisoc
hyperact
help("NLSY")
50
40
30
20
10
0
math
15
20
25
30
35
40
read



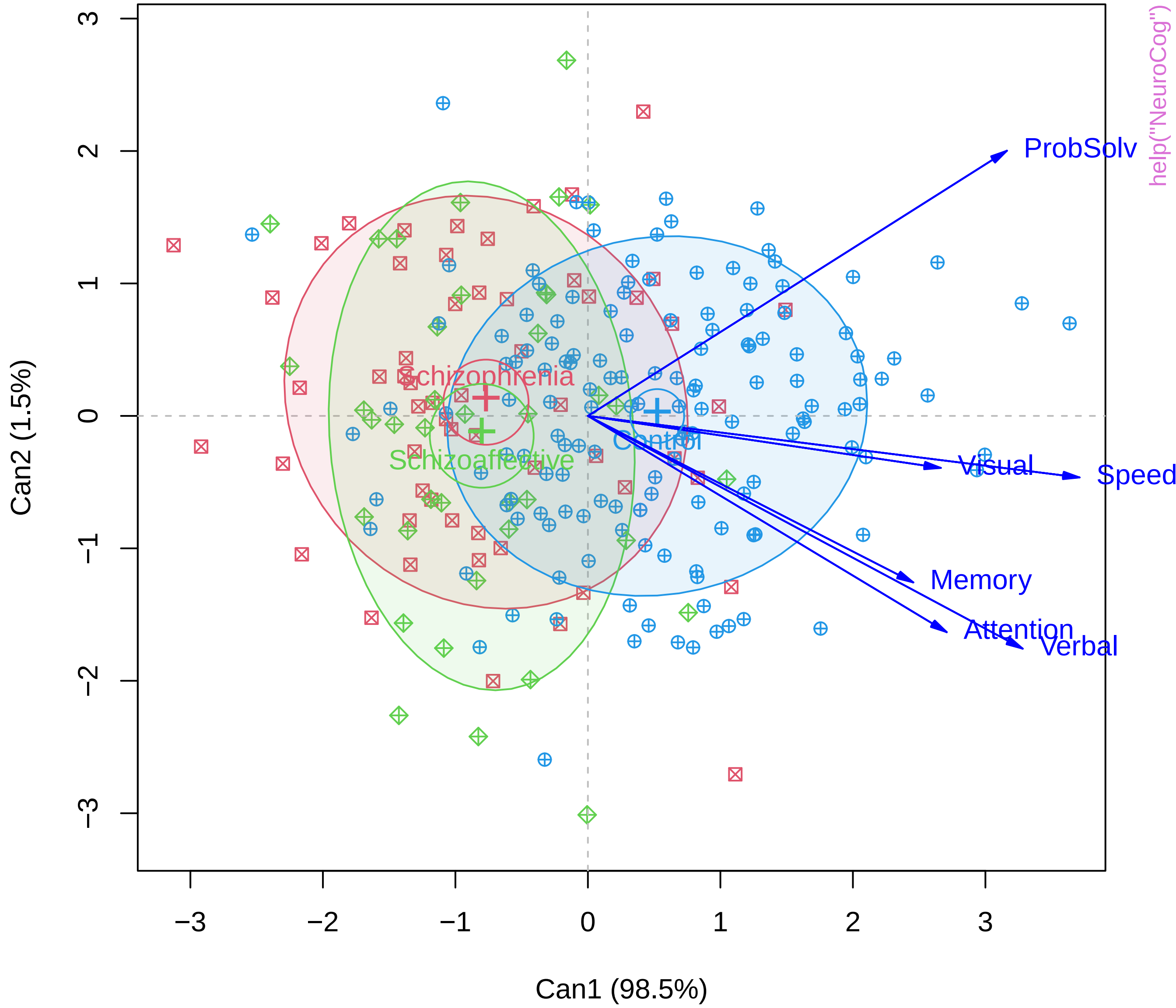
educ
income
Error
mod2|mod1
antisoc
hyperact
help("NLSY")
math
read
10
20
30
40
15
20
25
30
35
40



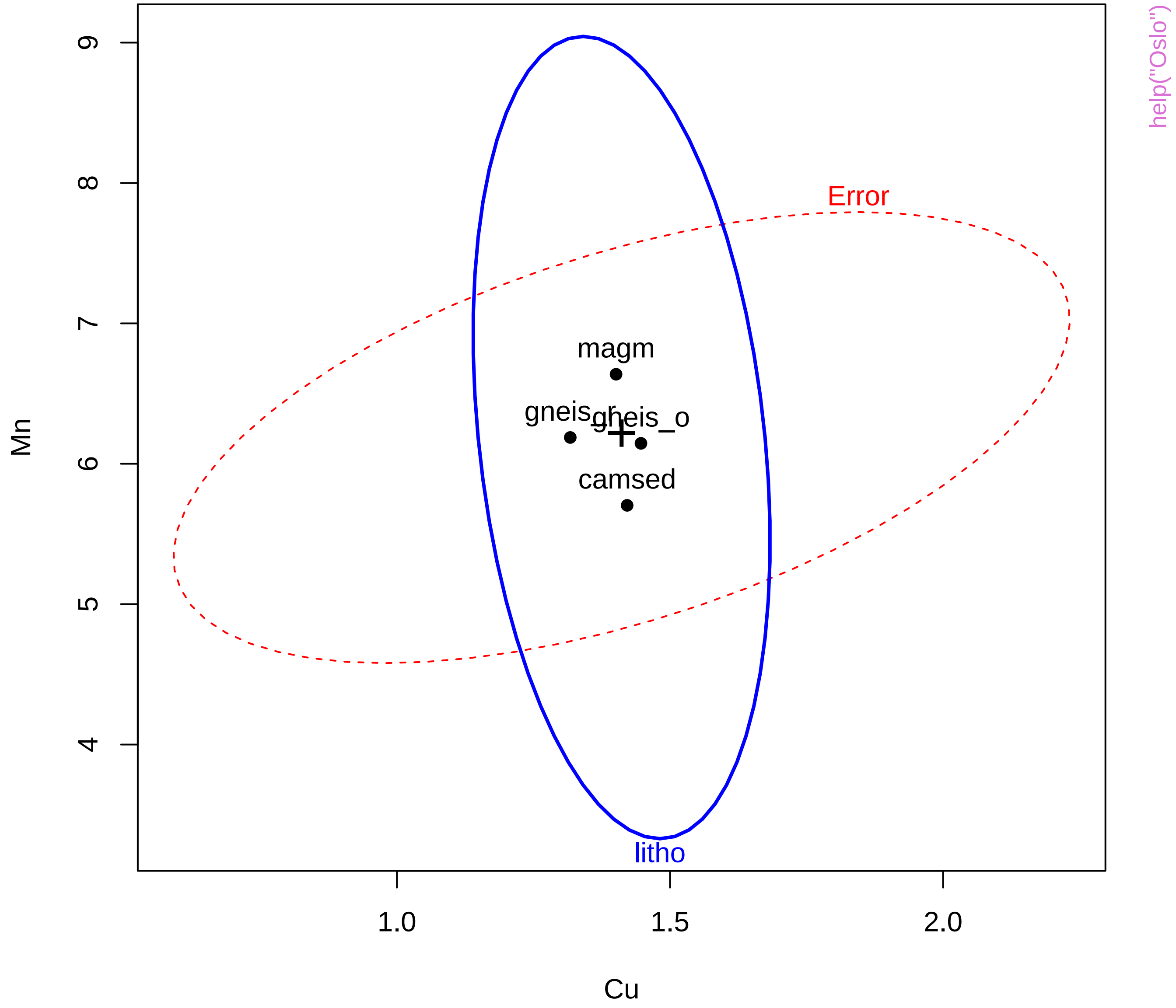
Chi−Square Q−Q Plot of NLSY.mod2
Squared Mahalanobis Distance
χ^2_2 Quantile
0
10
20
30
40
50
0
2
4
6
8
10
12
142
12
102
124
152
help("NLSY")

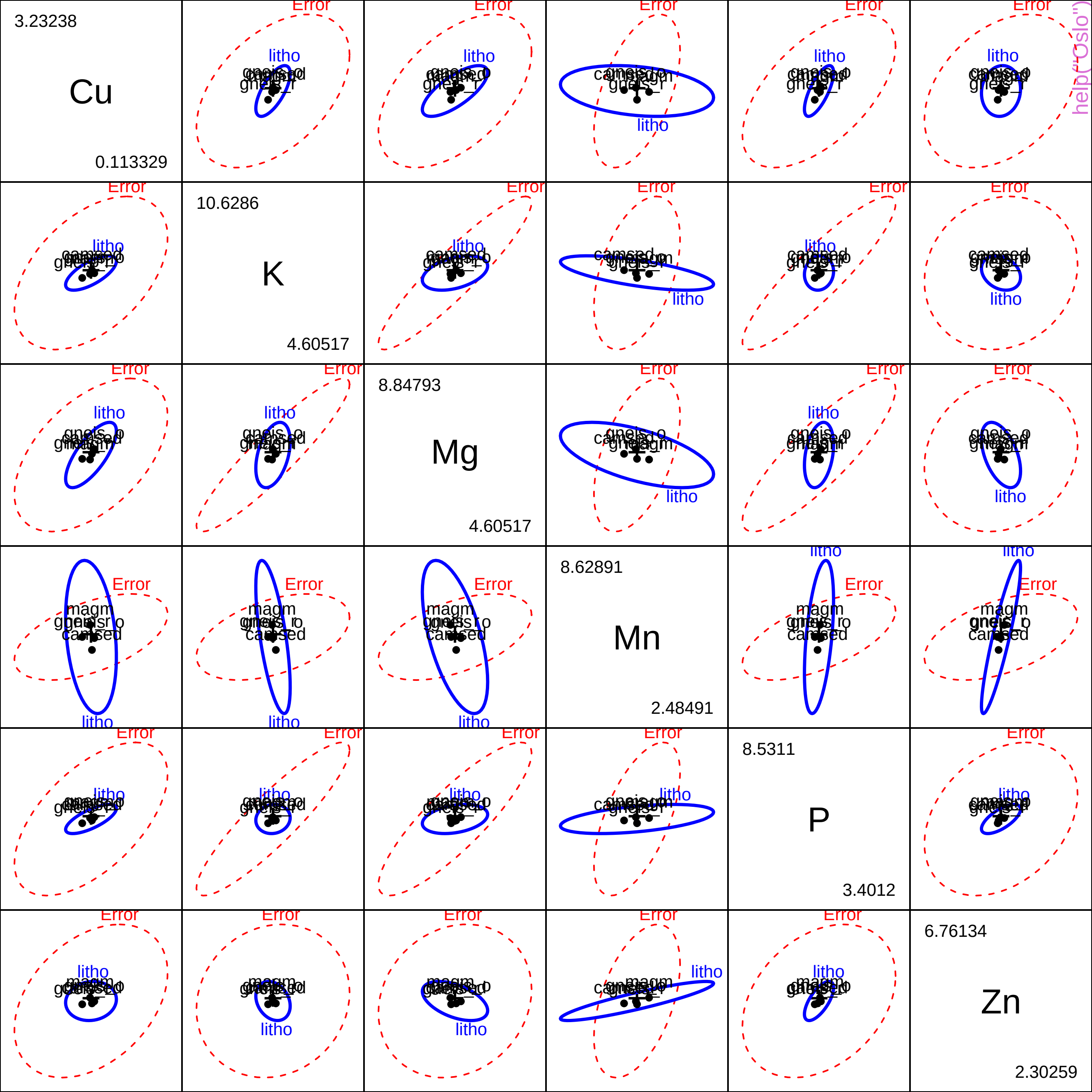


Speed
74
−1
Attention
67
1
Memory
71
3
Verbal
78
20
Visual
65
12
ProbSolv
65
29
DX
Error
Control
Schizoaffective
Schizophrenia
help("NeuroCog")

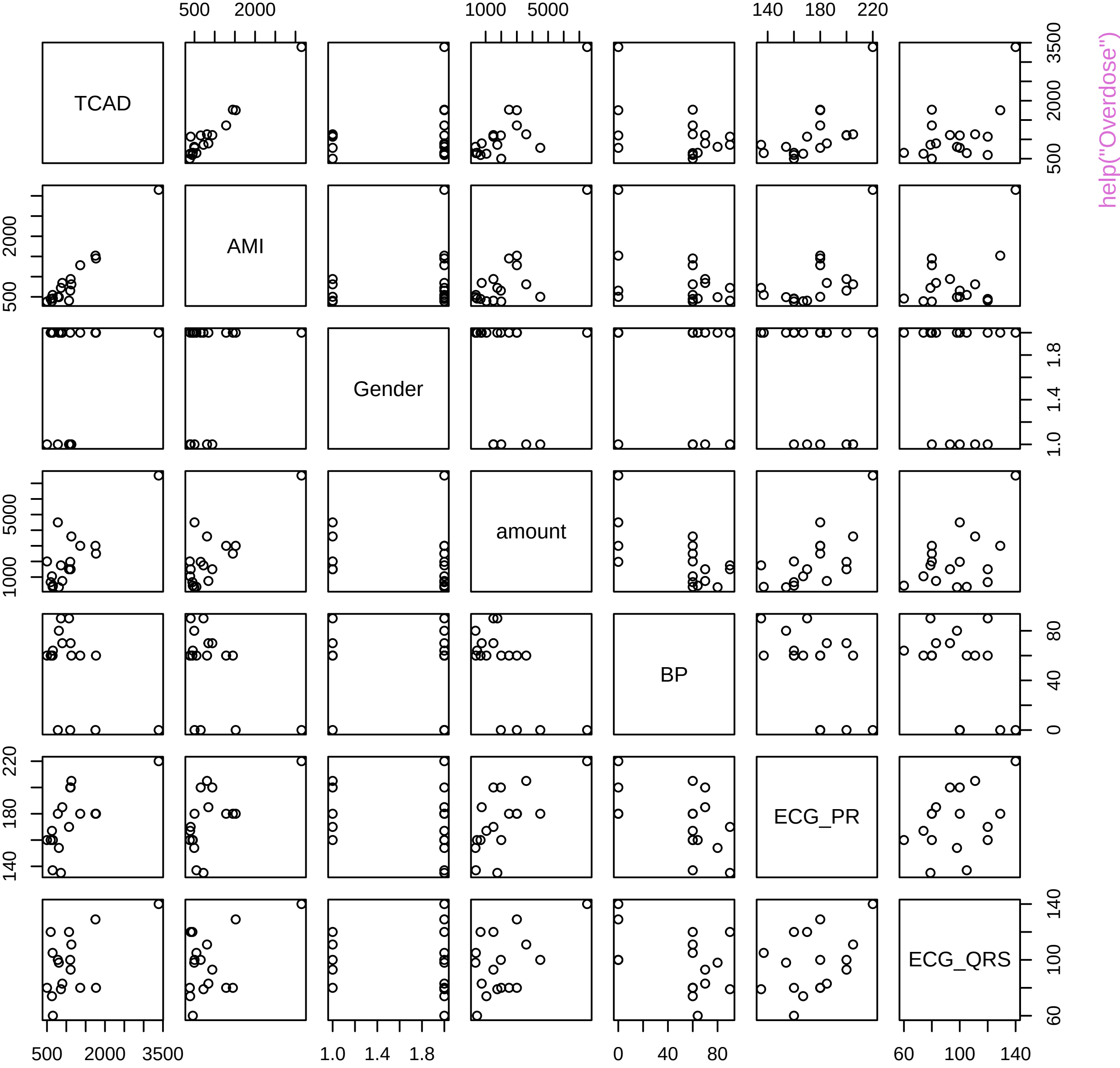


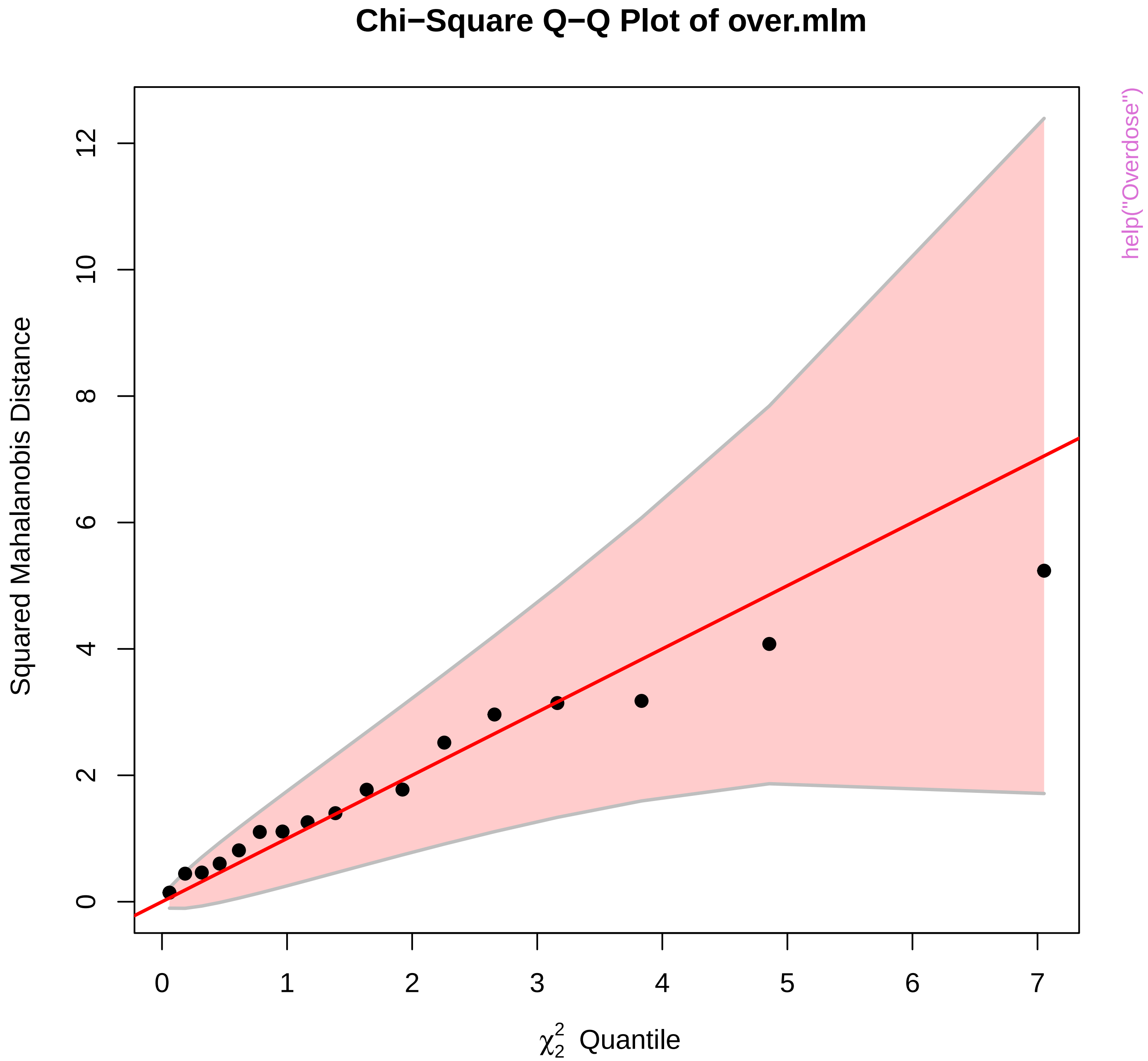
ProbSolv
Visual
Speed
Memory
Attention
Verbal
Schizophrenia
Schizoaffective
Control
help("NeuroCog")
Can1 (98.5%)
Can2 (1.5%)
−3
−2
−1
0
1
2
3


help("Oslo")
Error
magm
gneis_r
gneis_o
camsed
litho
9
8
7
6
5
4
Mn
1.0
1.5
2.0
Cu

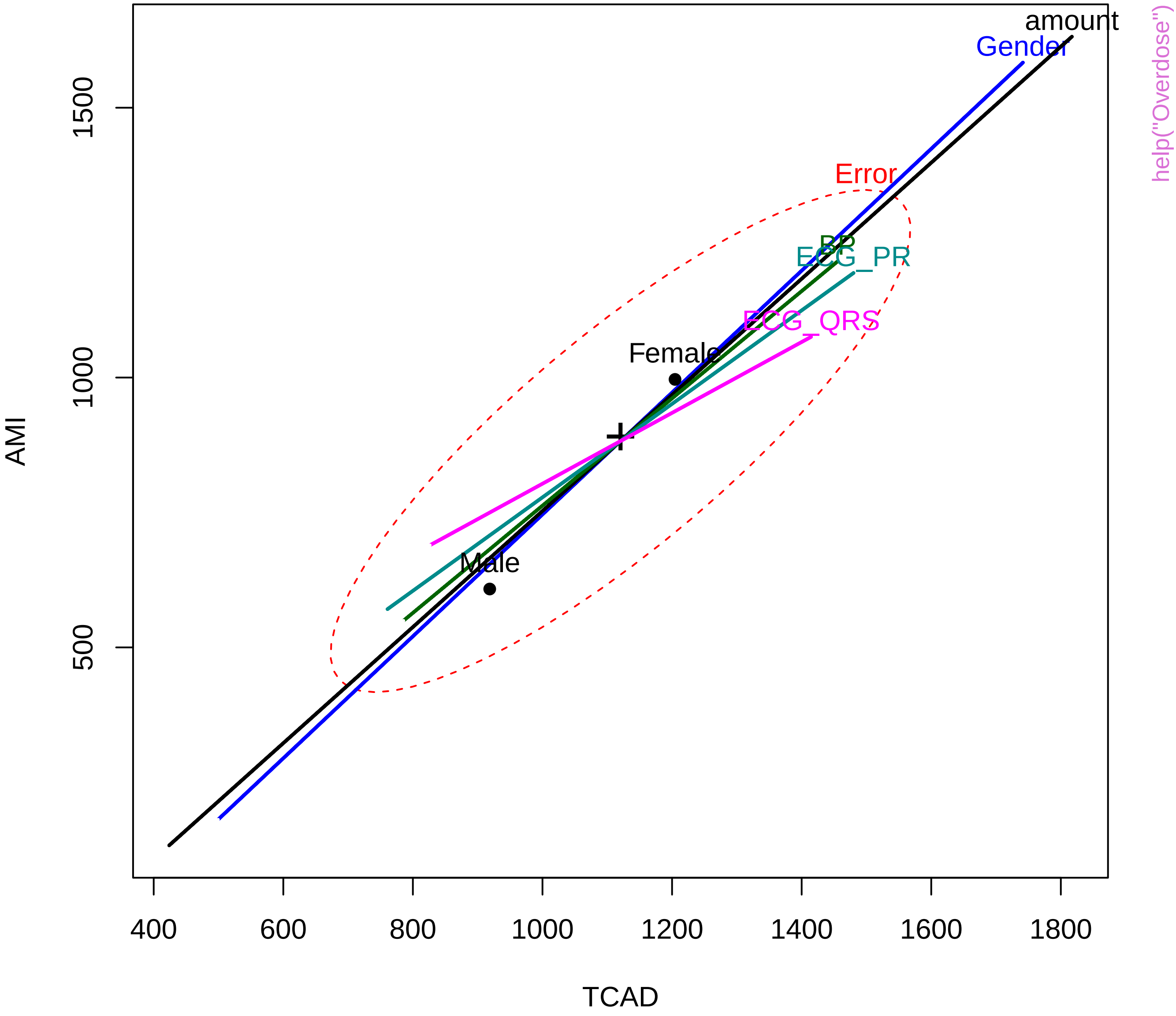


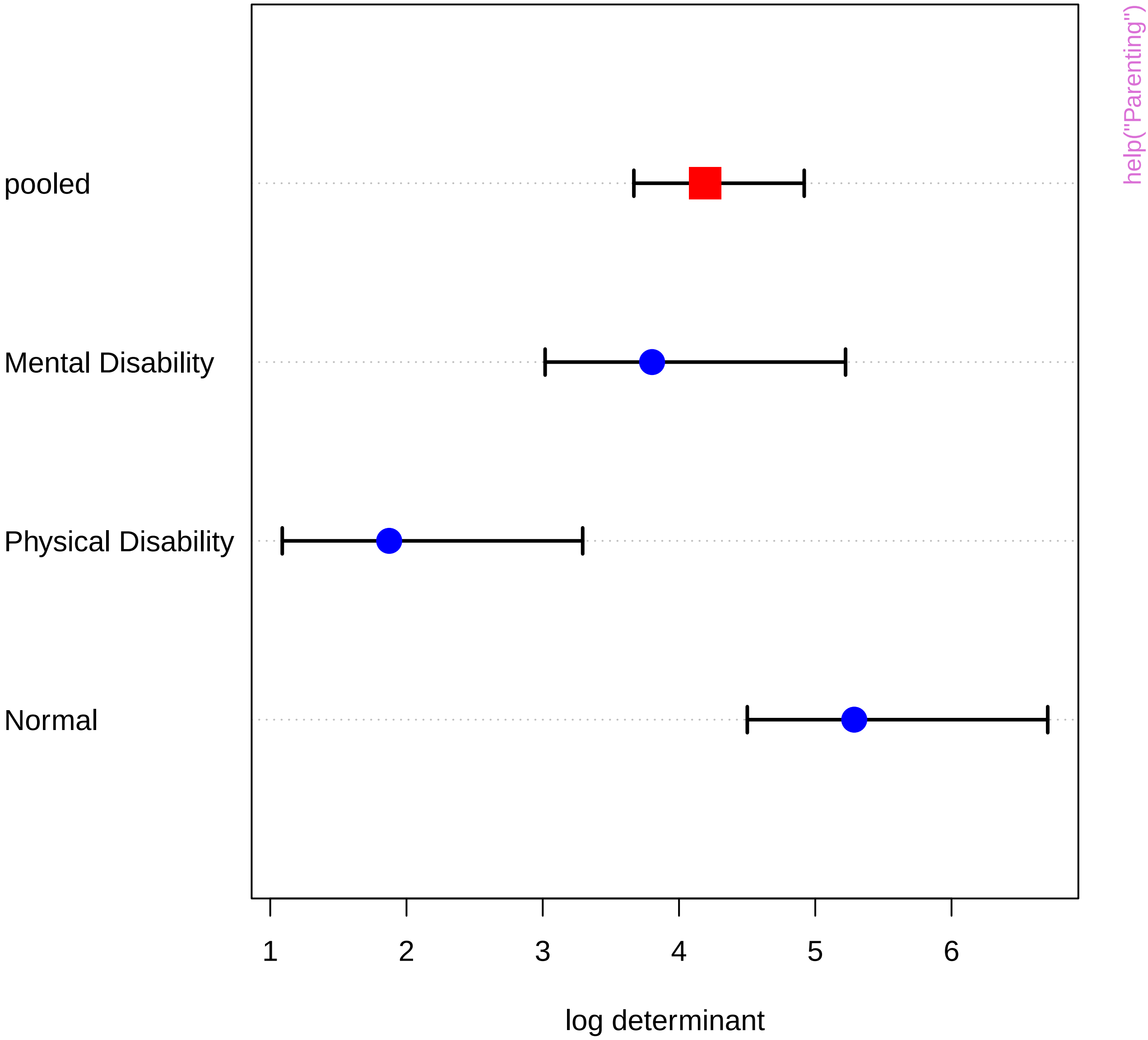
3.23238
Cu
0.113329
10.6286
K
4.60517
8.84793
Mg
4.60517
8.62891
Mn
2.48491
8.5311
P
3.4012
6.76134
Zn
2.30259
Error
litho
magm
help("Oslo")

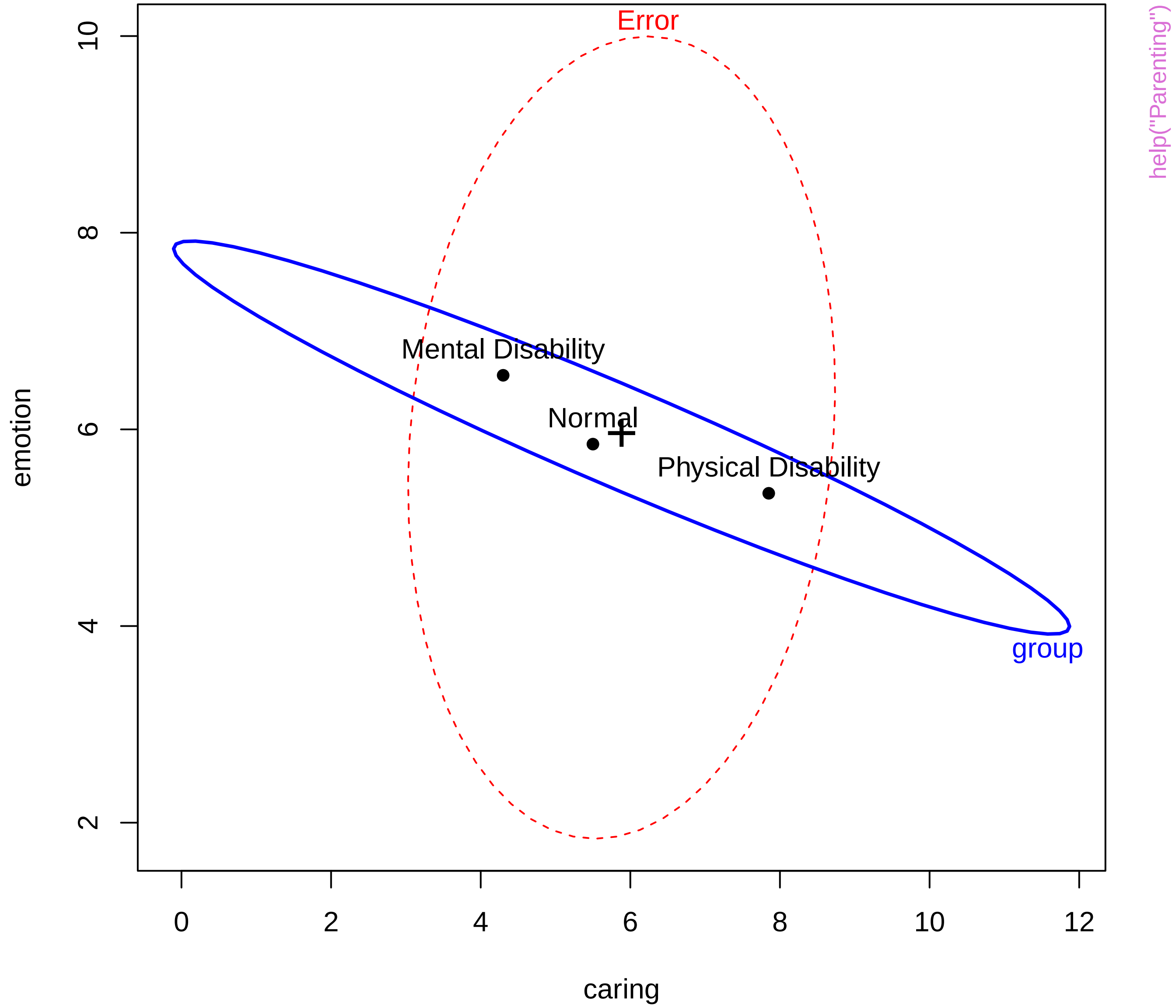

TCAD
AMI
Gender
amount
BP
ECG_PR
ECG_QRS
help("Overdose")

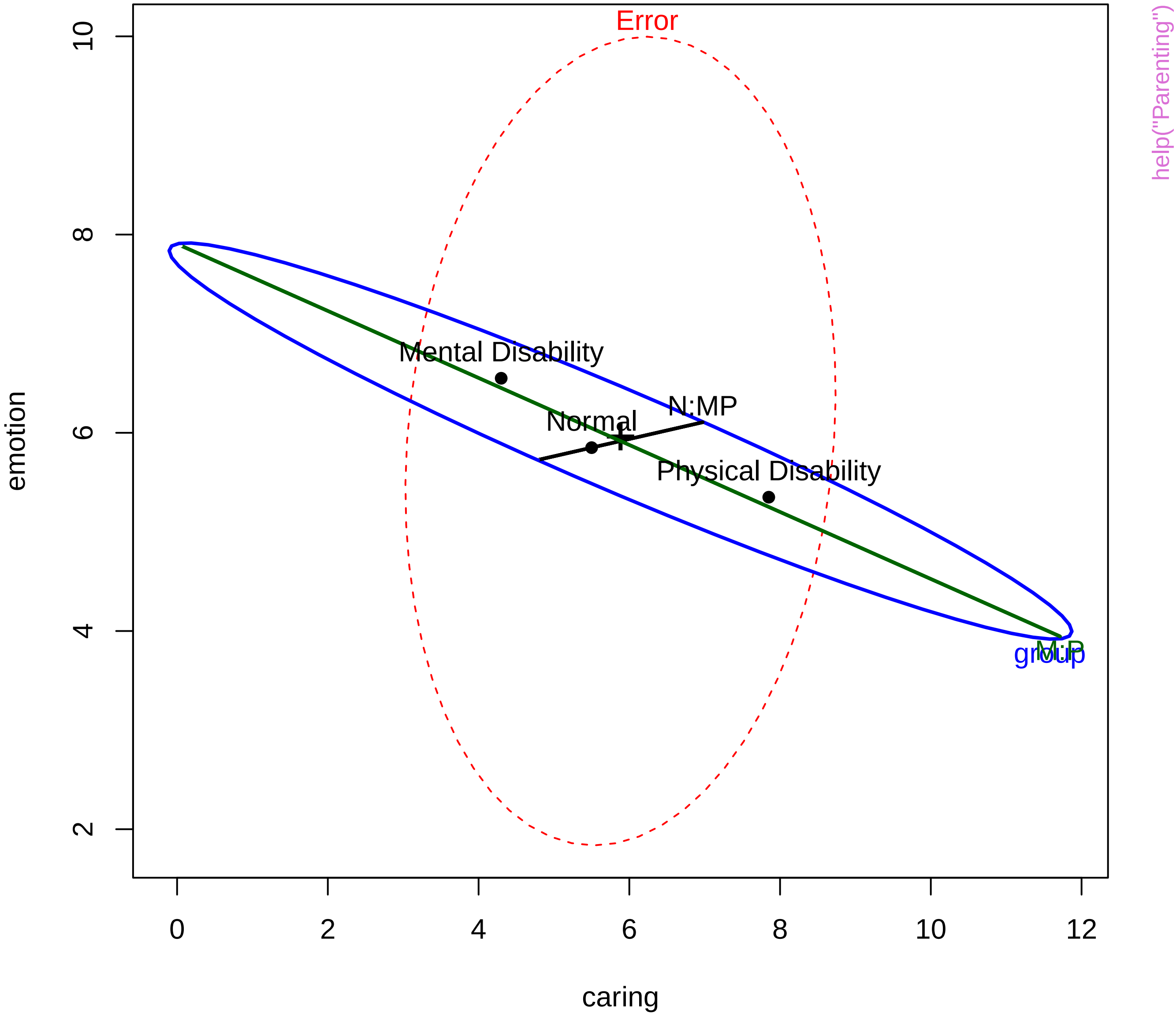


Chi−Square Q−Q Plot of over.mlm
Squared Mahalanobis Distance
χ^2_2 Quantile
help("Overdose")
0
2
4
6
8
10
12
0
1
2
3
4
5
6
7

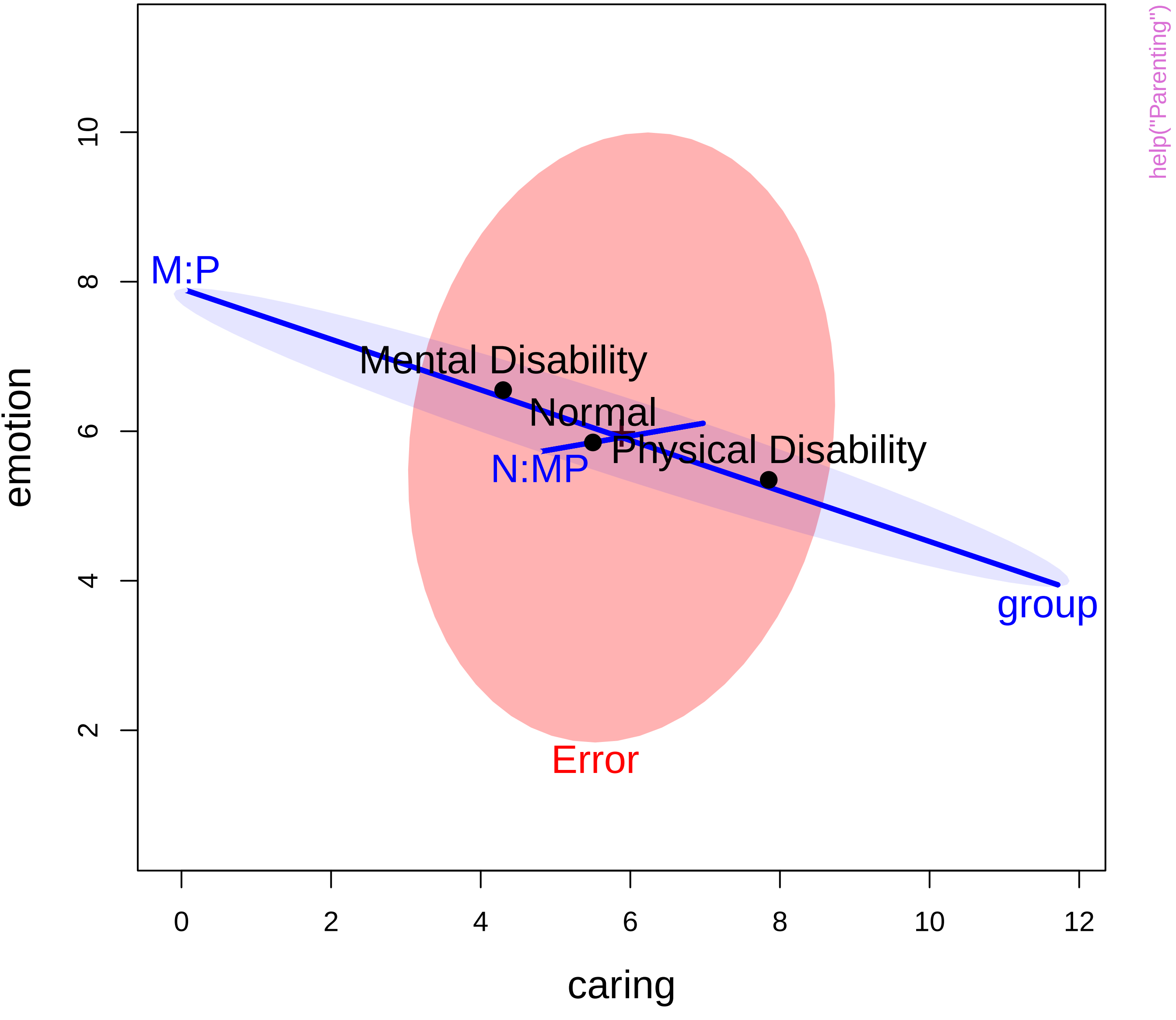

amount
Gender
Error
BP
ECG_PR
ECG_QRS
Female
Male
1500
1000
500
AMI
400
600
800
1000
1200
1400
1600
1800
TCAD
help("Overdose")


pooled
Mental Disability
Physical Disability
Normal
1
2
3
4
5
6
log determinant
help("Parenting")

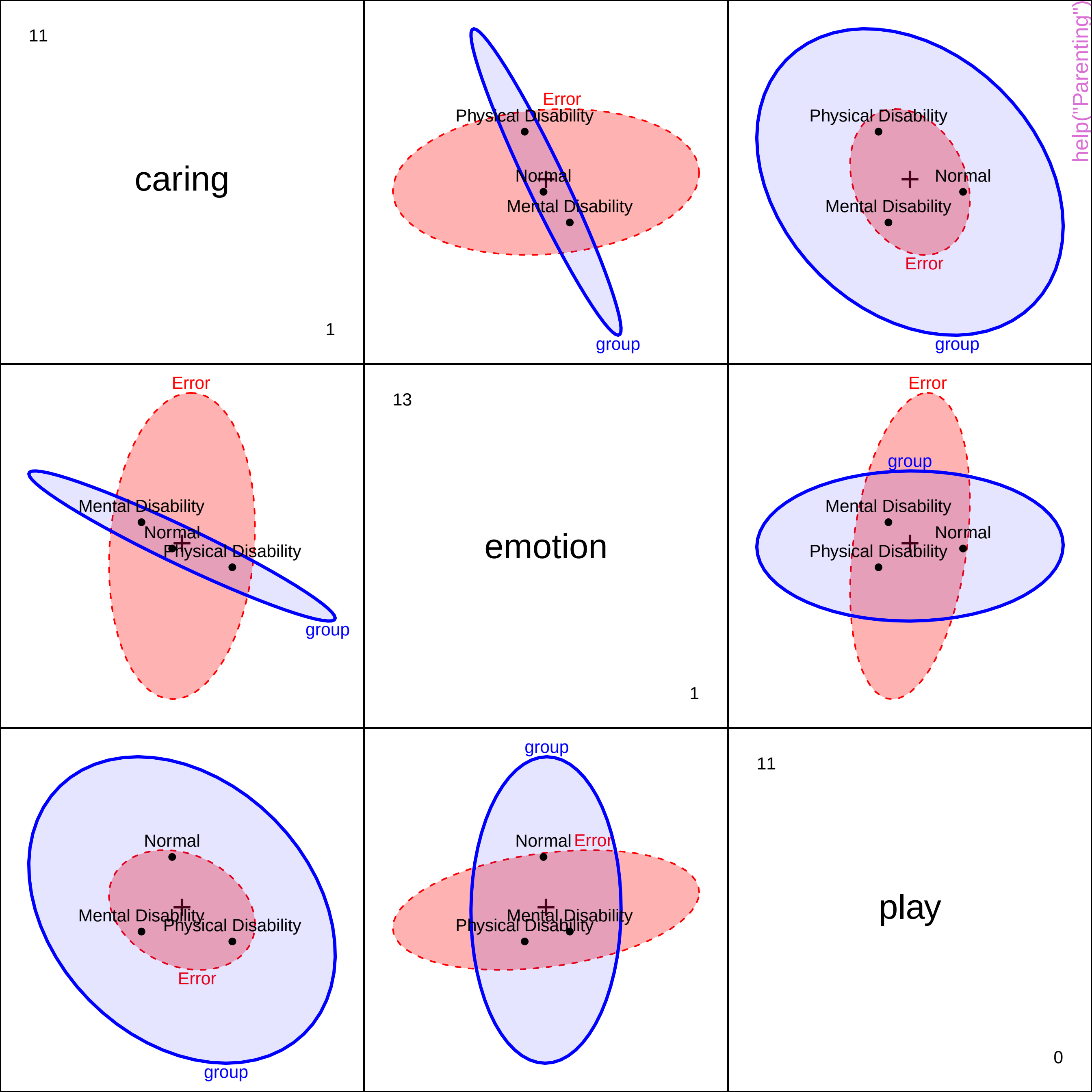

Error
help("Parenting")
Mental Disability
Normal
Physical Disability
group
emotion
caring
10
8
6
4
2
0
2
4
6
8
10
12



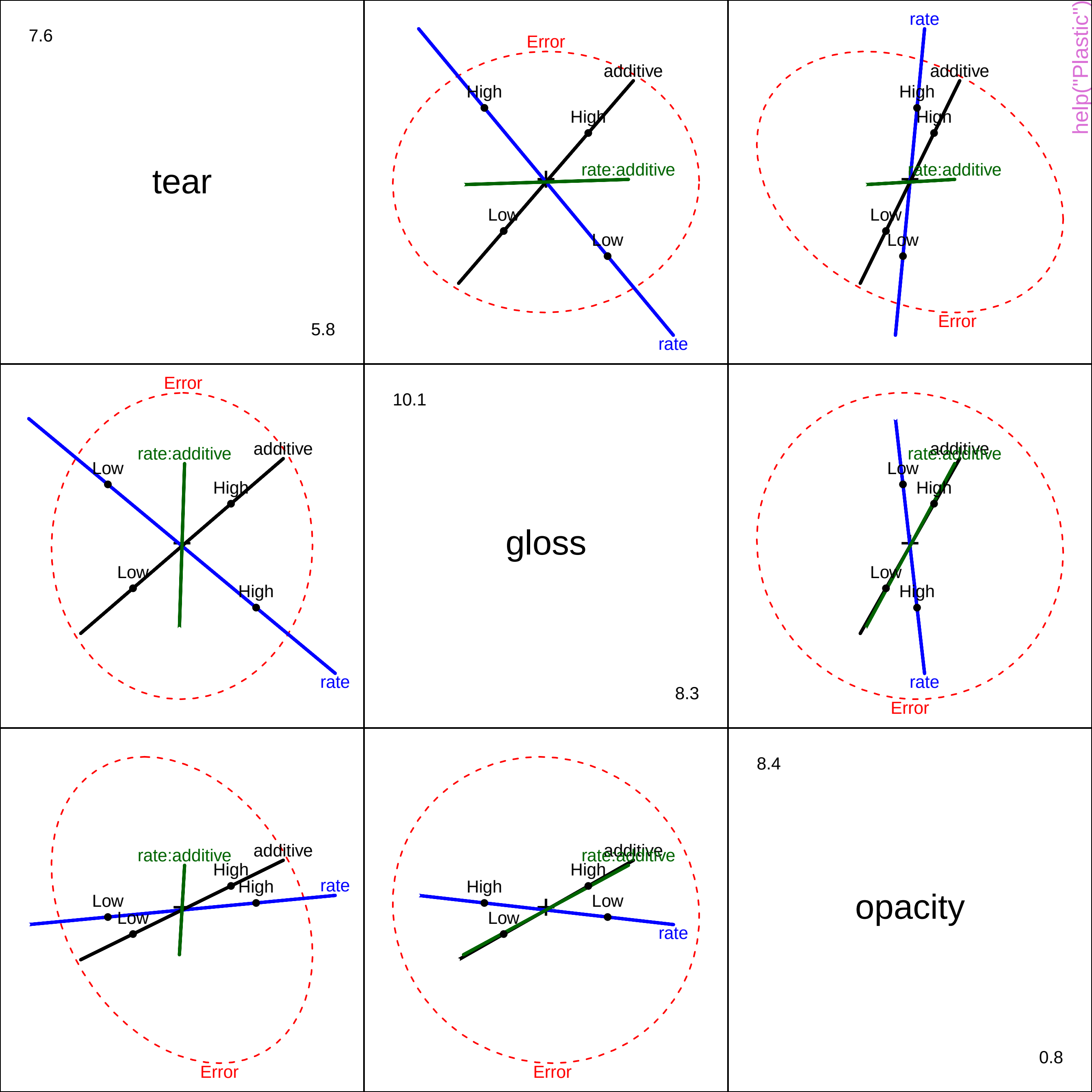
Error
Mental Disability
N:MP
Normal
Physical Disability
M:P
group
help("Parenting")
emotion
caring
0
2
4
6
8
10
12
2
4
6
8
10



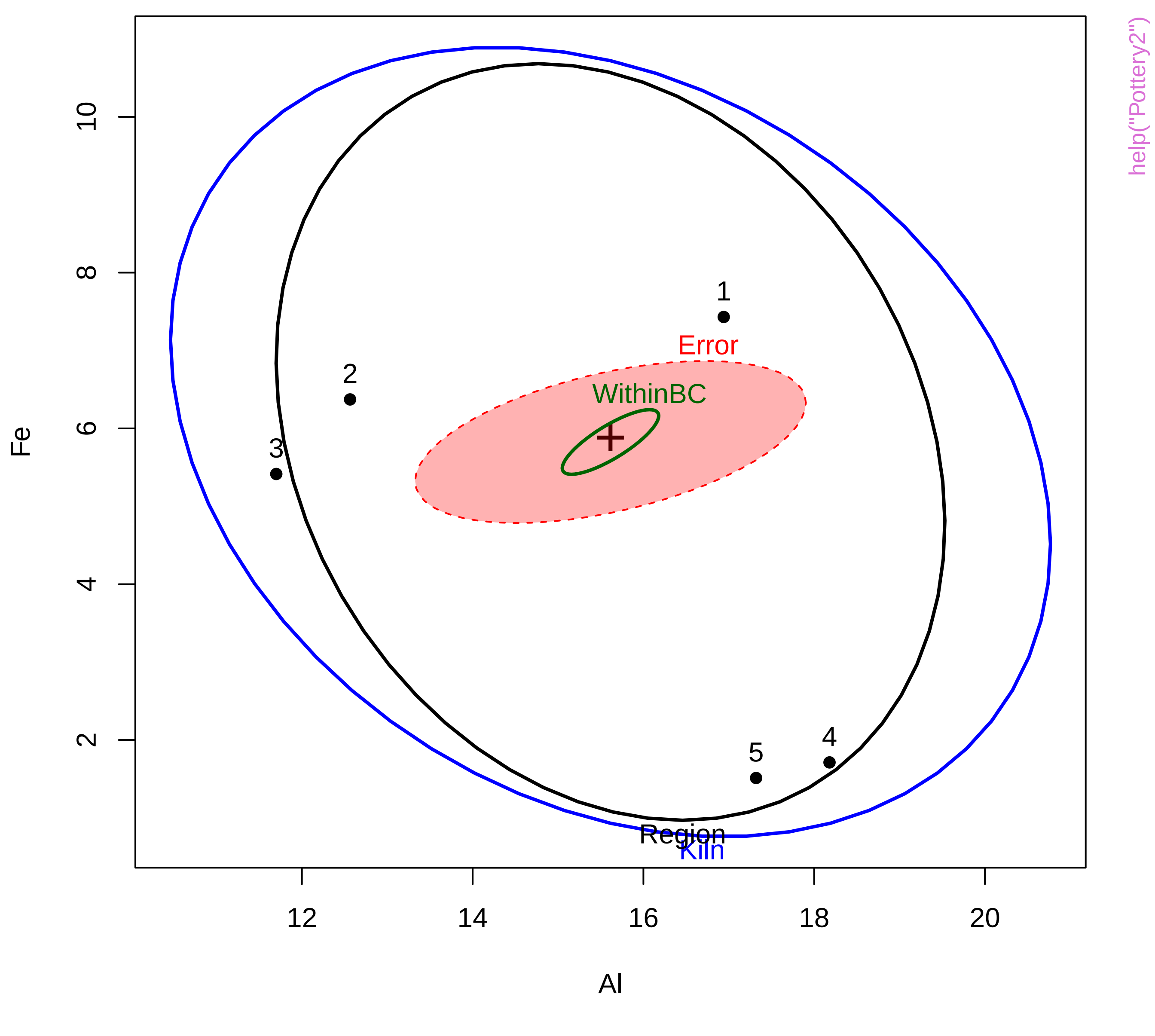
M:P
Mental Disability
Normal
Physical Disability
N:MP
group
Error
emotion
caring
help("Parenting")



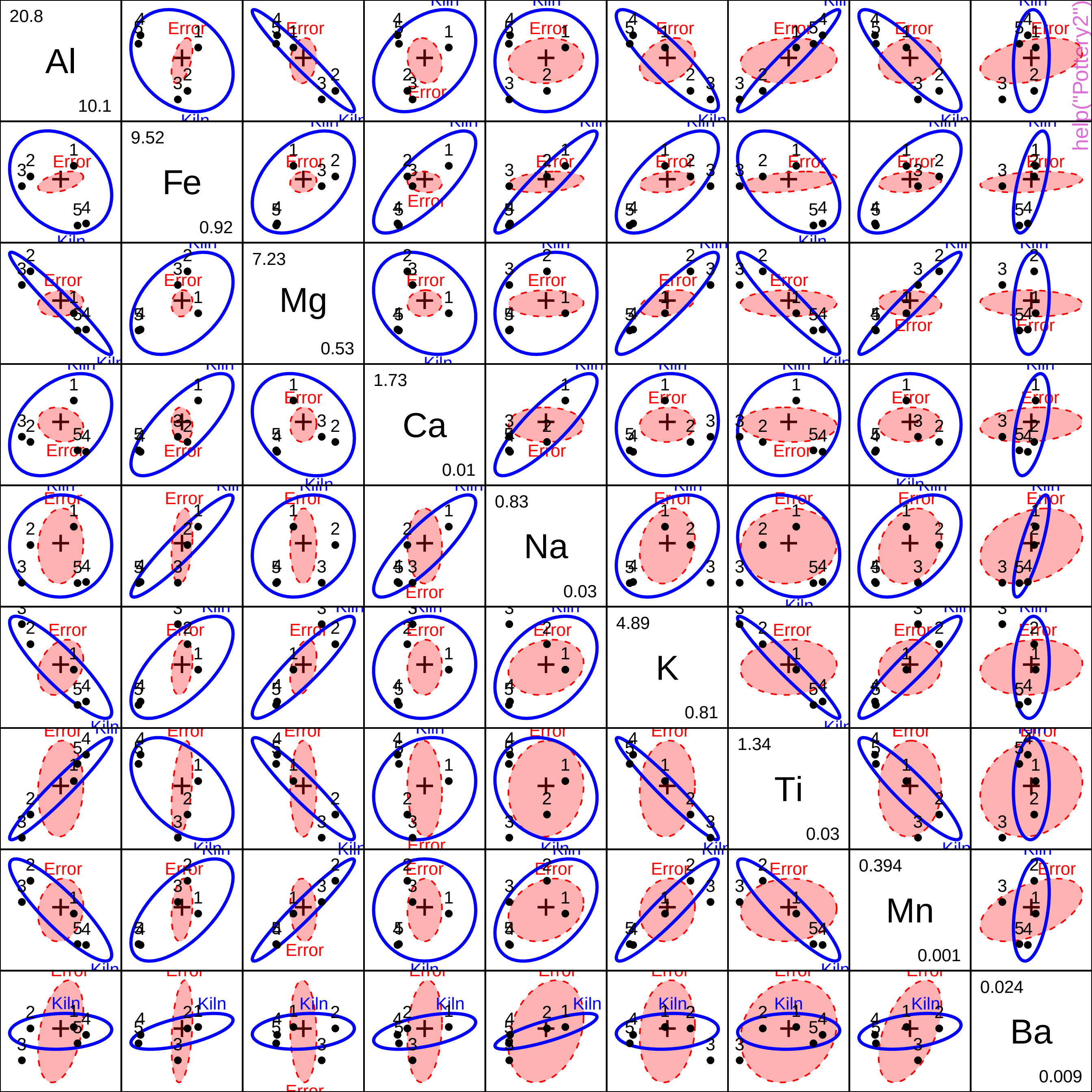
caring
11
1
emotion
13
1
play
11
0
Error
group
Physical Disability
Normal
Mental Disability
help("Parenting")

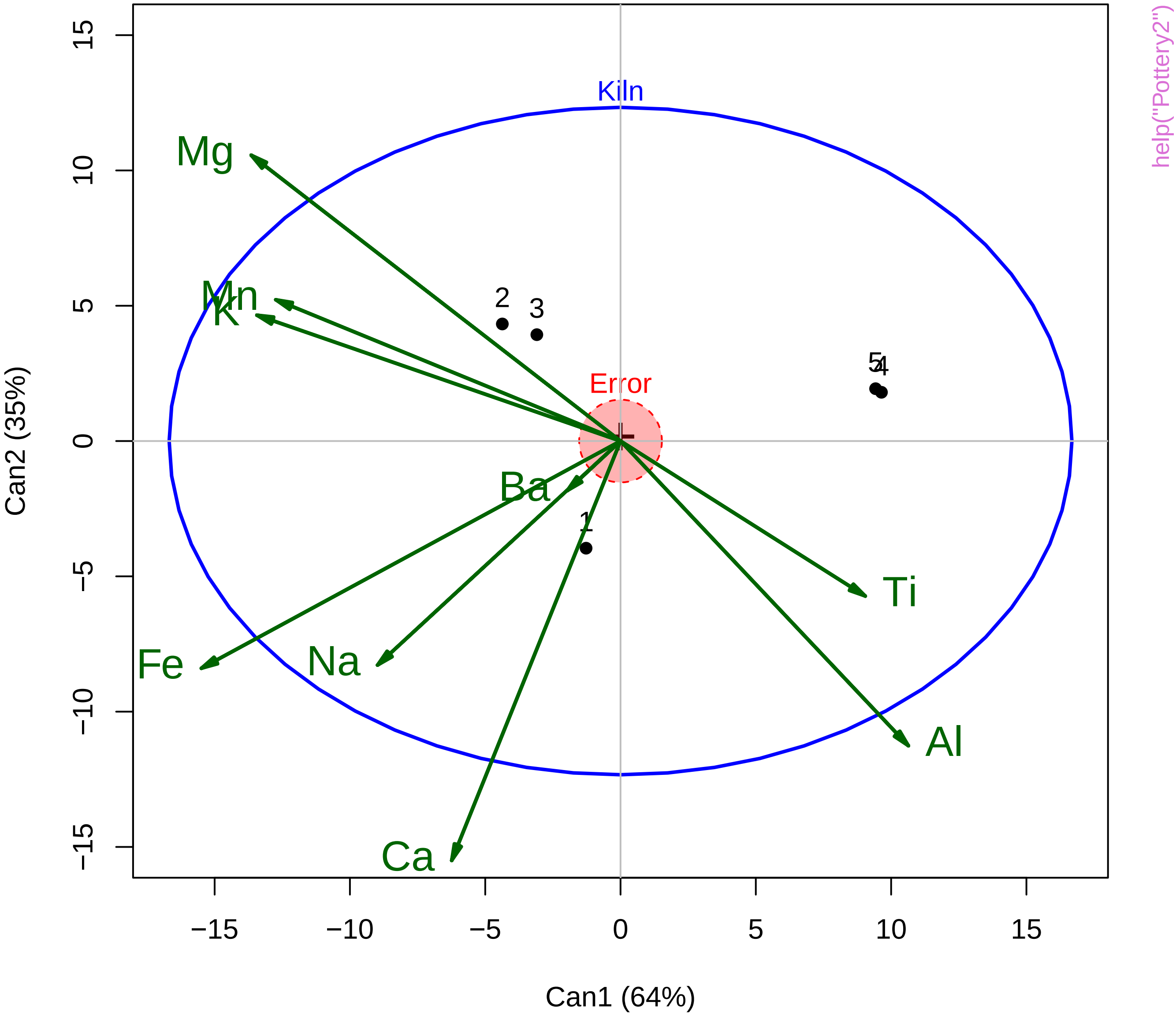


7.6
tear
5.8
Error
additive
High
High
rate:additive
Low
Low
rate
rate
additive
High
High
rate:additive
Low
Low
Error
help("Plastic")
Error
rate:additive
additive
Low
High
Low
High
rate
10.1
gloss
8.3
rate:additive
additive
Low
High
Low
High
rate
Error
rate:additive
additive
High
High
rate
Low
Low
Error
additive
rate:additive
High
High
Low
Low
rate
Error
8.4
opacity
0.8

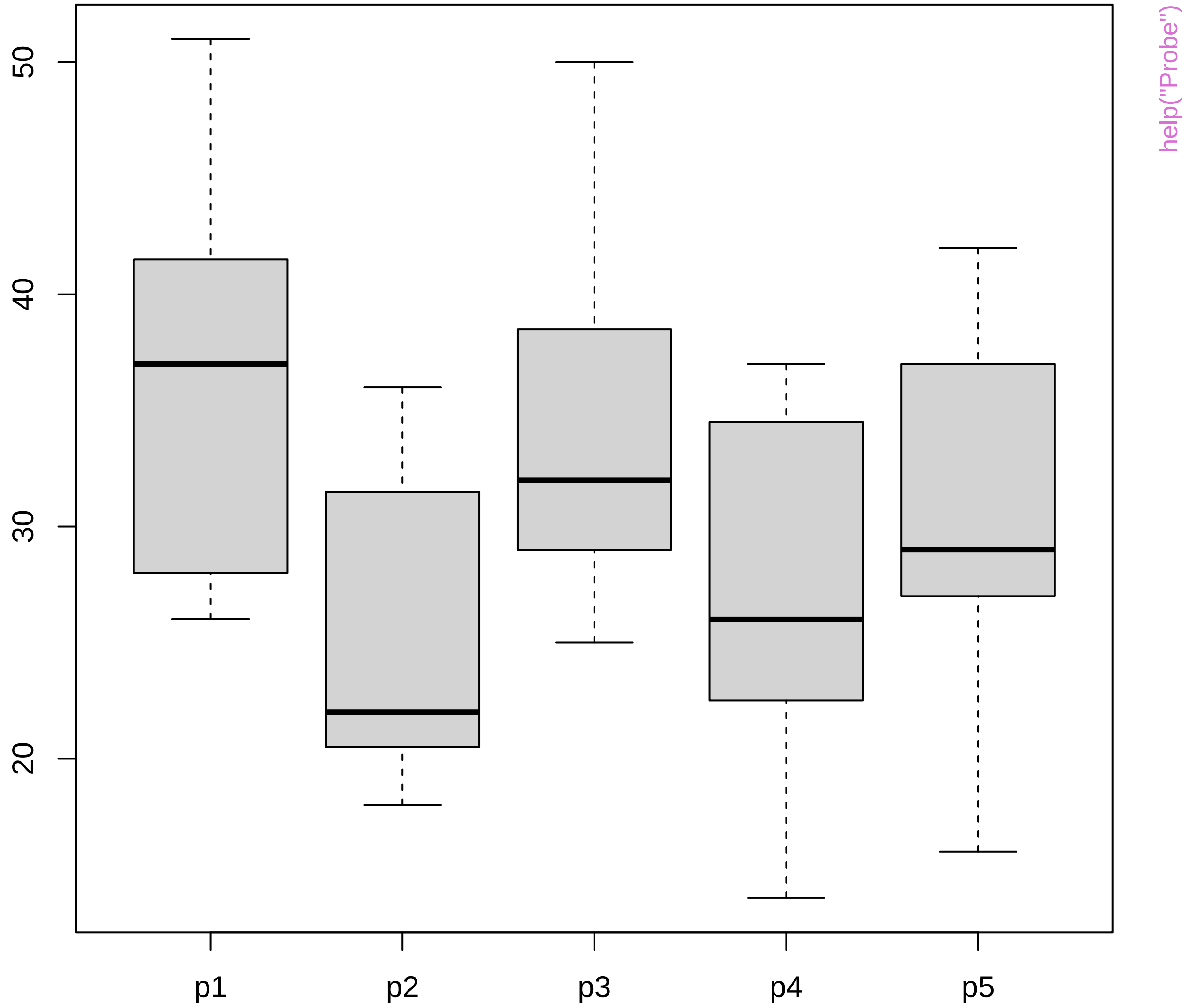


help("Pottery2")
1
2
3
4
5
Error
WithinBC
Region
Kiln
Fe
Al
2
4
6
8
10
12
14
16
18
20

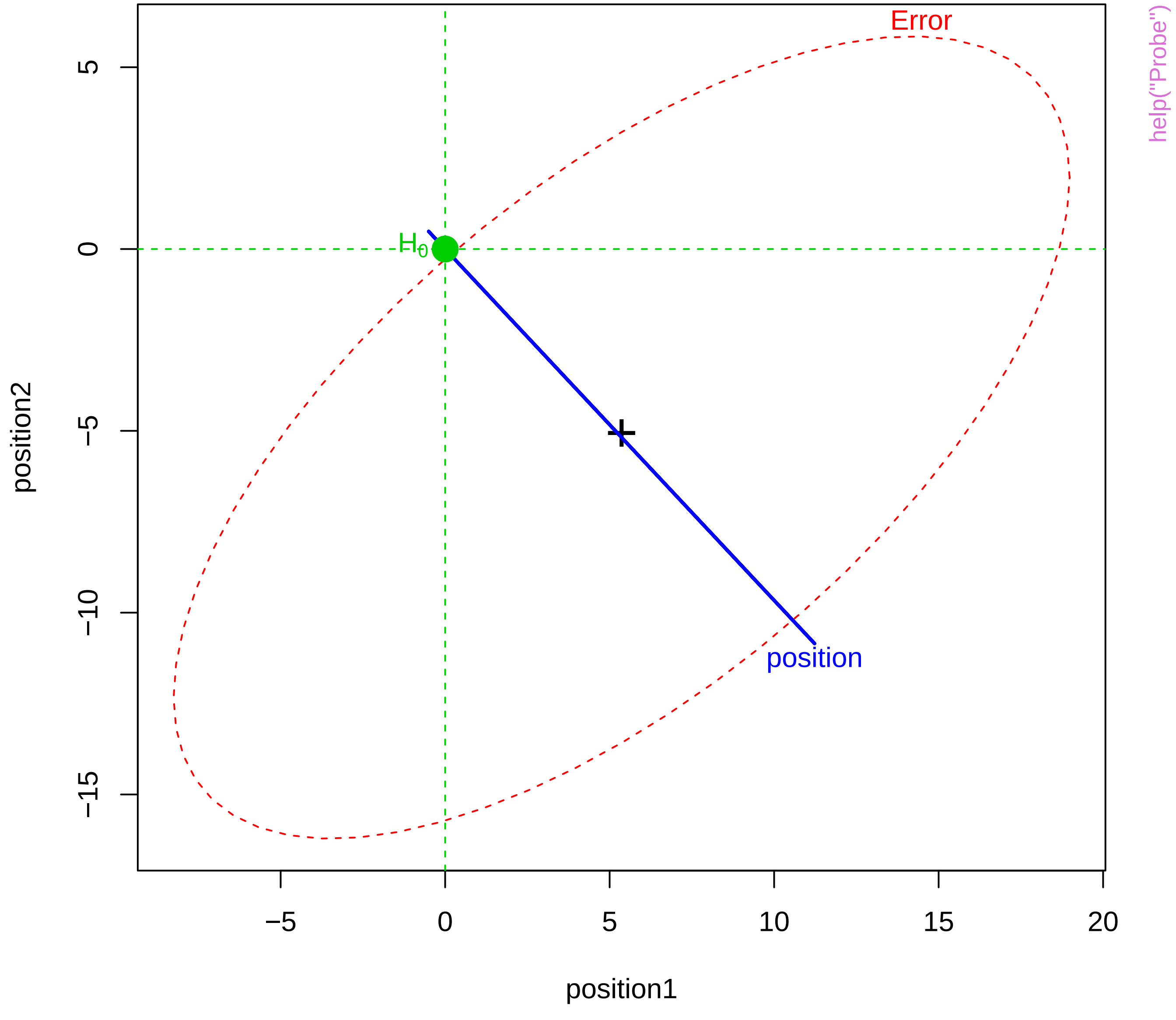

20.8
Al
10.1
9.52
Fe
0.92
7.23
Mg
0.53
1.73
Ca
0.01
0.83
Na
0.03
4.89
K
0.81
1.34
Ti
0.03
0.394
Mn
0.001
0.024
Ba
0.009
Error
Kiln
help("Pottery2")



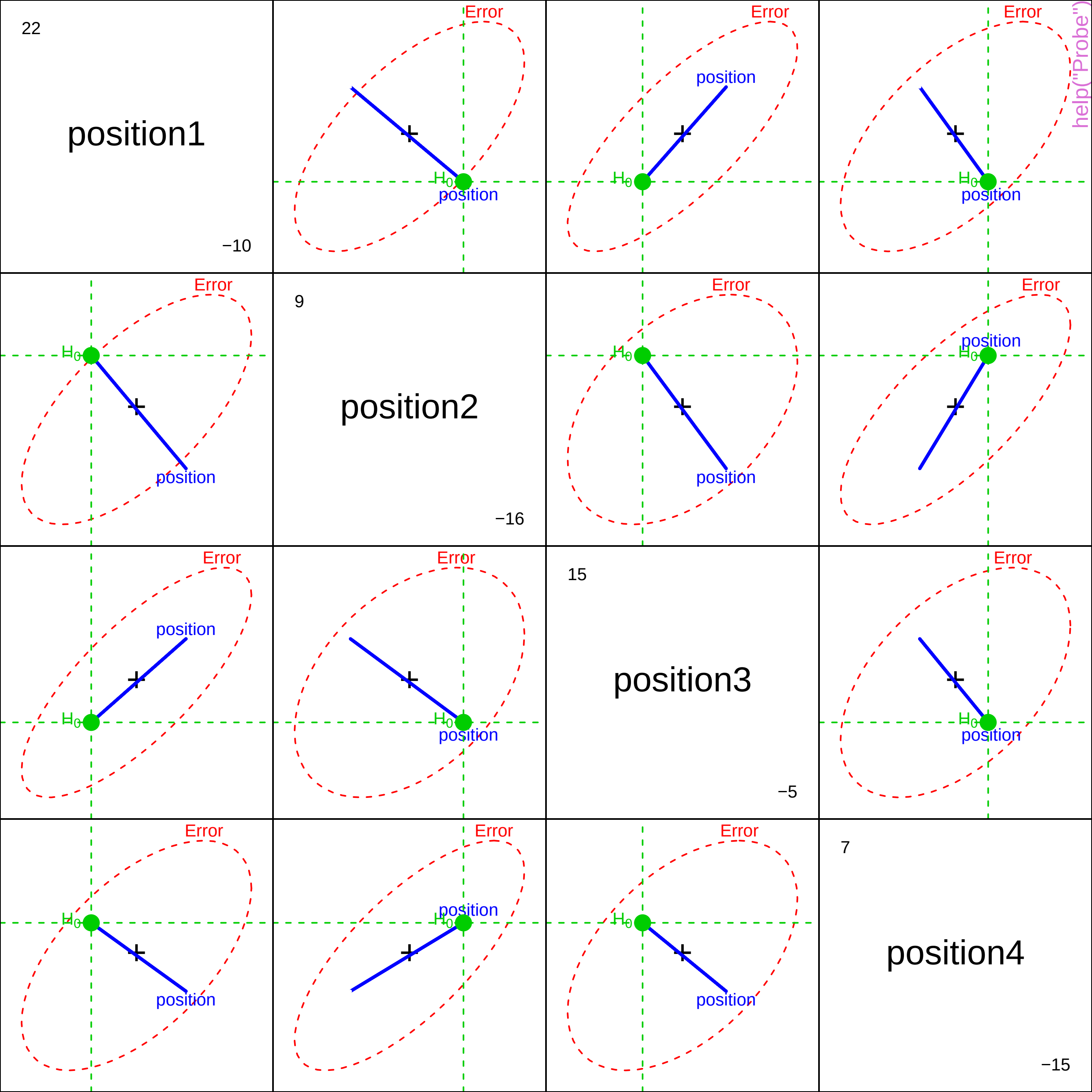
Kiln
Error
Mg
Mn
K
Ba
Fe
Na
Ca
Ti
Al
1
2
3
4
5
15
10
5
0
−5
−10
−15
Can2 (35%)
Can1 (64%)
help("Pottery2")



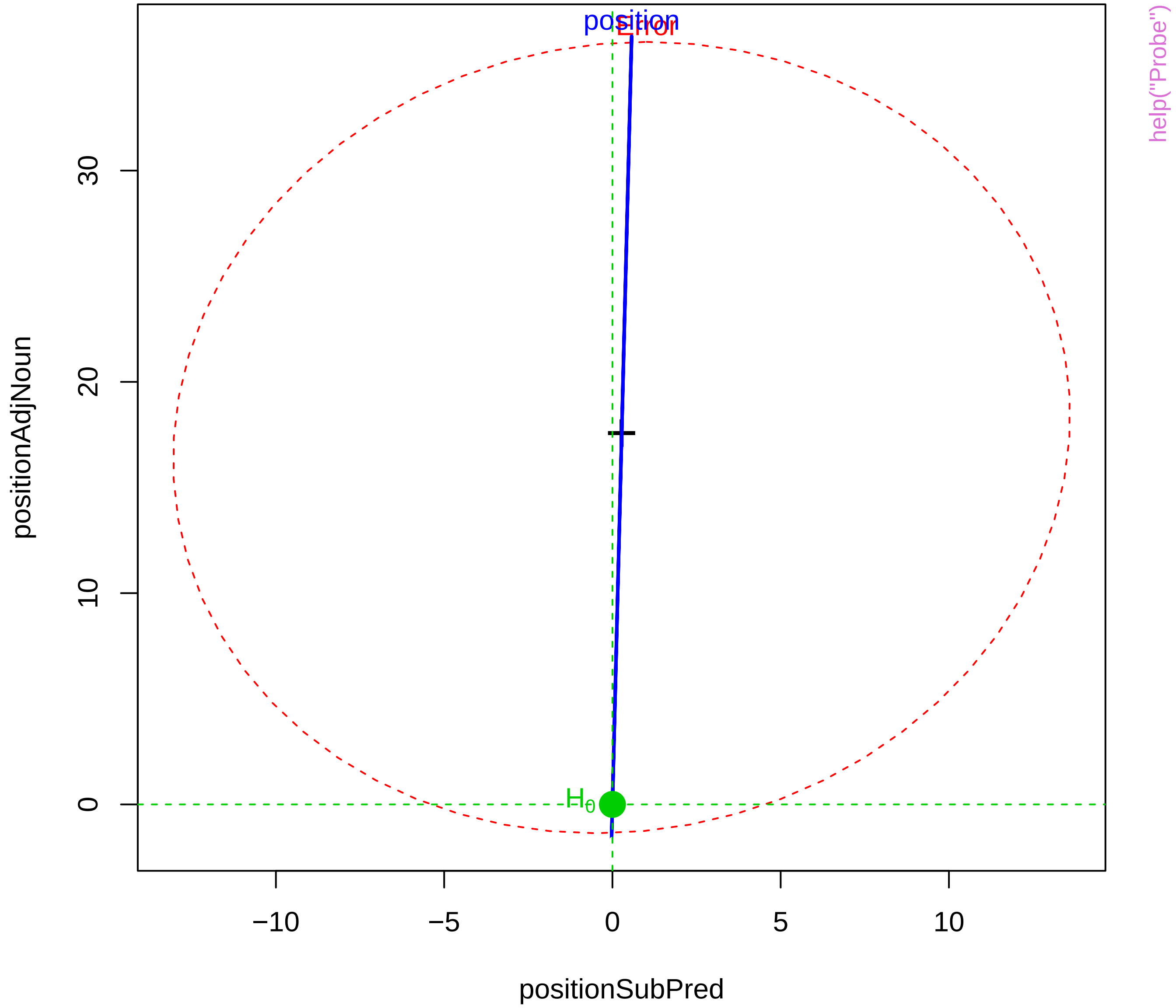
50
40
30
20
p1
p2
p3
p4
p5
help("Probe")



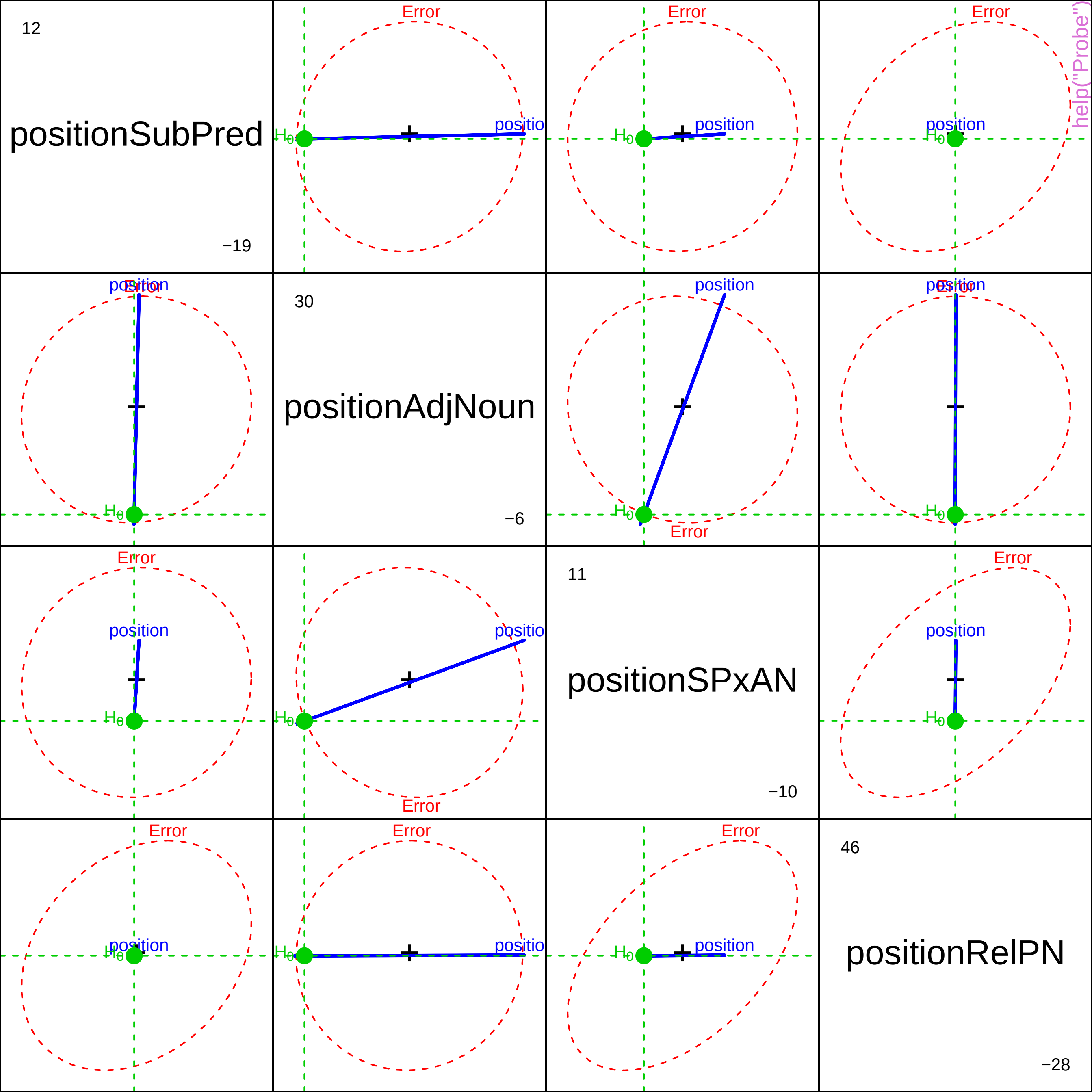
Error
help("Probe")
H0
position
position1
position2
5
0
−5
−10
−15
−5
0
5
10
15
20



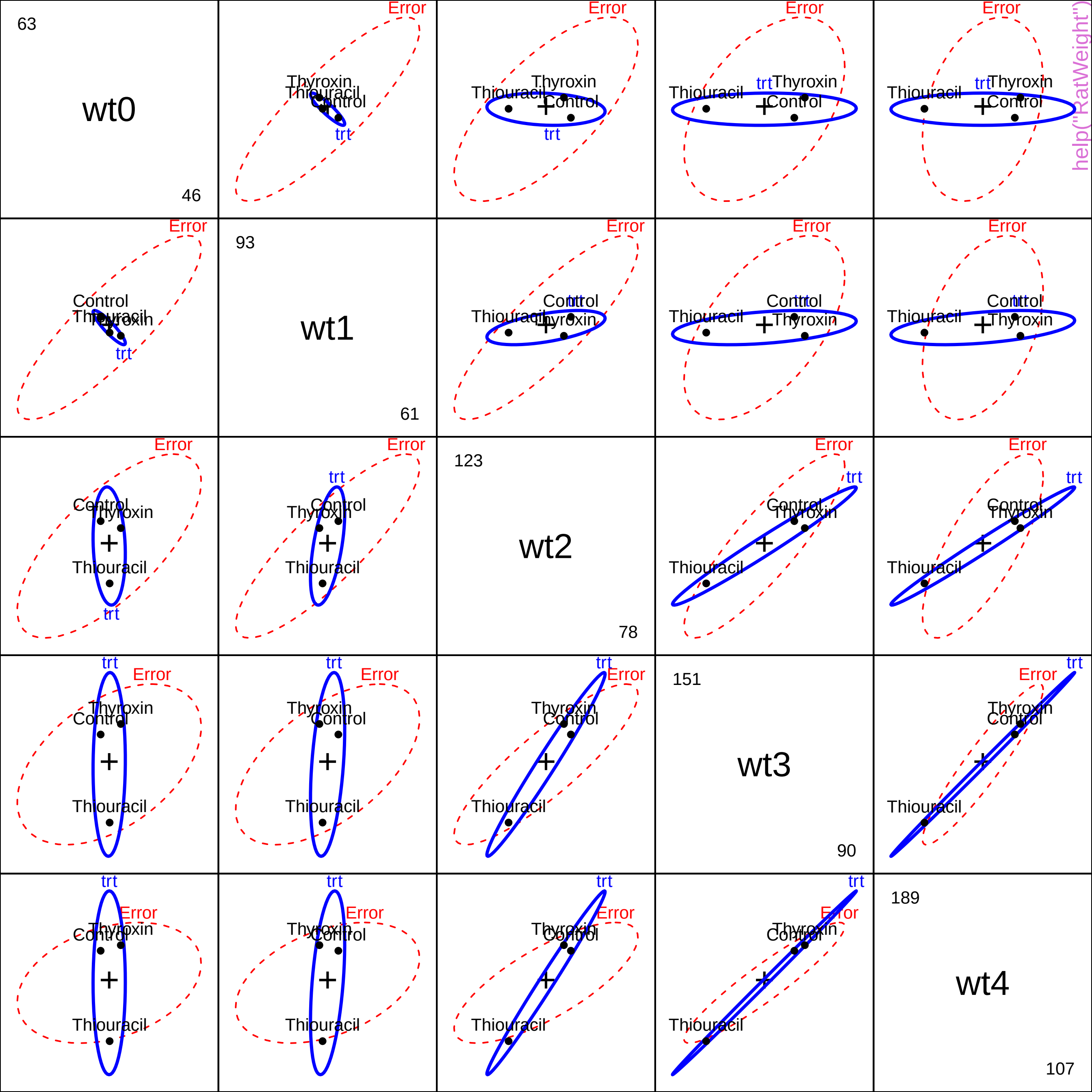
position1
22
−10
position2
9
−16
position3
15
−5
position4
7
−15
Error
position
H_0
help("Probe")



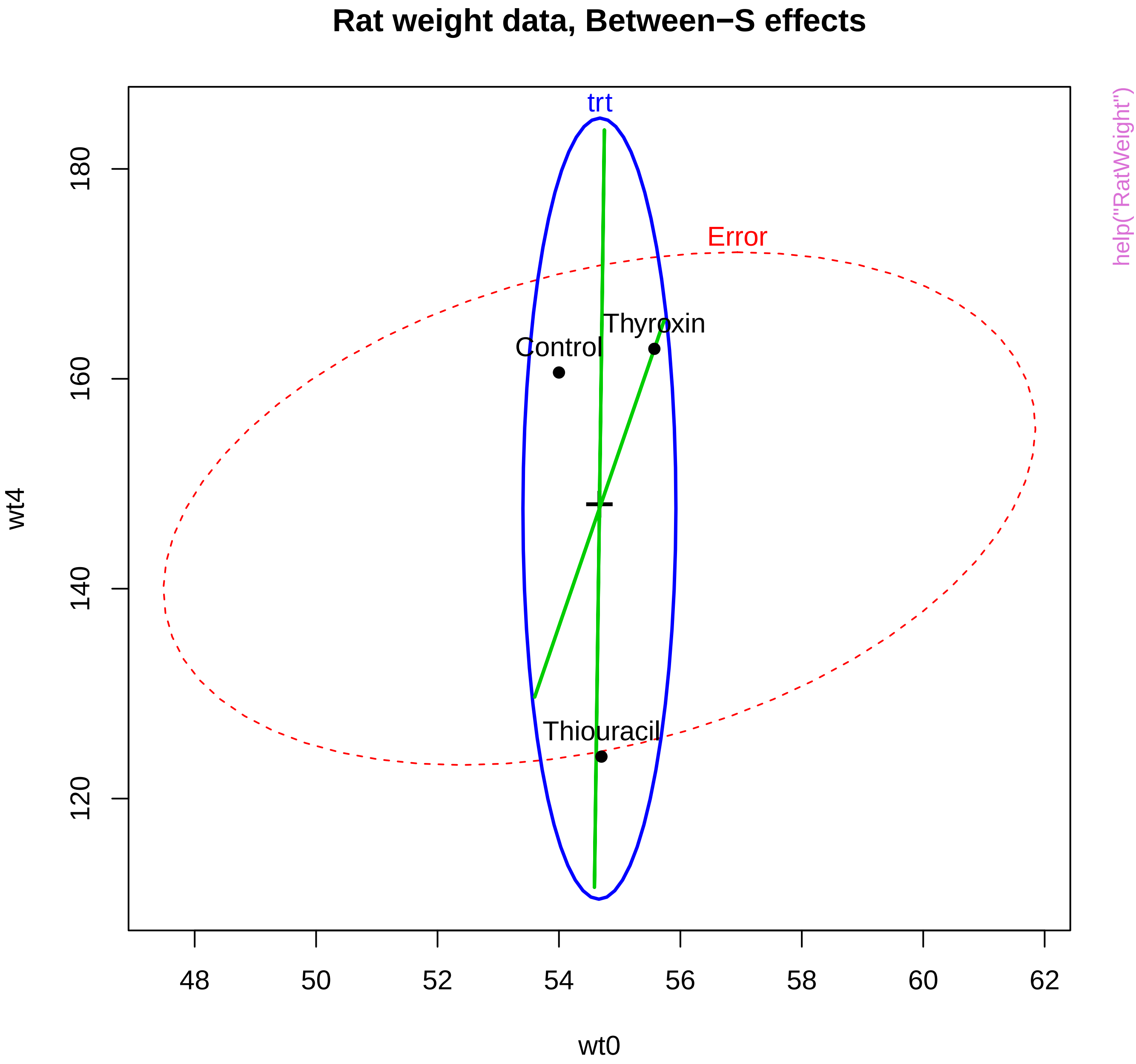
position
Error
H0
help("Probe")
positionAdjNoun
positionSubPred
0
10
20
30
−10
−5
0
5
10

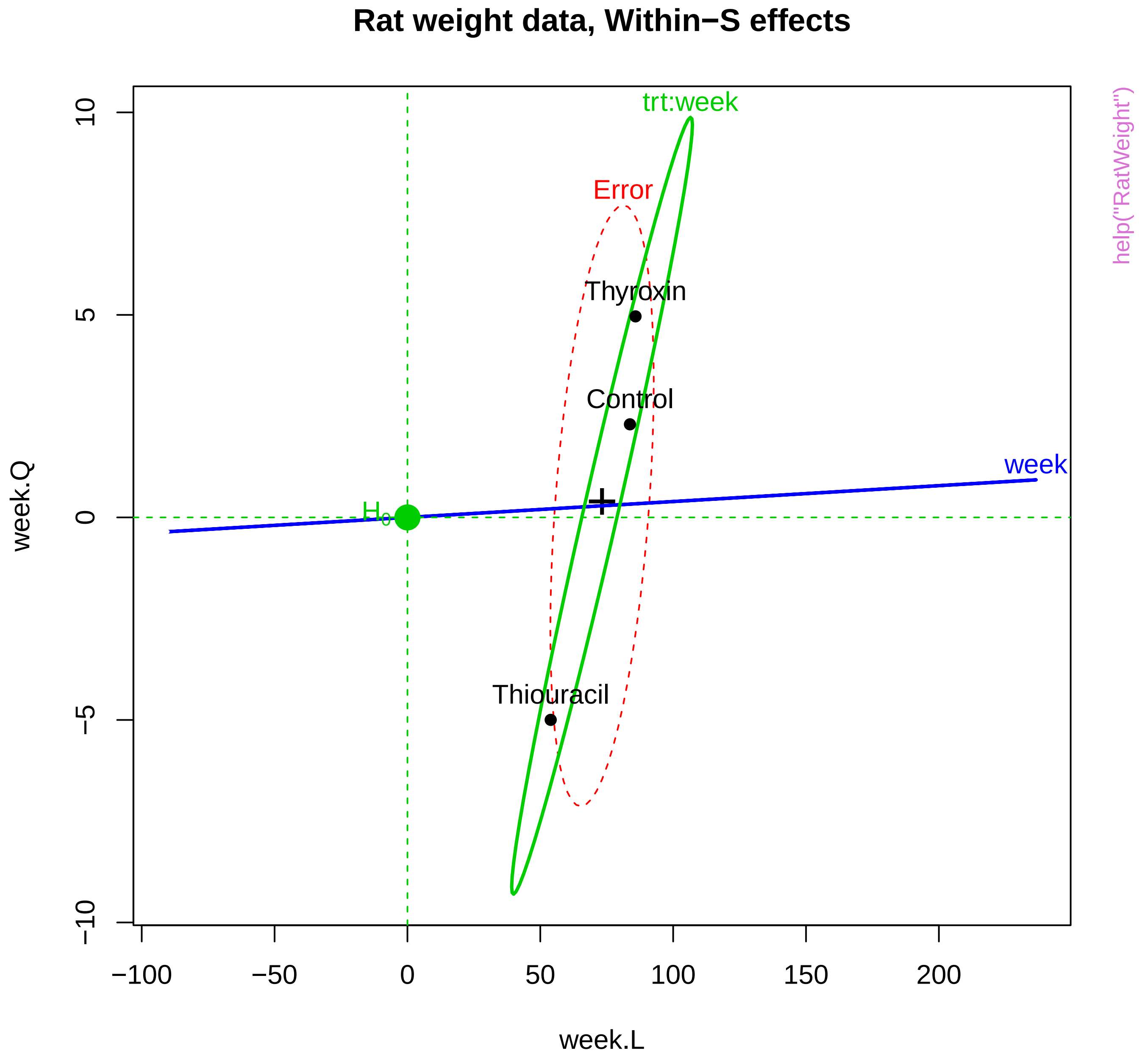


12
positionSubPred
−19
Error
position
H0
help("Probe")
30
positionAdjNoun
−6
11
positionSPxAN
−10
46
positionRelPN
−28

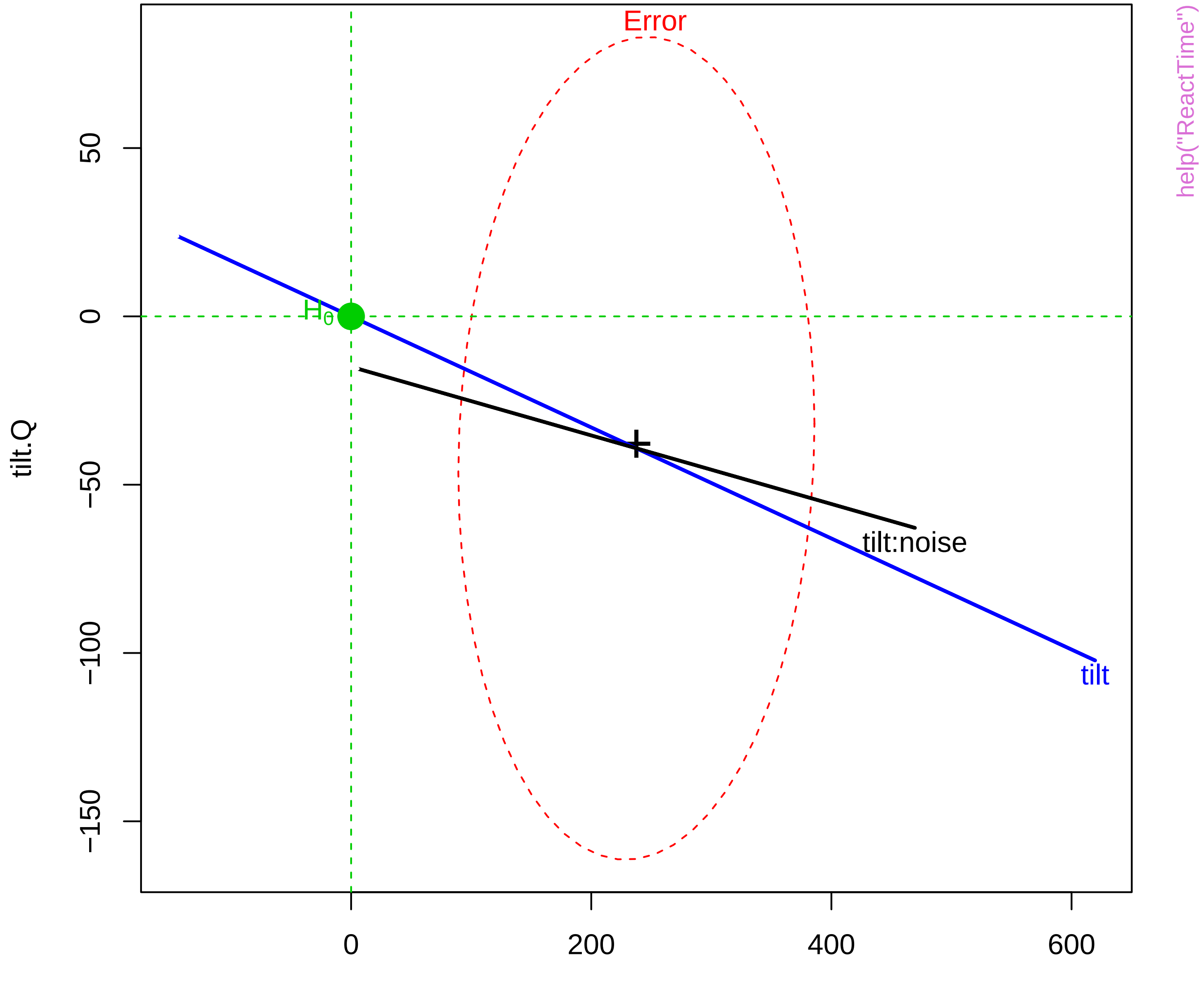


63
wt0
46
93
wt1
61
123
wt2
78
151
wt3
90
189
wt4
107
Error
trt
Thyroxin
Thiouracil
Control
help("RatWeight")

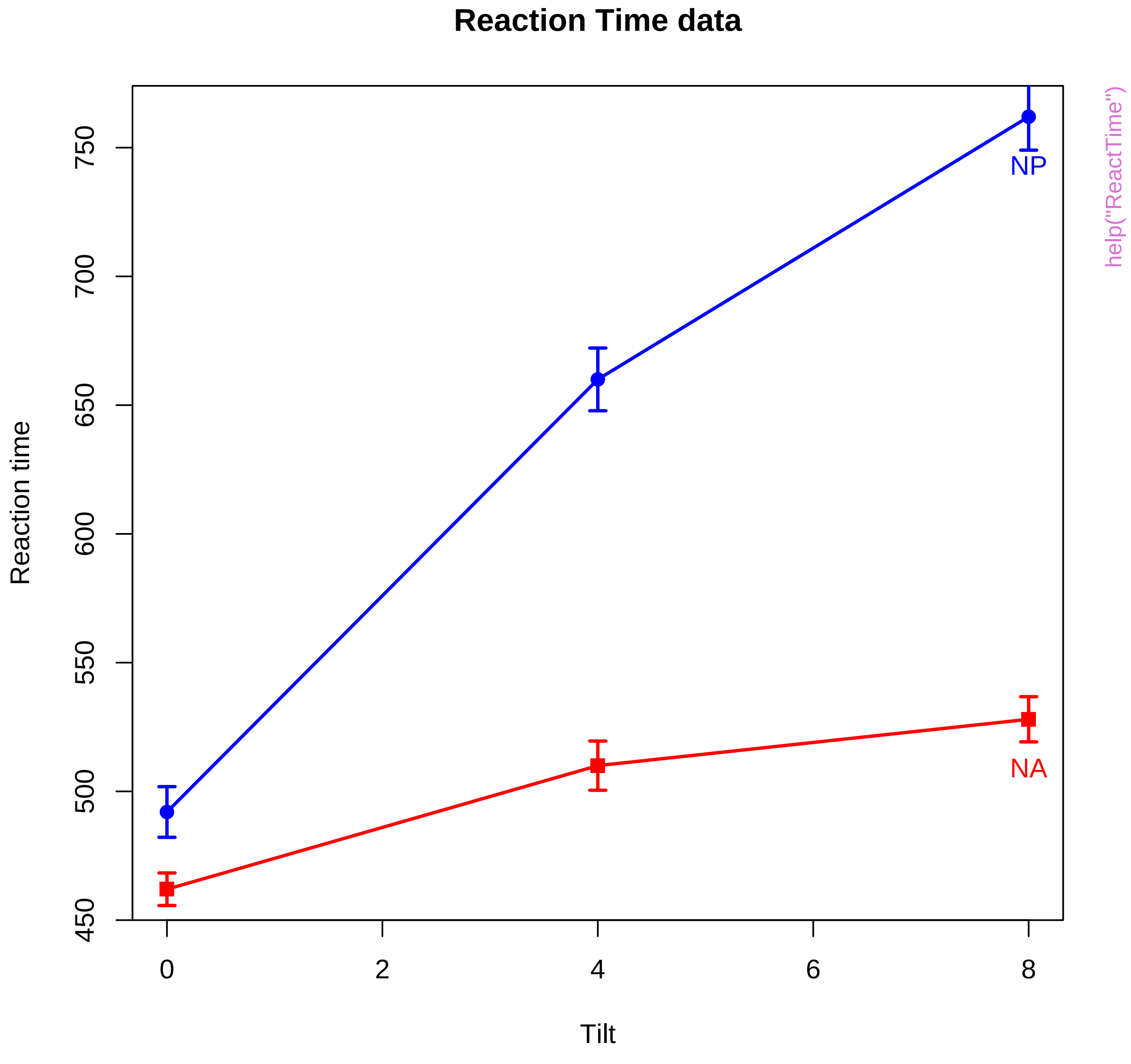

Rat weight data, Between−S effects
trt
Error
Thyroxin
Control
Thiouracil
180
160
140
120
wt4
48
50
52
54
56
58
60
62
wt0
help("RatWeight")

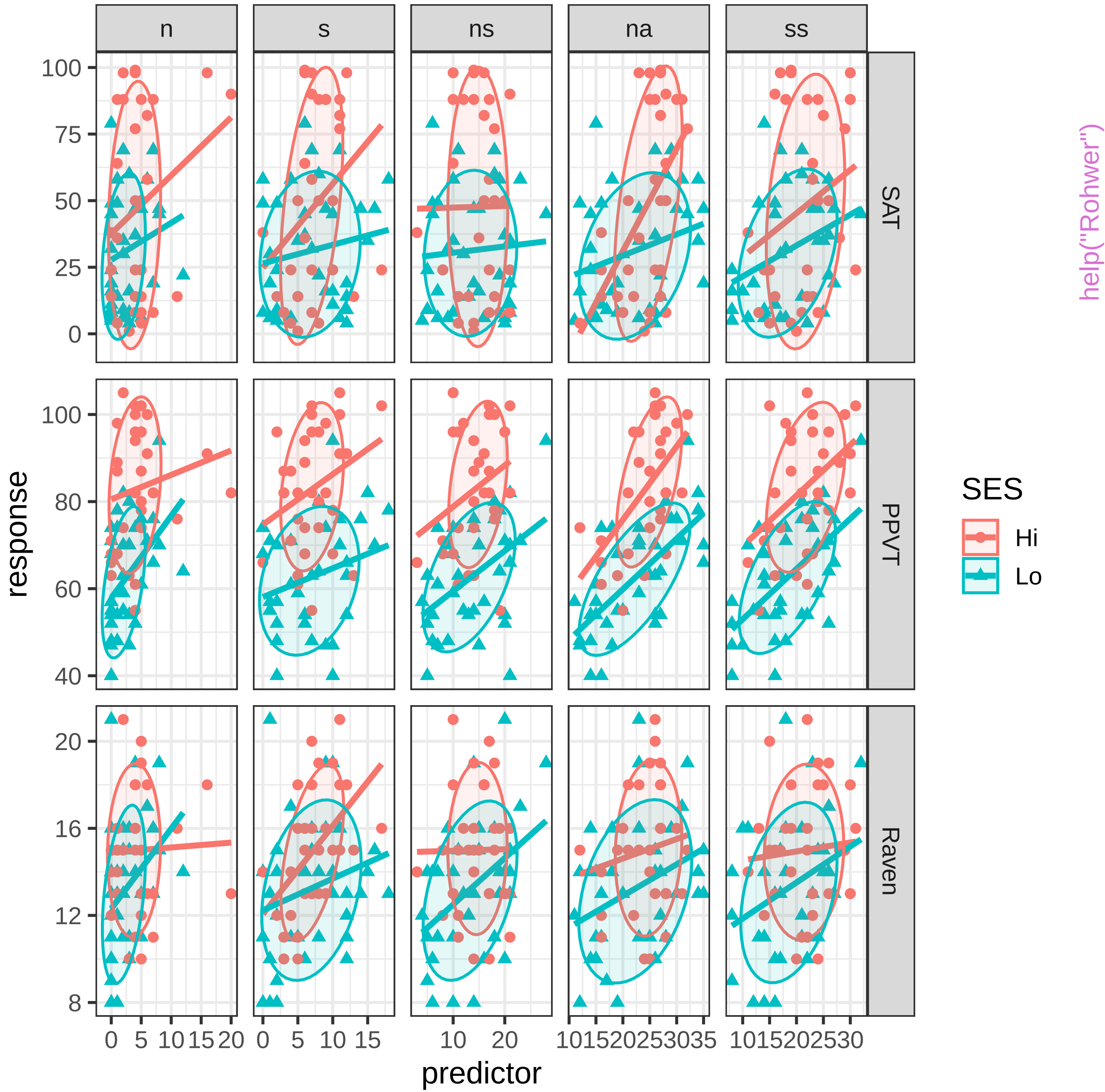


Rat weight data, Within−S effects
trt:week
Error
Thyroxin
Control
week
H0
Thiouracil
10
5
0
−5
−10
−100
−50
0
50
100
150
200
week.Q
week.L
help("RatWeight")

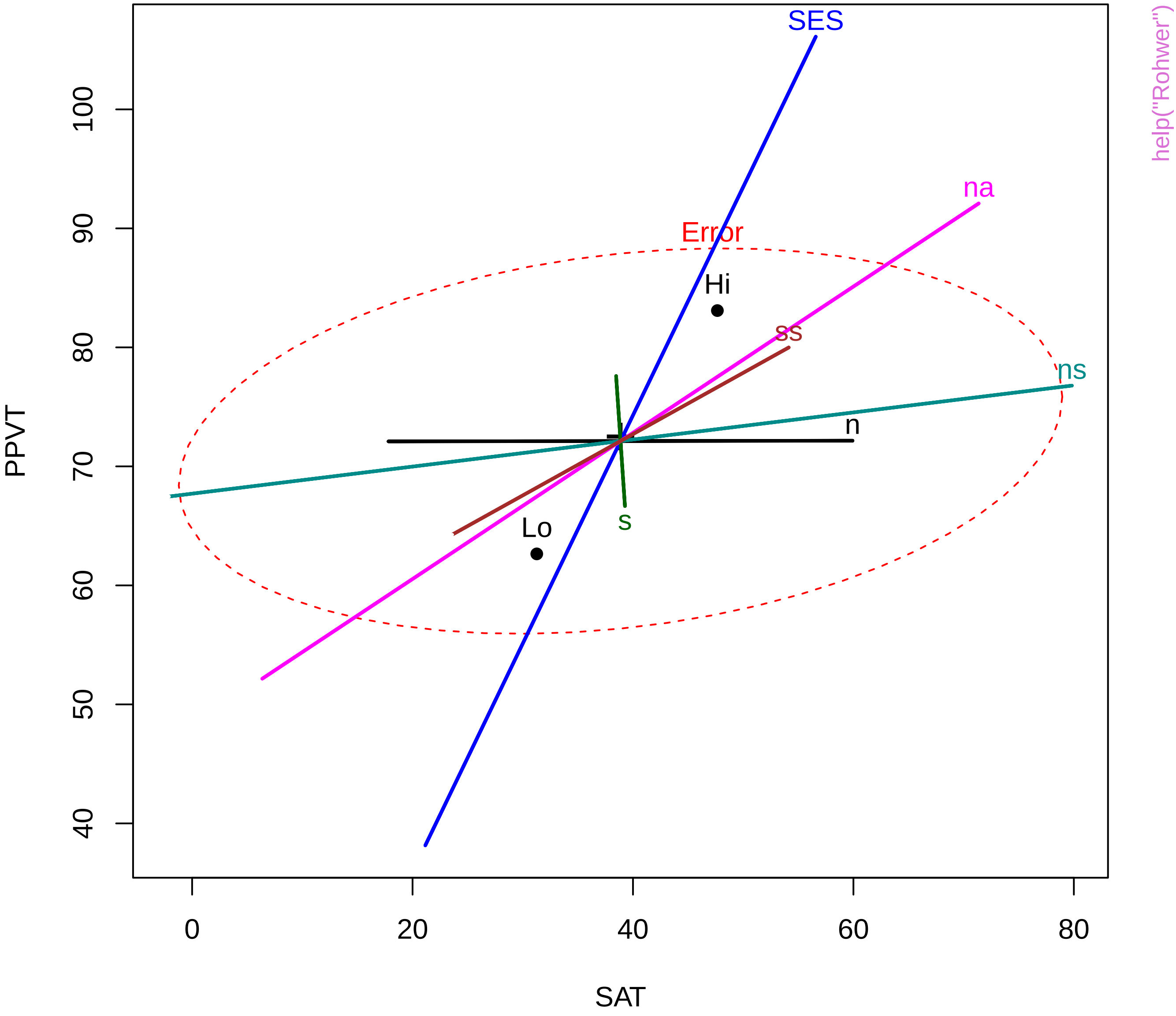


Error
H0
tilt:noise
tilt
50
0
−50
−100
−150
0
200
400
600
tilt.L
tilt.Q
help("ReactTime")

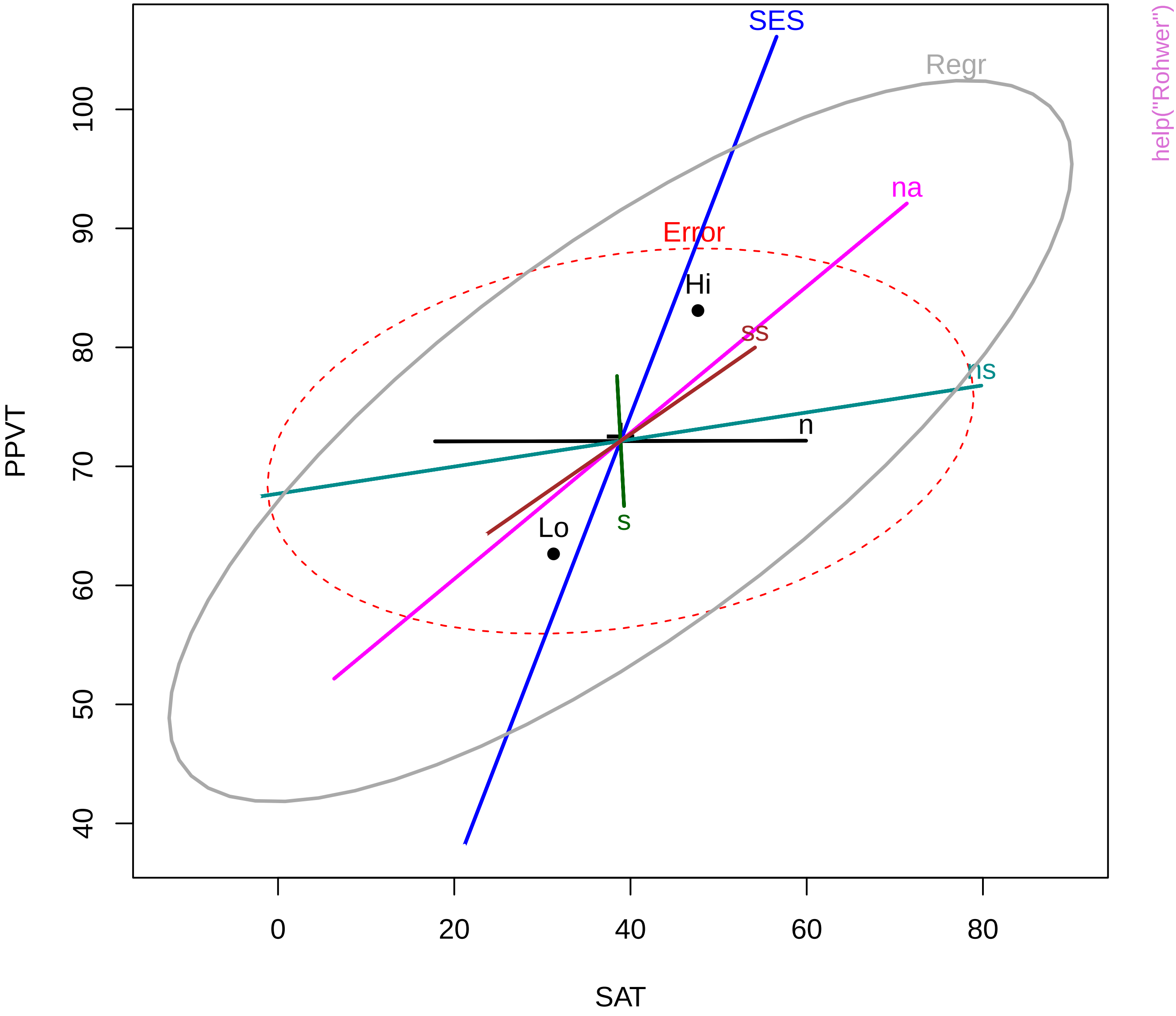

Reaction Time data
Reaction time
450
500
550
600
650
700
750
Tilt
0
2
4
6
8
NP
NA
help("ReactTime")



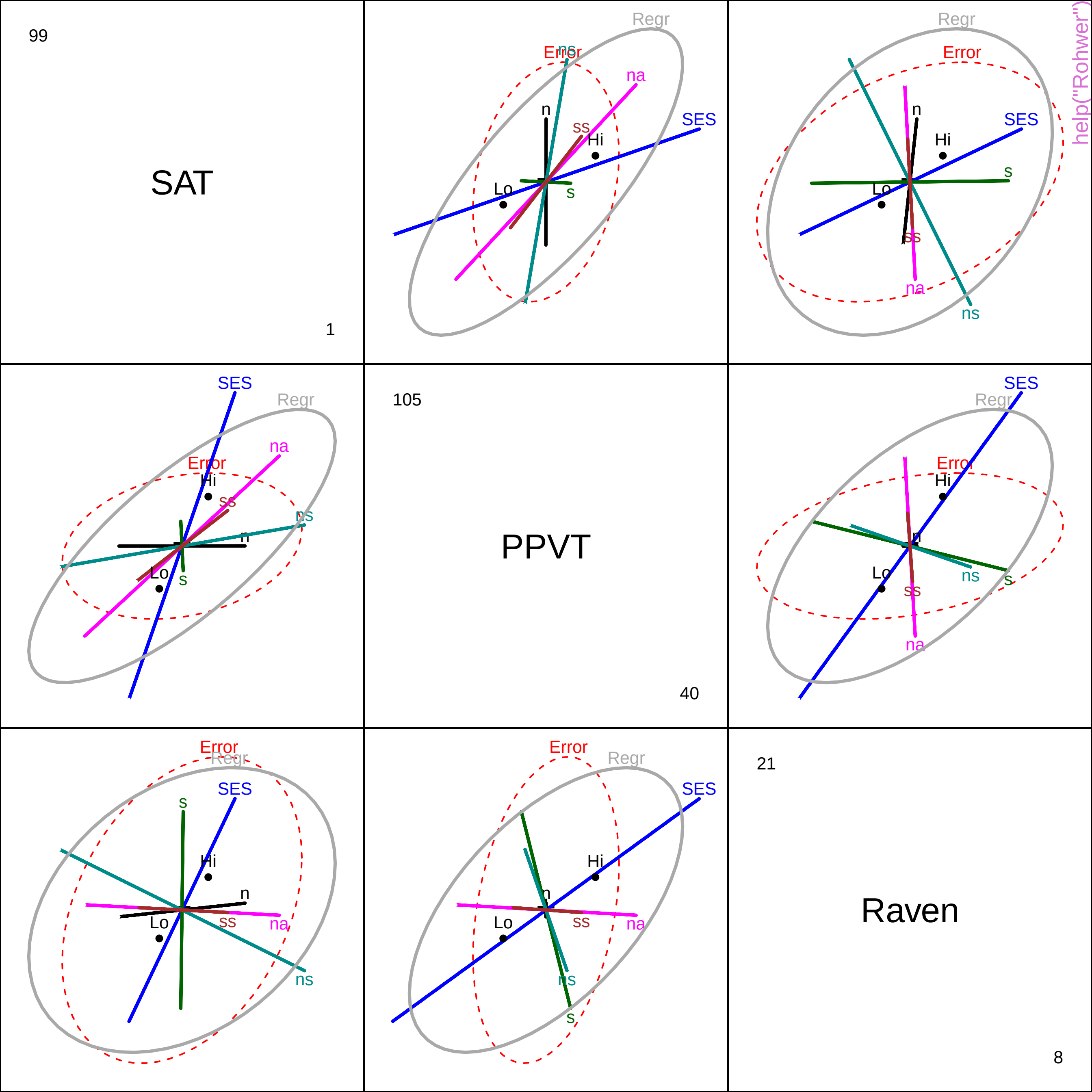
n
s
ns
na
ss
SAT
PPVT
Raven
response
predictor
SES
Hi
Lo
help("Rohwer")

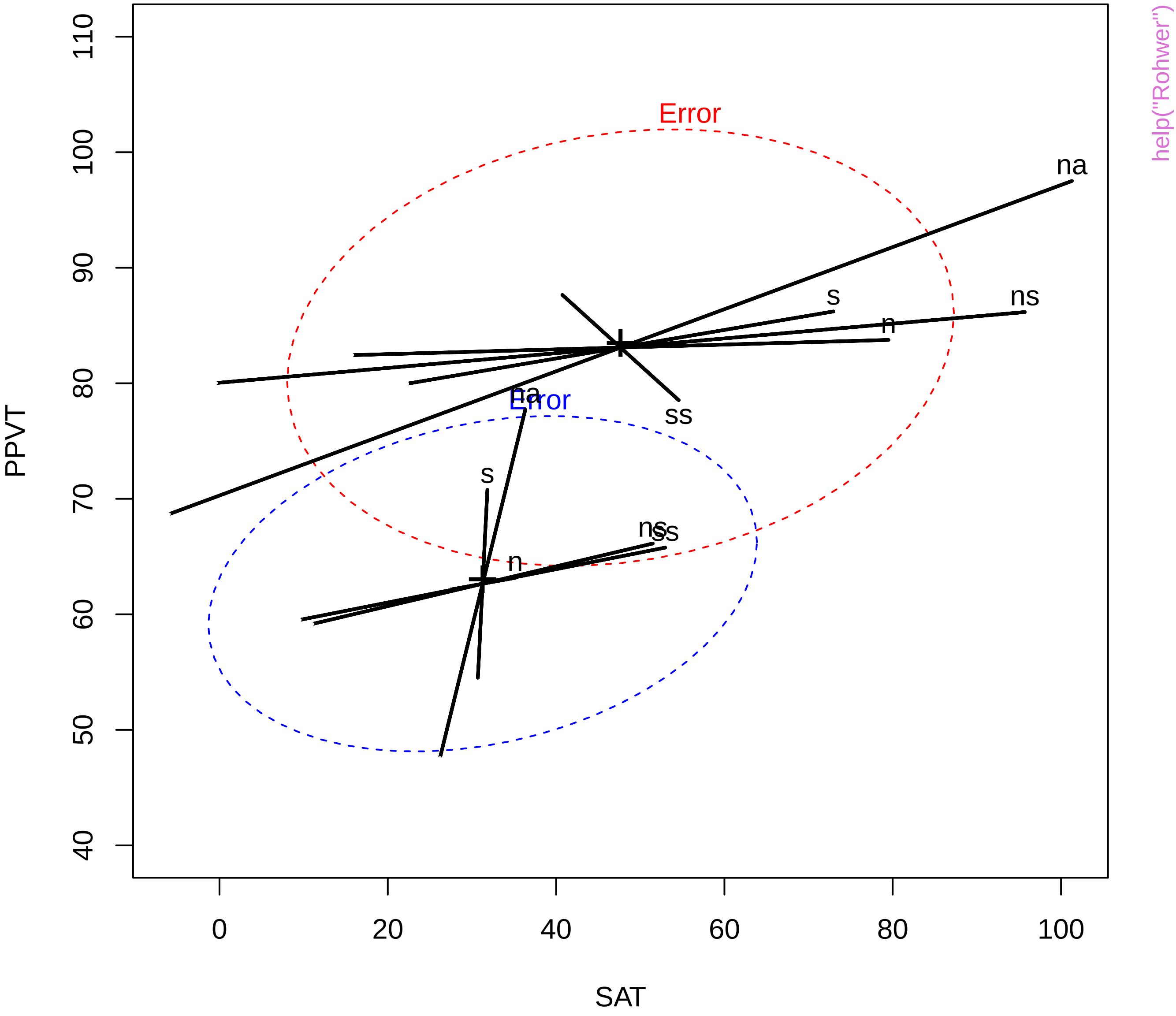


SES
na
Error
Hi
ss
ns
n
s
Lo
PPVT
SAT
100
90
80
70
60
50
40
0
20
40
60
80
help("Rohwer")

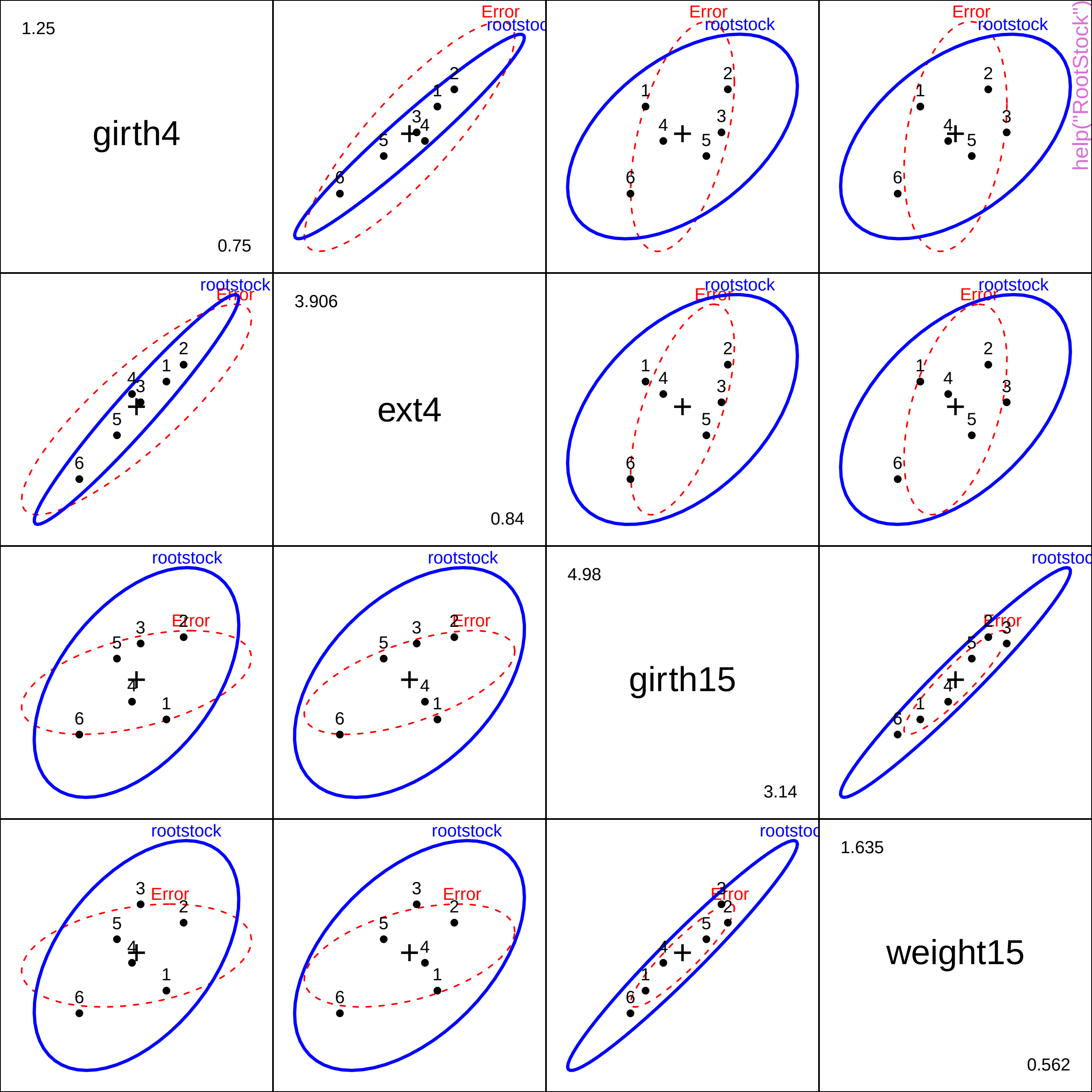


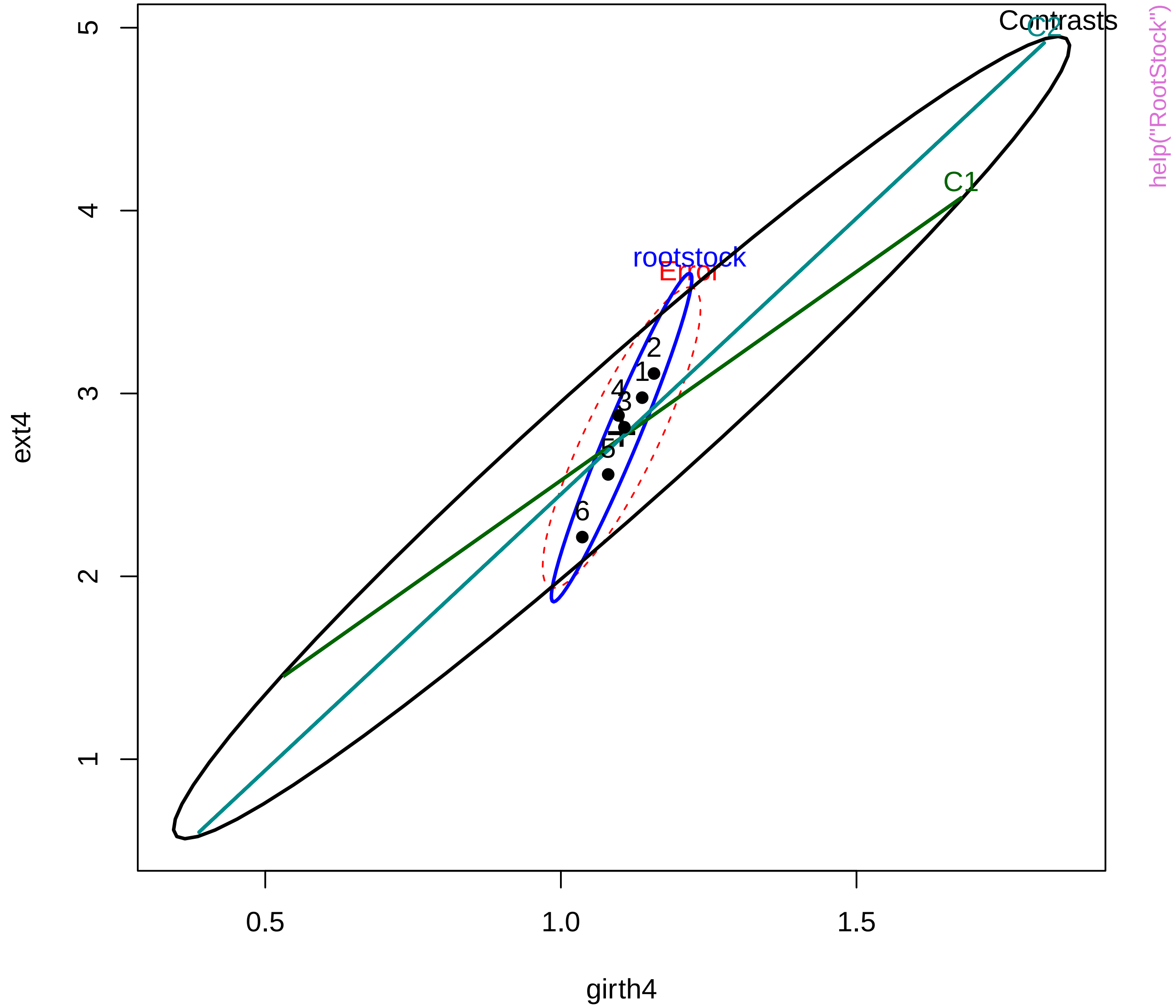
SES
Regr
na
Error
Hi
ss
ns
n
s
Lo
100
90
80
70
60
50
40
PPVT
0
20
40
60
80
SAT
help("Rohwer")


99
SAT
1
Regr
Error
ns
na
n
ss
SES
Hi
Lo
s
help("Rohwer")
105
PPVT
40
21
Raven
8

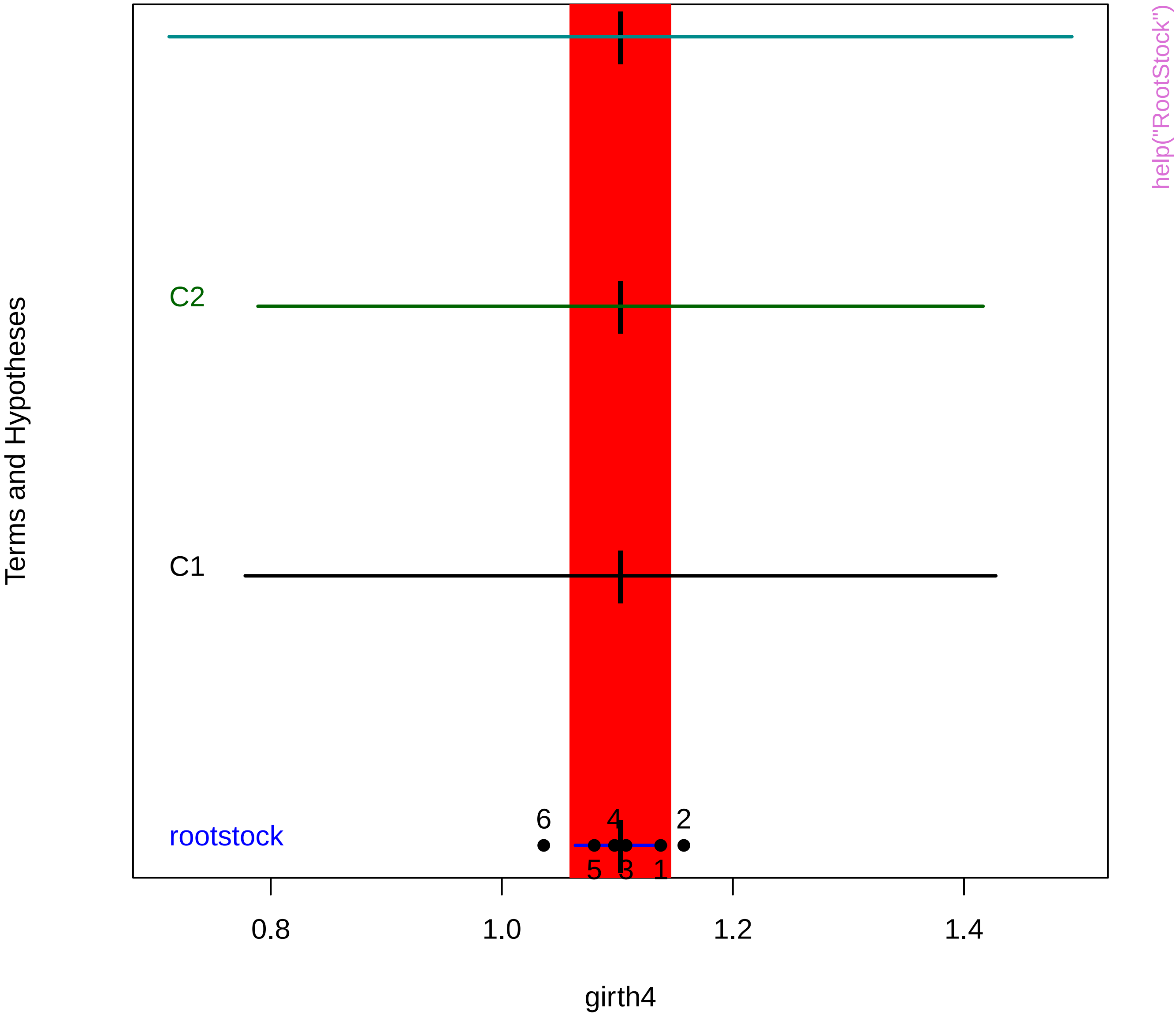


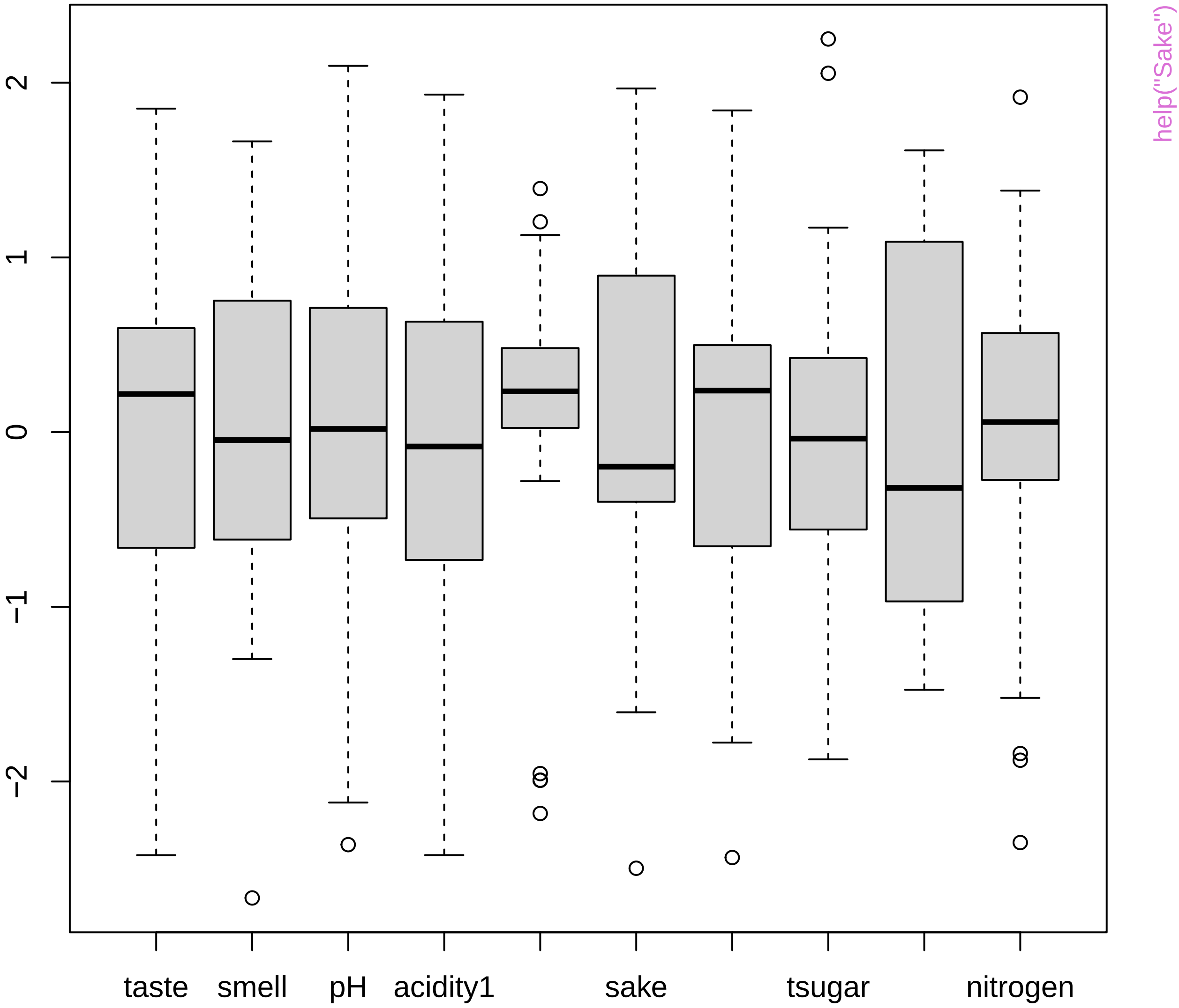
Error
Error
na
s
ns
n
ss
na
s
ns
ss
n
PPVT
SAT
help("Rohwer")
0
20
40
60
80
100
40
50
60
70
80
90
100
110

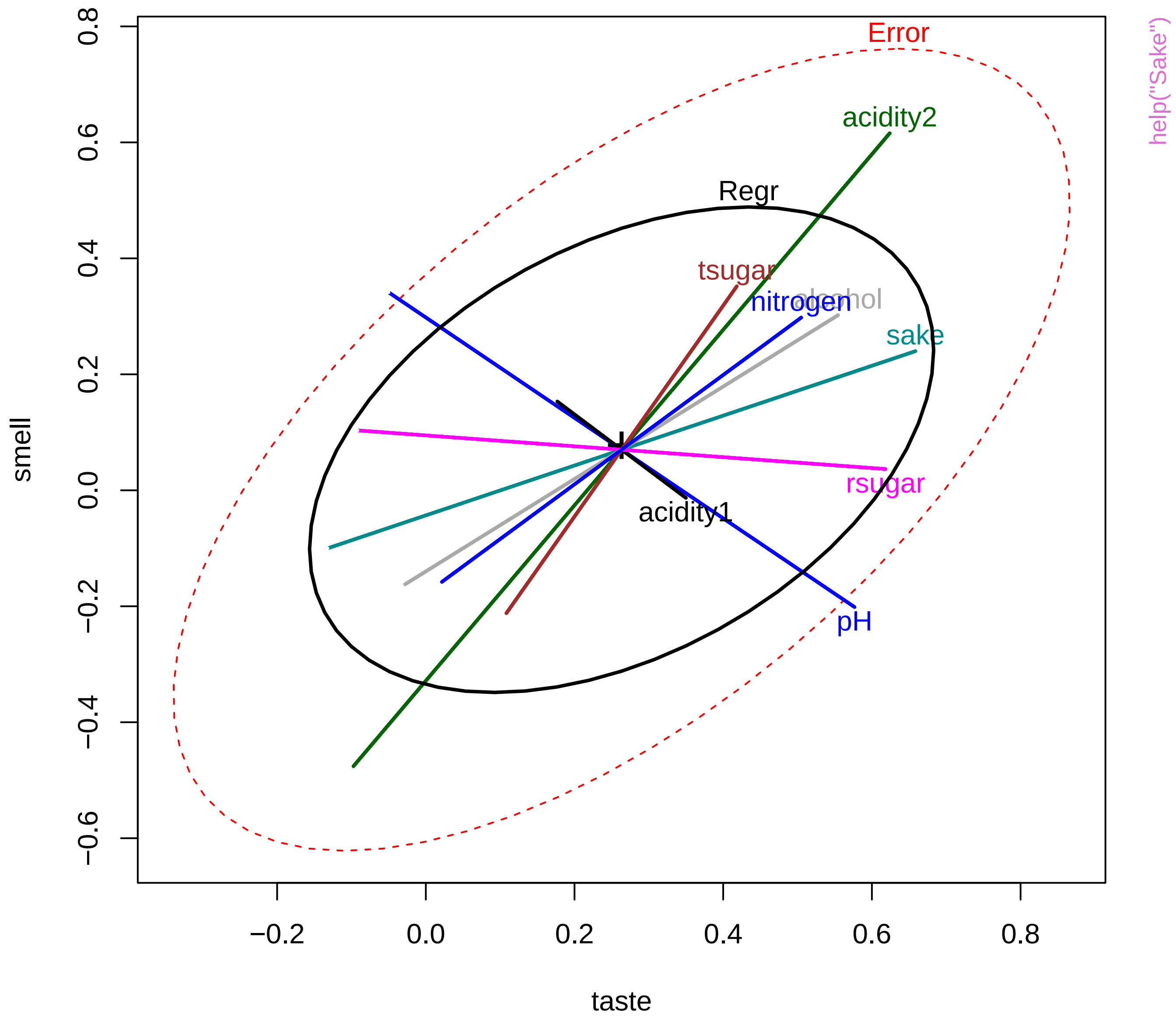

girth4
ext4
girth15
weight15
1.25
0.75
3.906
0.84
4.98
3.14
1.635
0.562
rootstock
Error
help("RootStock")

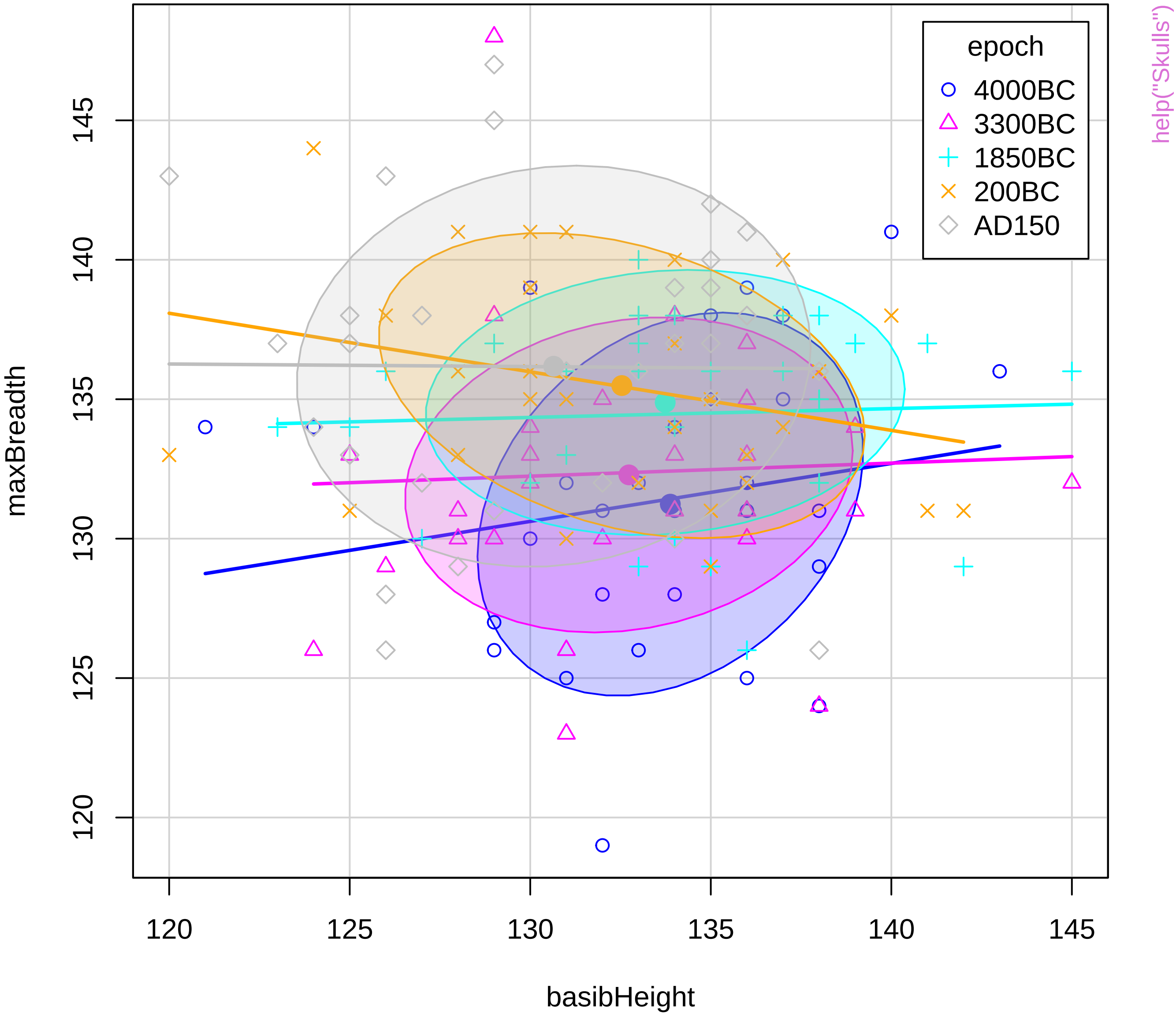


Contrasts
C2
C1
rootstock
Error
1
2
3
4
5
6
help("RootStock")
ext4
girth4
0.5
1.0
1.5
1
2
3
4
5

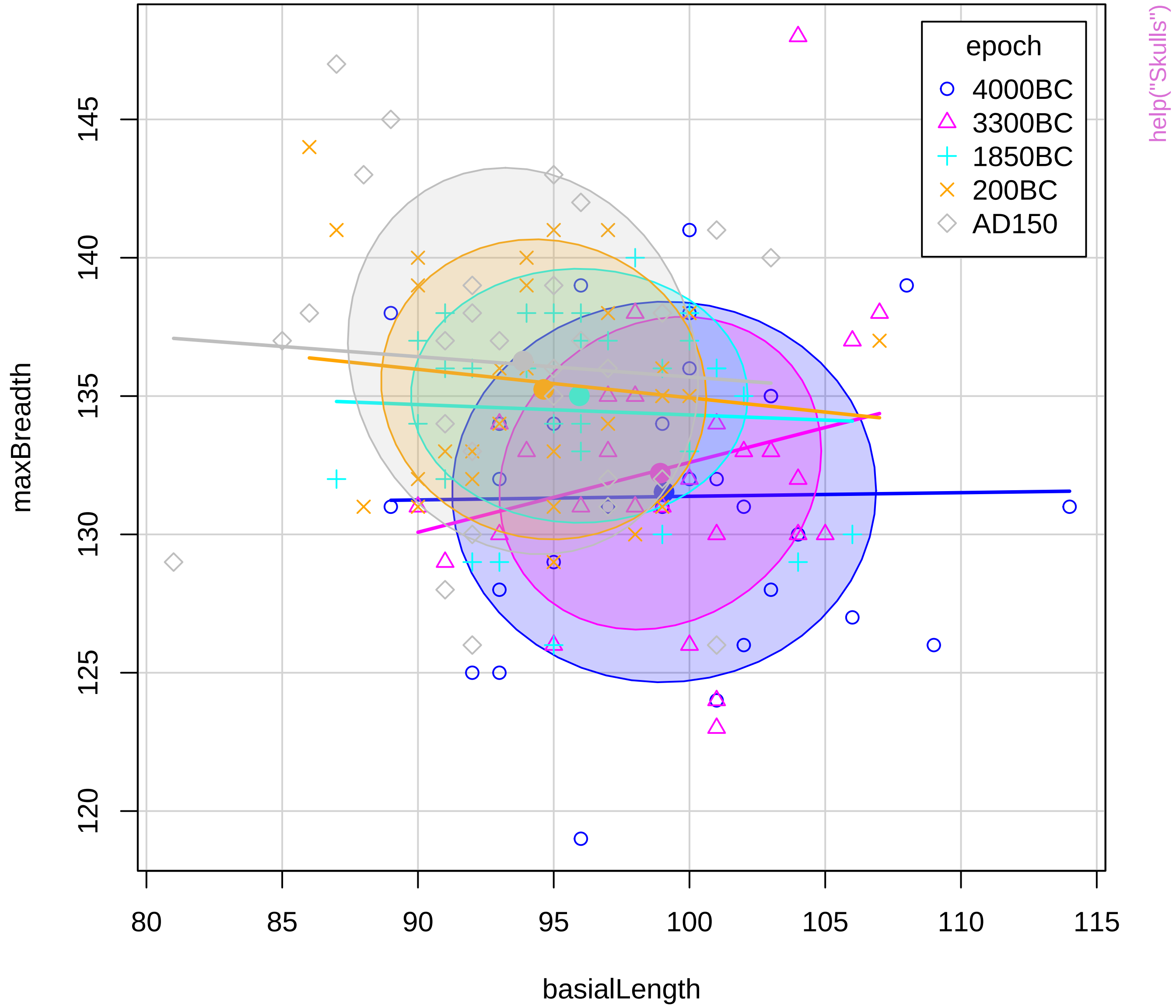

help("RootStock")
C2
C1
rootstock
6
4
2
5
3
1
0.8
1.0
1.2
1.4
girth4
Terms and Hypotheses


2
1
0
−1
−2
taste
smell
pH
acidity1
sake
tsugar
nitrogen
help("Sake")

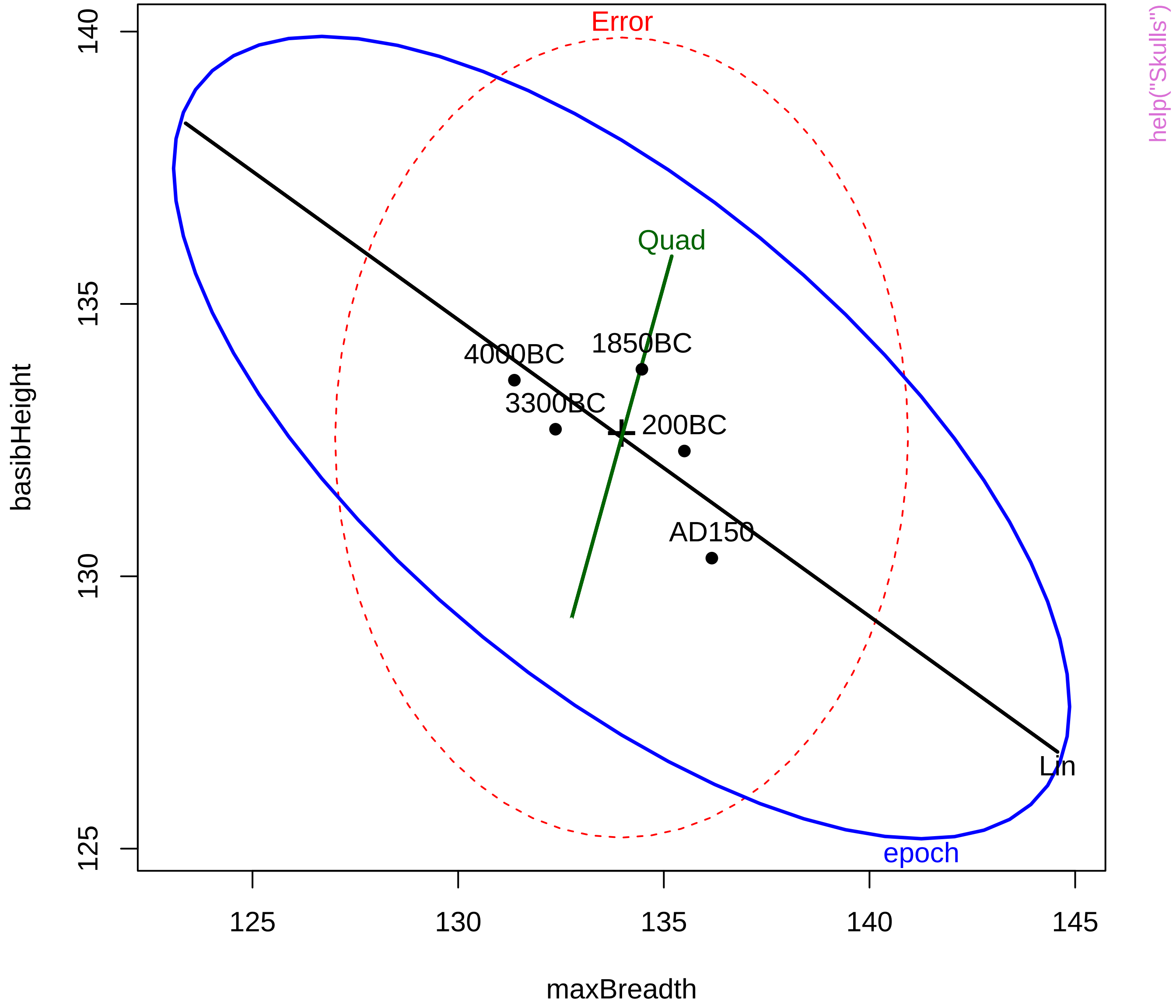

Error
acidity2
Regr
tsugar
alcohol
nitrogen
sake
rsugar
acidity1
pH
help("Sake")
smell
taste
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.2
0.0
0.2
0.4
0.6
0.8

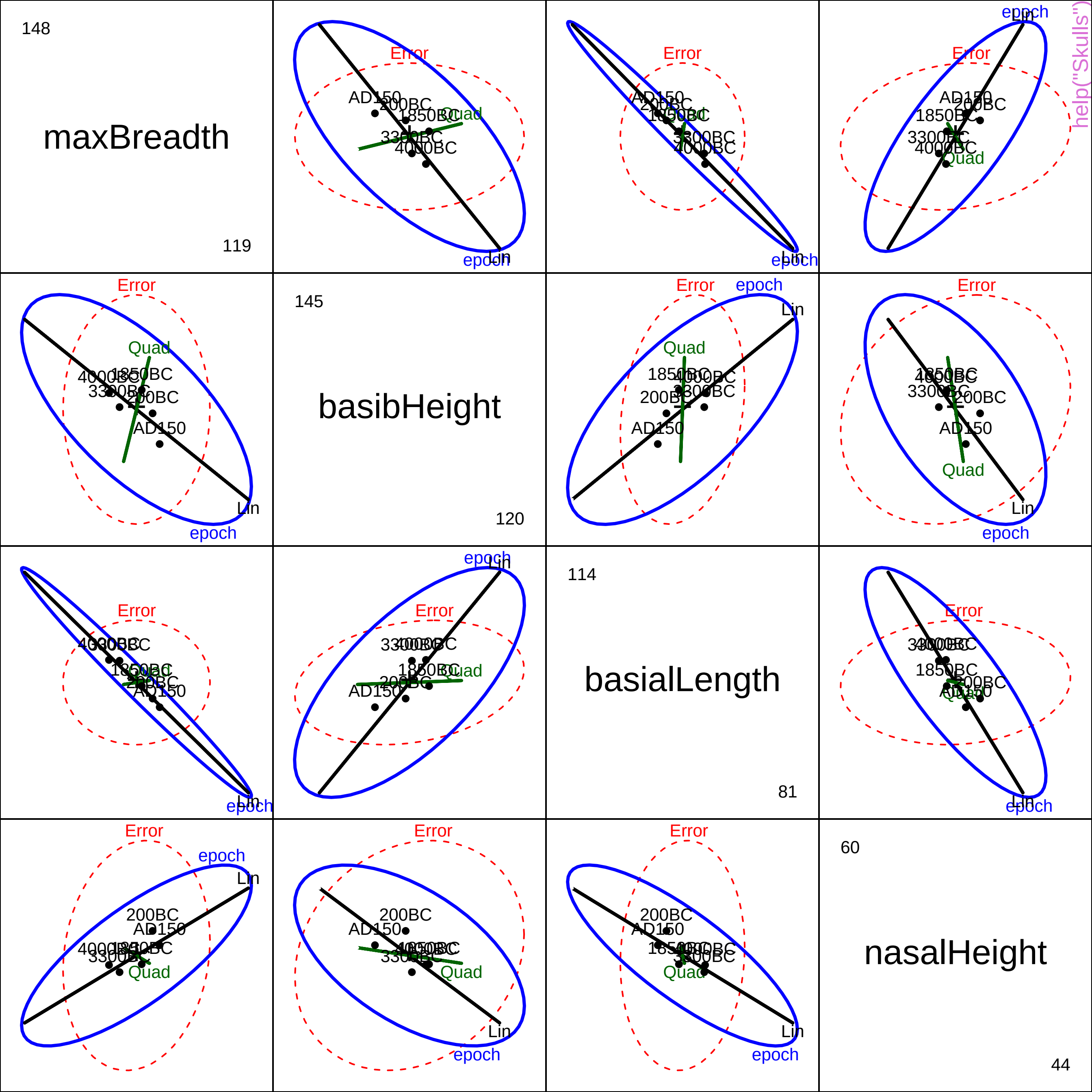


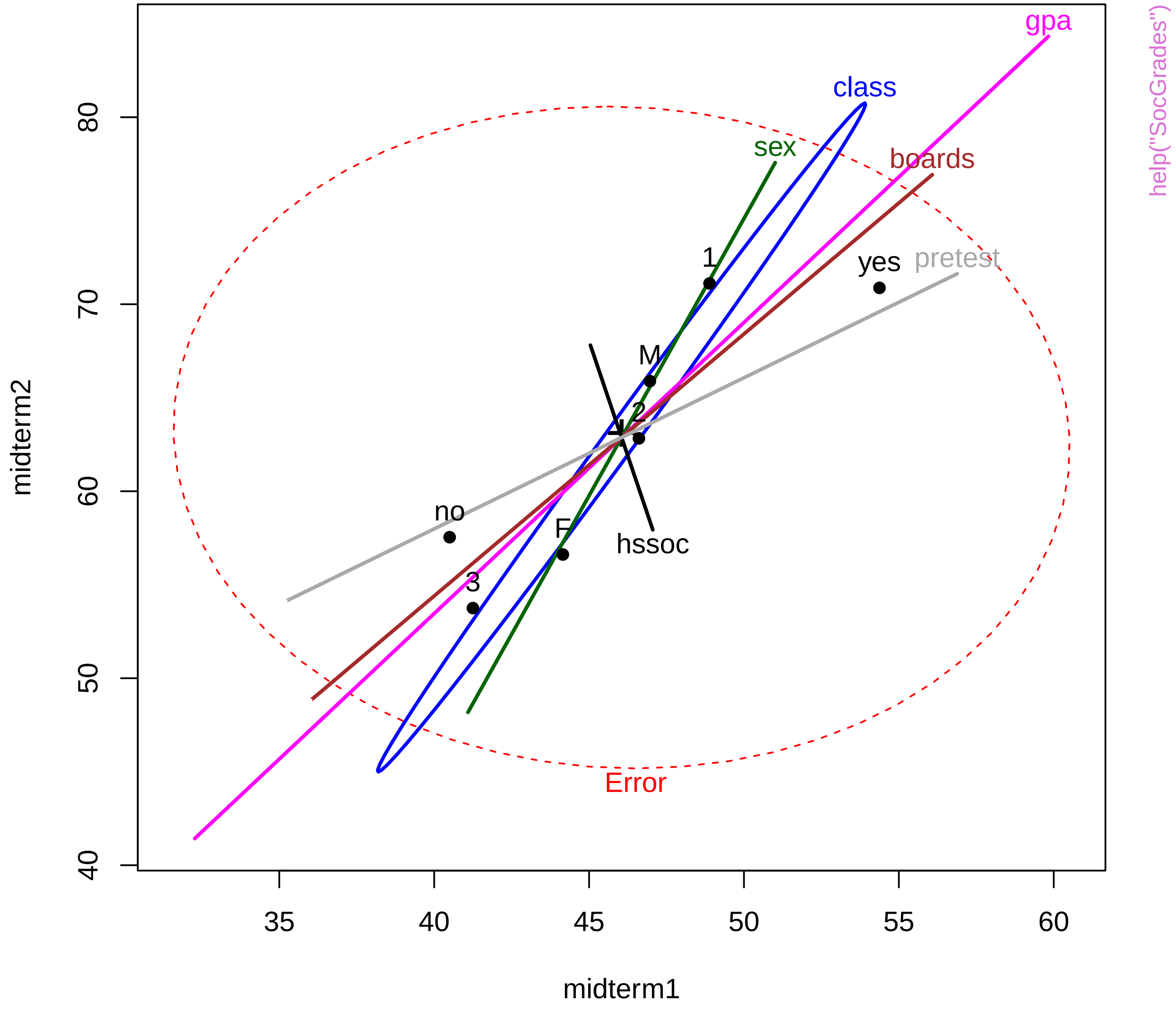
epoch
4000BC
3300BC
1850BC
200BC
AD150
maxBreadth
basibHeight
120
125
130
135
140
145
help("Skulls")



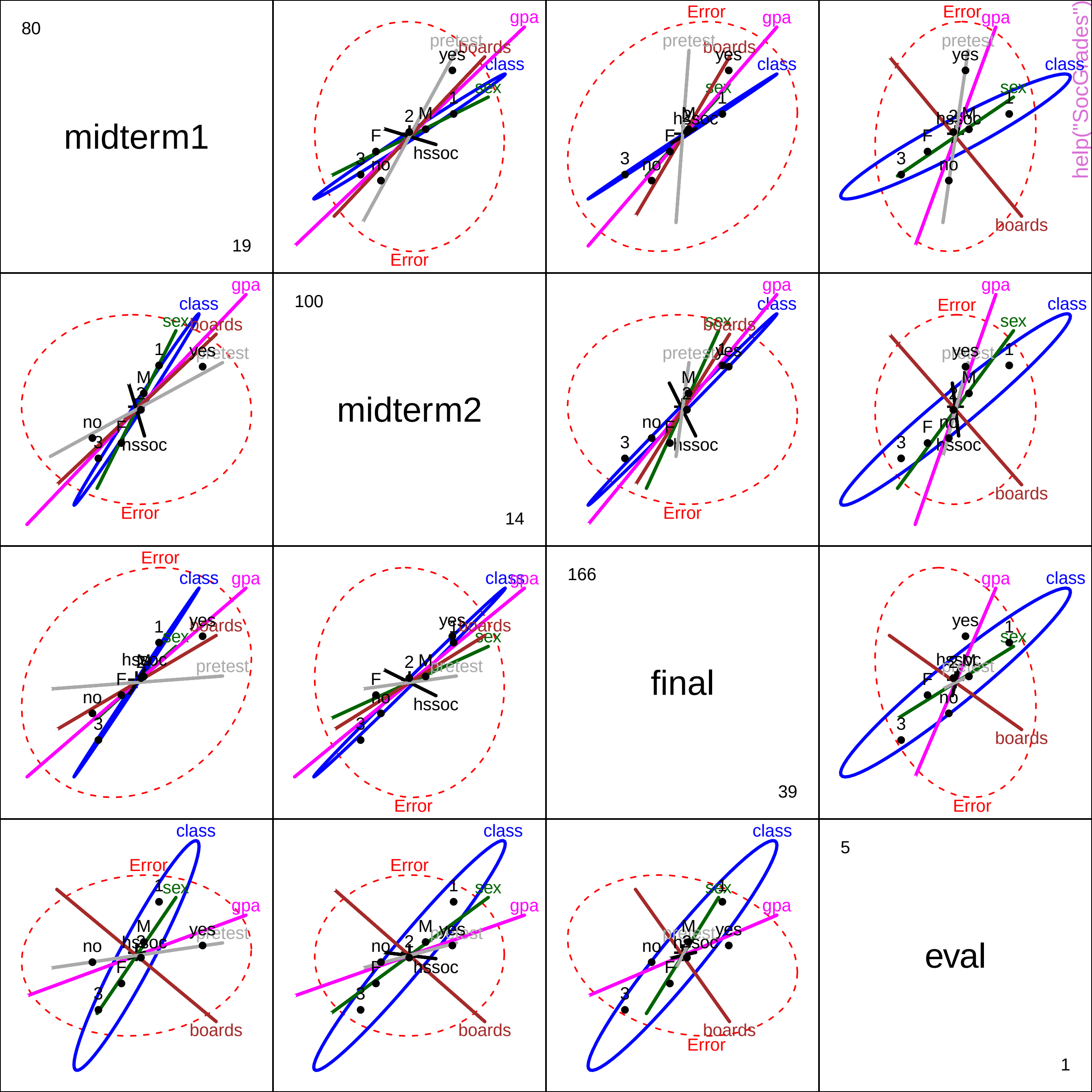
epoch
4000BC
3300BC
1850BC
200BC
AD150
help("Skulls")
maxBreadth
basialLength
120
125
130
135
140
145
80
85
90
95
100
105
110
115


Error
Quad
4000BC
1850BC
3300BC
200BC
AD150
Lin
epoch
basibHeight
maxBreadth
125
130
135
140
145
help("Skulls")

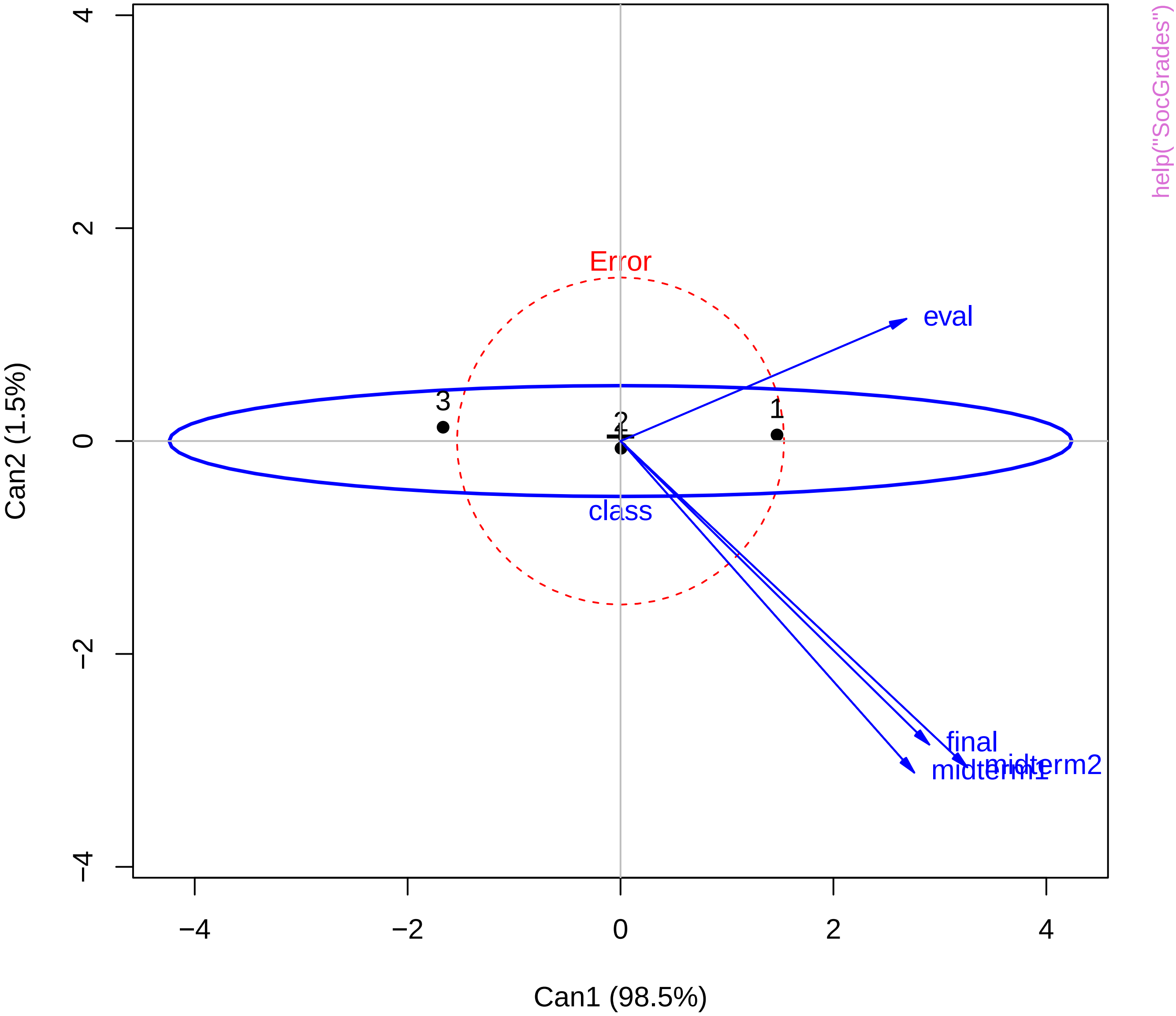

148
maxBreadth
119
145
basibHeight
120
114
basialLength
81
60
nasalHeight
44
help("Skulls")
Error
epoch
Lin
Quad
AD150
200BC
1850BC
3300BC
4000BC



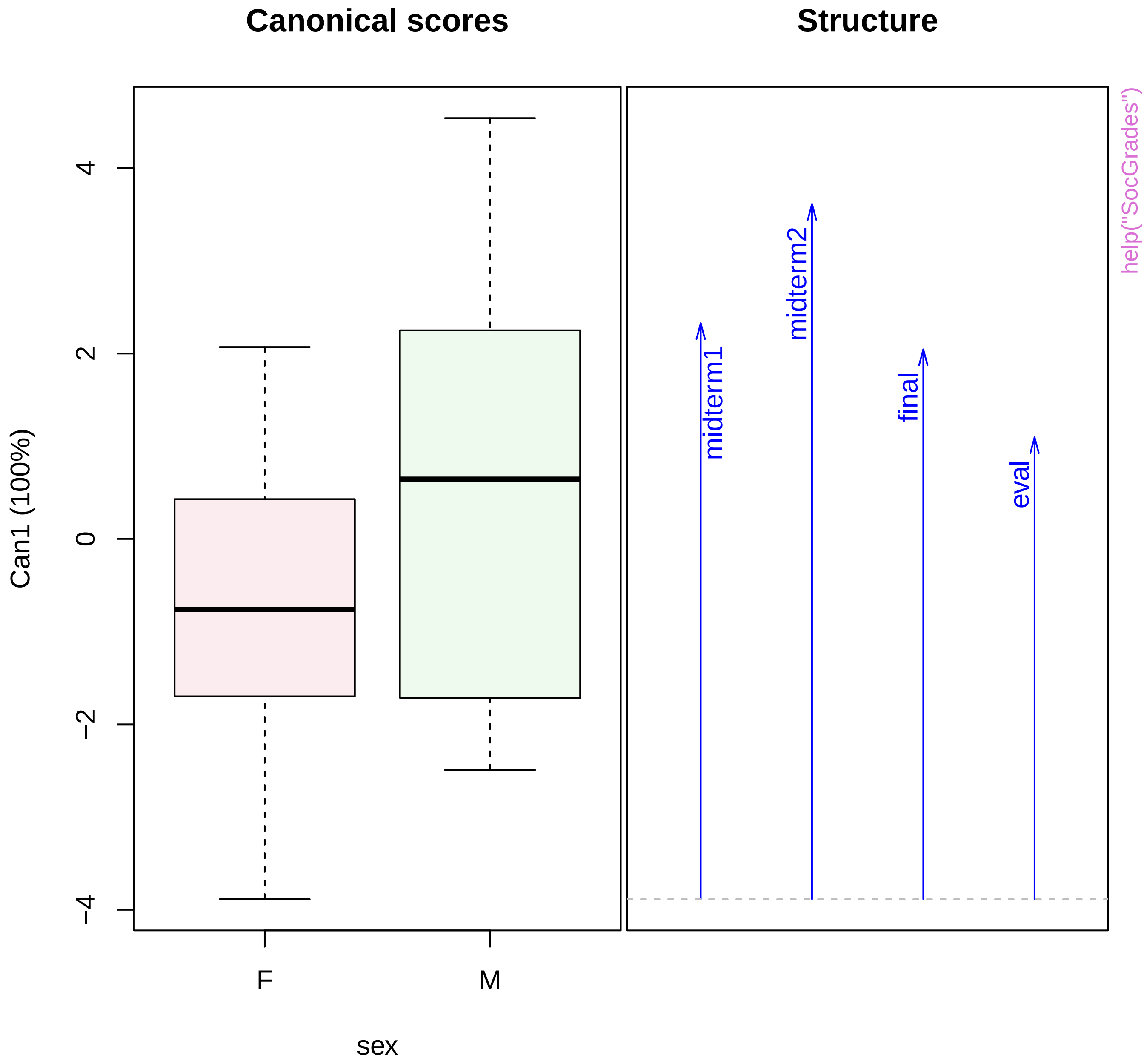
gpa
class
sex
boards
pretest
yes
1
M
2
no
F
hssoc
3
Error
help("SocGrades")
80
70
60
50
40
35
40
45
50
55
60
midterm1
midterm2

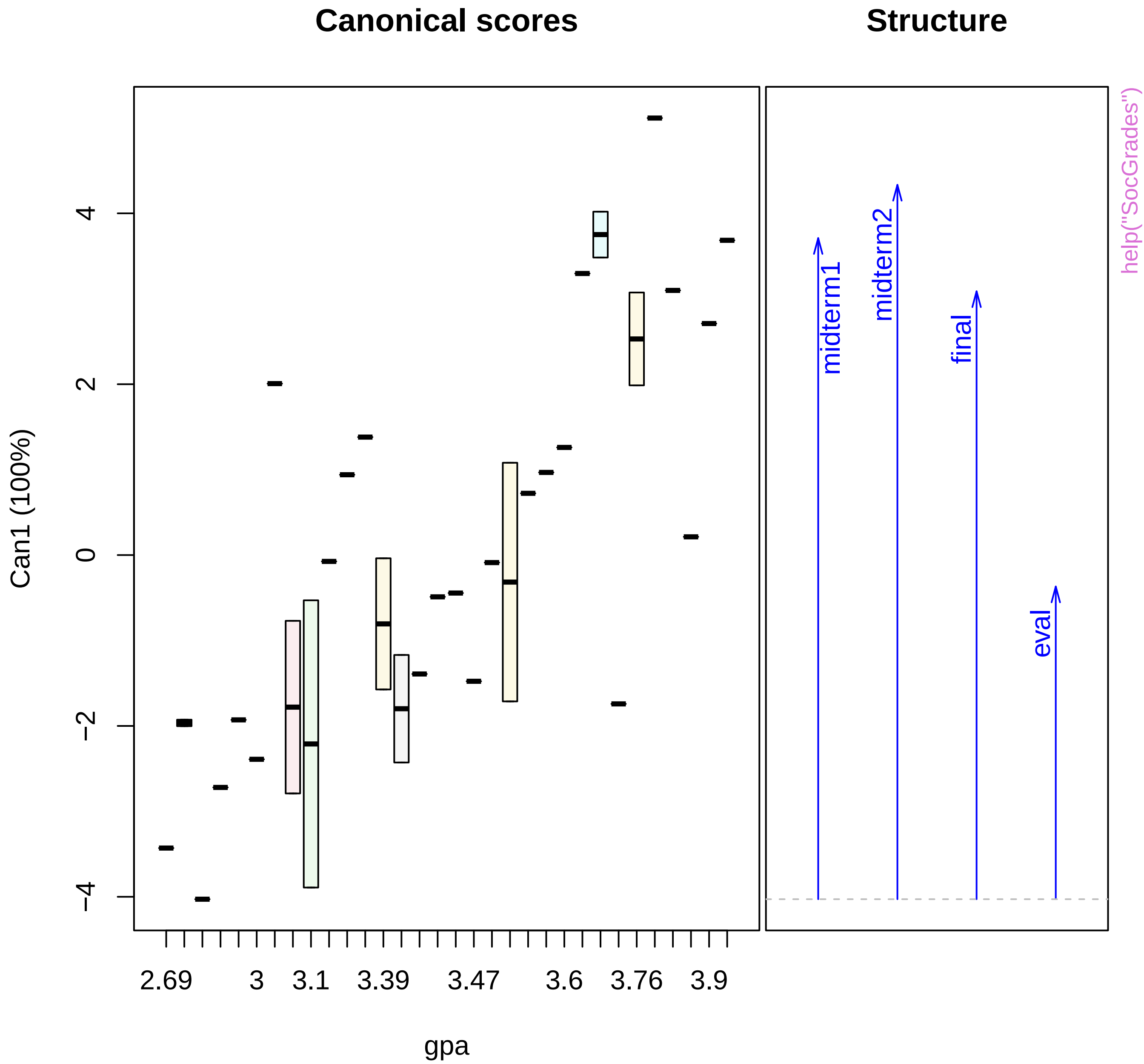


80
midterm1
19
100
midterm2
14
166
final
39
5
eval
1
help("SocGrades")
gpa
class
sex
boards
pretest
Error
yes
no
hssoc
M
F
1
2
3

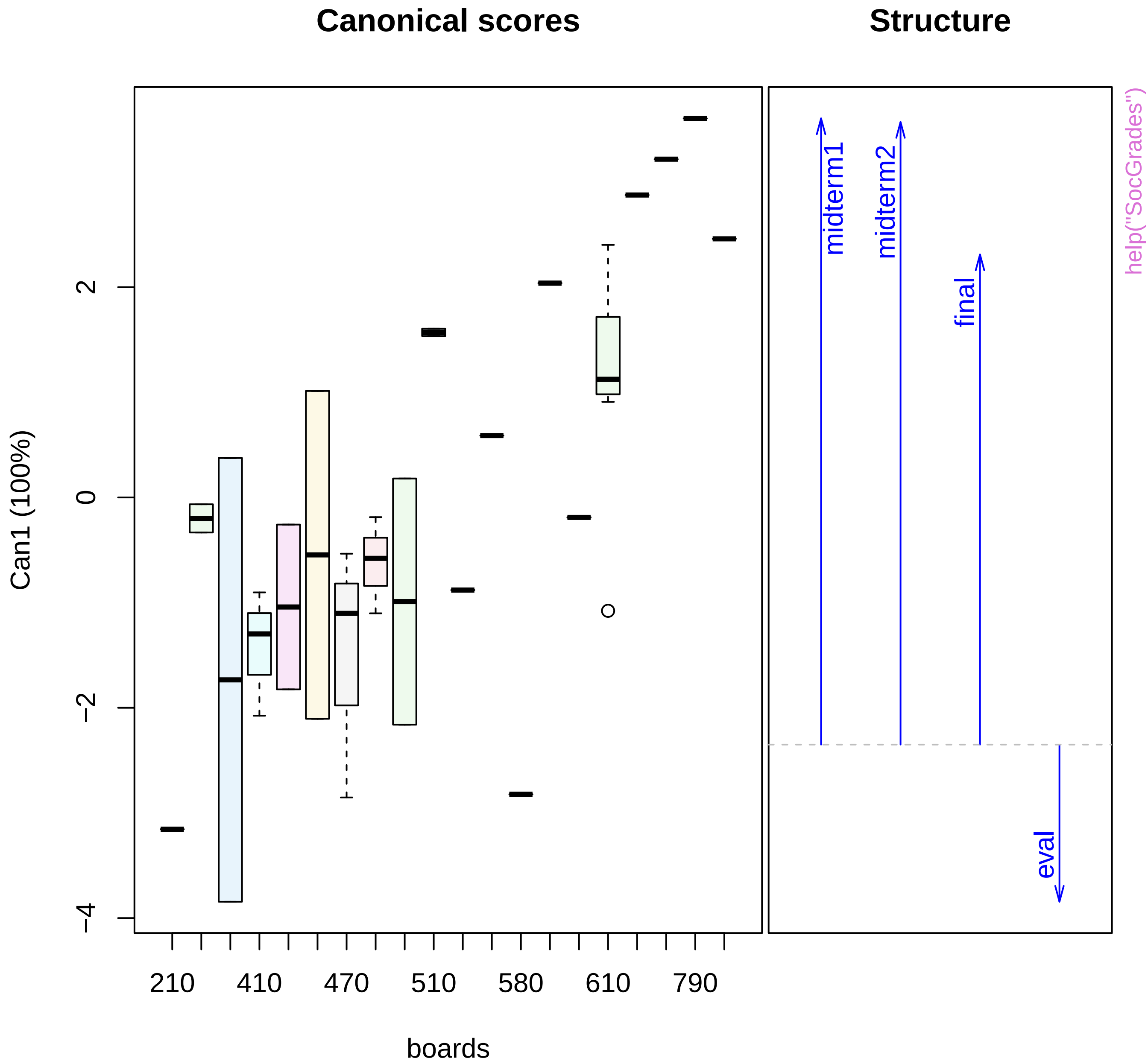


Error
eval
class
final
midterm2
midterm1
1
2
3
Can1 (98.5%)
Can2 (1.5%)
help("SocGrades")
−4
−2
0
2
4

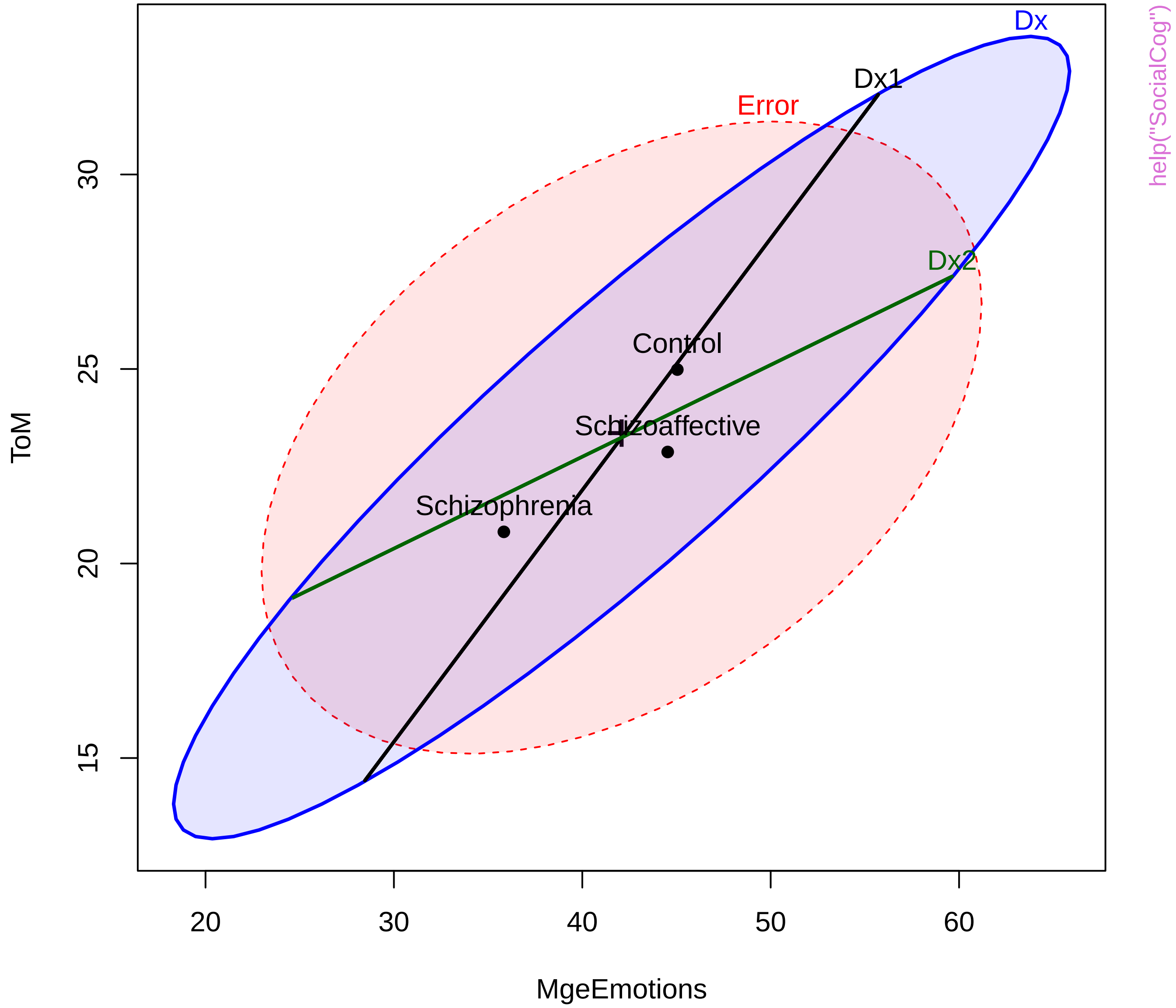

Canonical scores
Structure
Can1 (100%)
4
2
0
−2
−4
F
M
sex
midterm1
midterm2
final
eval
help("SocGrades")

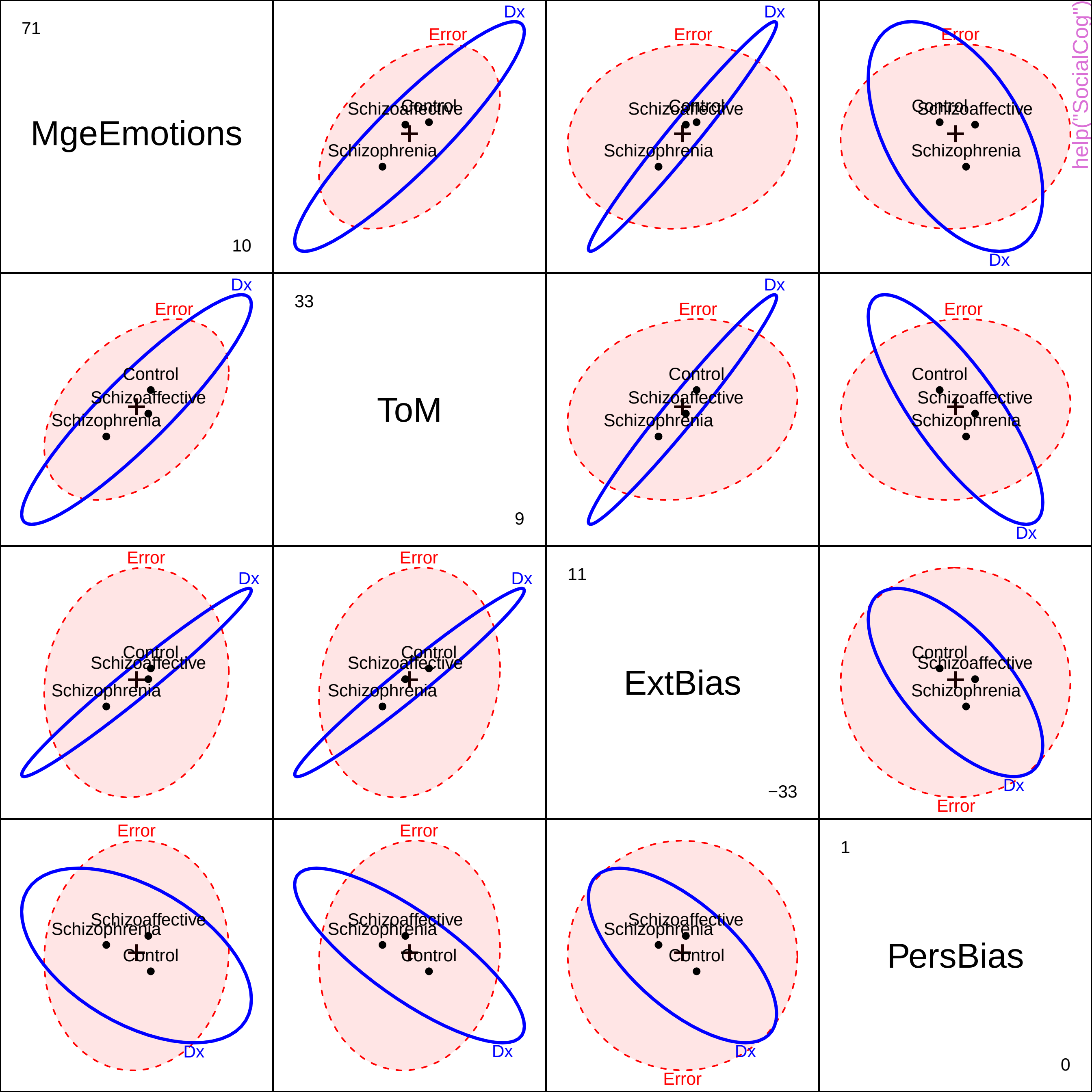


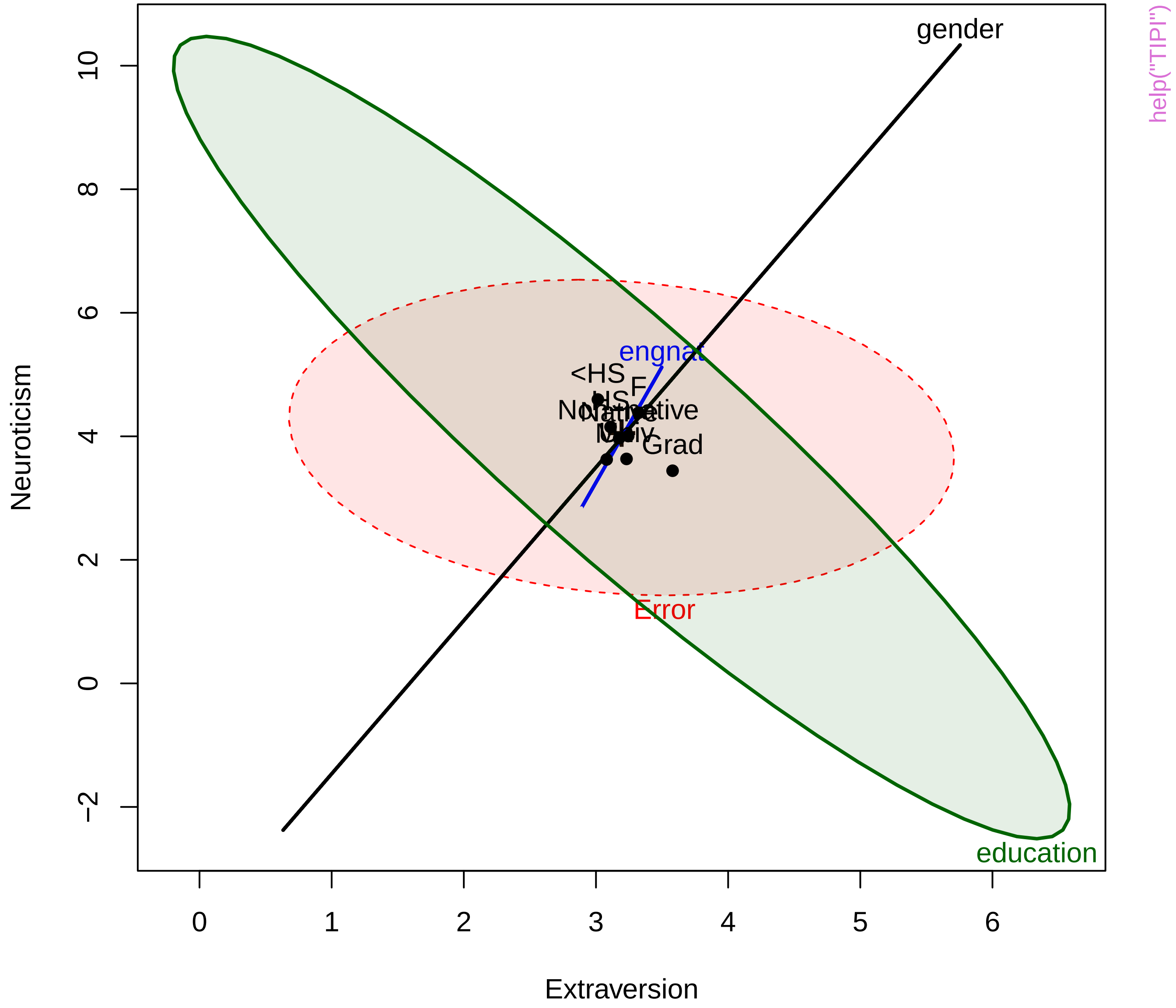
Canonical scores
Structure
Can1 (100%)
4
2
0
−2
−4
2.69
3
3.1
3.39
3.47
3.6
3.76
3.9
gpa
midterm1
midterm2
final
eval
help("SocGrades")



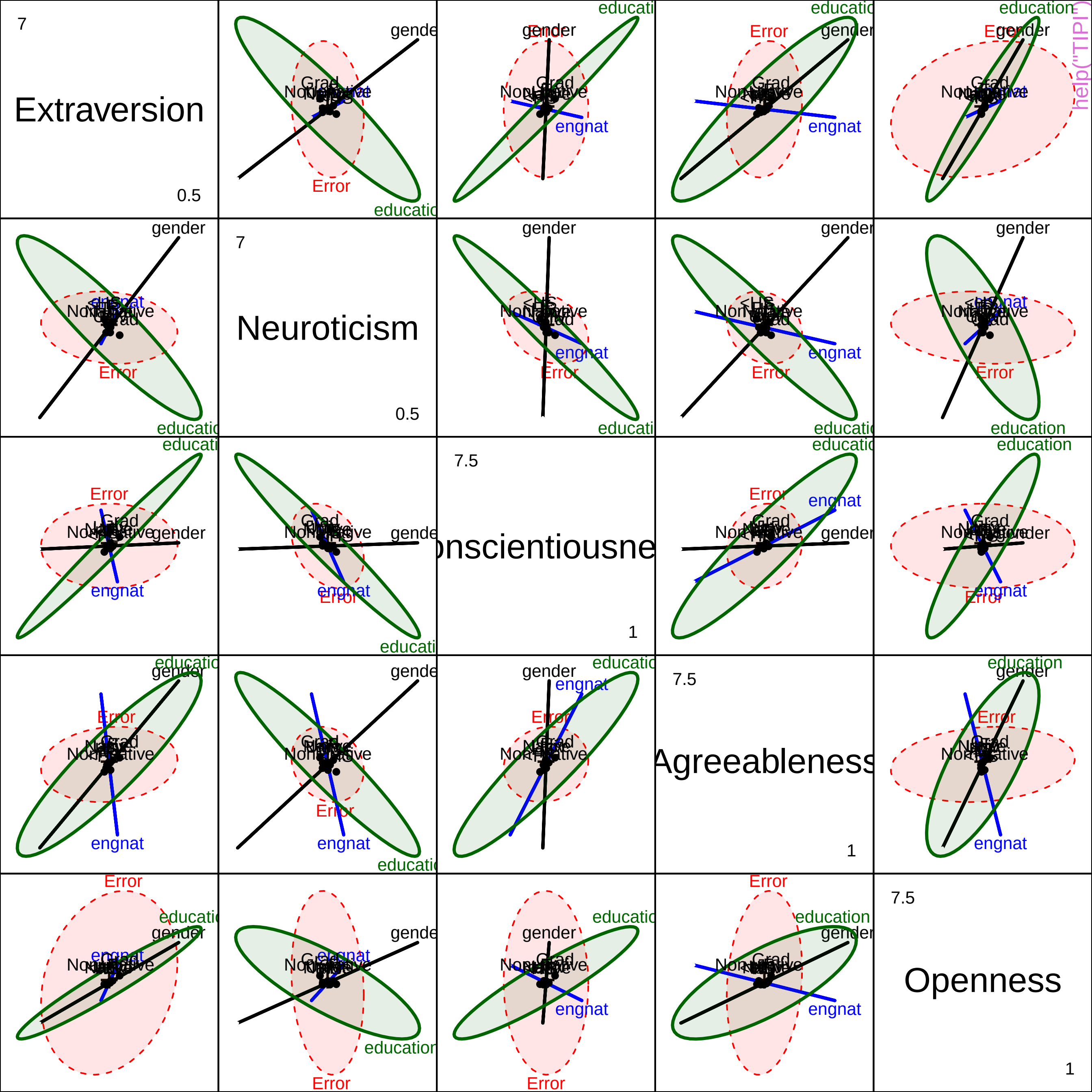
Canonical scores
Structure
Can1 (100%)
2
0
−2
−4
210
410
470
510
580
610
790
boards
midterm1
midterm2
final
eval
help("SocGrades")


Dx
Dx1
Error
Dx2
Control
Schizoaffective
Schizophrenia
30
25
20
15
ToM
20
30
40
50
60
MgeEmotions
help("SocialCog")

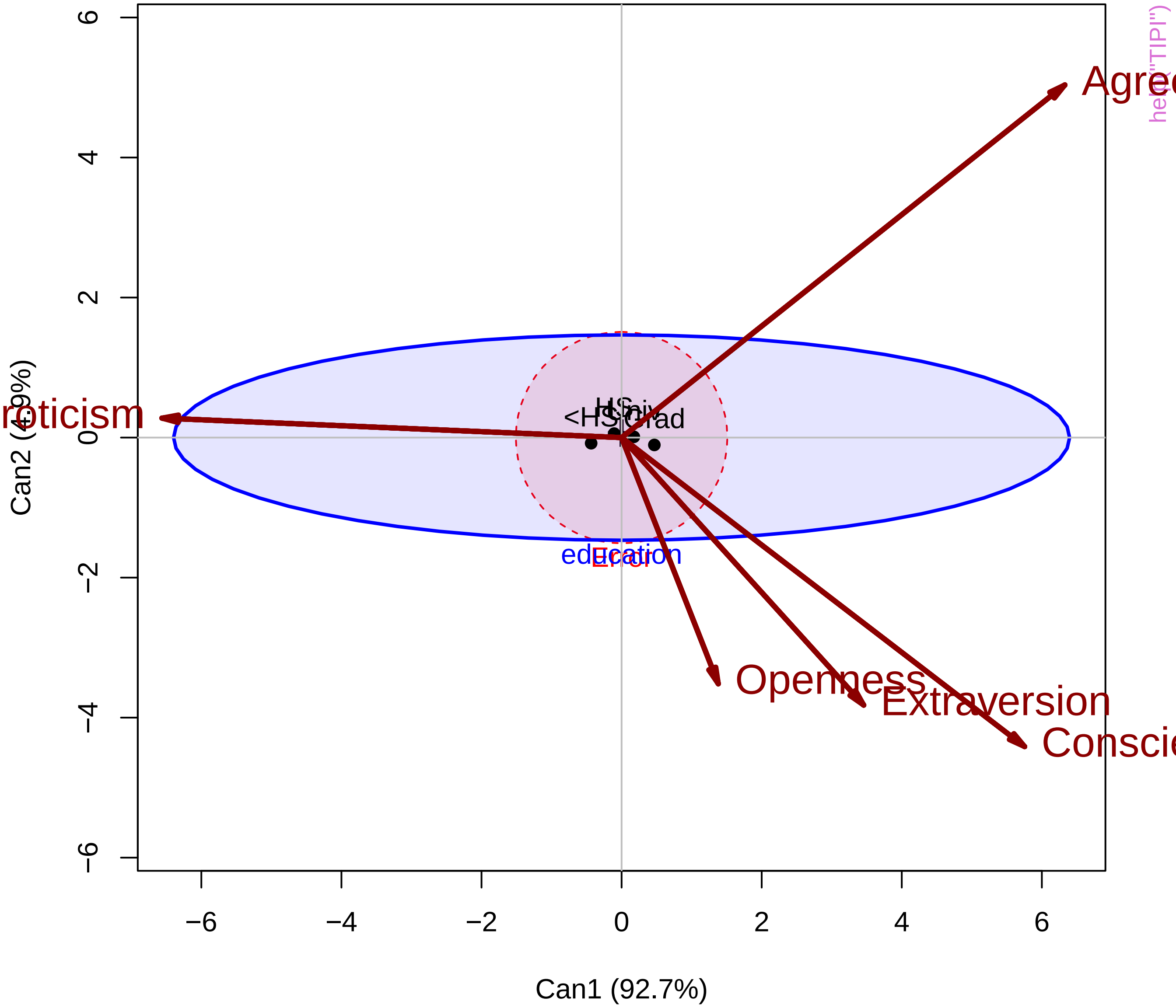

71
MgeEmotions
10
33
ToM
9
11
ExtBias
−33
1
PersBias
0
Error
Dx
Control
Schizoaffective
Schizophrenia
help("SocialCog")



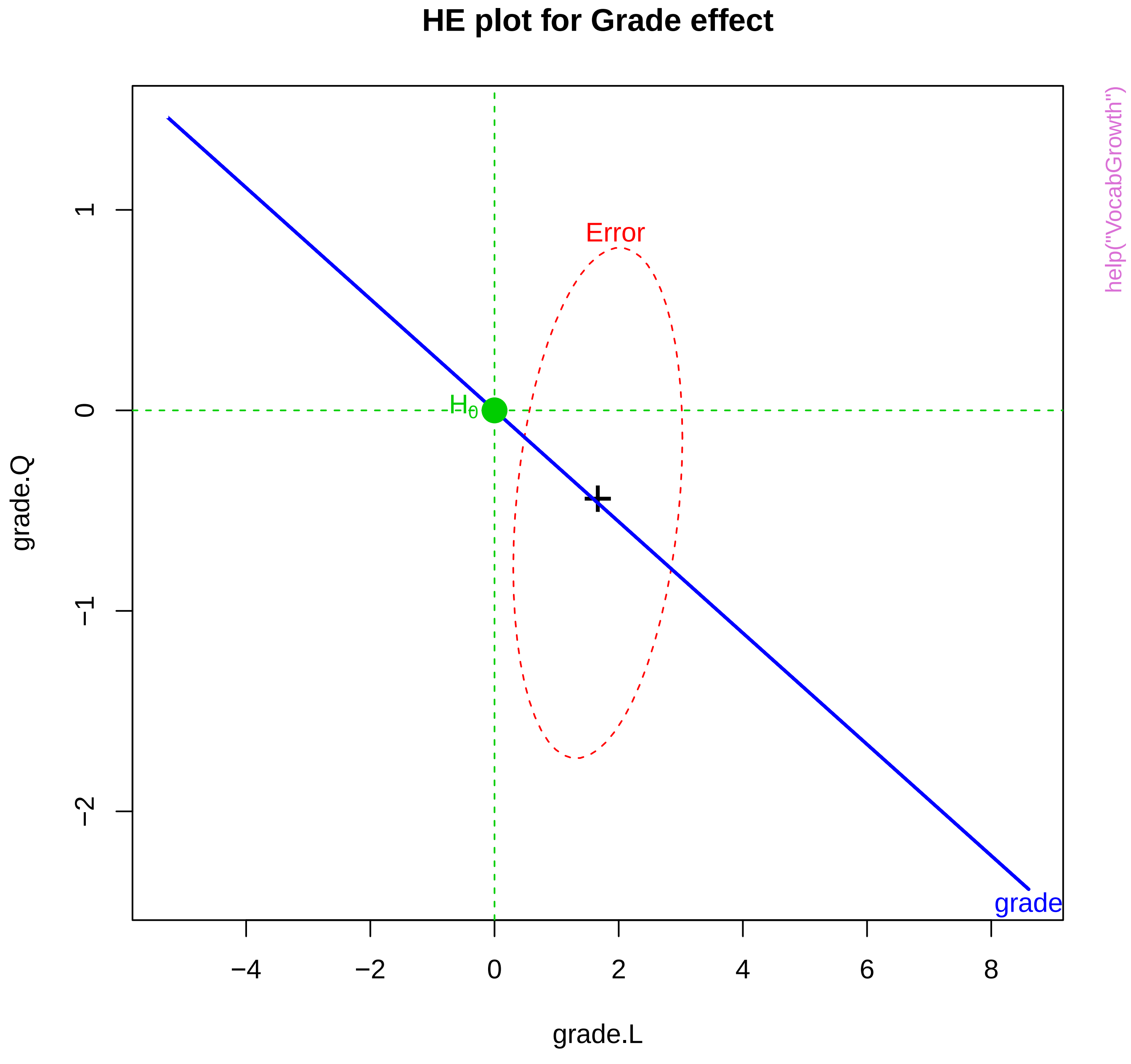
gender
engnat
<HS
F
HS
Nonnative
Native
M
Univ
Grad
Error
education
Neuroticism
Extraversion
help("TIPI")
10
8
6
4
2
0
−2
0
1
2
3
4
5
6

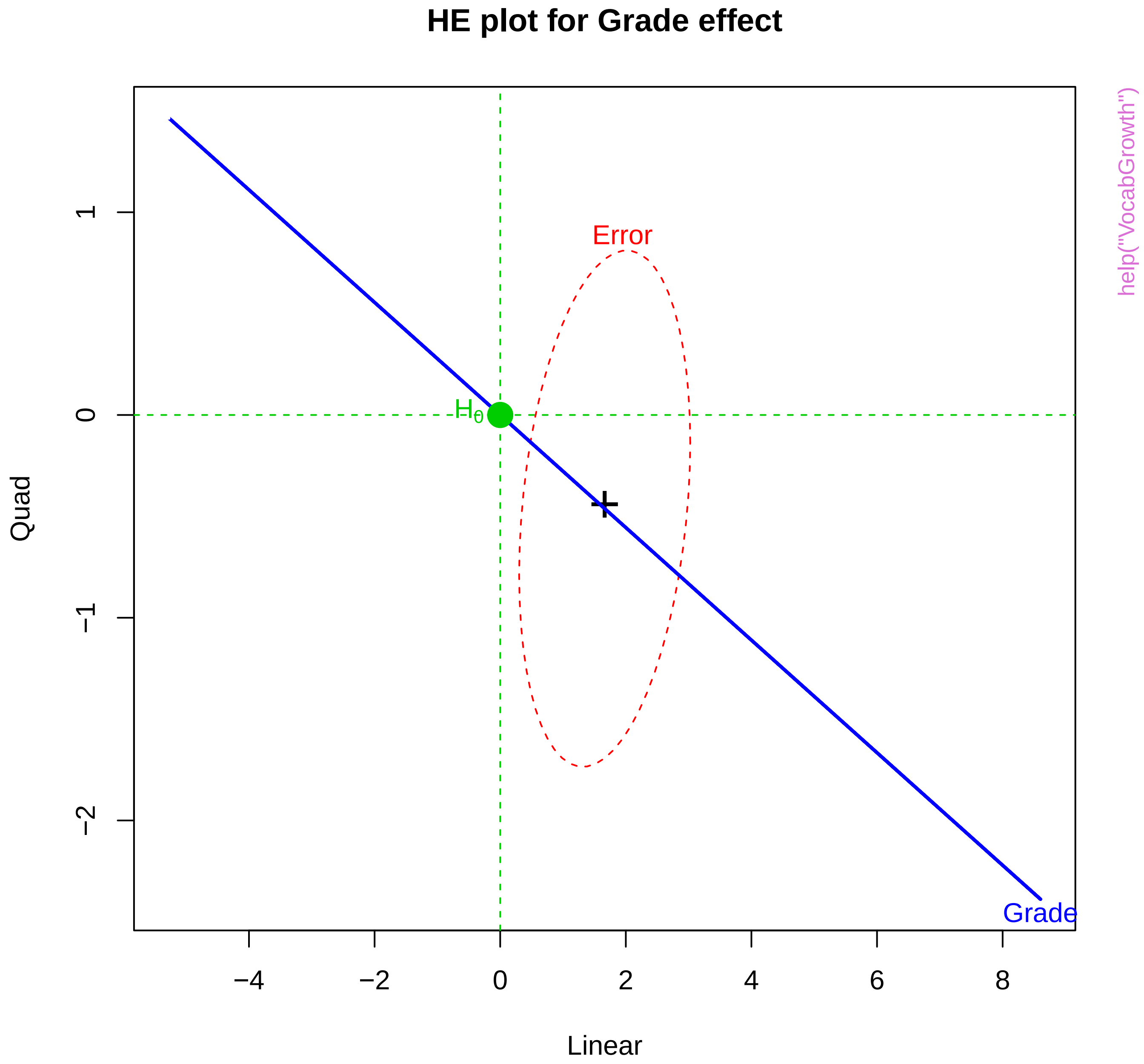


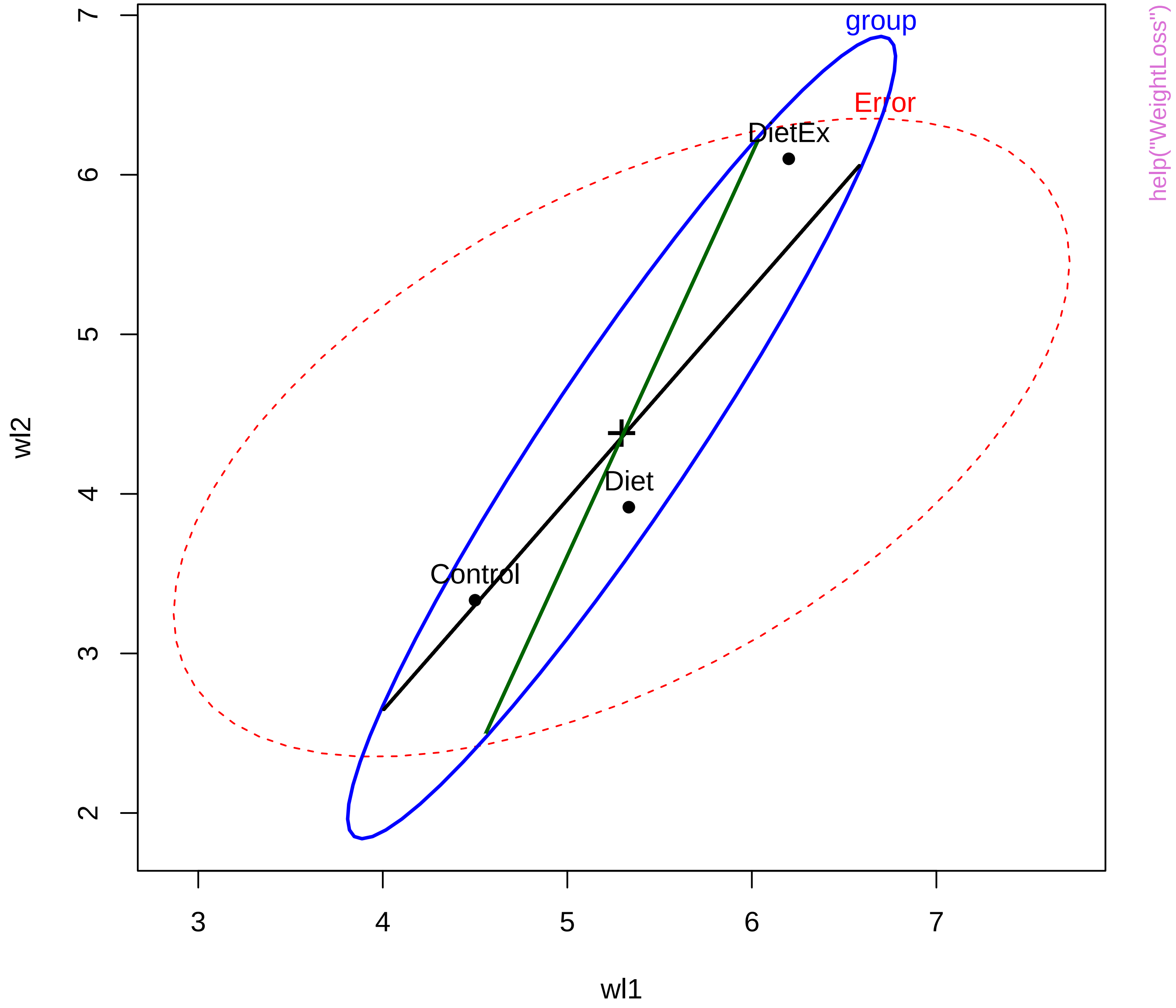
Extraversion
7
0.5
Neuroticism
7
0.5
nscientiousne
7.5
1
Agreeableness
7.5
1
Openness
7.5
1
help("TIPI")
gender
education
Error
engnat



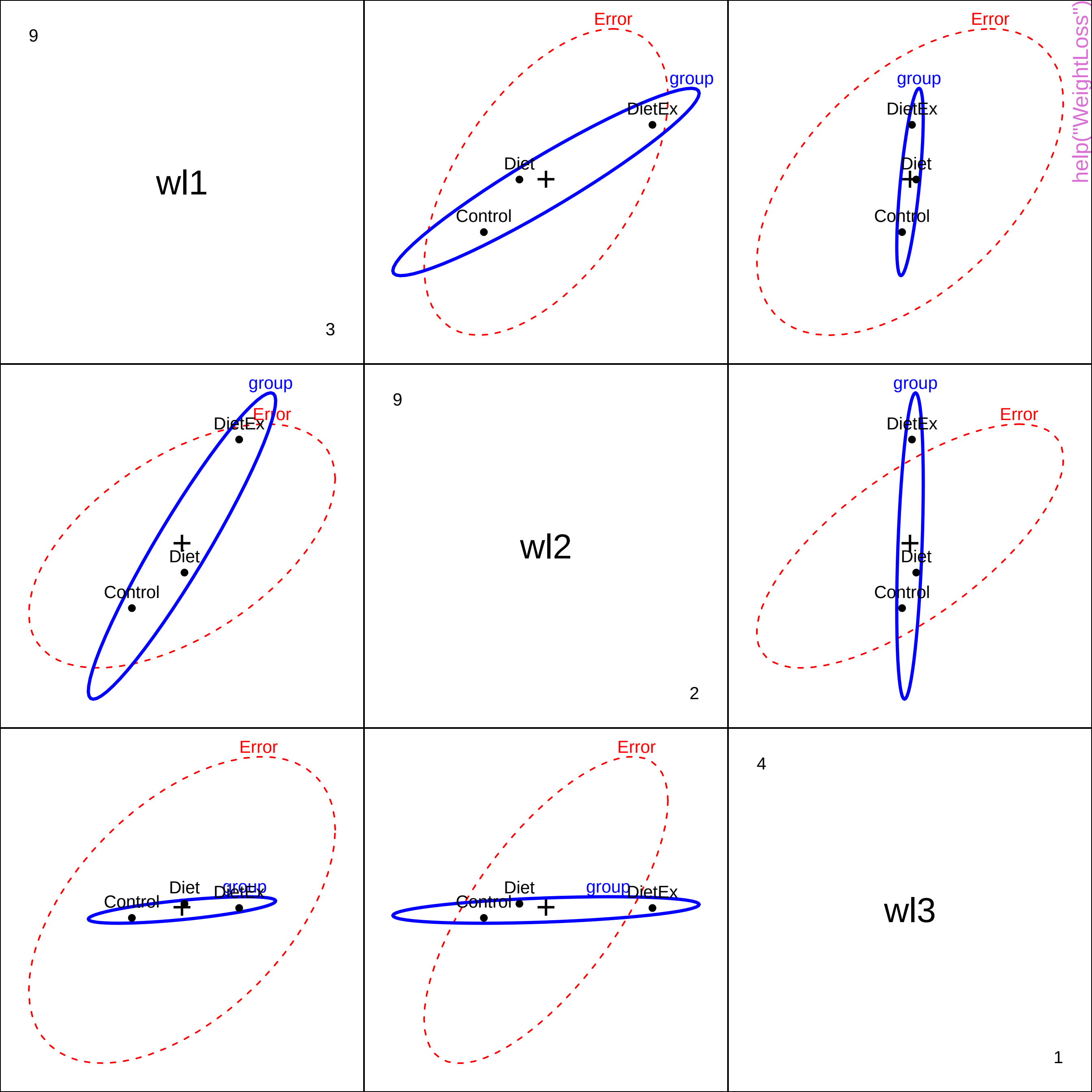
Agree
roticism
HS
Univ
<HS
Grad
education
Error
Openness
Extraversion
Conscie
6
4
2
0
−2
−4
−6
Can2 (4.9%)
−6
−4
−2
0
2
4
6
Can1 (92.7%)


HE plot for Grade effect
Error
H_0
grade
grade.L
grade.Q
−4
−2
0
2
4
6
8
1
0
−1
−2
help("VocabGrowth")

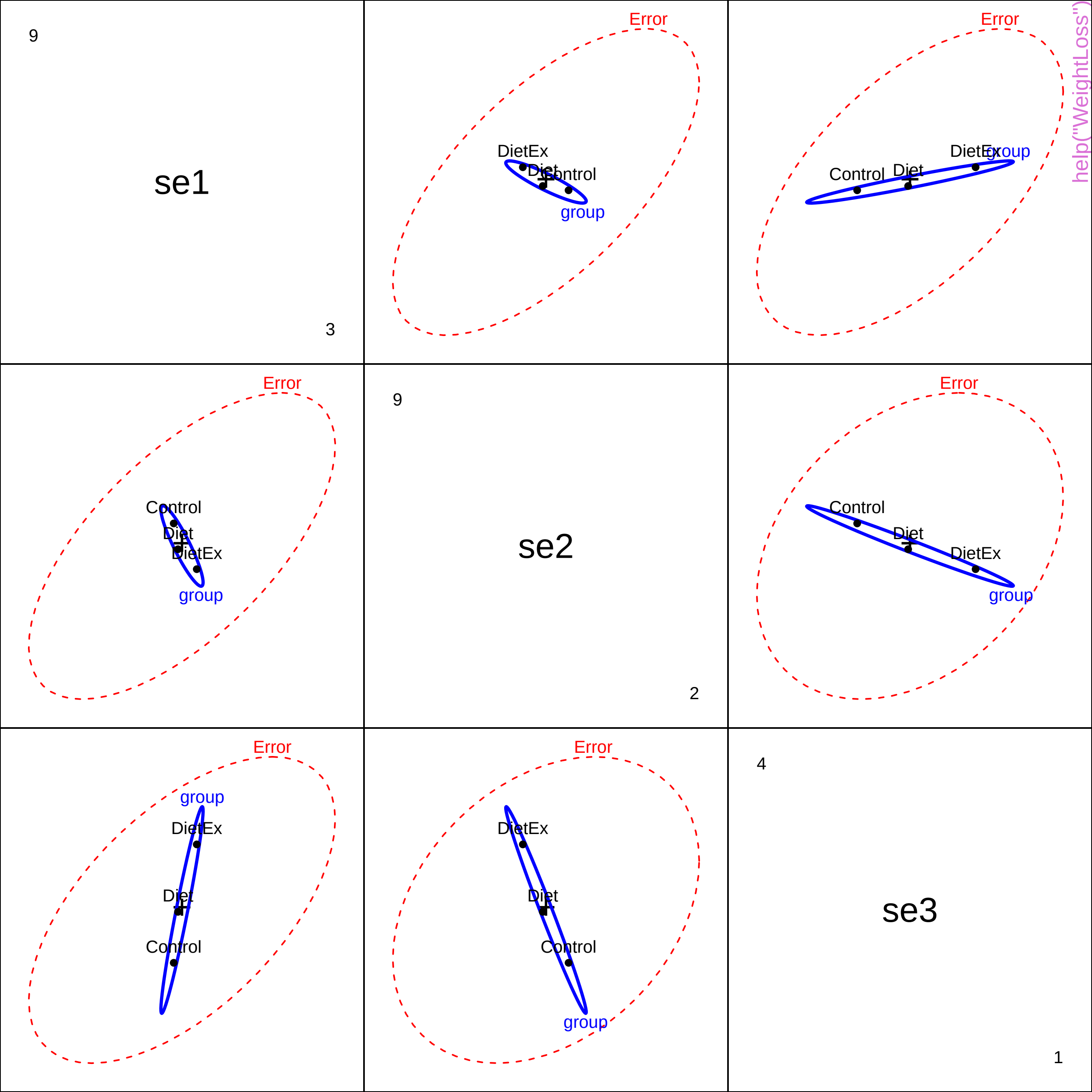

HE plot for Grade effect
Error
H_0
Grade
Quad
Linear
1
0
−1
−2
−4
−2
0
2
4
6
8
help("VocabGrowth")



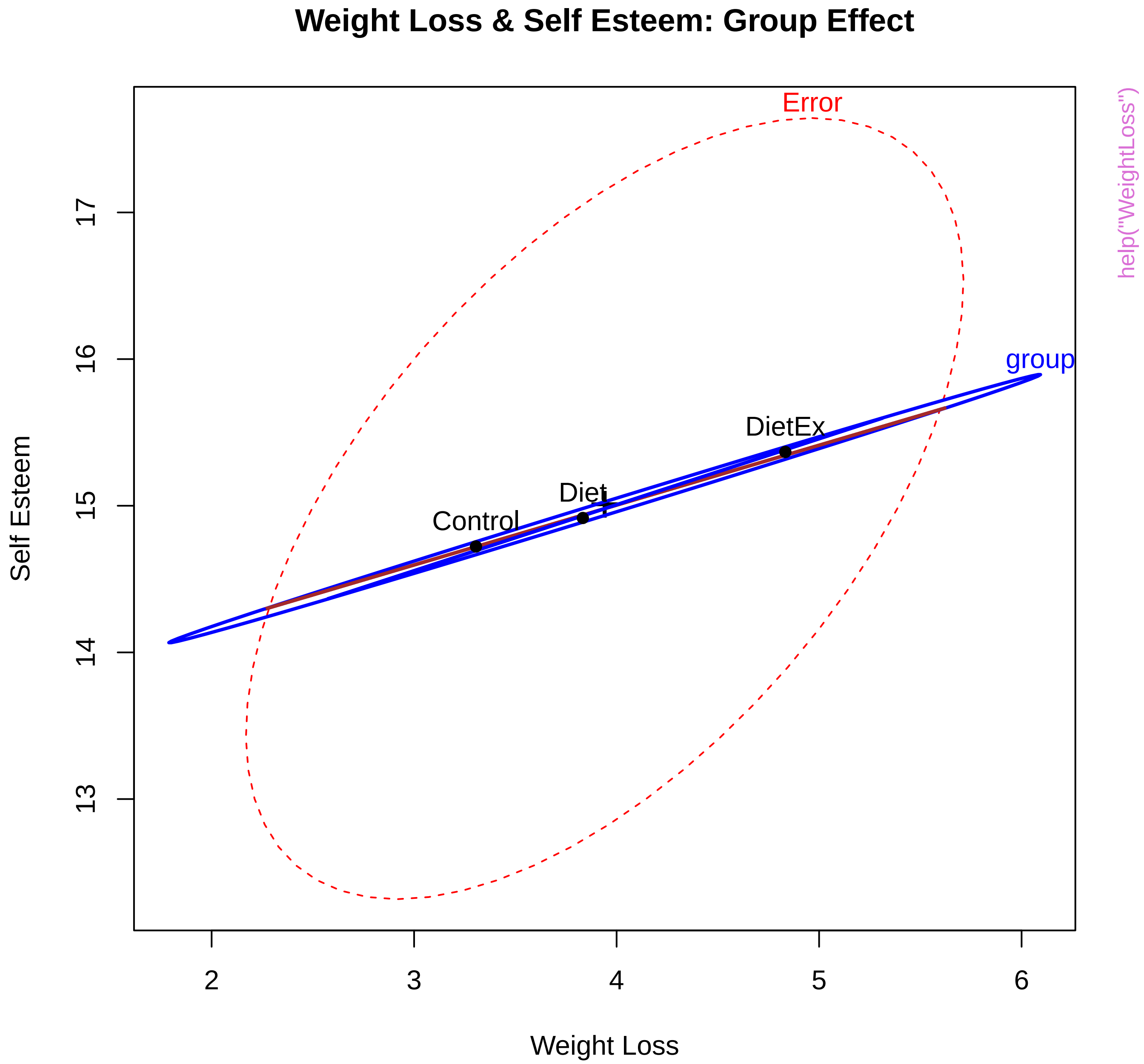
group
Error
DietEx
Diet
Control
wl1
wl2
3
4
5
6
7
2
help("WeightLoss")



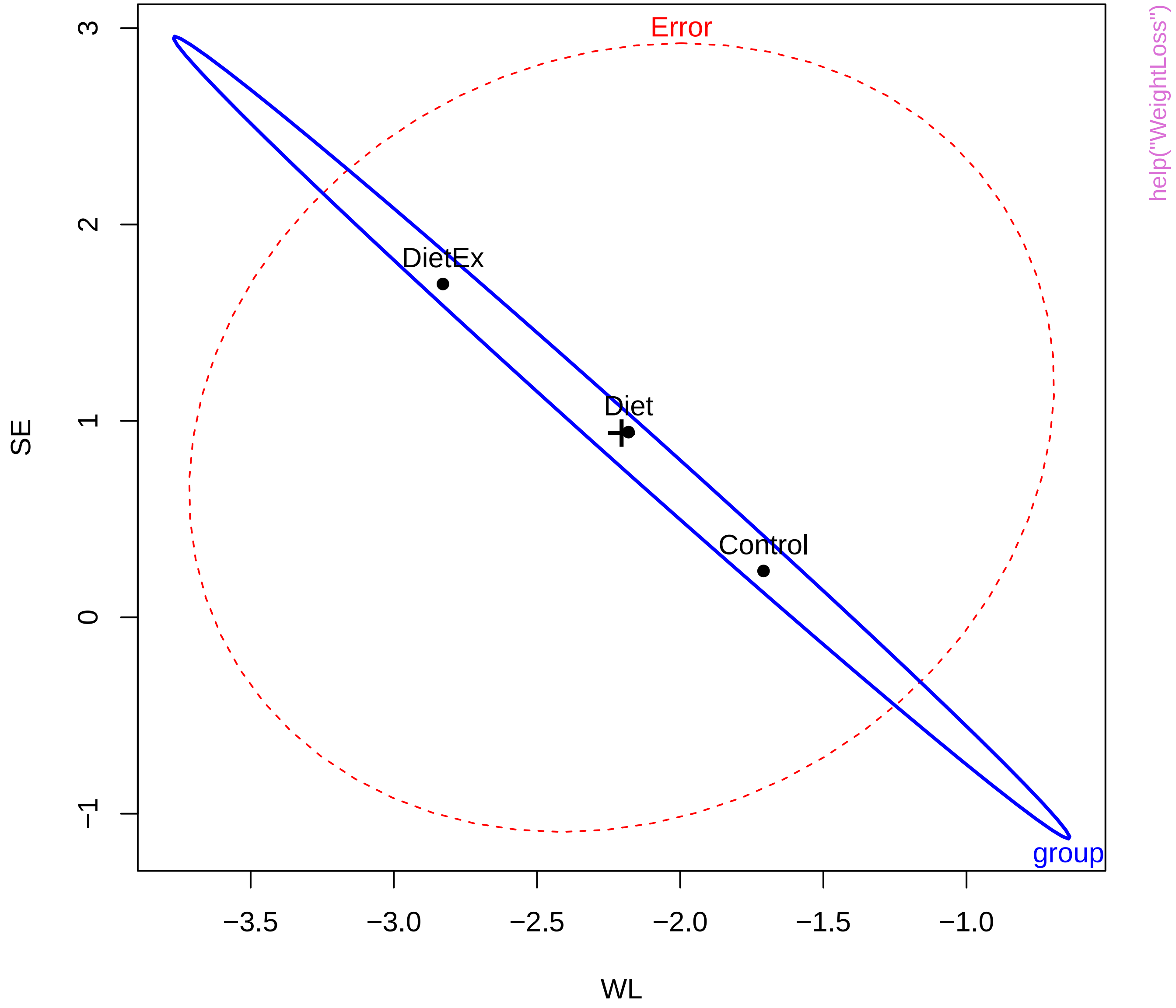
wl1
wl2
wl3
9
3
9
2
4
1
Error
group
DietEx
Diet
Control
help("WeightLoss")



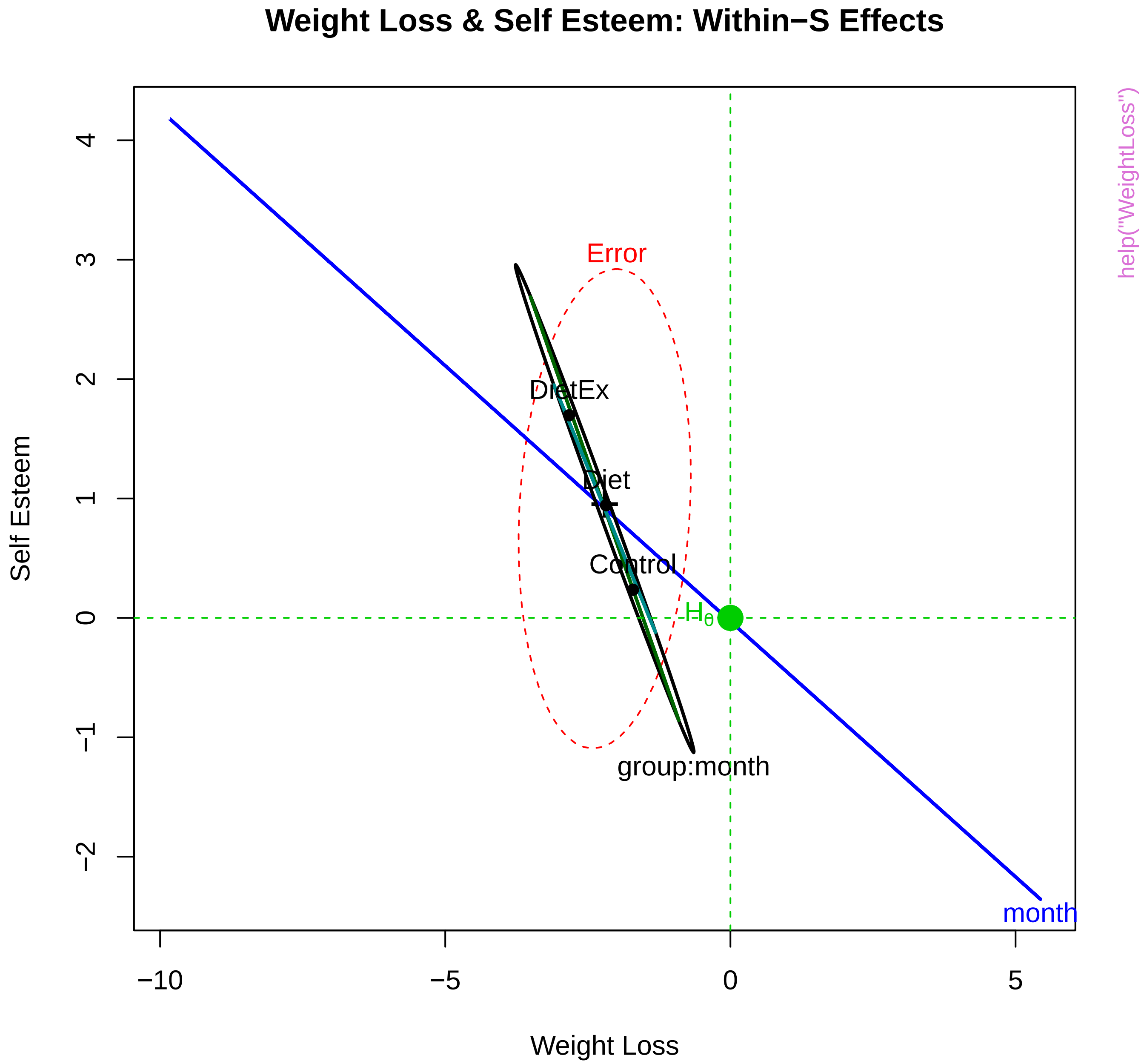
se1
se2
se3
Error
group
Control
Diet
DietEx
9
3
2
4
1
help("WeightLoss")

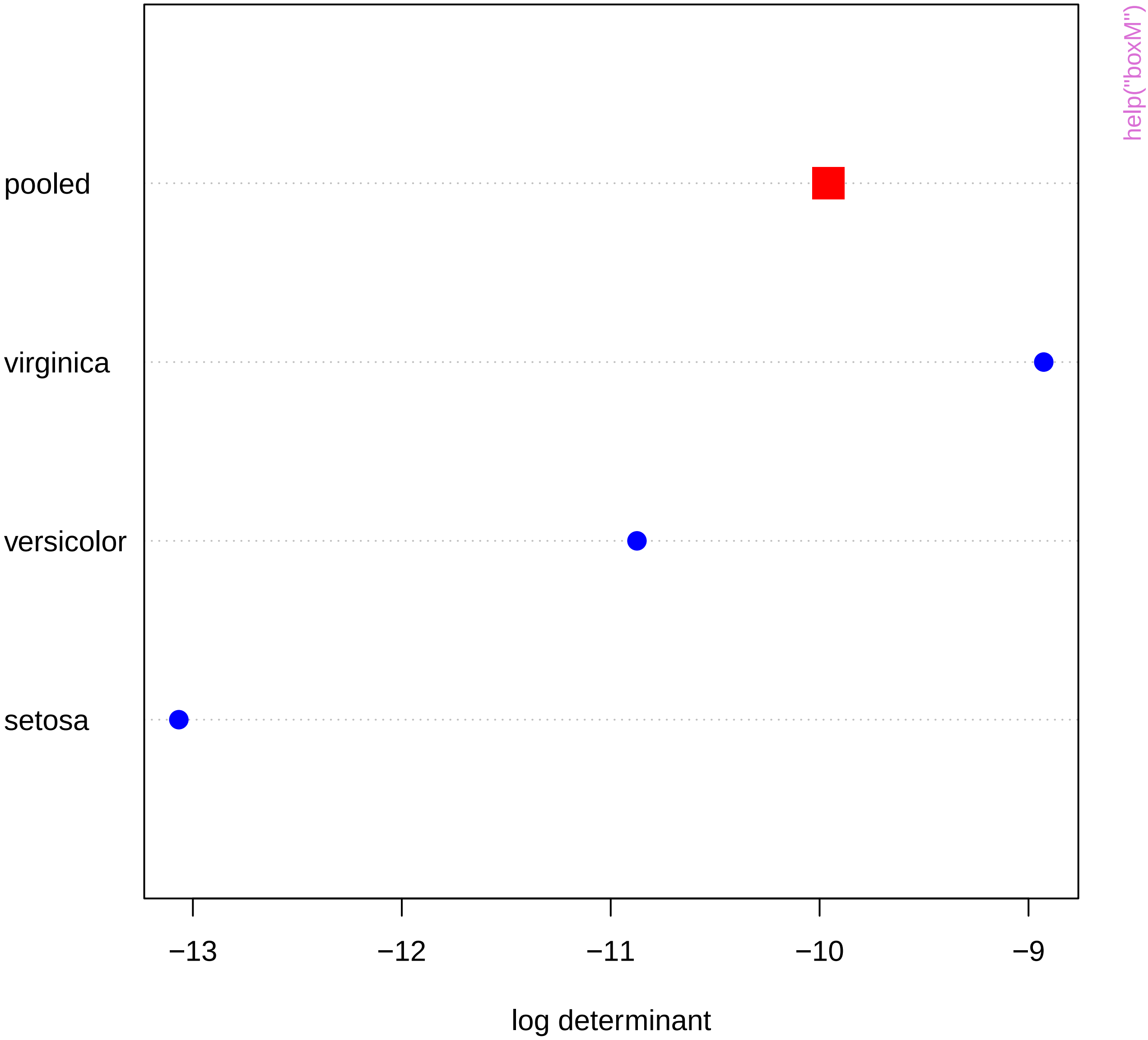


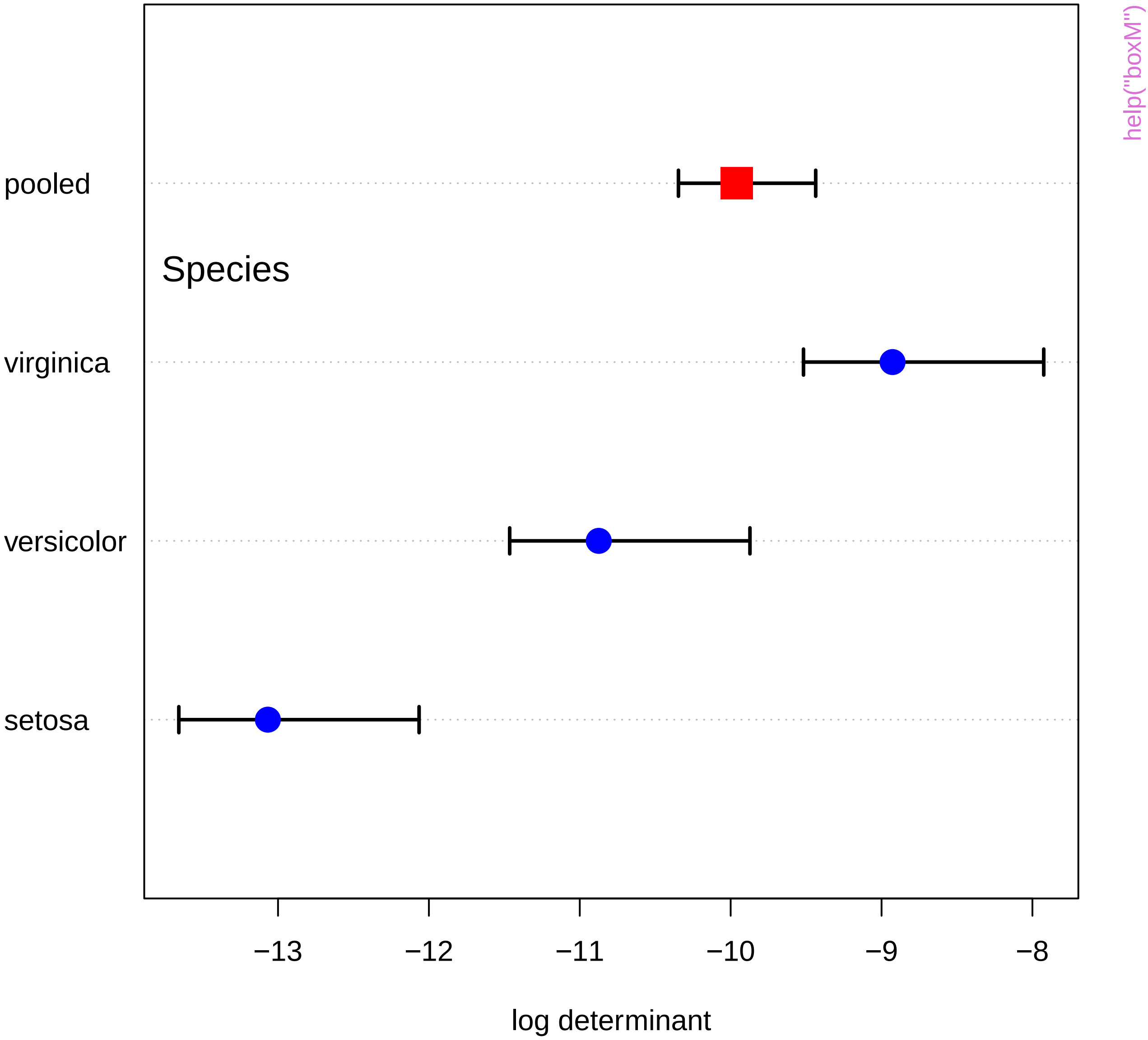
Weight Loss & Self Esteem: Group Effect
Error
group
DietEx
Diet
Control
17
16
15
14
13
Self Esteem
2
3
4
5
6
Weight Loss
help("WeightLoss")

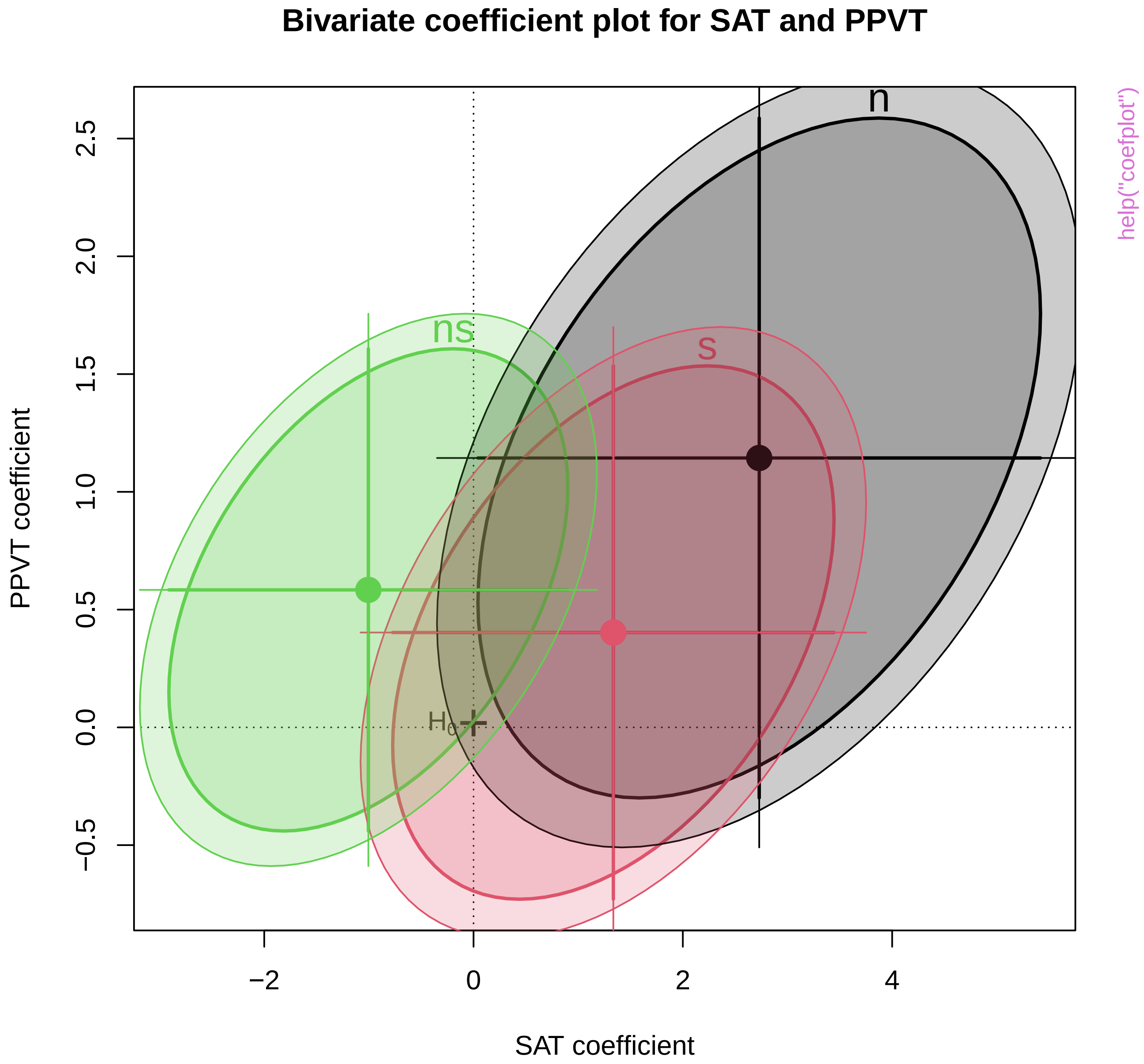


Error
DietEx
Diet
Control
group
SE
WL
3
2
1
0
−1
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
help("WeightLoss")

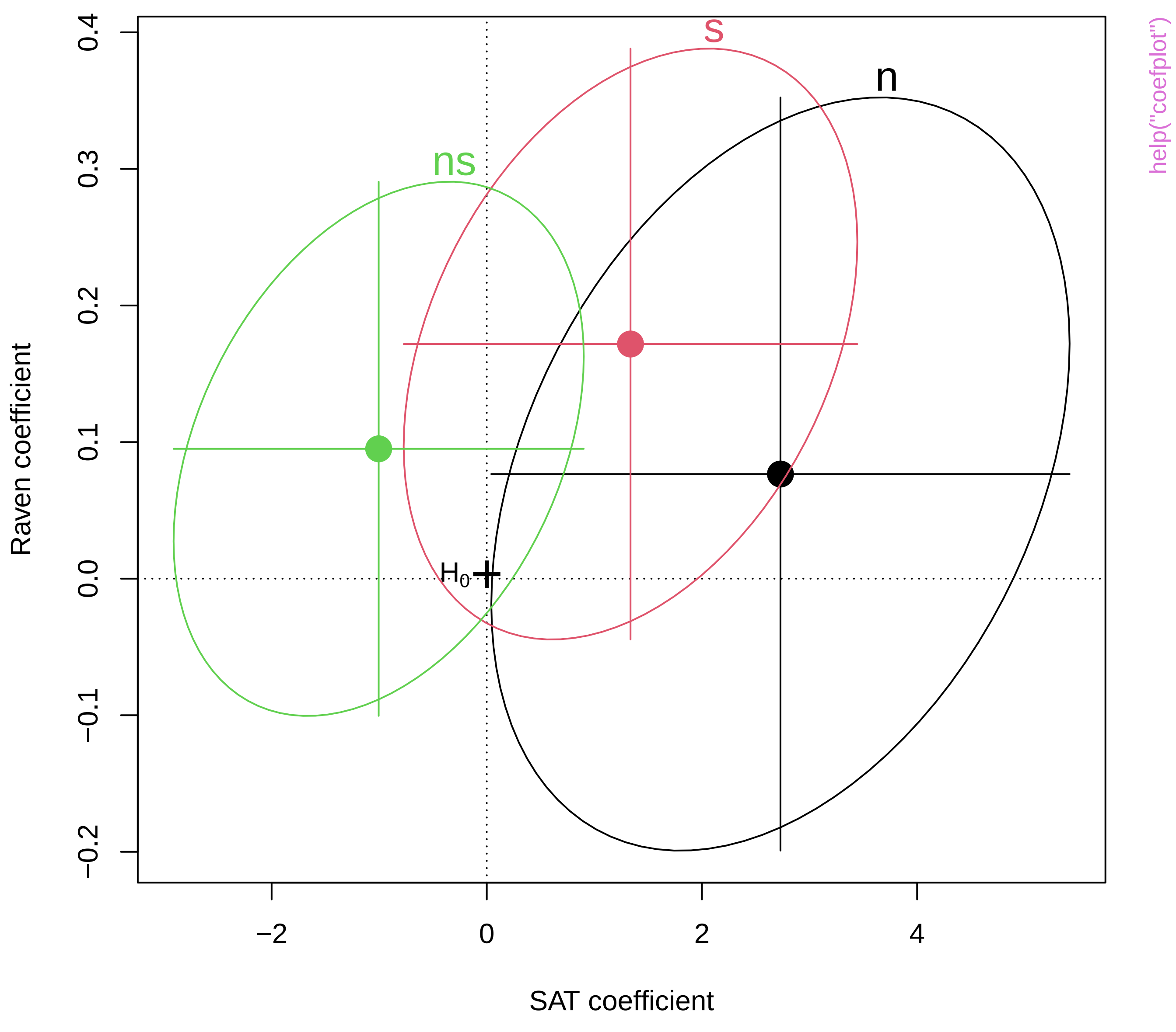

Weight Loss & Self Esteem: Within−S Effects
Self Esteem
Weight Loss
Error
DietEx
Diet
Control
group:month
H_0
month
help("WeightLoss")
−10
−5
0
5
−2
−1
0
1
2
3
4


pooled
virginica
versicolor
setosa
−13
−12
−11
−10
−9
log determinant
help("boxM")

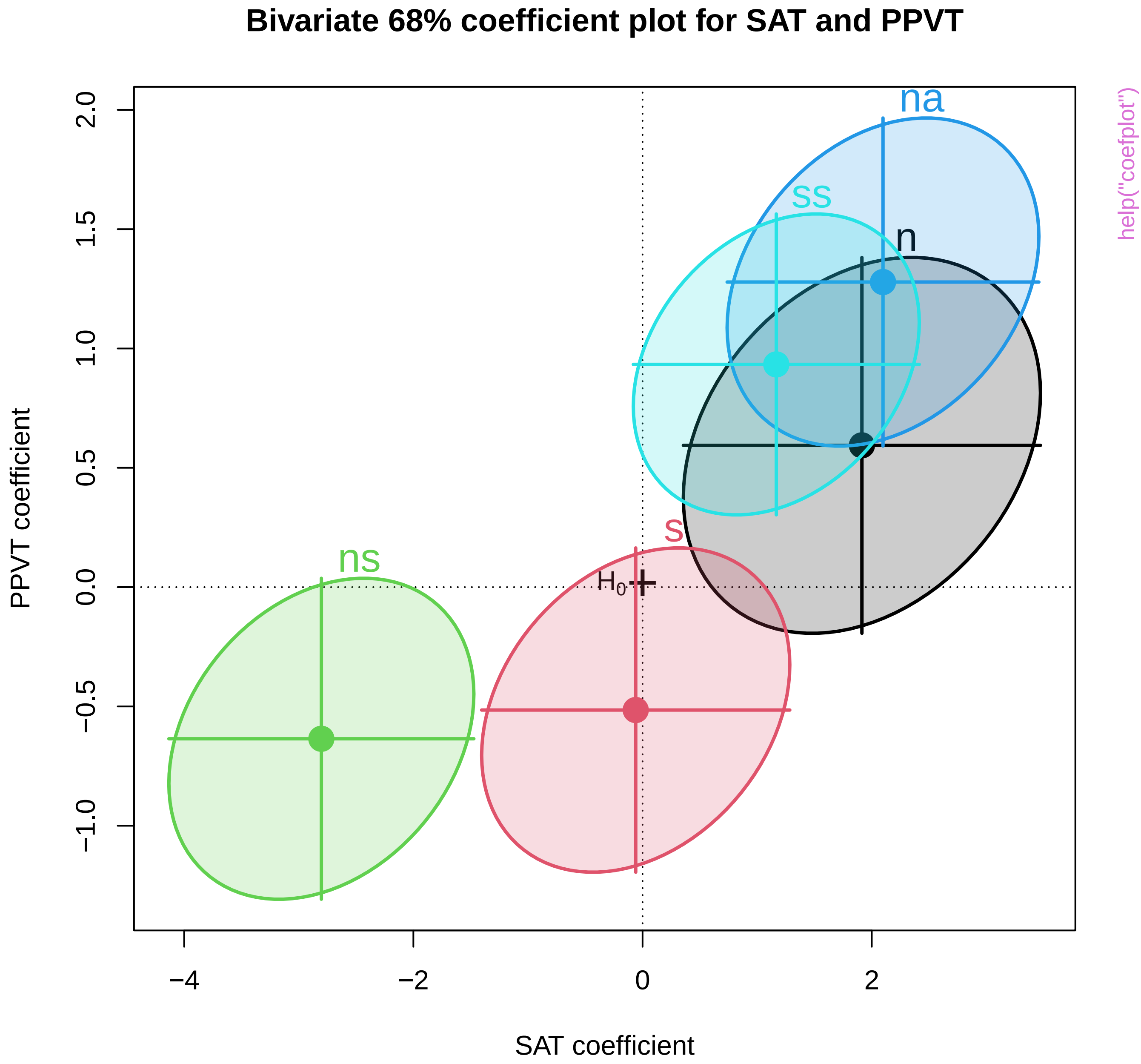

pooled
Species
virginica
versicolor
setosa
−13
−12
−11
−10
−9
−8
log determinant
help("boxM")



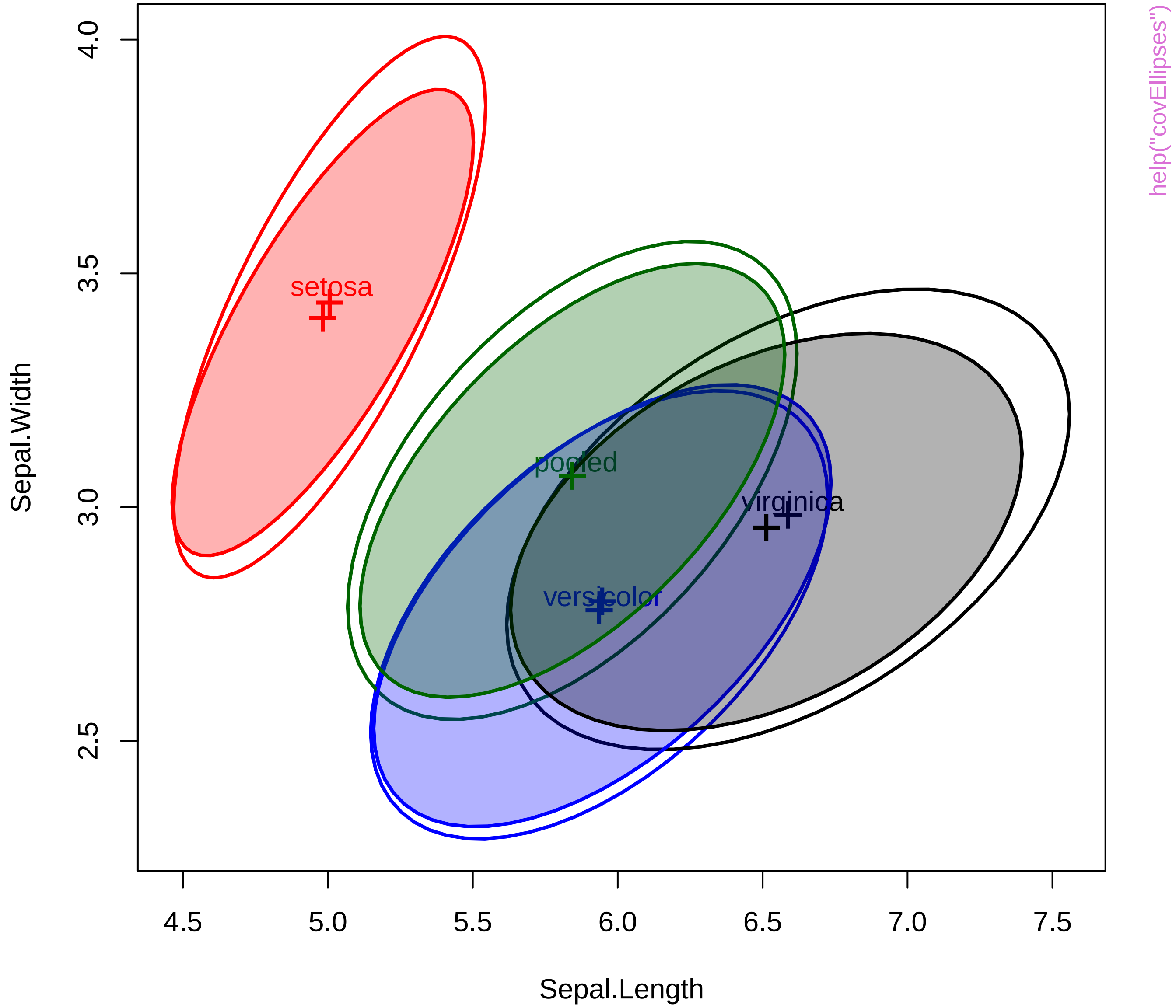
Bivariate coefficient plot for SAT and PPVT
PPVT coefficient
SAT coefficient
ns
s
n
H0
help("coefplot")
−2
0
2
4
−0.5
0.0
0.5
1.0
1.5
2.0
2.5

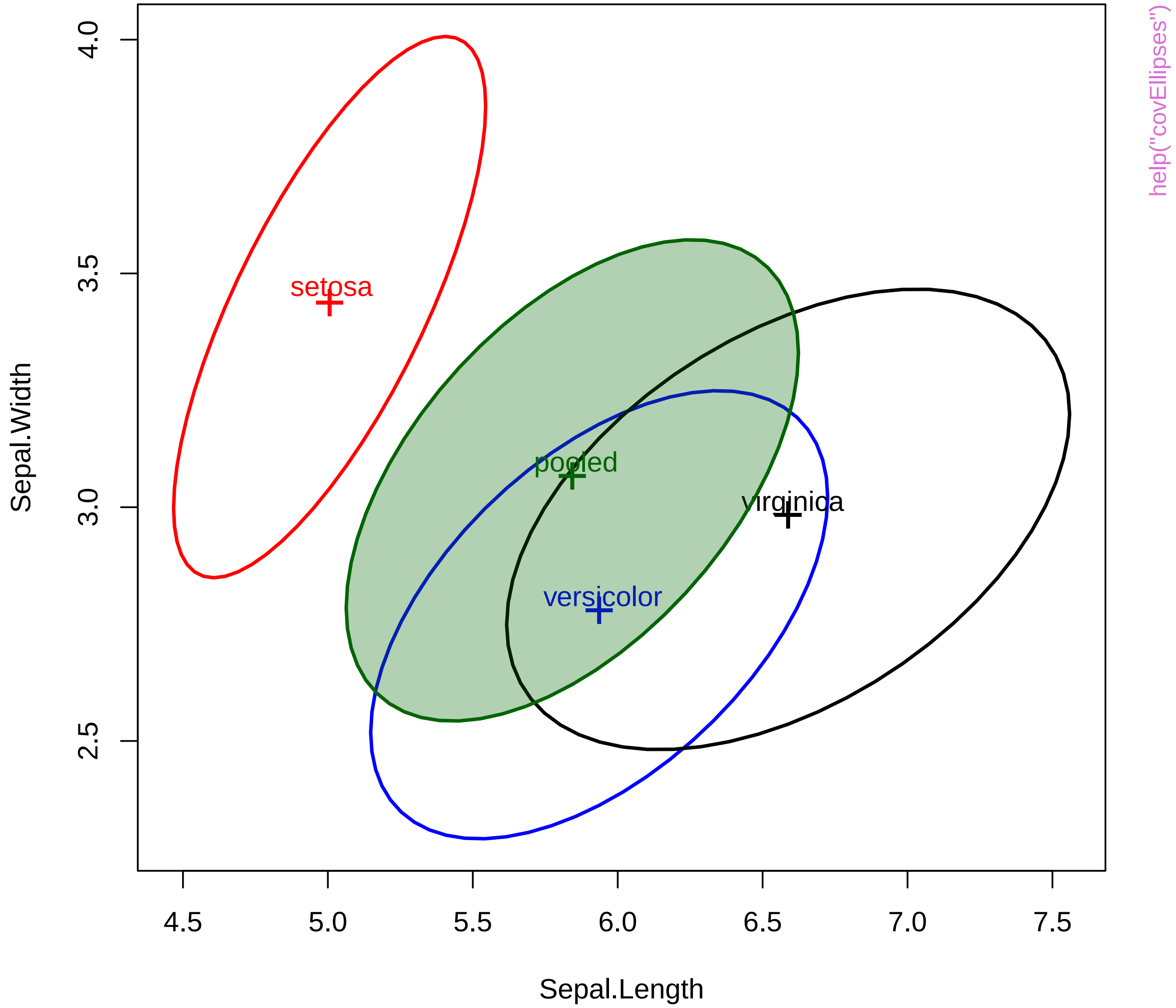


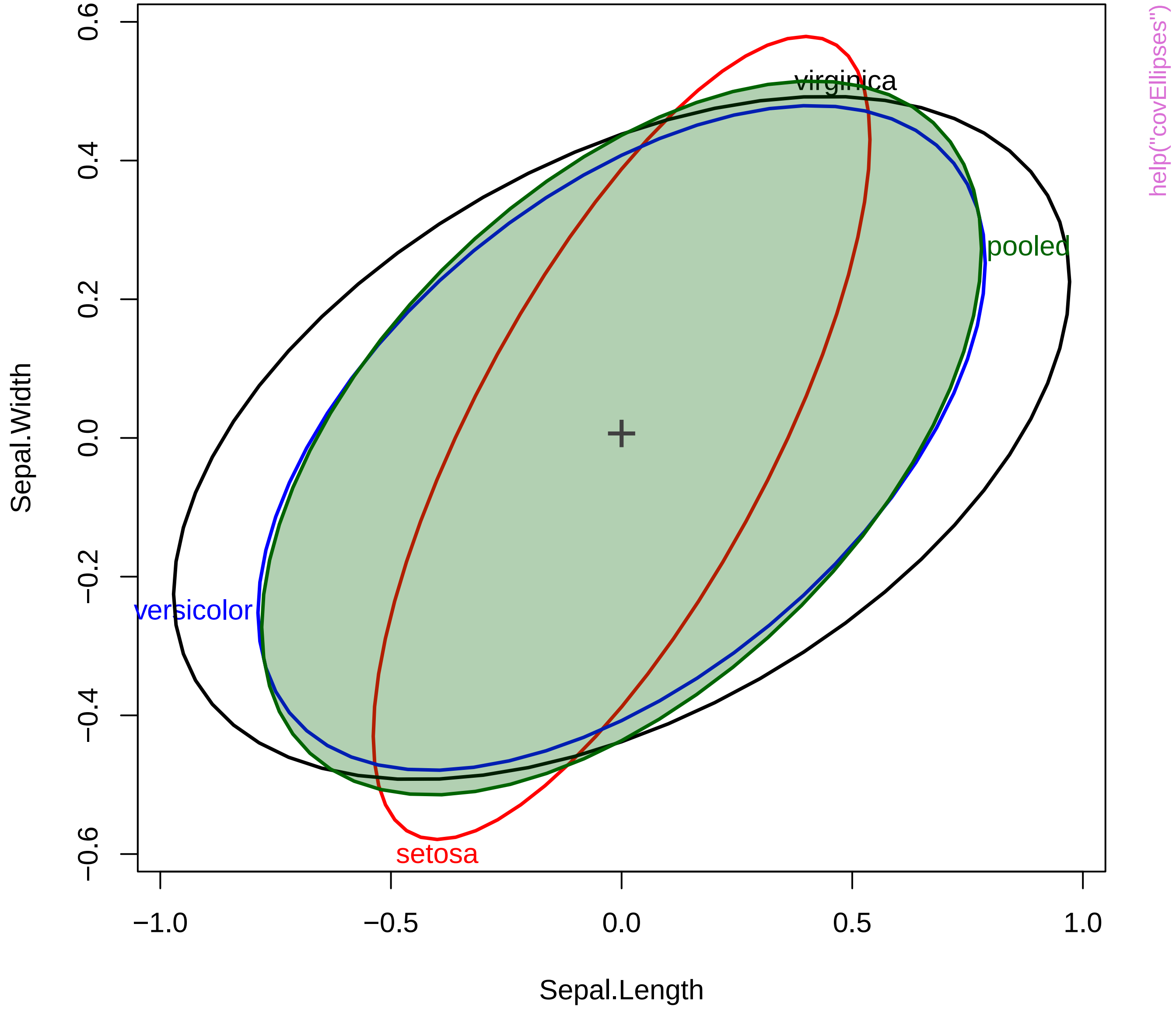
ns
s
n
H0
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−2
0
2
4
Raven coefficient
SAT coefficient
help("coefplot")



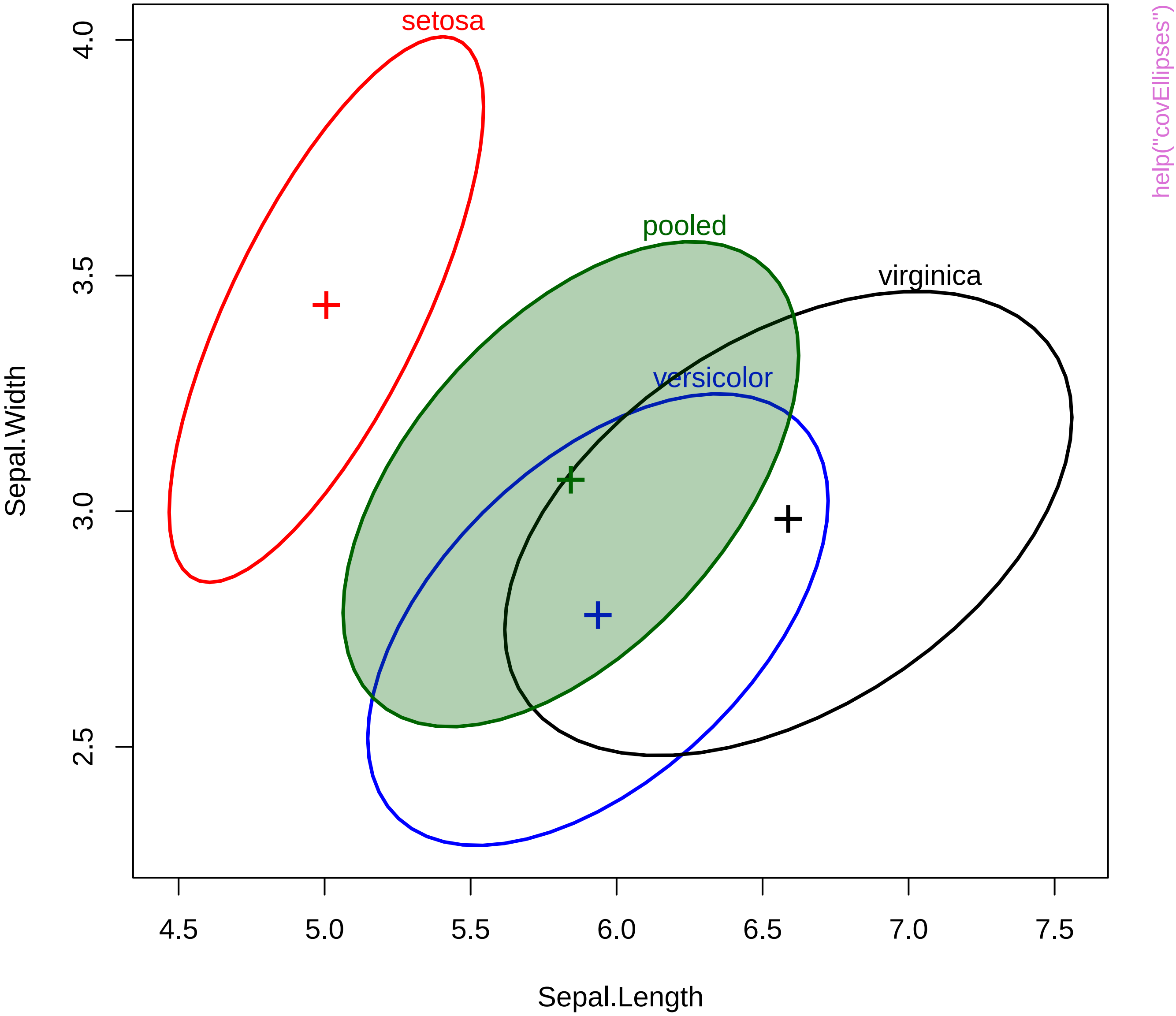
Bivariate 68% coefficient plot for SAT and PPVT
PPVT coefficient
SAT coefficient
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−4
−2
0
2
na
ss
n
s
ns
H_0
help("coefplot")

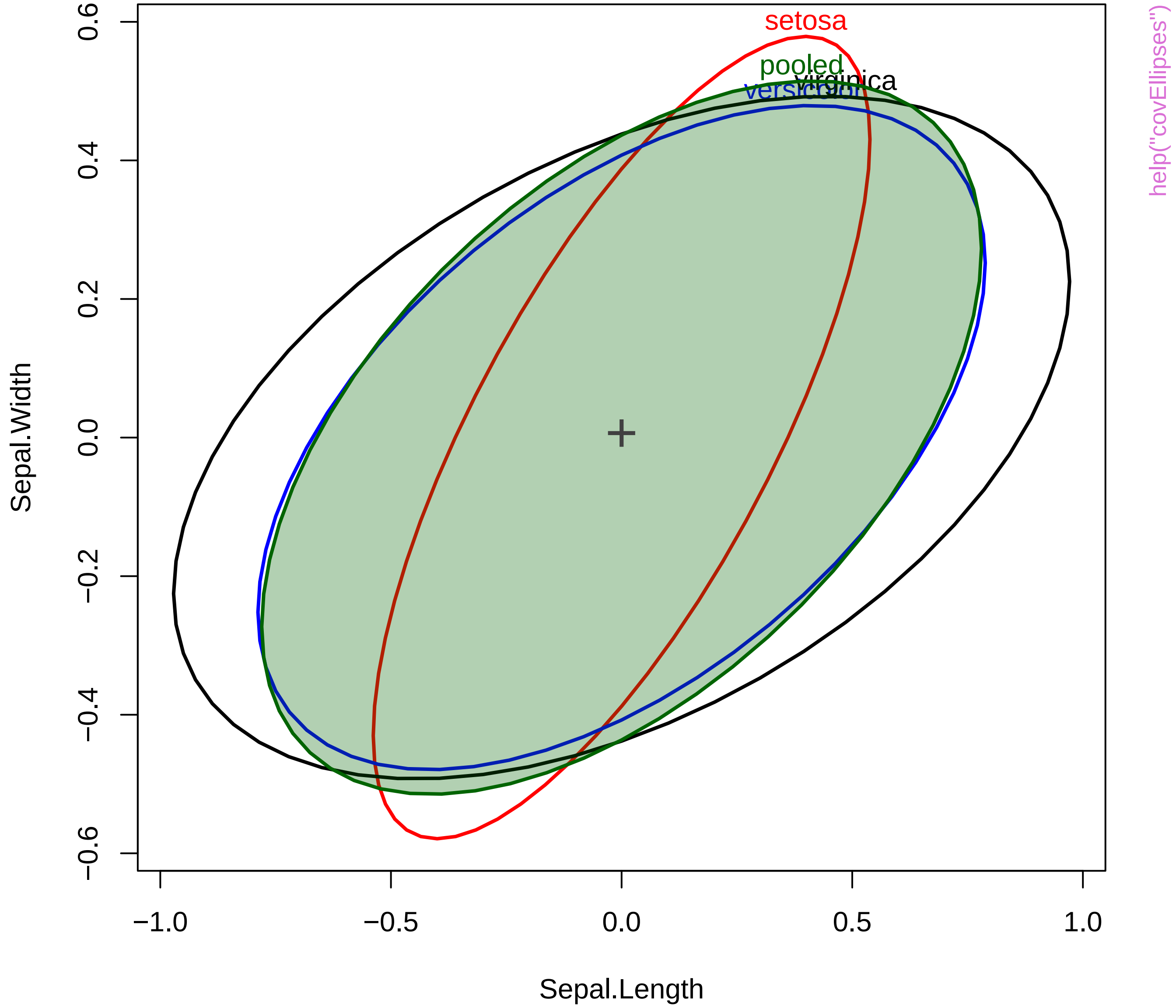

setosa
pooled
virginica
versicolor
4.0
3.5
3.0
2.5
4.5
5.0
5.5
6.0
6.5
7.0
7.5
Sepal.Width
Sepal.Length
help("covEllipses")

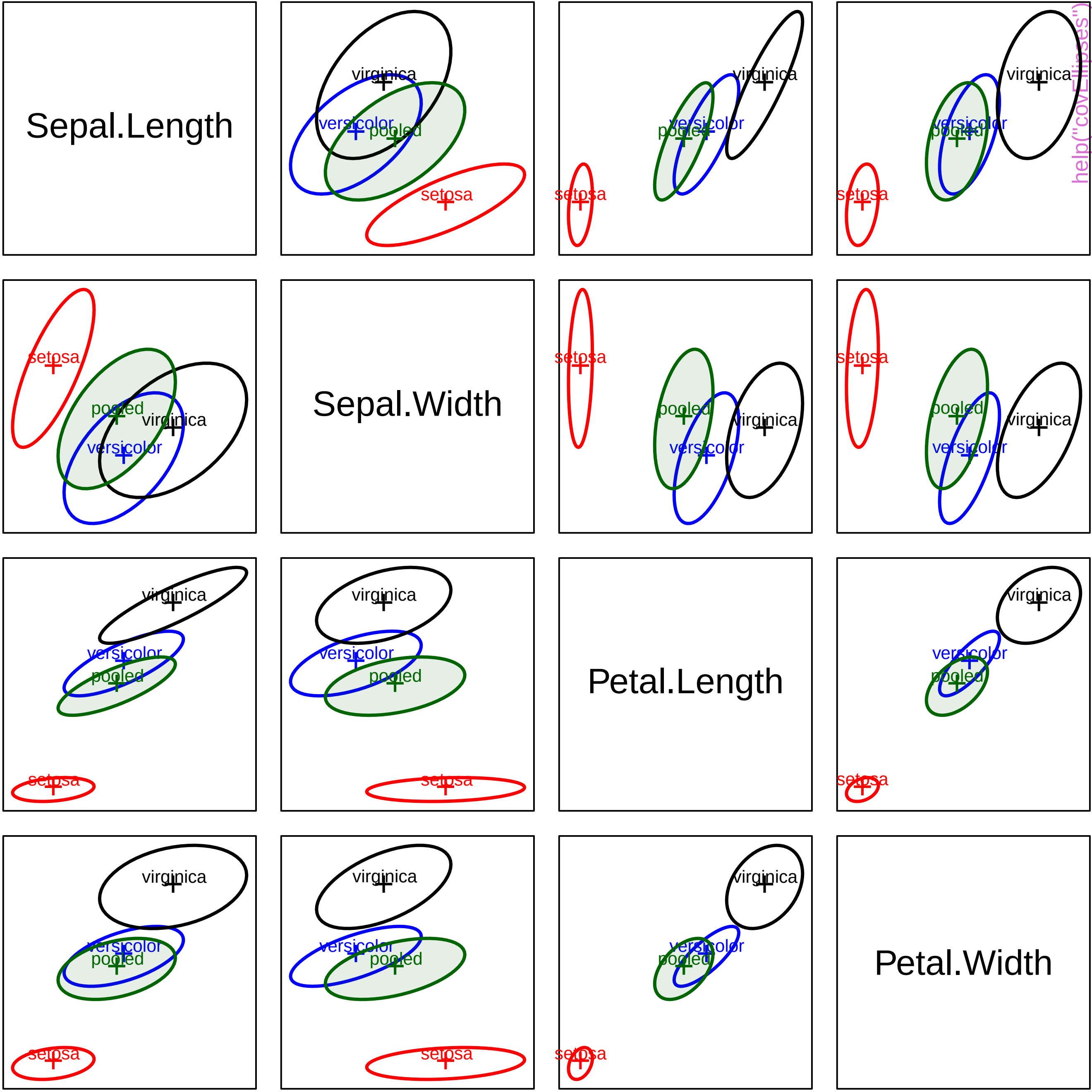

setosa
pooled
virginica
versicolor
4.0
3.5
3.0
2.5
4.5
5.0
5.5
6.0
6.5
7.0
7.5
Sepal.Width
Sepal.Length
help("covEllipses")

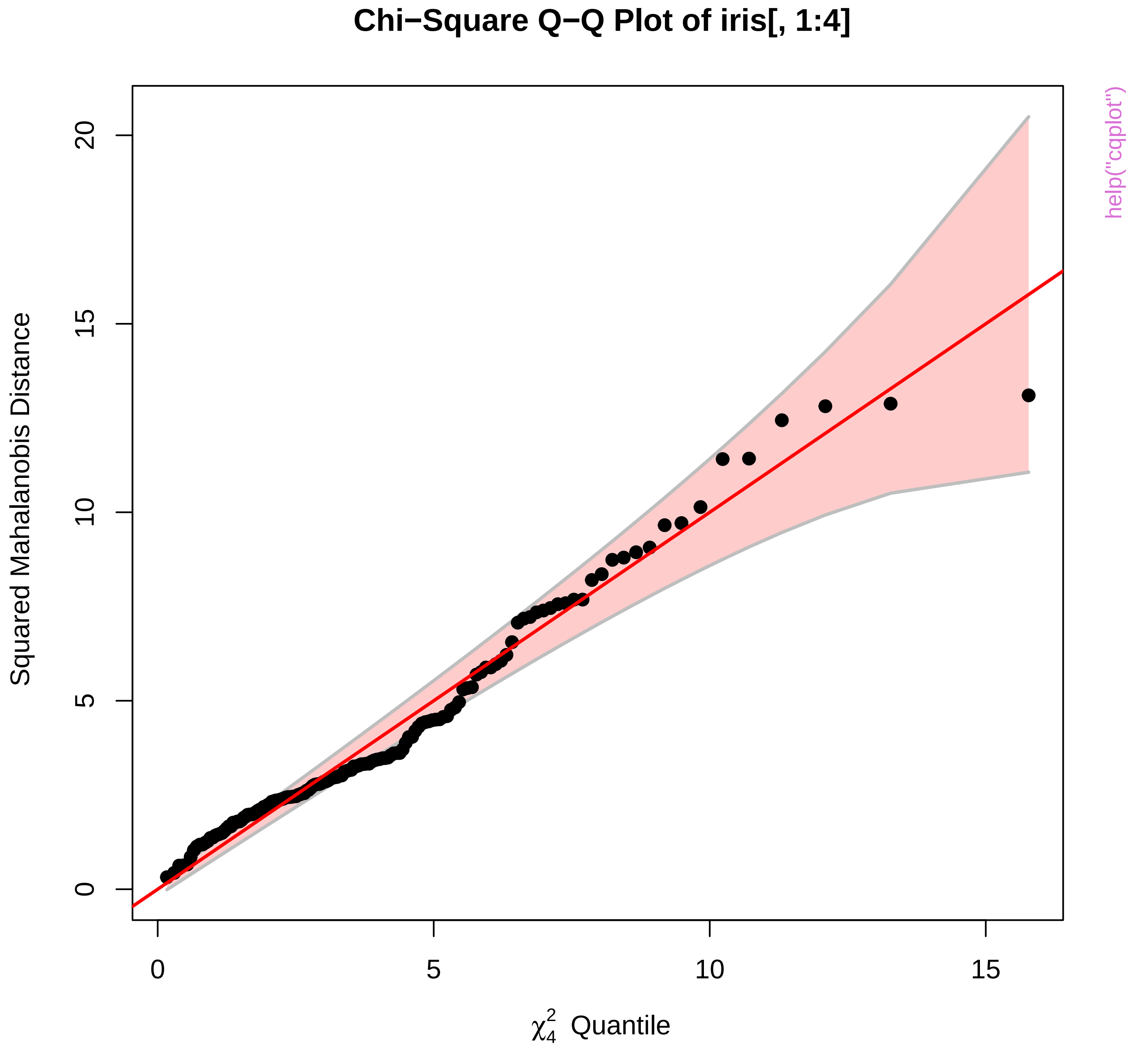


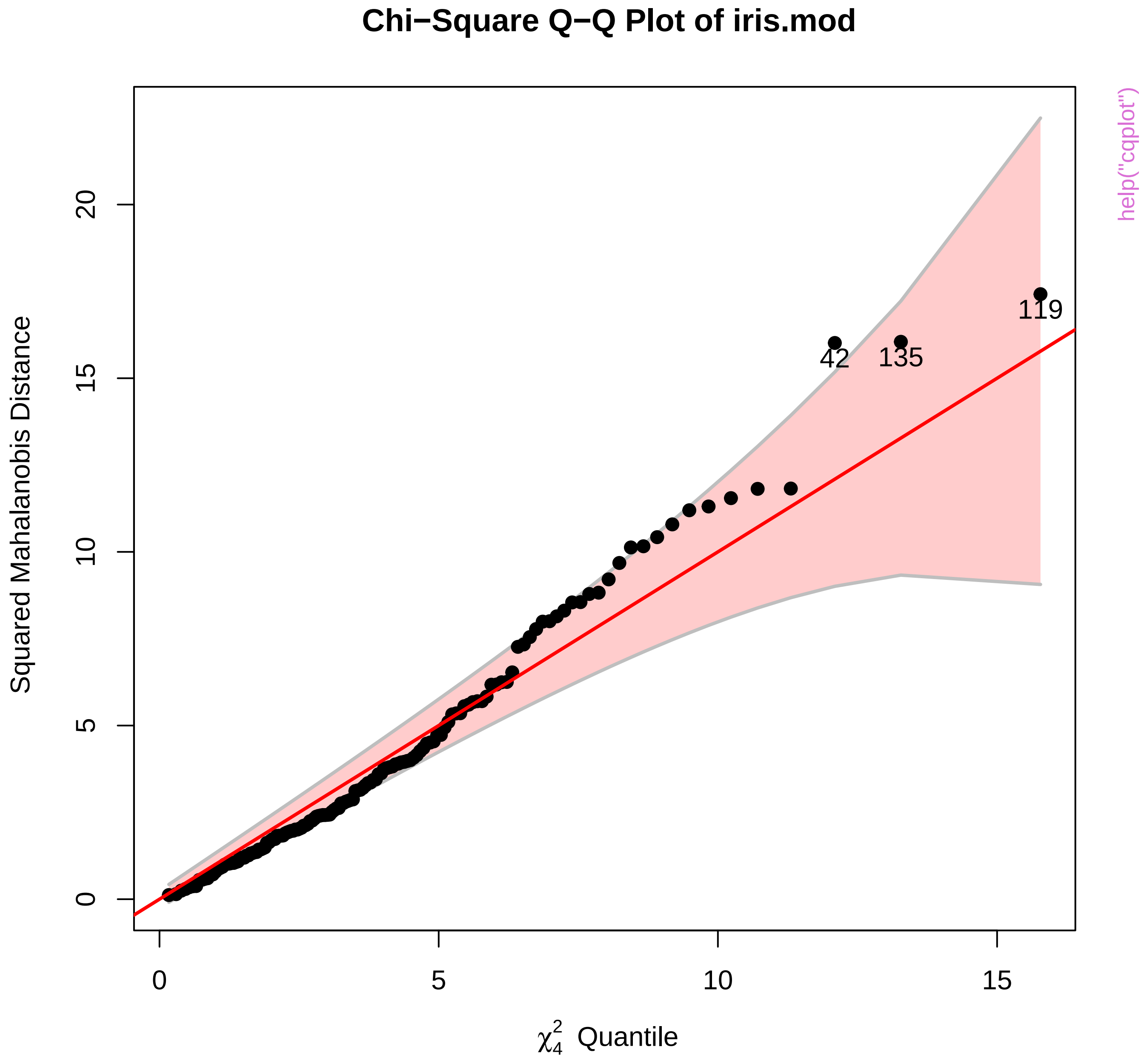
virginica
pooled
versicolor
setosa
help("covEllipses")
Sepal.Length
Sepal.Width
−1.0
−0.5
0.0
0.5
1.0
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


setosa
pooled
virginica
versicolor
4.0
3.5
3.0
2.5
Sepal.Width
4.5
5.0
5.5
6.0
6.5
7.0
7.5
Sepal.Length
help("covEllipses")

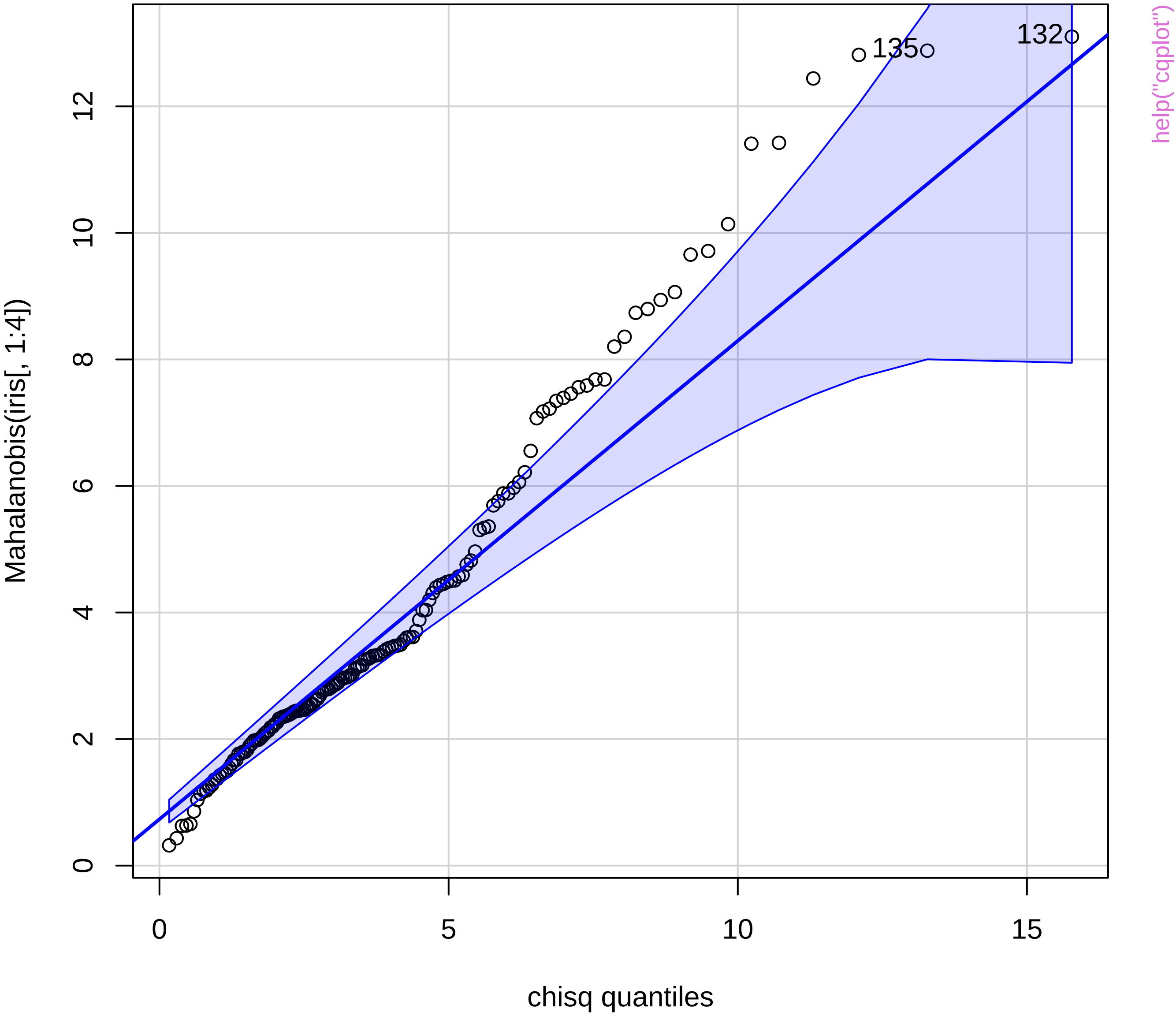

setosa
pooled
virginica
versicolor
help("covEllipses")
Sepal.Width
Sepal.Length
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−1.0
−0.5
0.0
0.5
1.0


Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
virginica
versicolor
pooled
setosa
help("covEllipses")

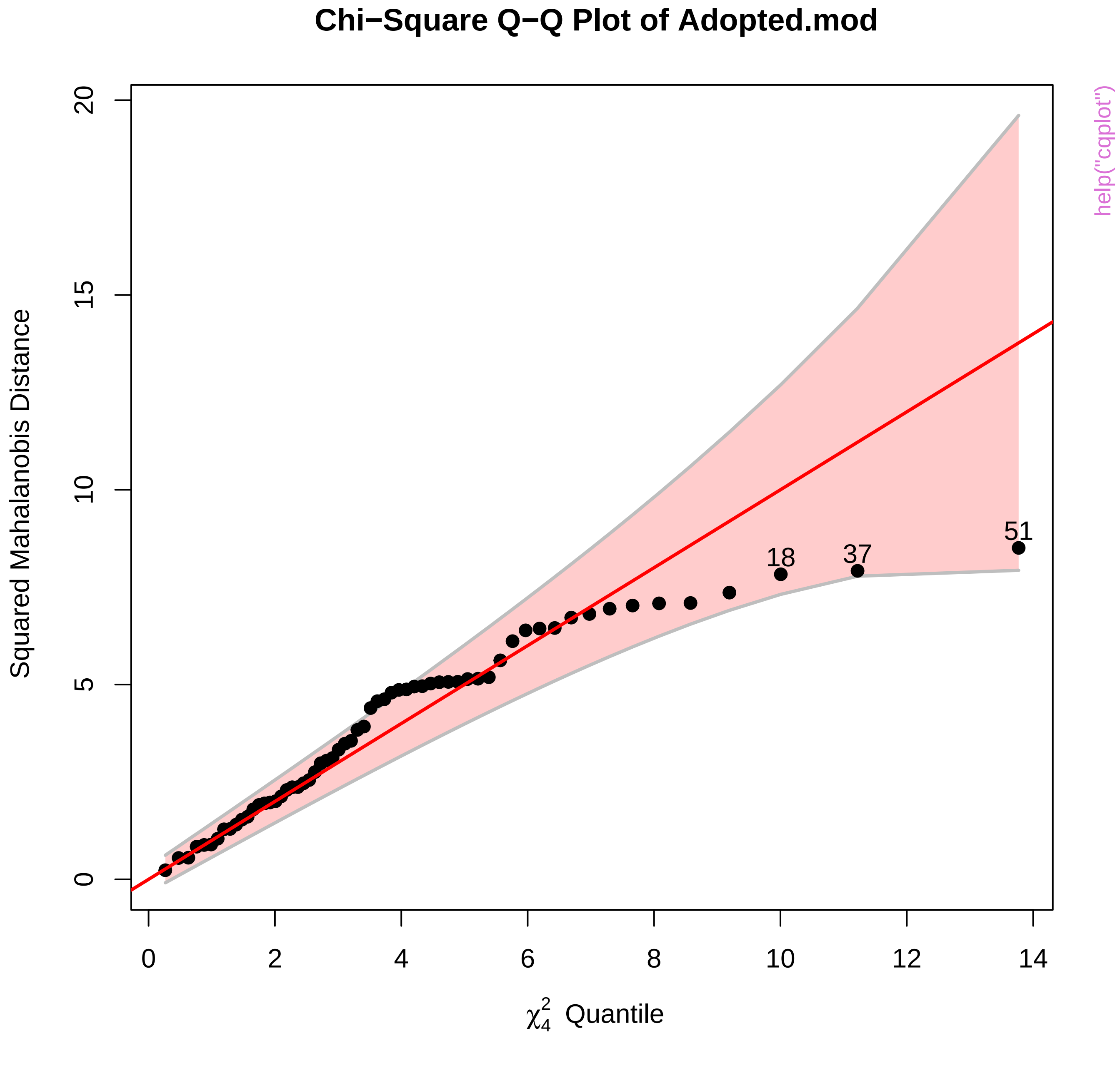

Chi−Square Q−Q Plot of iris[, 1:4]
Squared Mahalanobis Distance
χ^2_4 Quantile
help("cqplot")
0
5
10
15
20

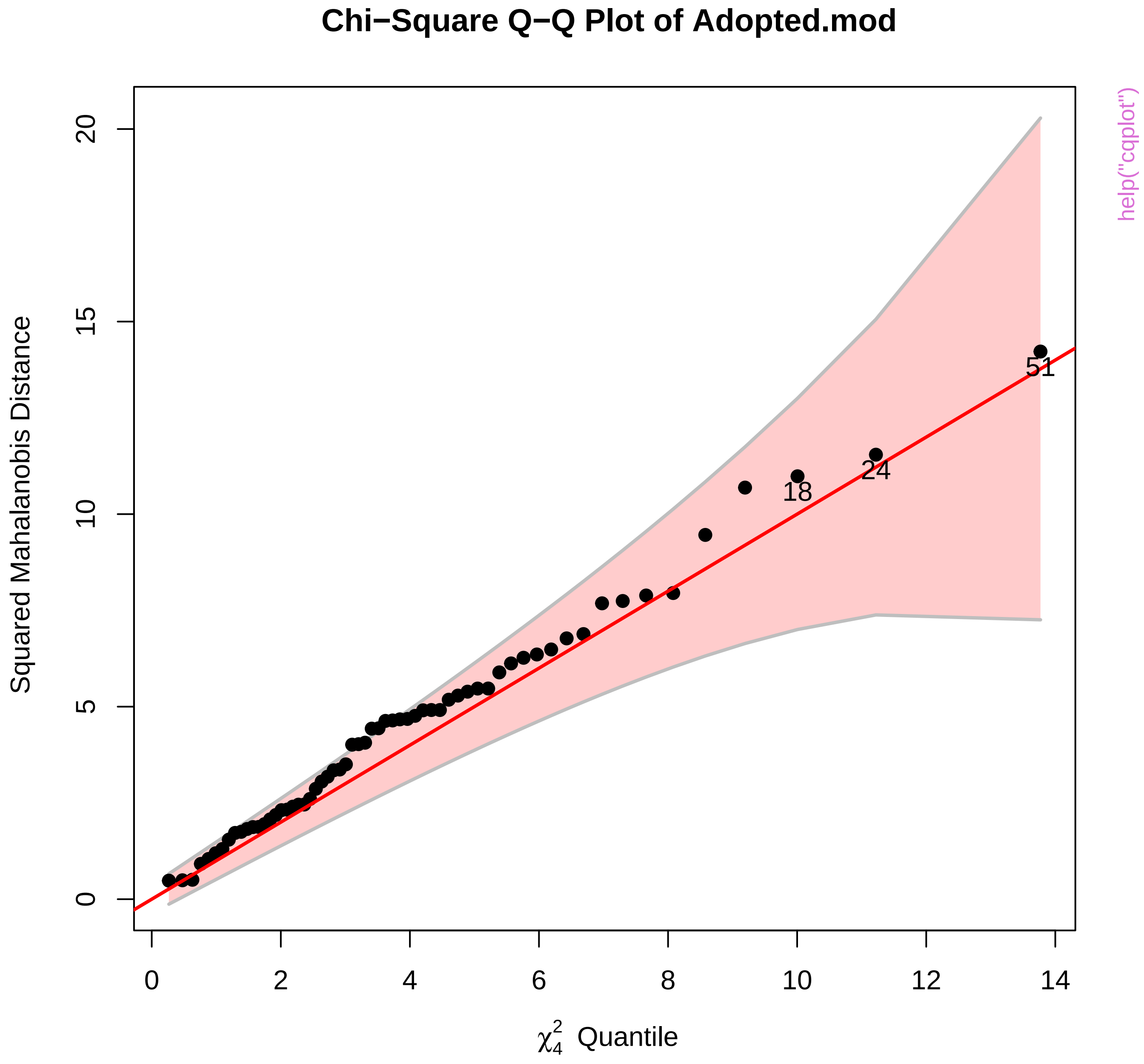


Chi−Square Q−Q Plot of iris.mod
Squared Mahalanobis Distance
0
5
10
15
20
0
5
10
15
χ^2_4 Quantile
42
135
119
help("cqplot")

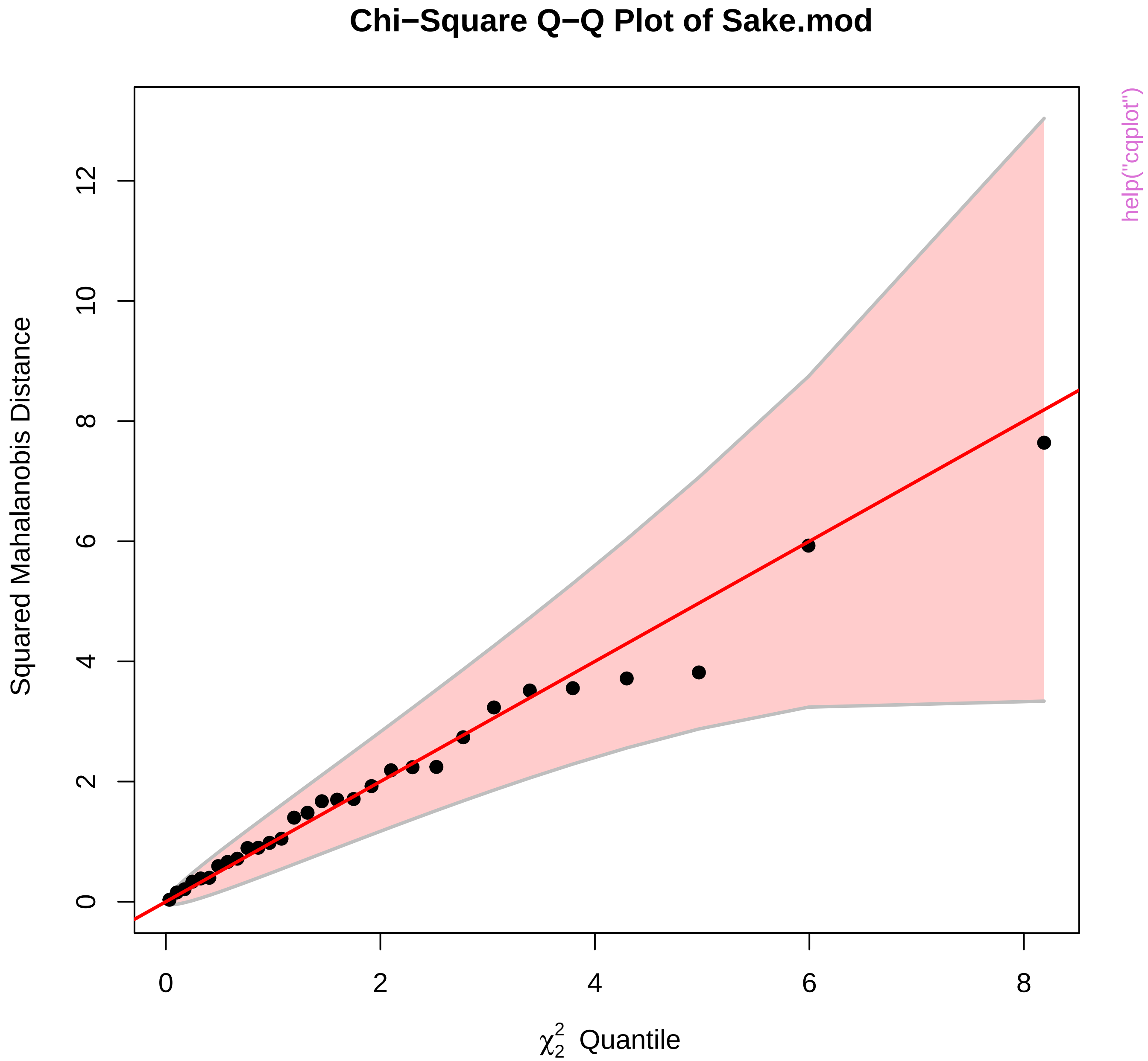


135
132
help("cqplot")
Mahalanobis(iris[, 1:4])
chisq quantiles
0
2
4
6
8
10
12
0
5
10
15

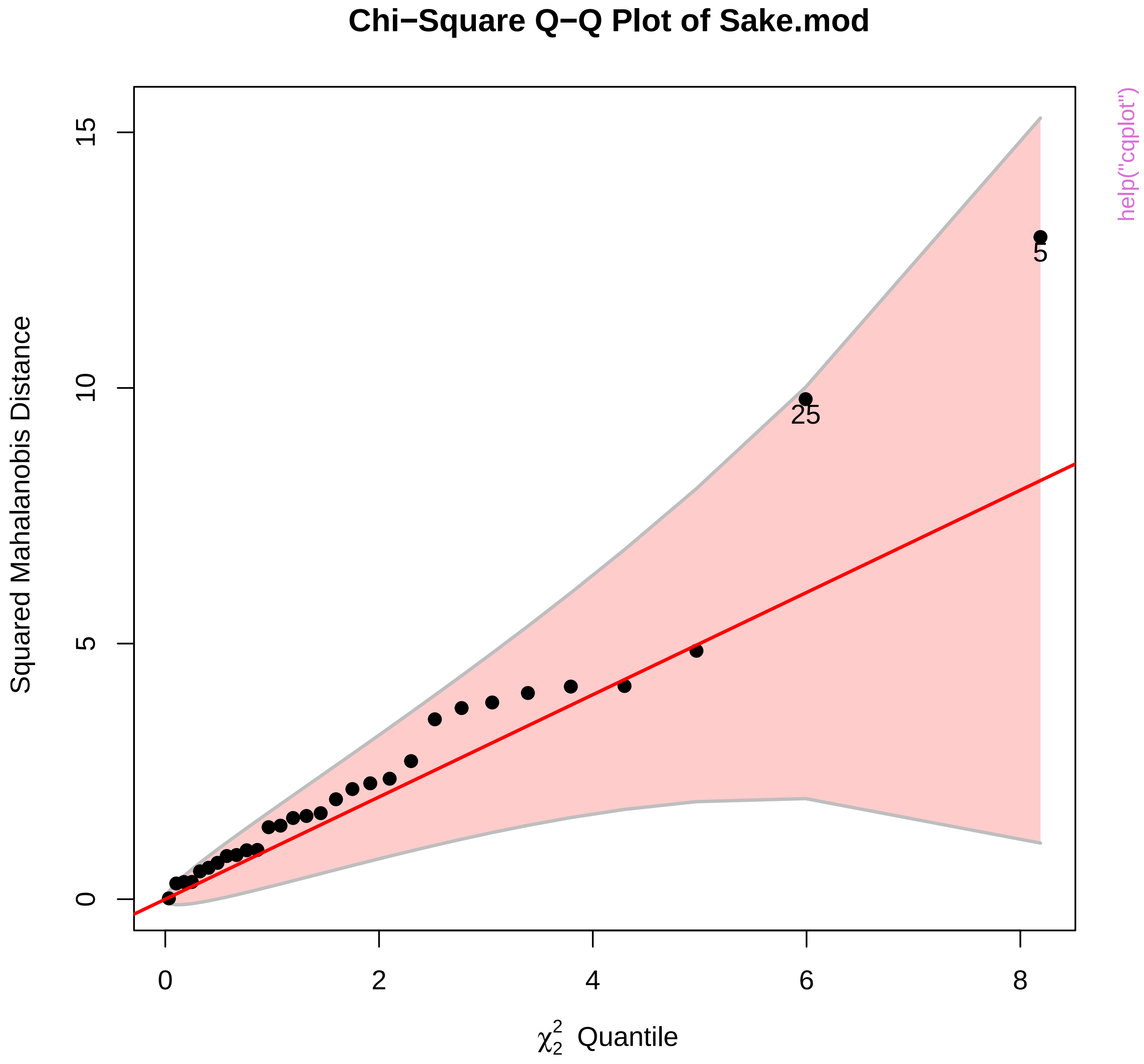

Chi−Square Q−Q Plot of Adopted.mod
Squared Mahalanobis Distance
χ^2_4 Quantile
0
5
10
15
20
0
2
4
6
8
10
12
14
18
37
51
help("cqplot")

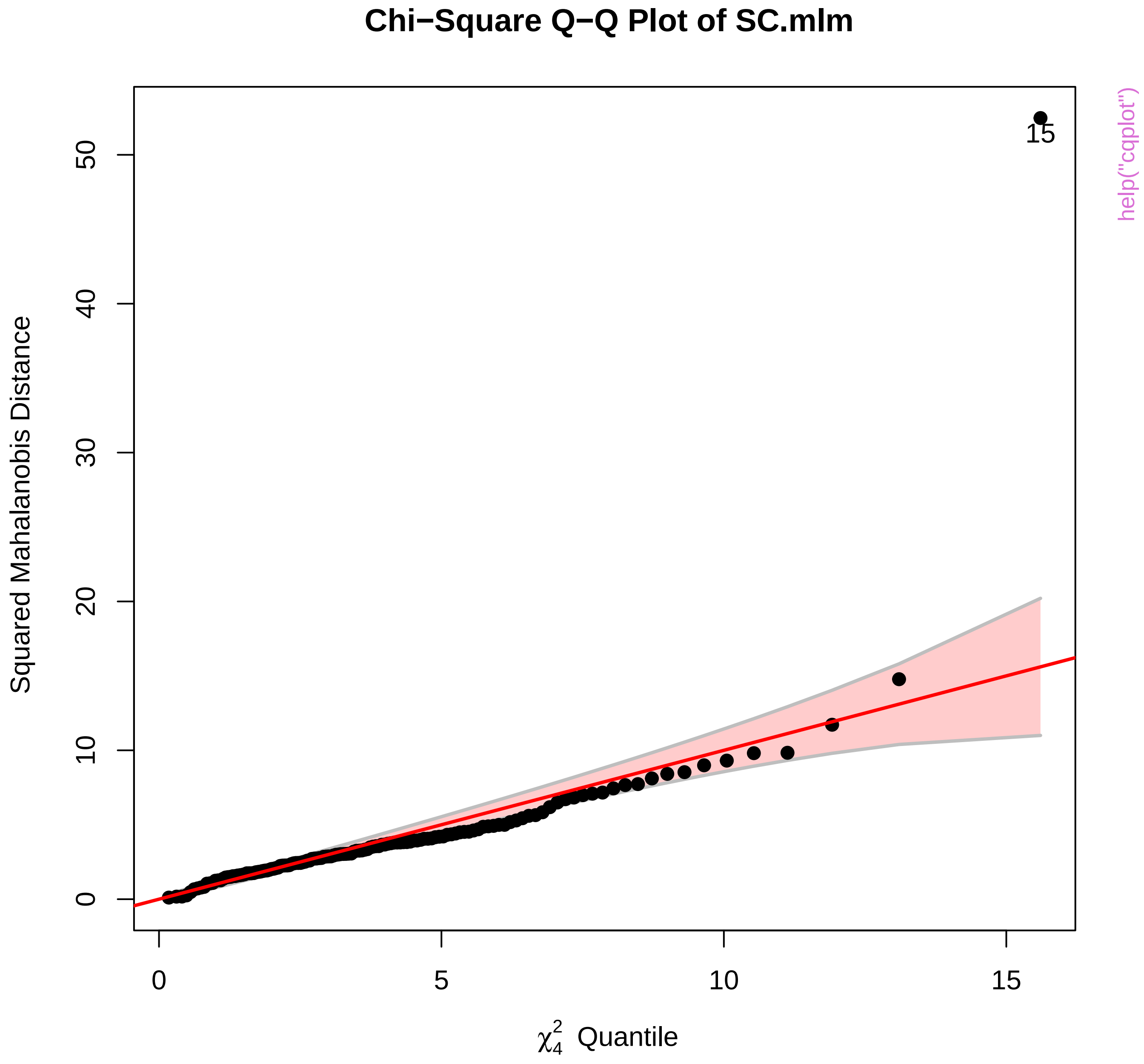


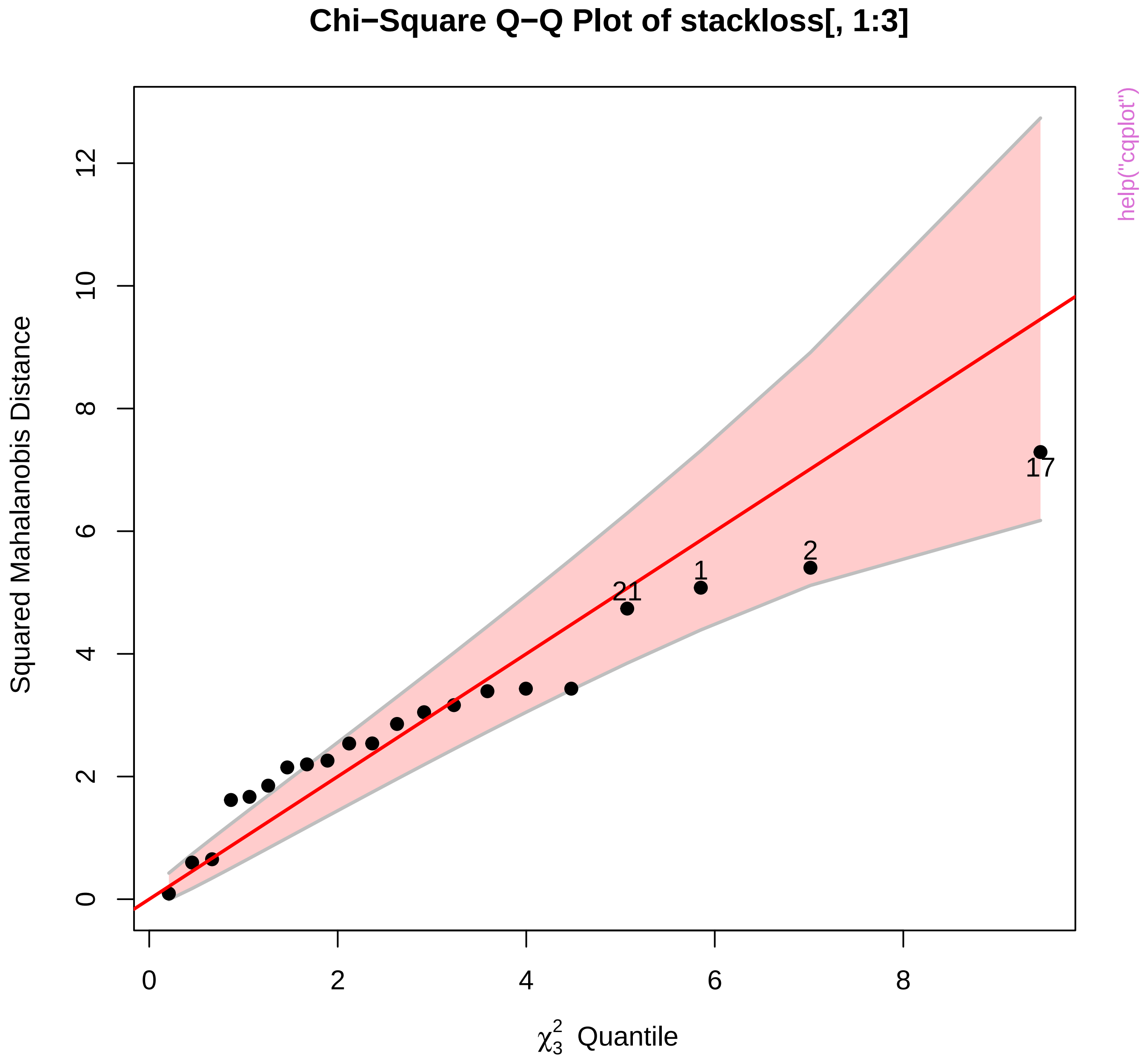
Chi−Square Q−Q Plot of Adopted.mod
Squared Mahalanobis Distance
0
5
10
15
20
0
2
4
6
8
10
12
14
χ^2_4 Quantile
18
24
51
help("cqplot")

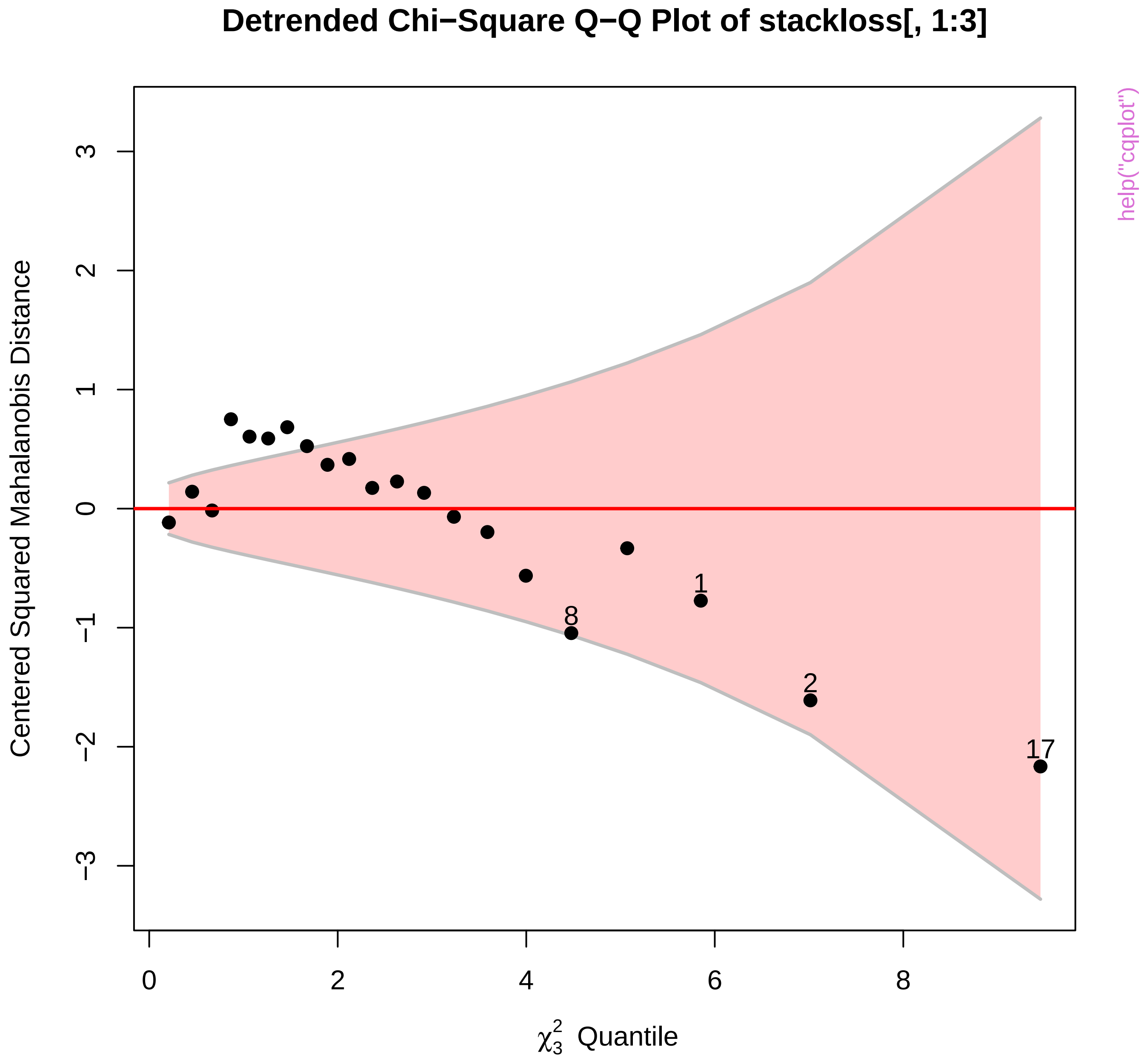


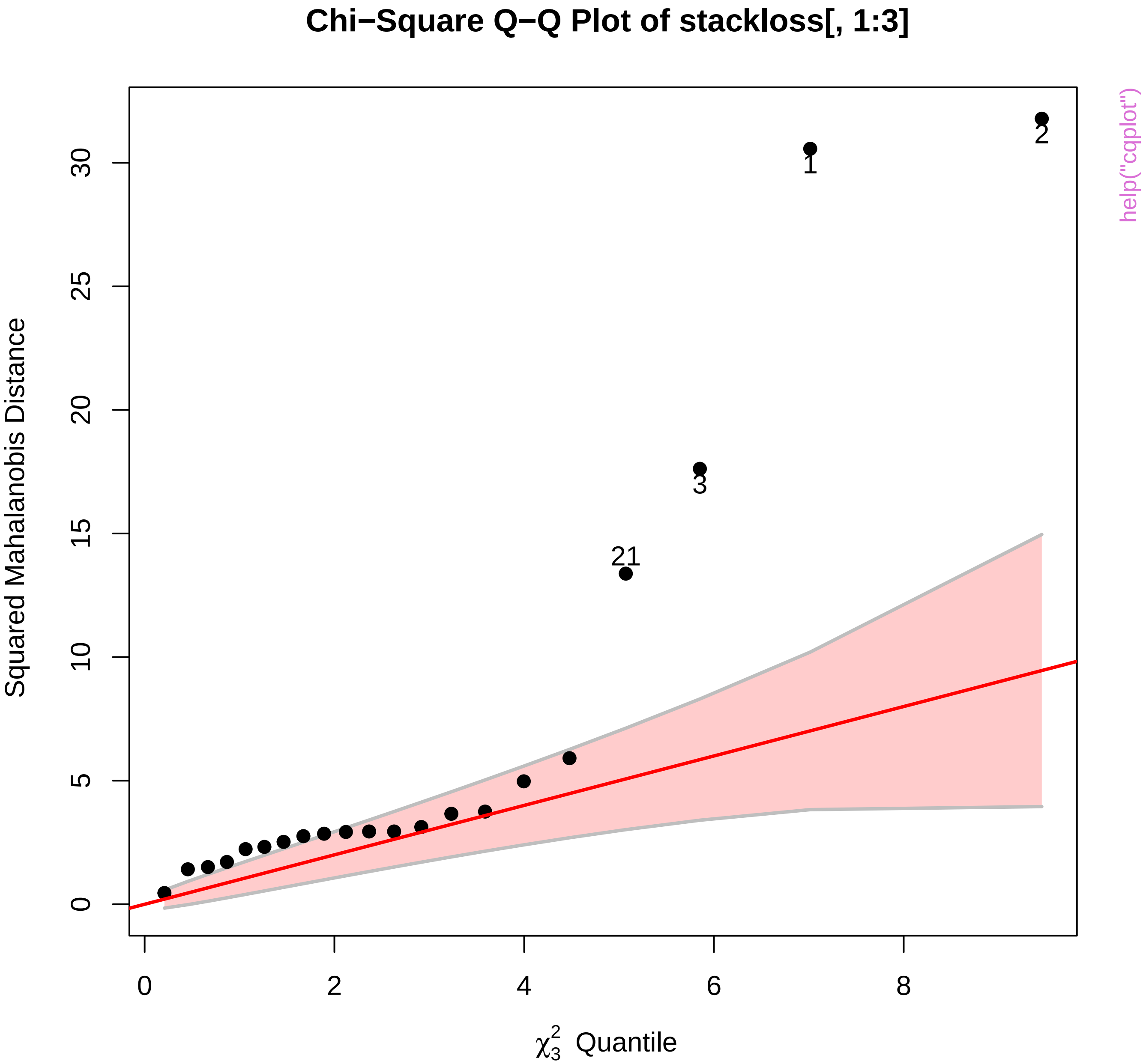
Chi−Square Q−Q Plot of Sake.mod
Squared Mahalanobis Distance
0
2
4
6
8
10
12
0
2
4
6
8
χ^2_2 Quantile
help("cqplot")


Chi−Square Q−Q Plot of Sake.mod
Squared Mahalanobis Distance
χ^2_2 Quantile
help("cqplot")
0
5
10
15
0
2
4
6
8
25
5

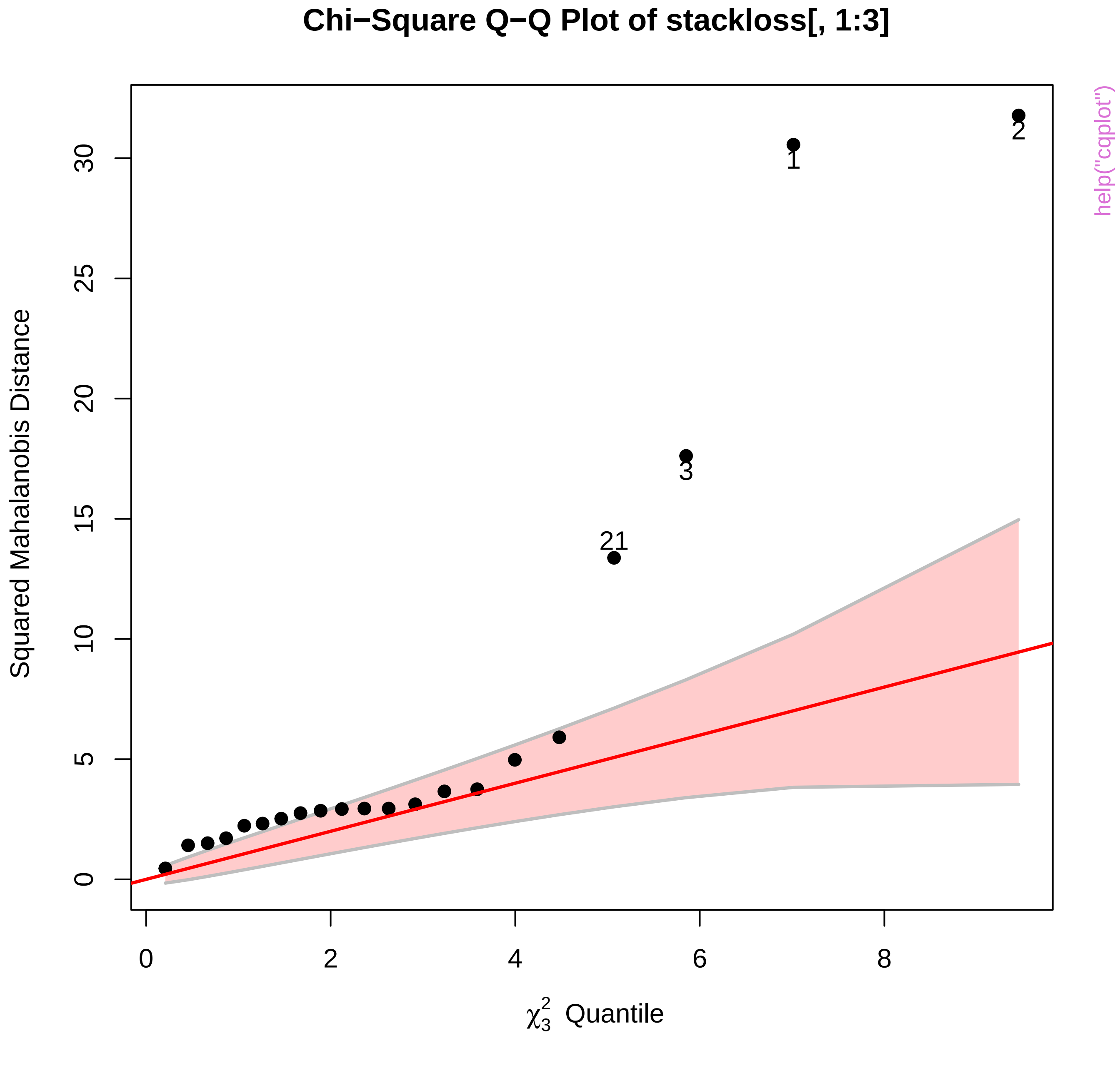

Chi−Square Q−Q Plot of SC.mlm
Squared Mahalanobis Distance
0
10
20
30
40
50
0
5
10
15
χ^2_4 Quantile
15
help("cqplot")


Chi−Square Q−Q Plot of stackloss[, 1:3]
Squared Mahalanobis Distance
χ^2_3 Quantile
0
2
4
6
8
10
12
0
2
4
6
8
21
1
2
17
help("cqplot")

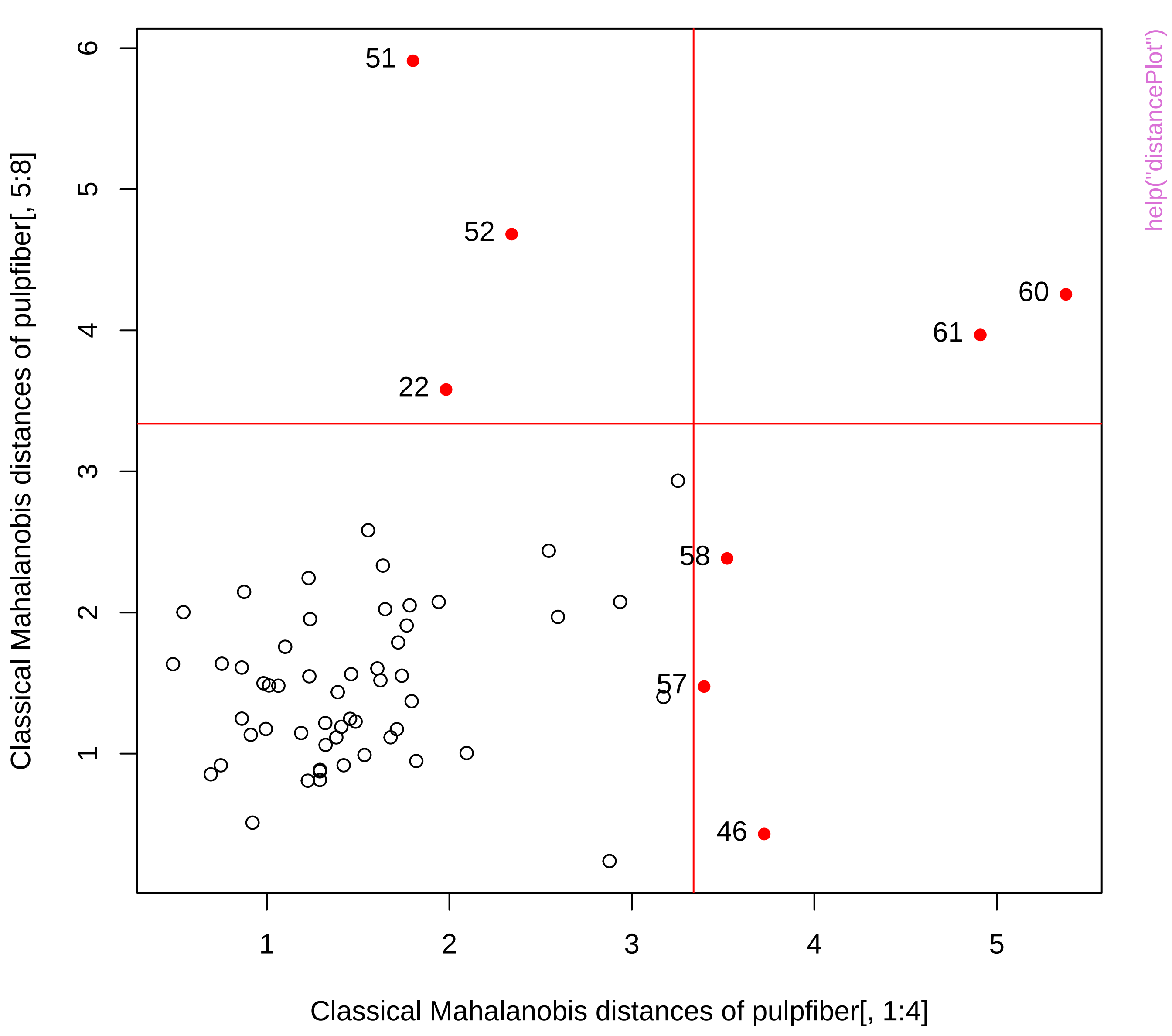

Detrended Chi−Square Q−Q Plot of stackloss[, 1:3]
Centered Squared Mahalanobis Distance
χ^2_3 Quantile
help("cqplot")
3
2
1
0
−1
−2
−3
0
2
4
6
8
8
1
2
17



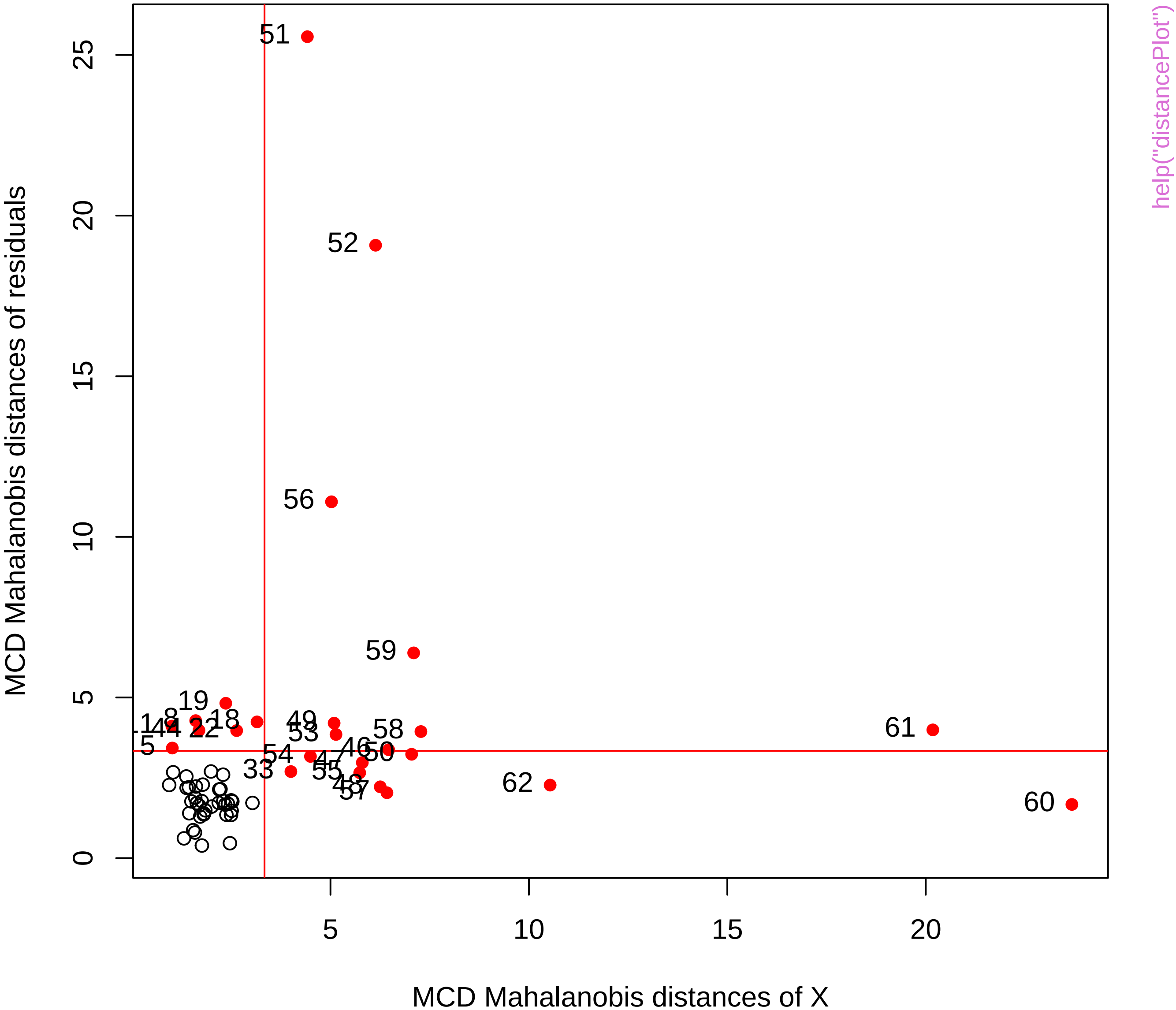
Chi−Square Q−Q Plot of stackloss[, 1:3]
Squared Mahalanobis Distance
0
5
10
15
20
25
30
0
2
4
6
8
χ^2_3 Quantile
1
2
3
21
help("cqplot")



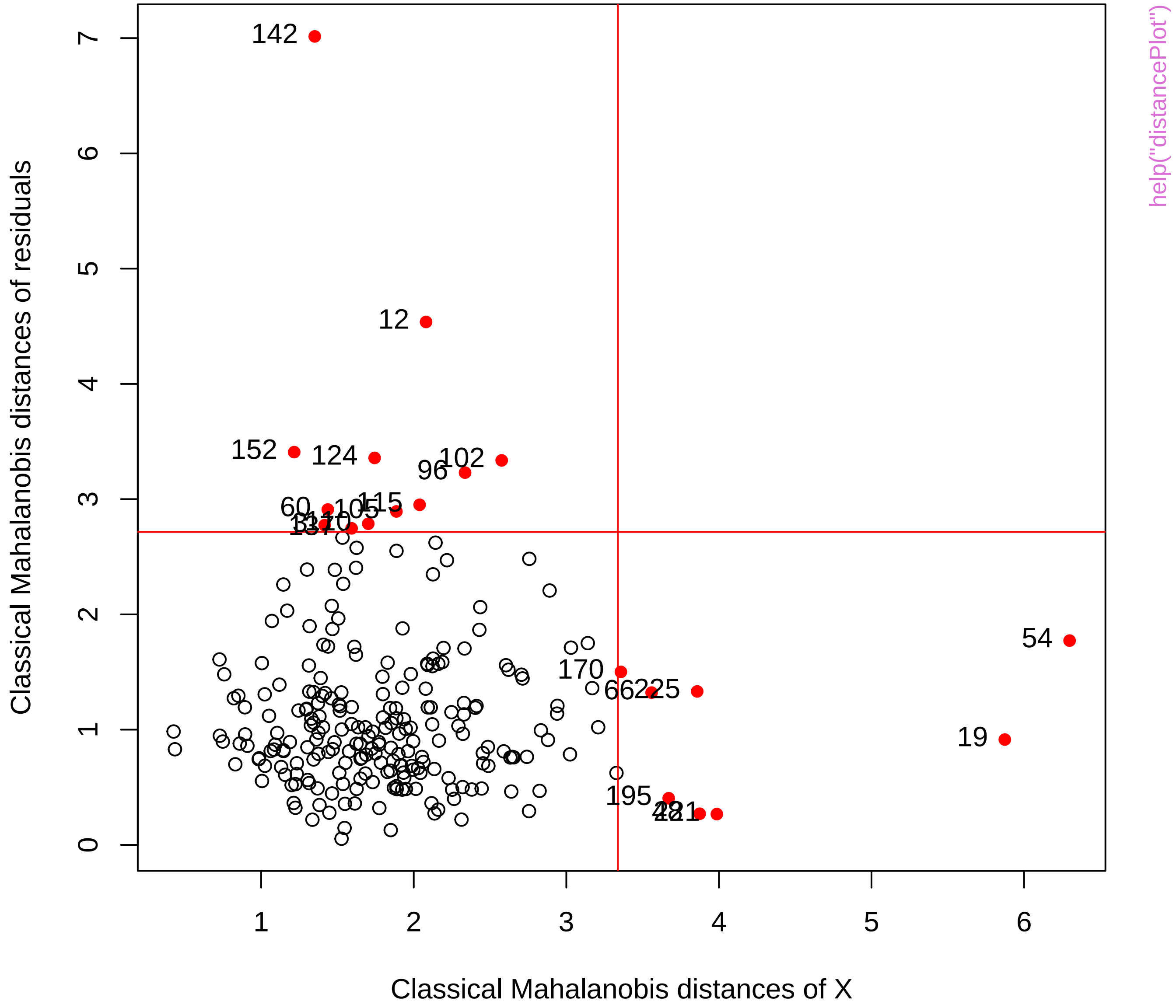
Chi−Square Q−Q Plot of stackloss[, 1:3]
Squared Mahalanobis Distance
χ^2_3 Quantile
0
5
10
15
20
25
30
0
2
4
6
8
1
2
3
21
help("cqplot")



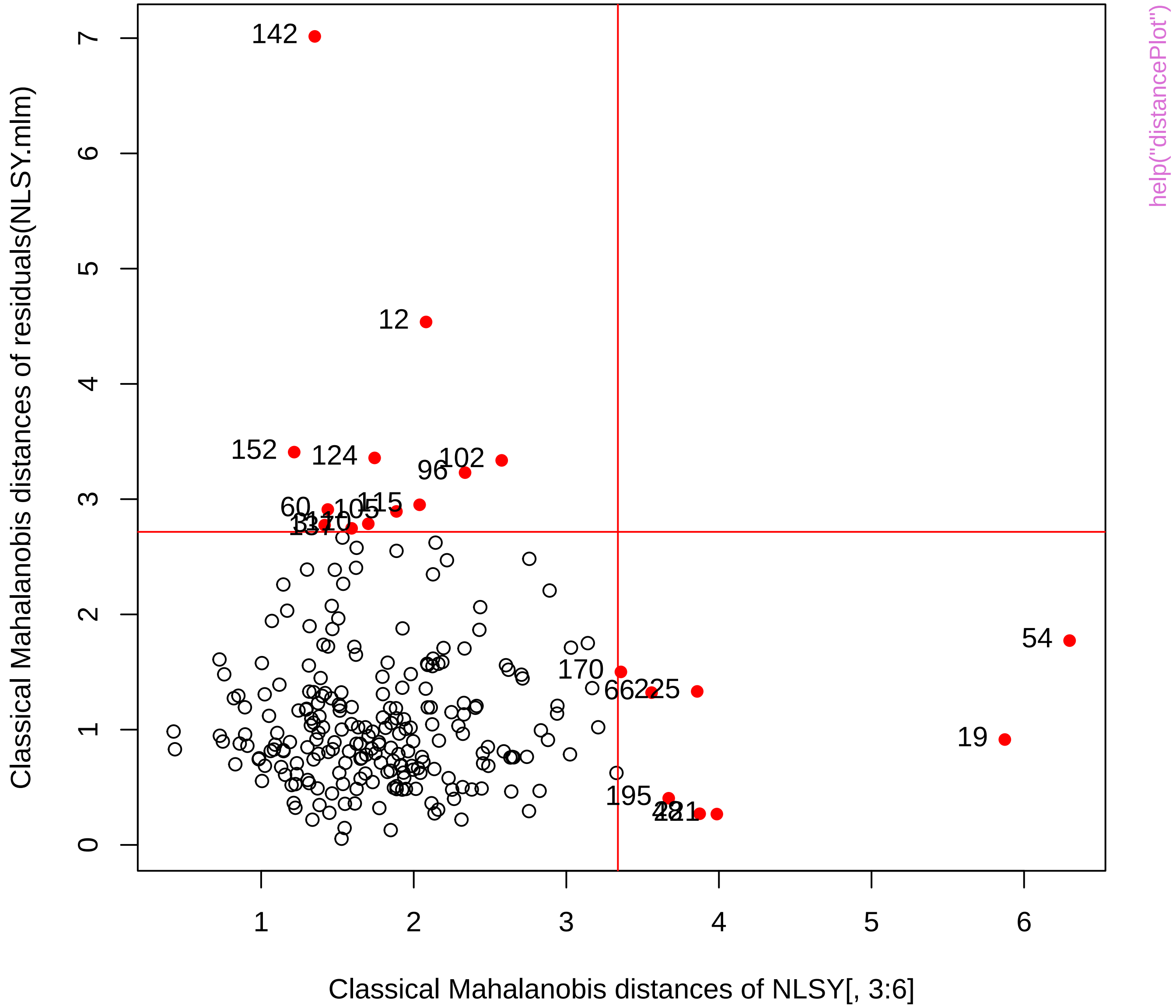
51
52
60
61
22
58
57
46
1
2
3
4
5
6
Classical Mahalanobis distances of pulpfiber[, 1:4]
Classical Mahalanobis distances of pulpfiber[, 5:8]
help("distancePlot")



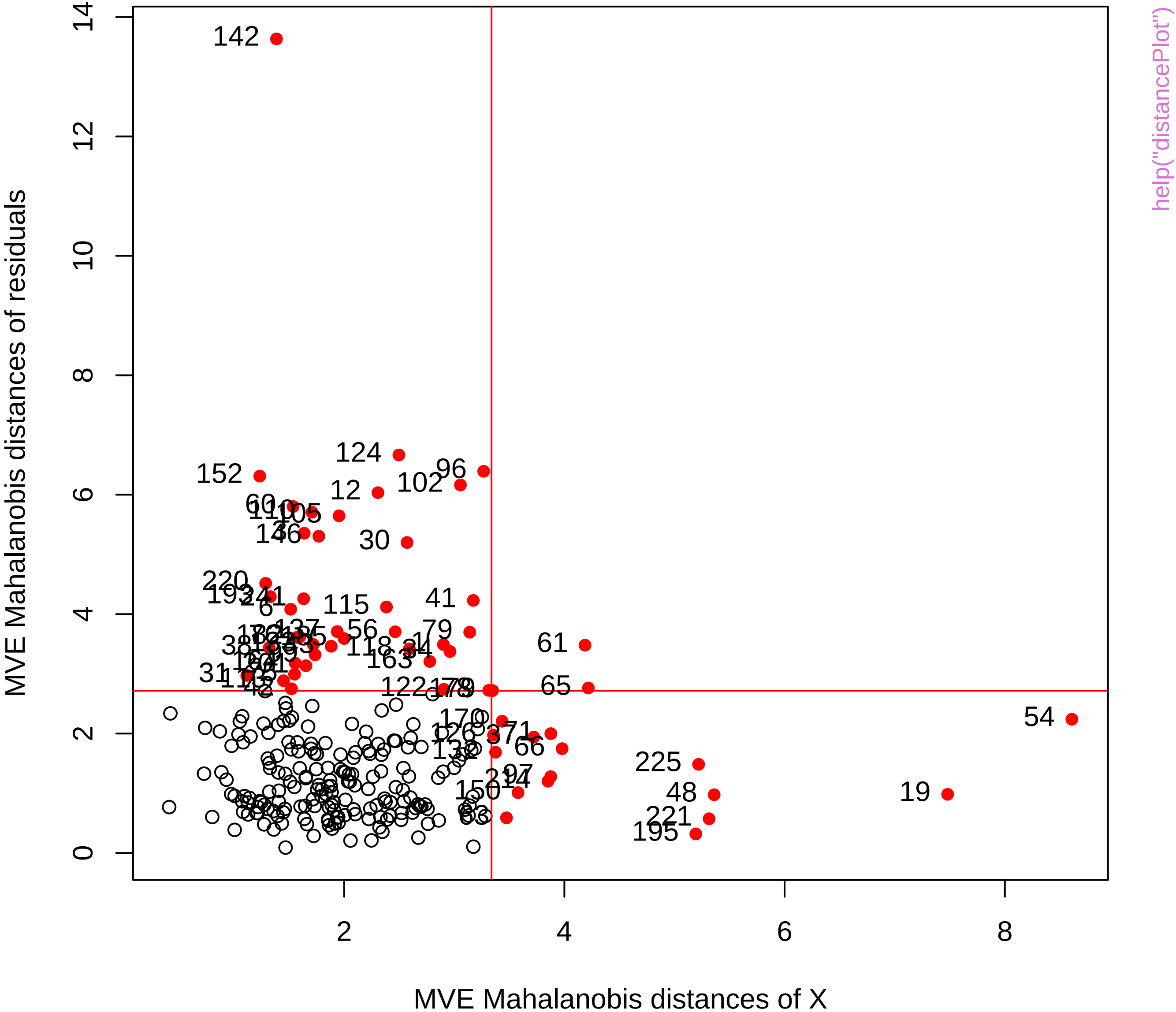
MCD Mahalanobis distances of residuals
MCD Mahalanobis distances of X
help("distancePlot")
0
5
10
15
20
25
5
10
15
20
51
52
56
59
19
18
22
49
53
58
61
54
33
46
50
55
48
57
62
60



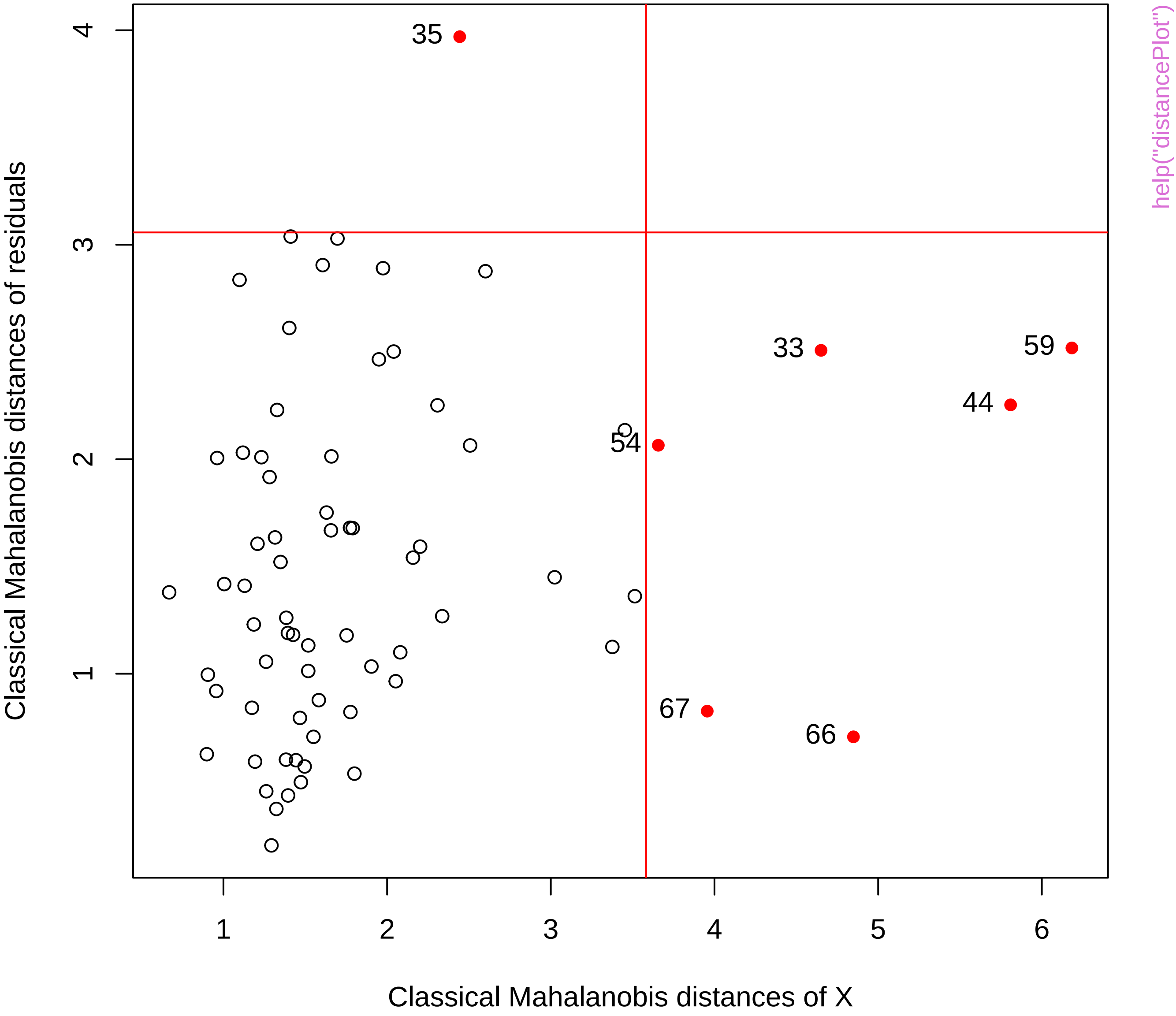
Classical Mahalanobis distances of residuals
Classical Mahalanobis distances of X
help("distancePlot")
142
12
152
124
102
96
60
105
115
170
66
225
54
19
195



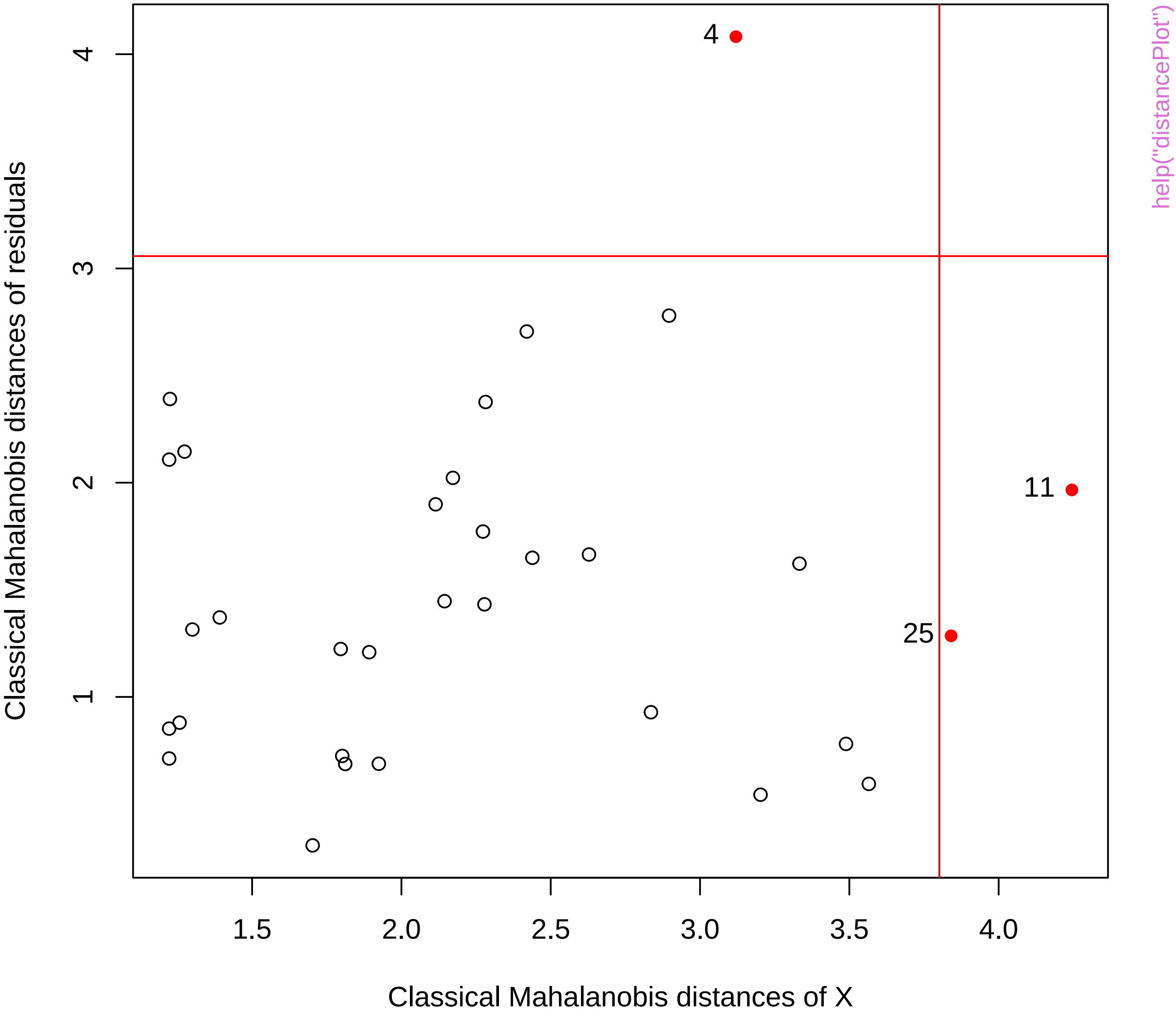
Classical Mahalanobis distances of residuals(NLSY.mlm)
Classical Mahalanobis distances of NLSY[, 3:6]
help("distancePlot")
142
12
152
124
102
96
60
105
115
54
170
66
225
19
195



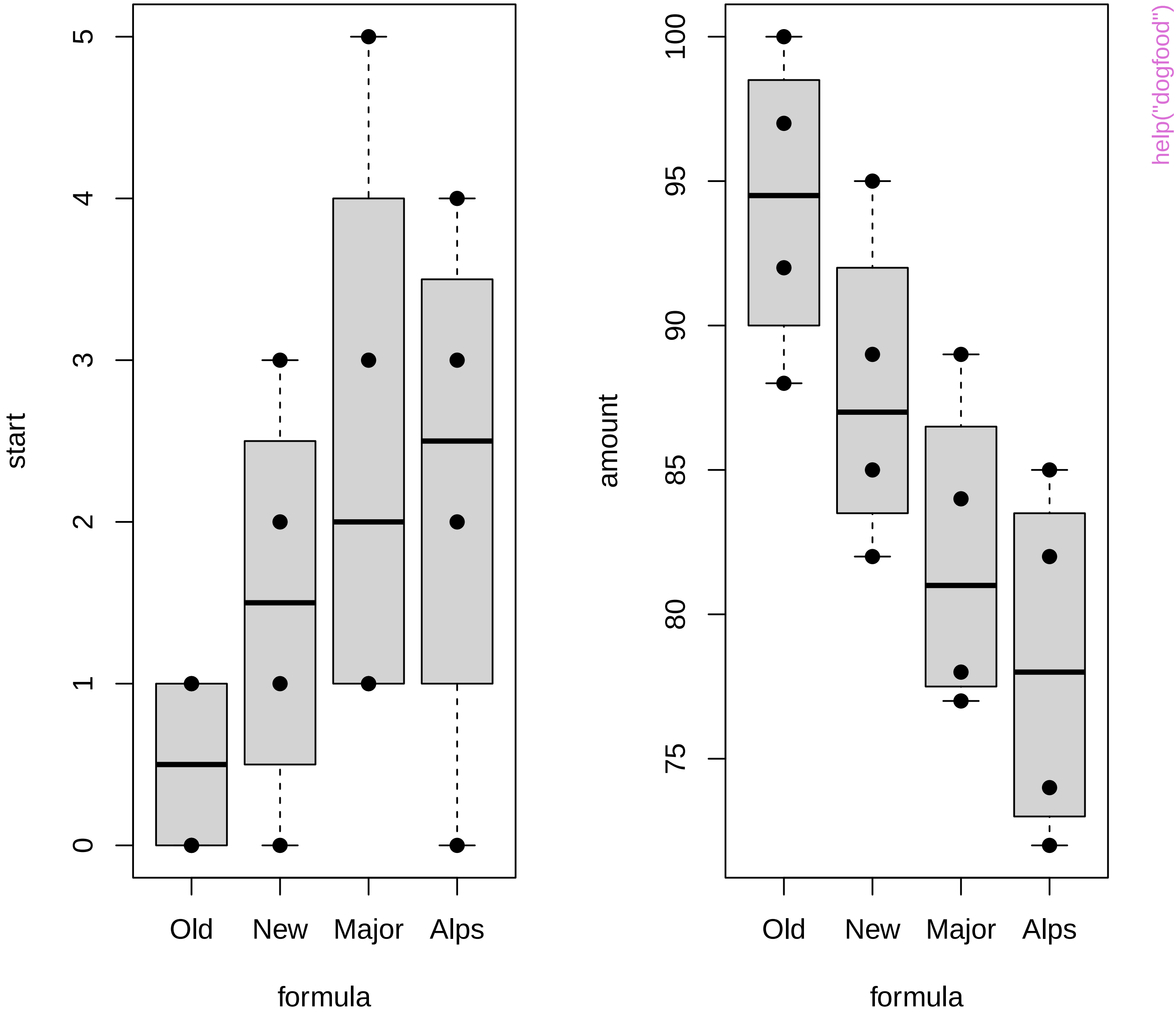
MVE Mahalanobis distances of residuals
MVE Mahalanobis distances of X
help("distancePlot")
0
2
4
6
8
10
12
14
2
4
6
8
142
124
96
152
102
12
30
220
115
41
56
79
61
65
54
225
48
19
221
195

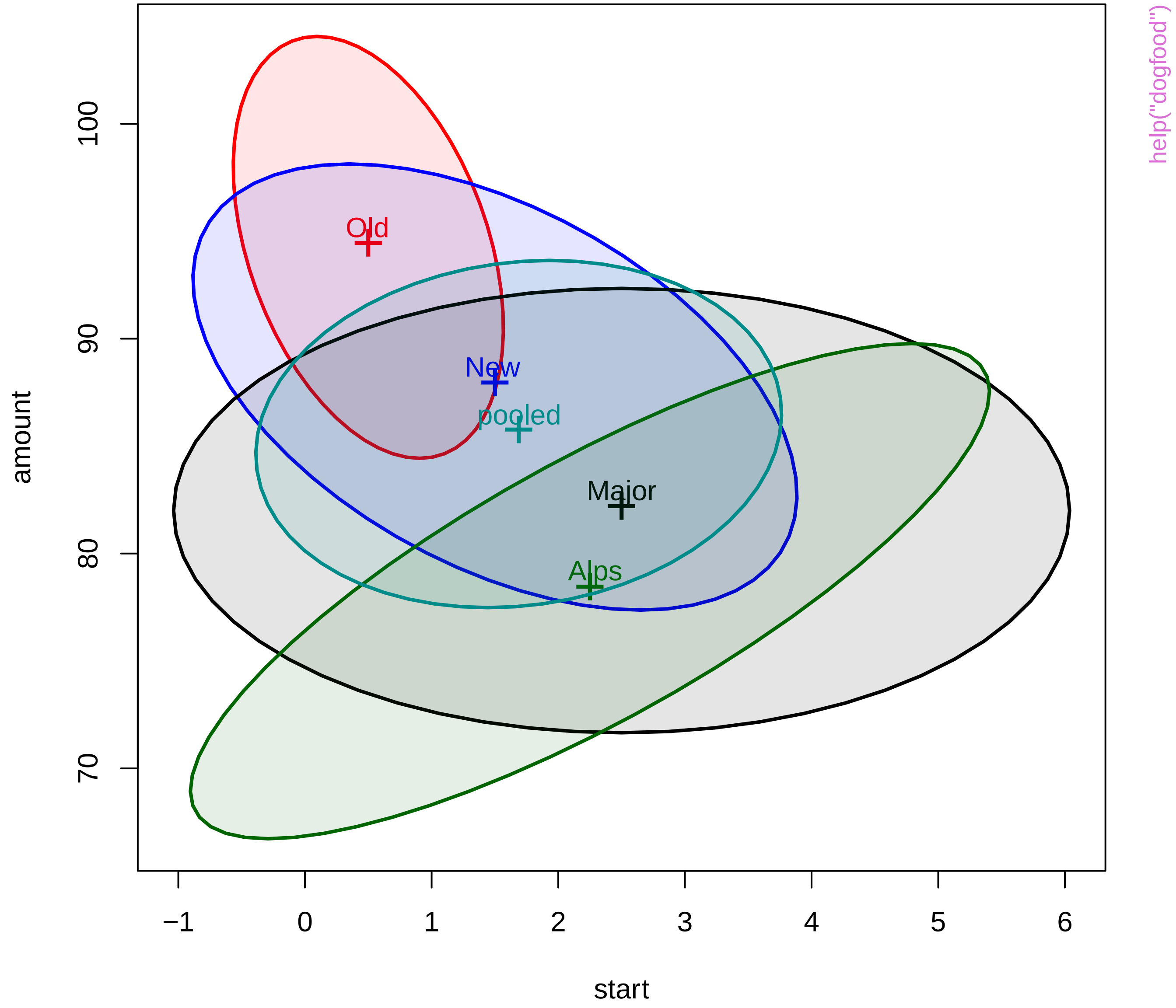


35
33
59
44
54
67
66
1
2
3
4
5
6
Classical Mahalanobis distances of X
Classical Mahalanobis distances of residuals
help("distancePlot")

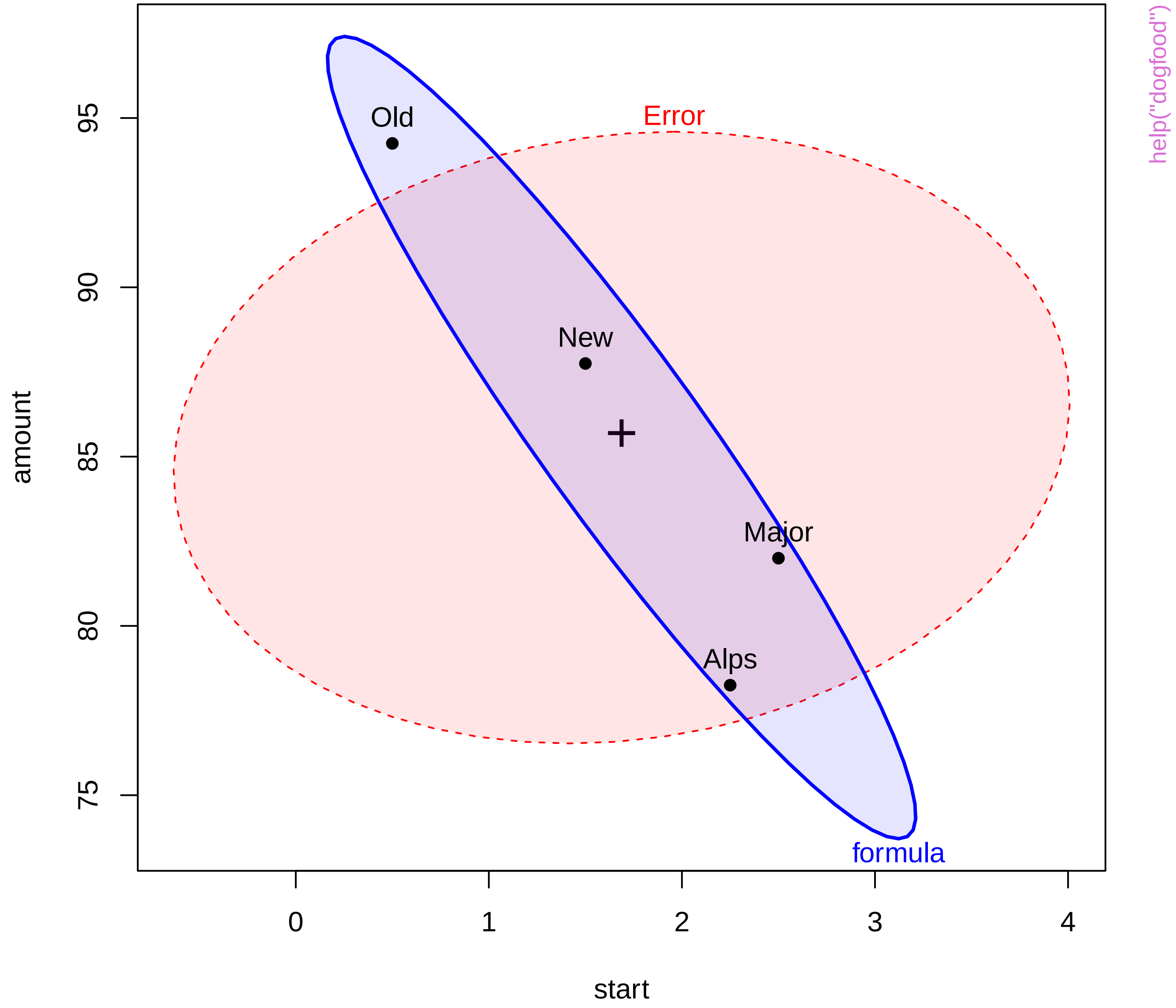


4
11
25
1.5
2.0
2.5
3.0
3.5
4.0
1
2
3
4
Classical Mahalanobis distances of X
Classical Mahalanobis distances of residuals
help("distancePlot")

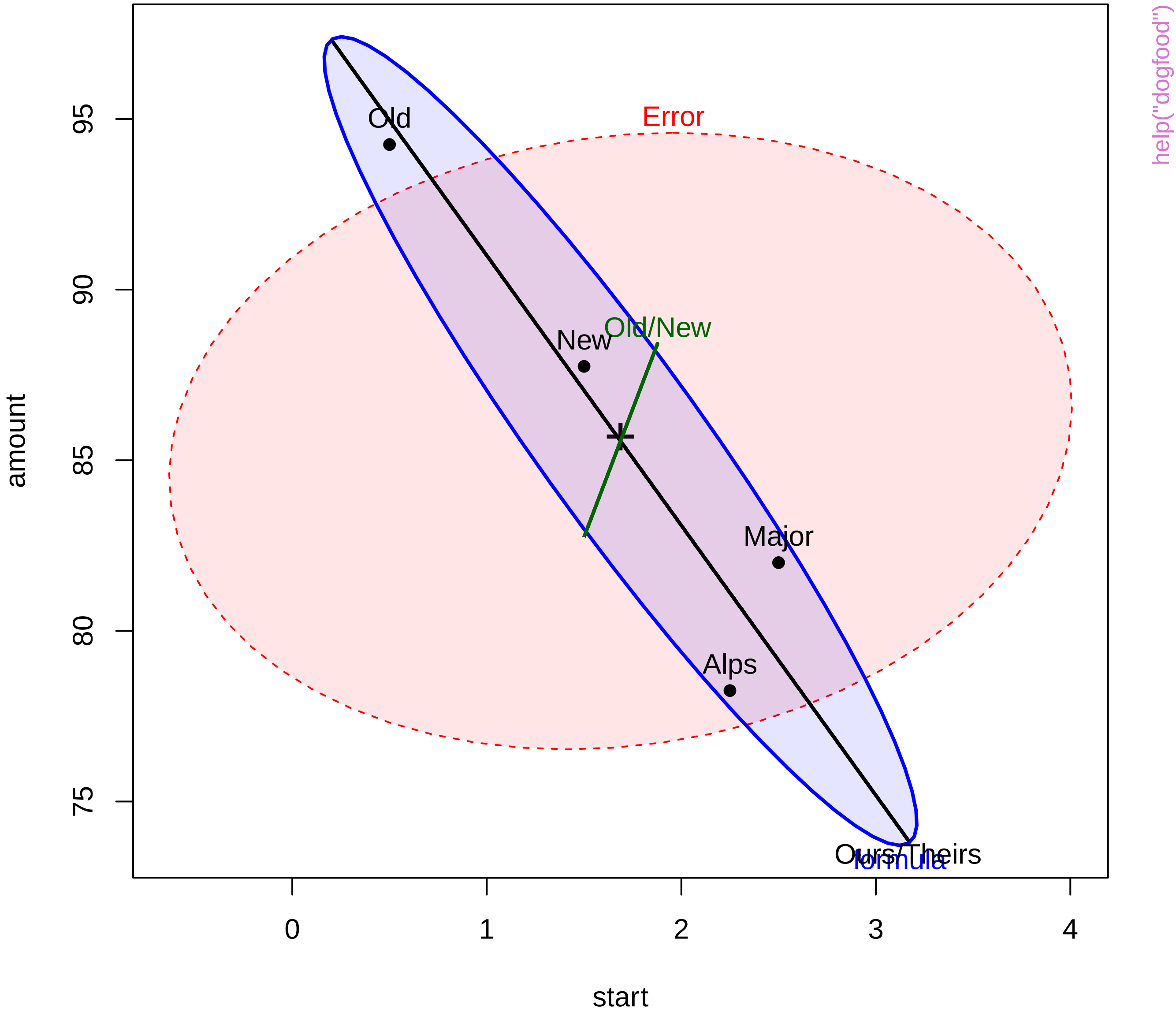

start
0
1
2
3
4
5
Old
New
Major
Alps
formula
amount
75
80
85
90
95
100
Old
New
Major
Alps
formula
help("dogfood")



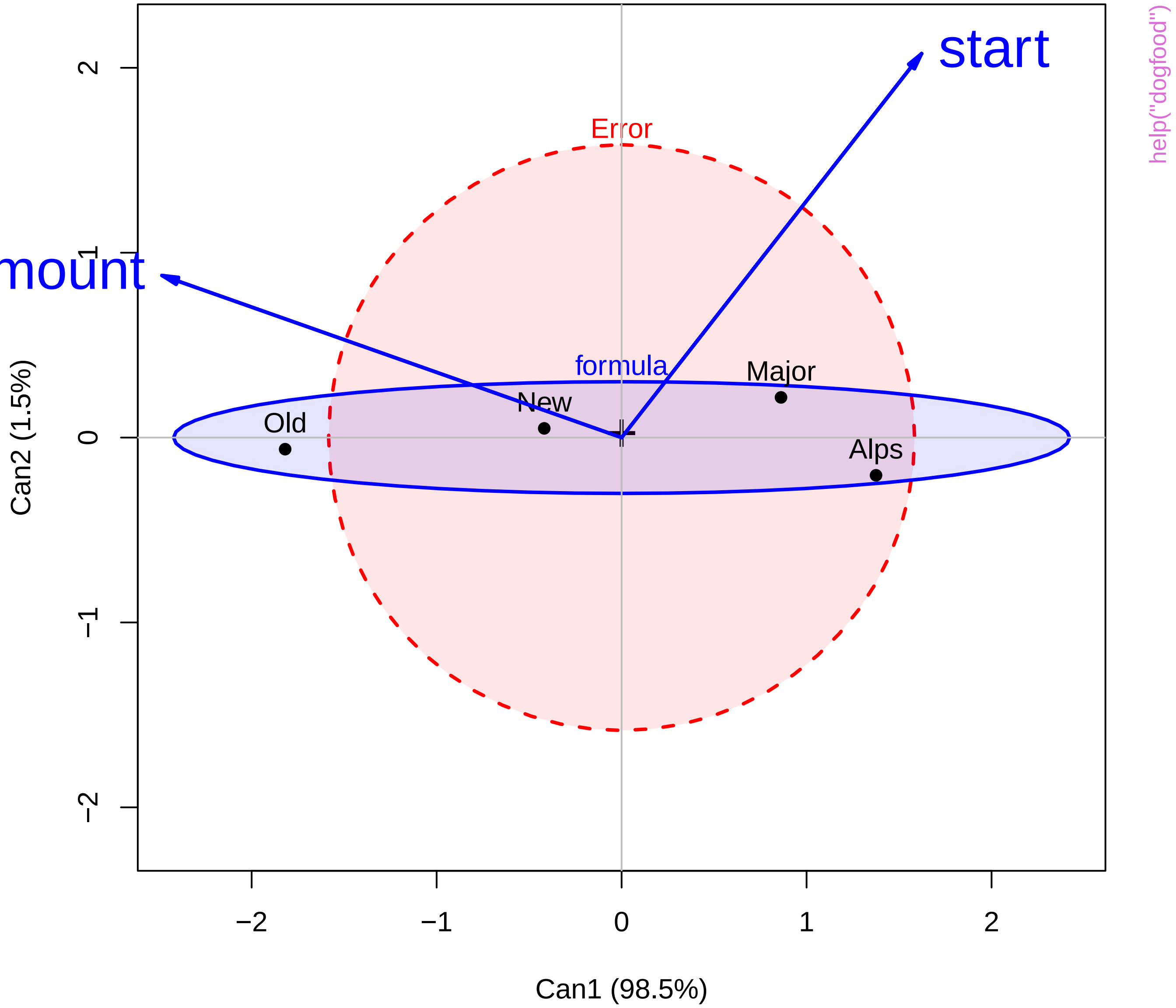
Old
New
pooled
Major
Alps
100
90
80
70
amount
−1
0
1
2
3
4
5
6
start
help("dogfood")



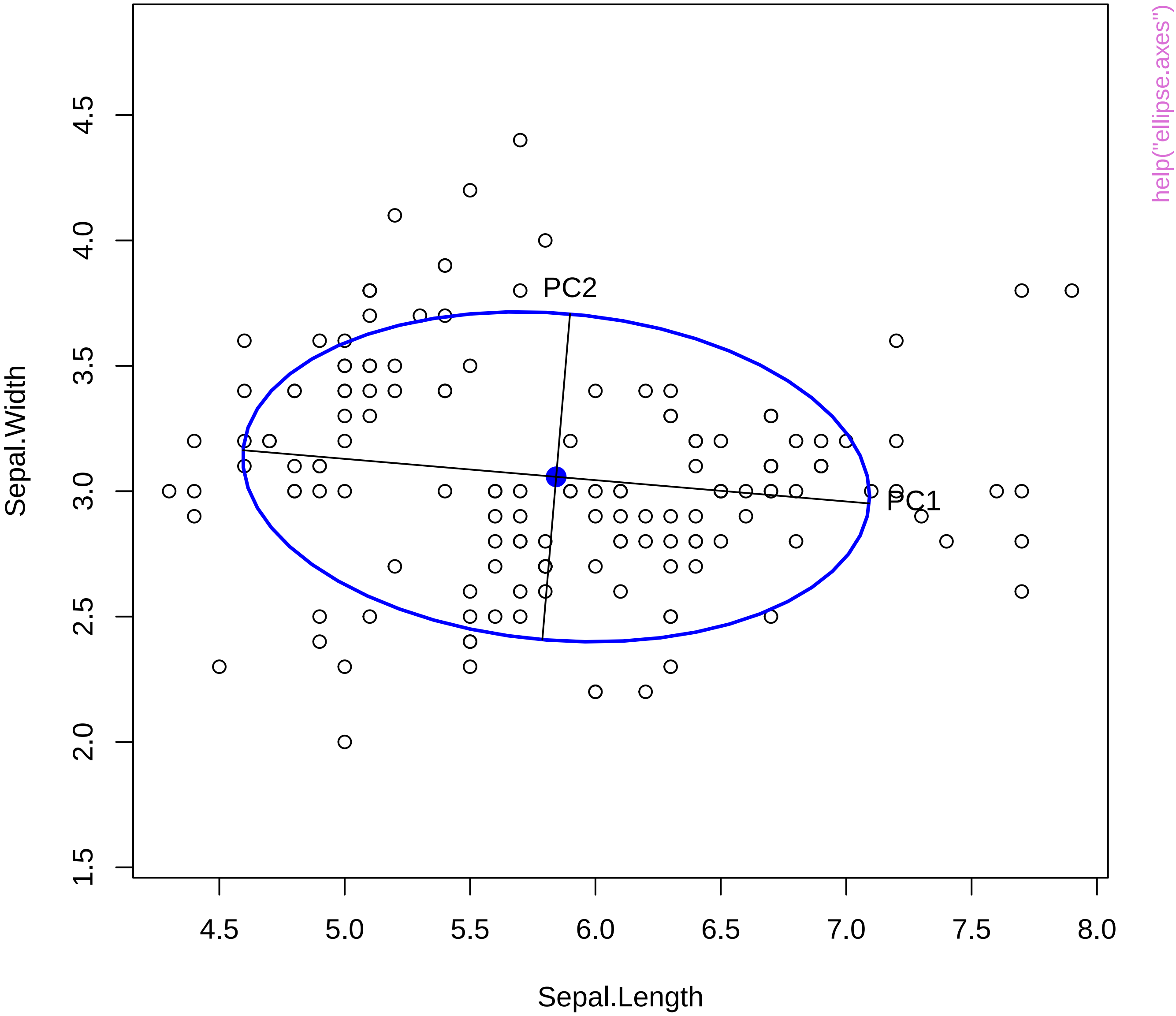
Old
Error
New
Major
Alps
formula
95
90
85
80
75
0
1
2
3
4
amount
start
help("dogfood")



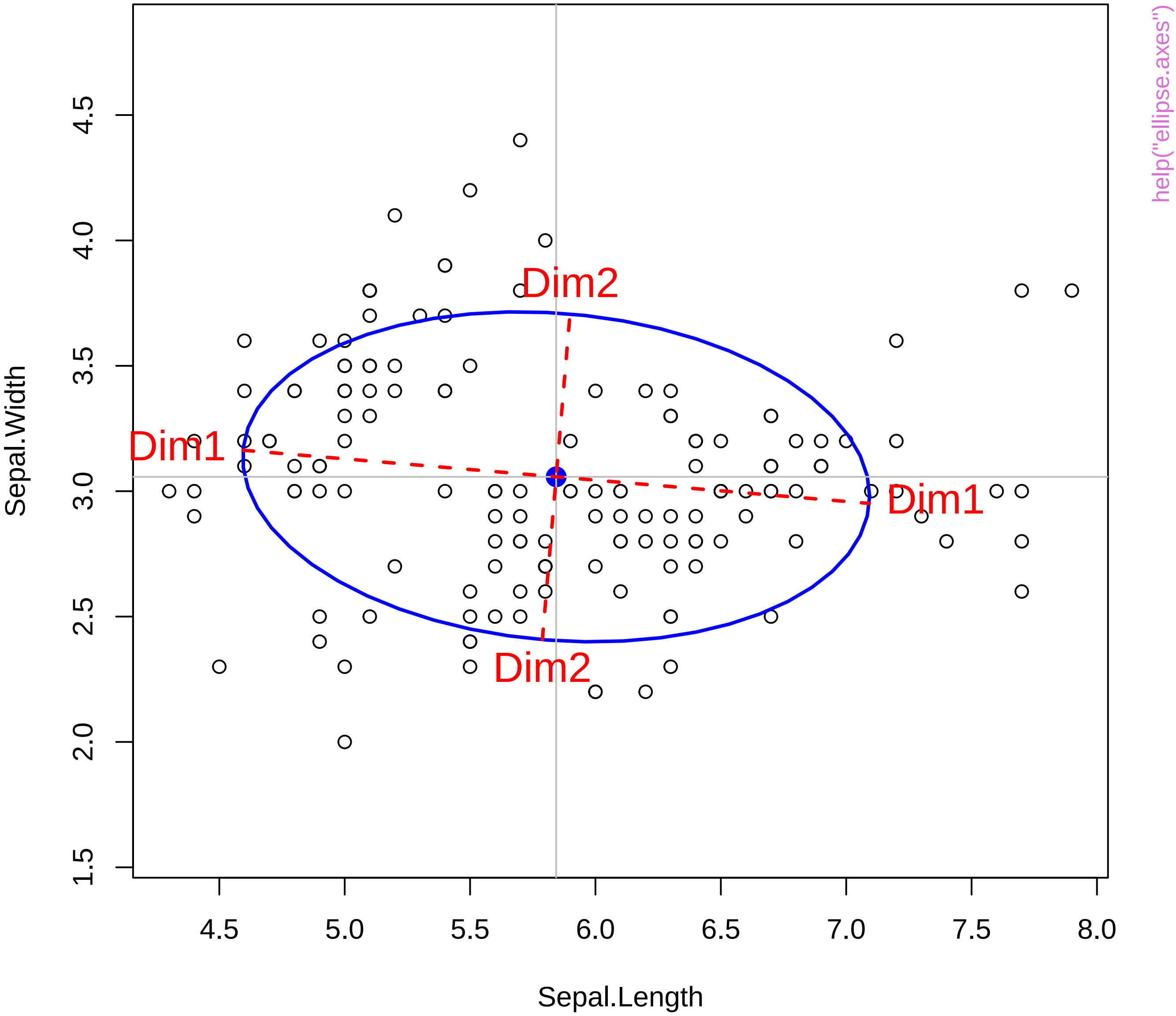
help("dogfood")
Error
Old
New
Old/New
Major
Alps
Ours/Theirs
formula
95
90
85
80
75
0
1
2
3
4
amount
start



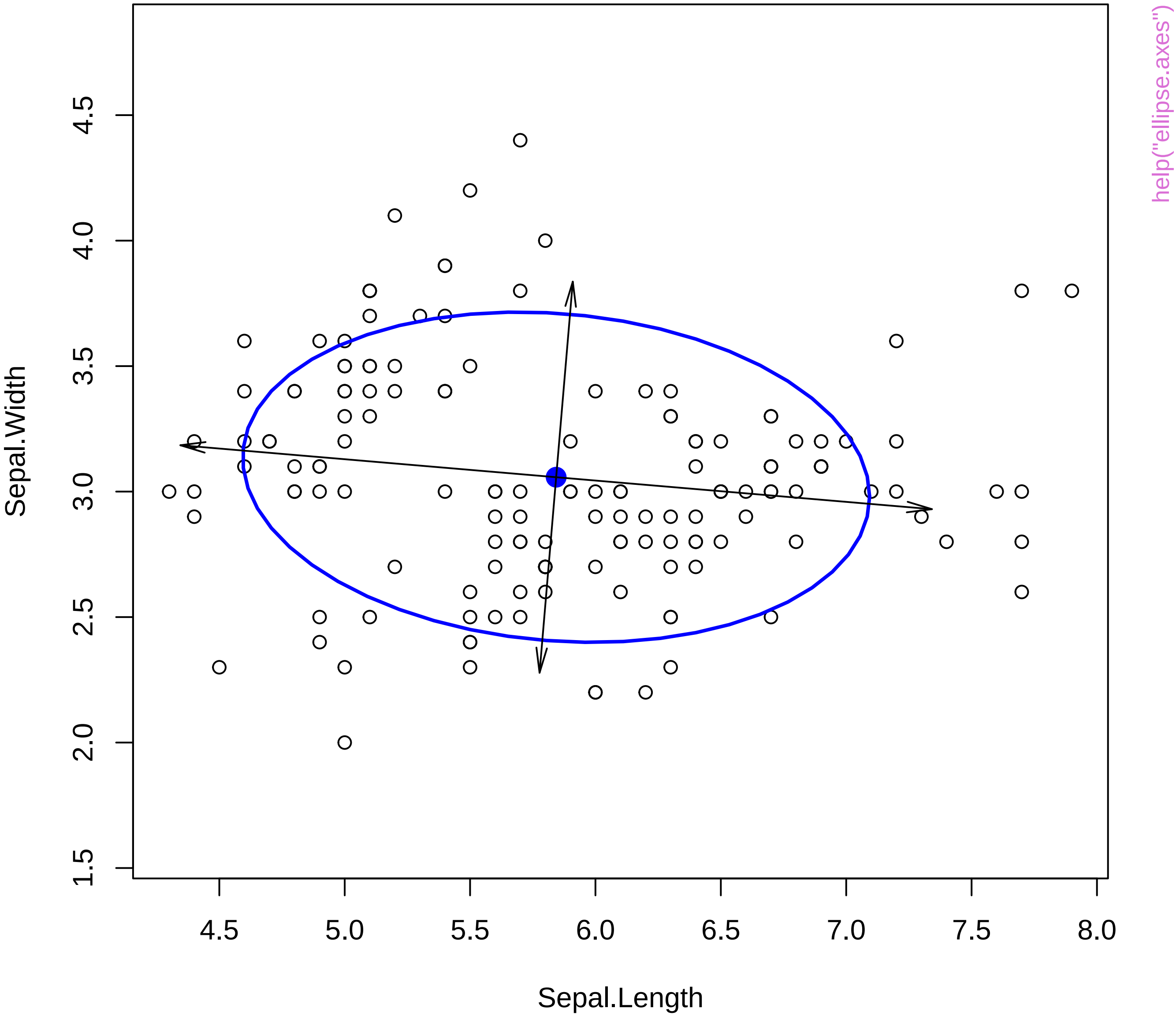
start
mount
Error
formula
Major
New
Old
Alps
help("dogfood")
2
1
0
−1
−2
Can2 (1.5%)
−2
−1
0
1
2
Can1 (98.5%)



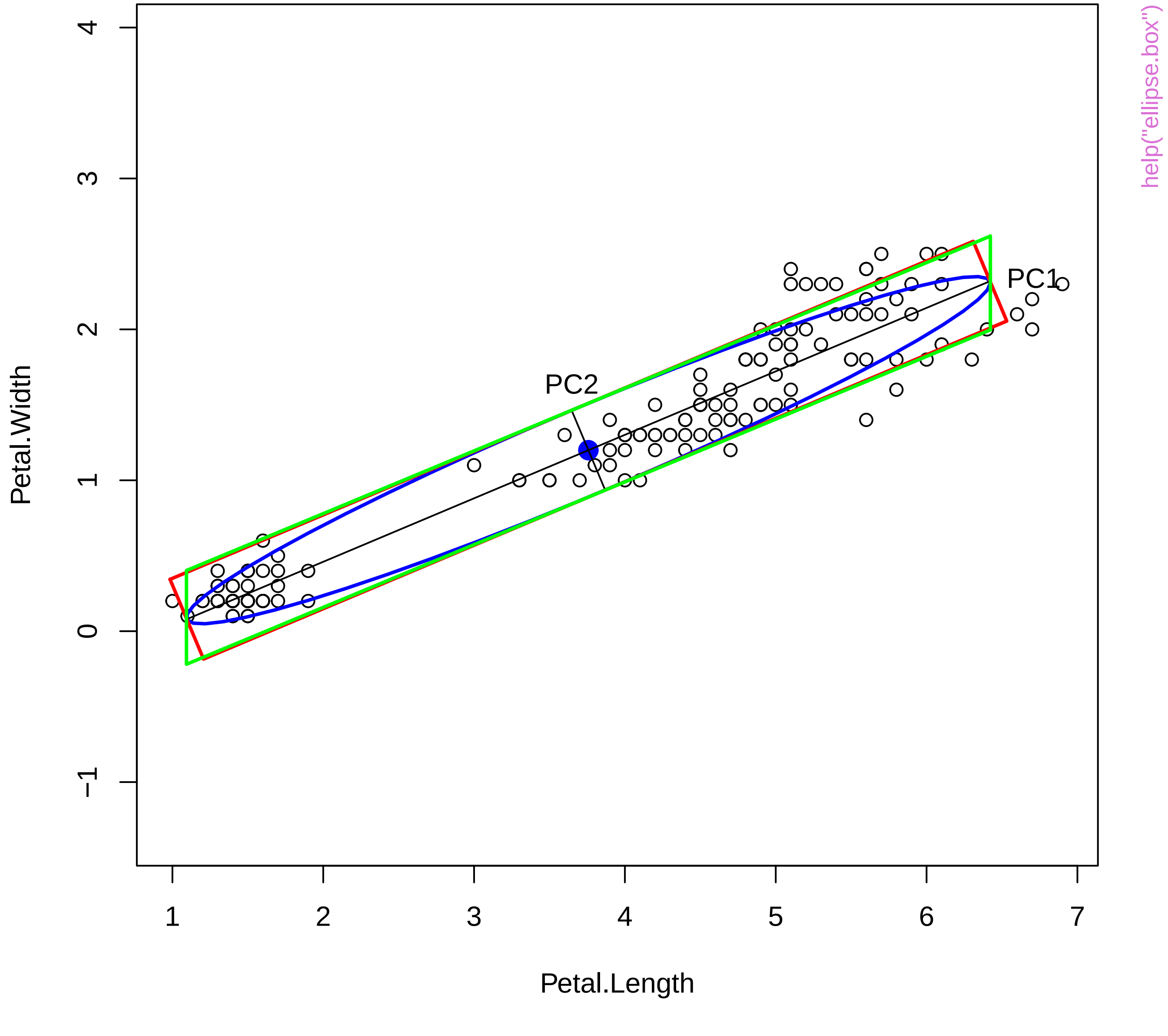
PC2
PC1
Sepal.Width
Sepal.Length
1.5
2.0
2.5
3.0
3.5
4.0
4.5
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
help("ellipse.axes")



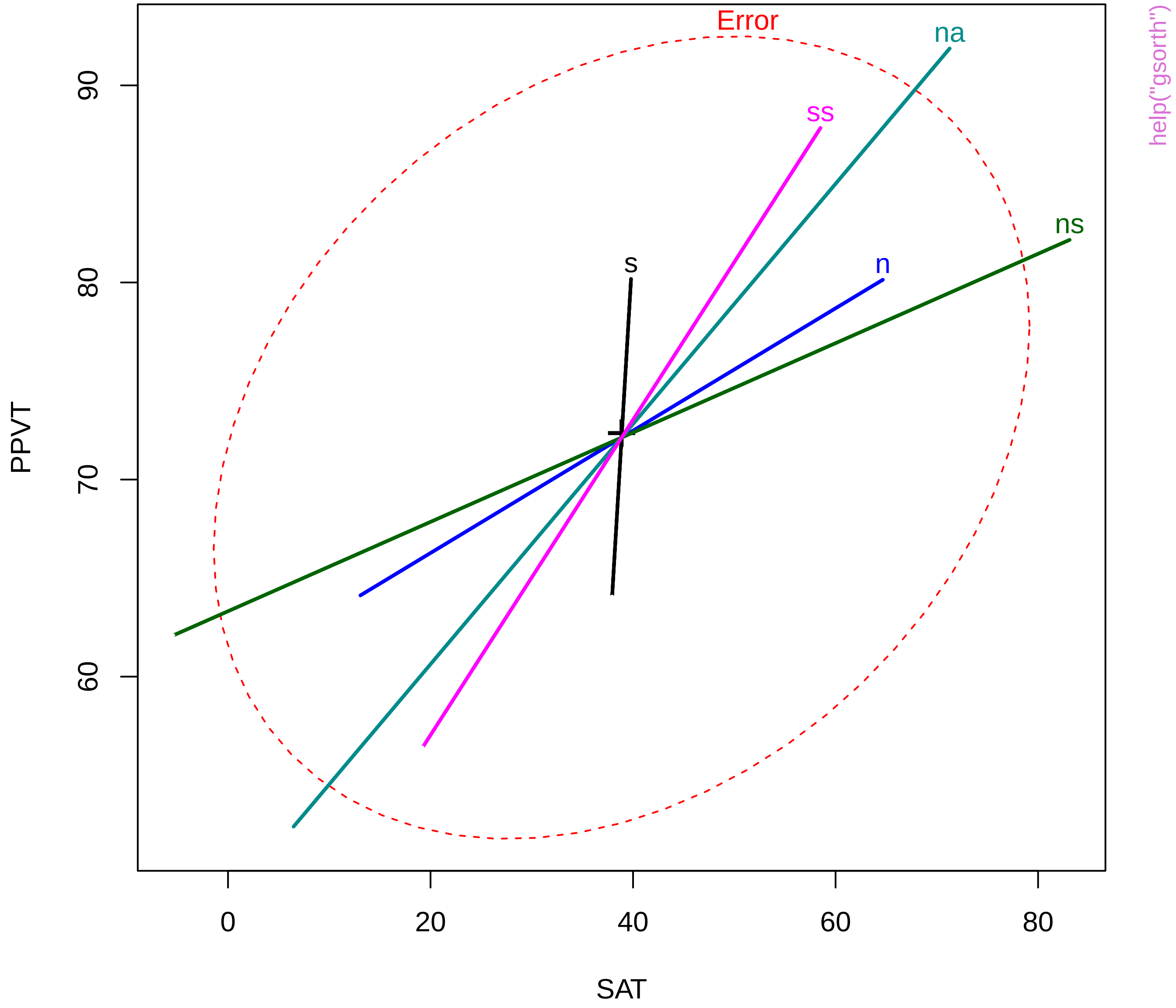
Dim1
Dim1
Dim2
Dim2
Sepal.Width
Sepal.Length
4.5
4.0
3.5
3.0
2.5
2.0
1.5
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
help("ellipse.axes")



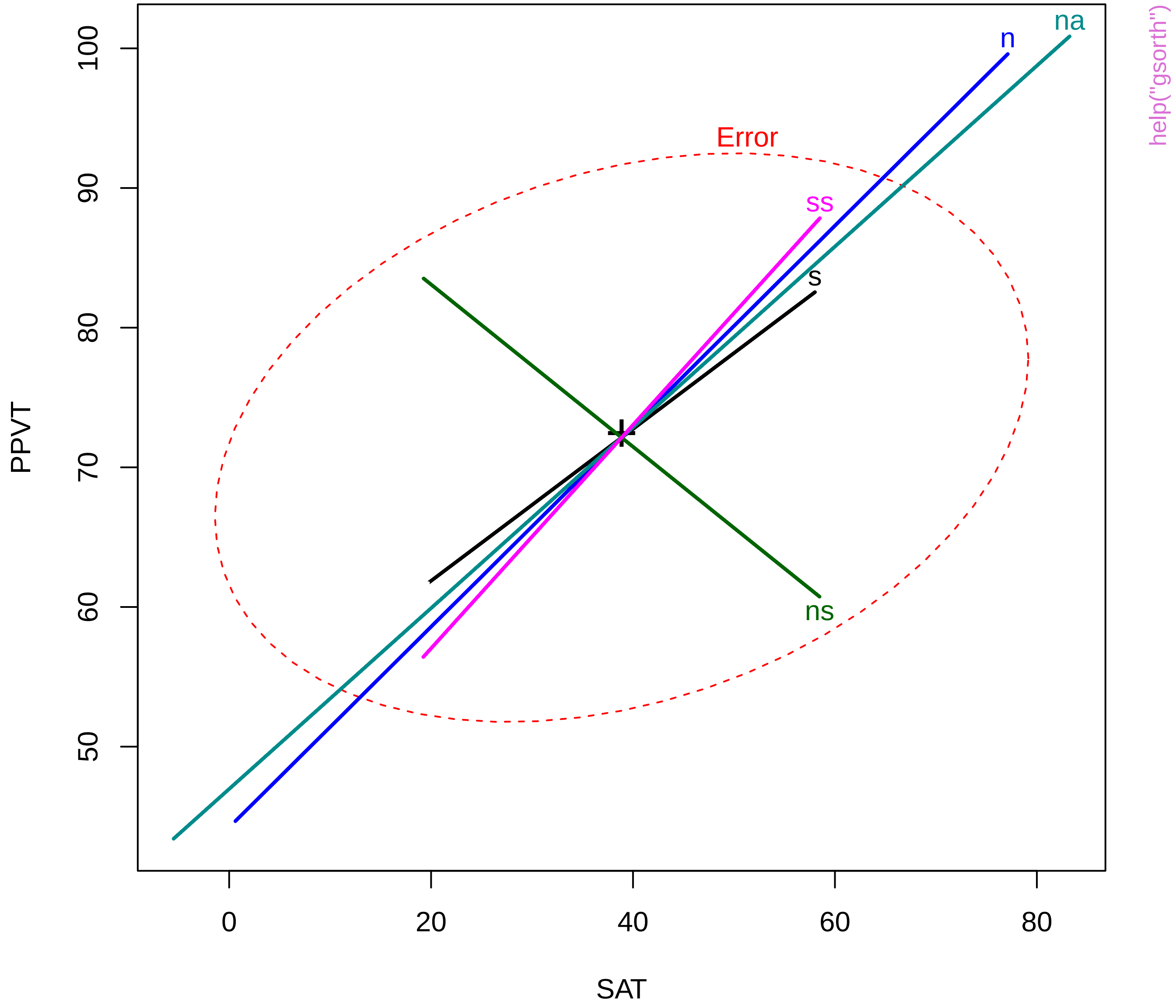
help("ellipse.axes")
Sepal.Width
Sepal.Length
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5



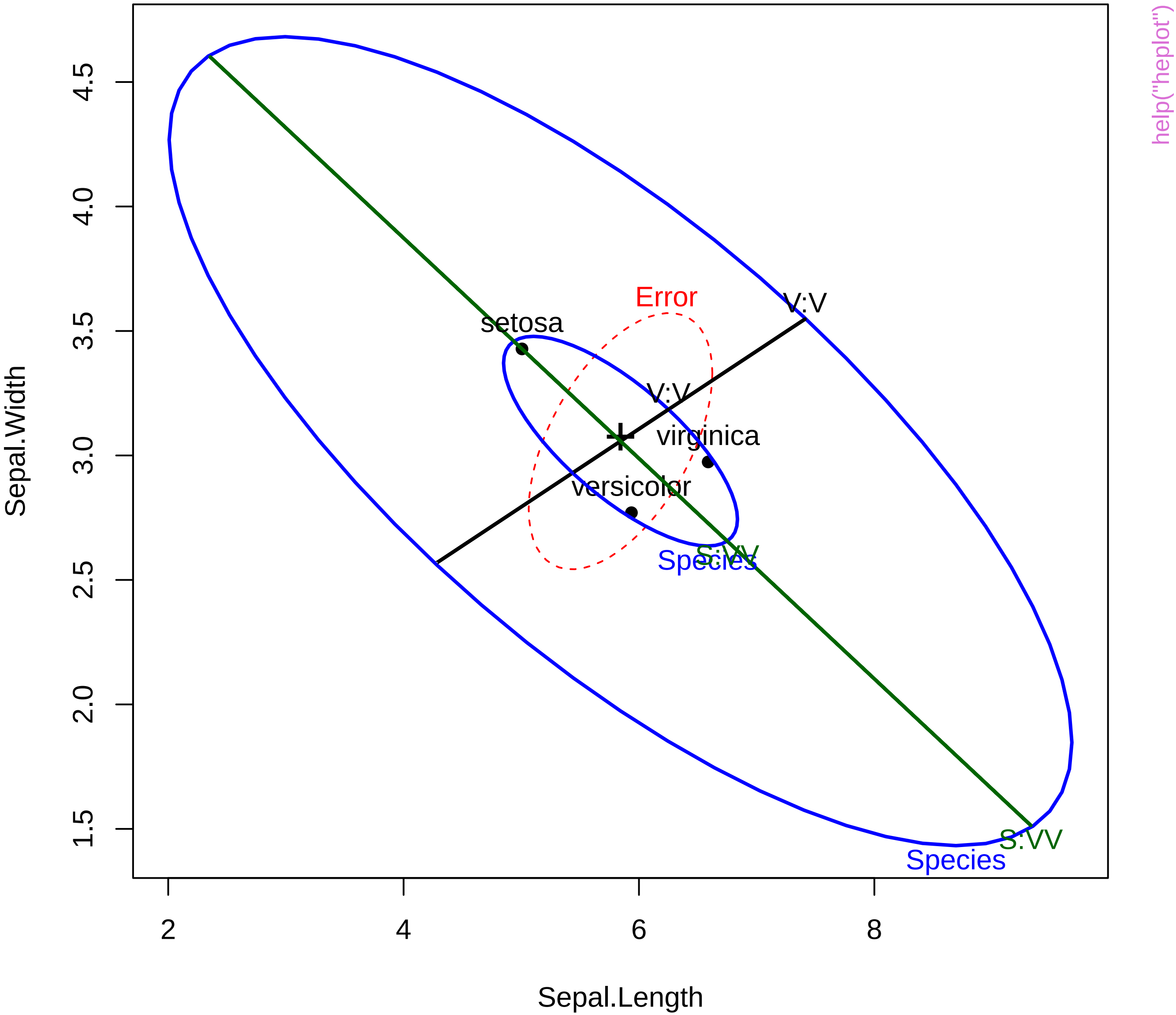
PC1
PC2
Petal.Length
Petal.Width
1
2
3
4
5
6
7
−1
0
1
2
3
4
help("ellipse.box")

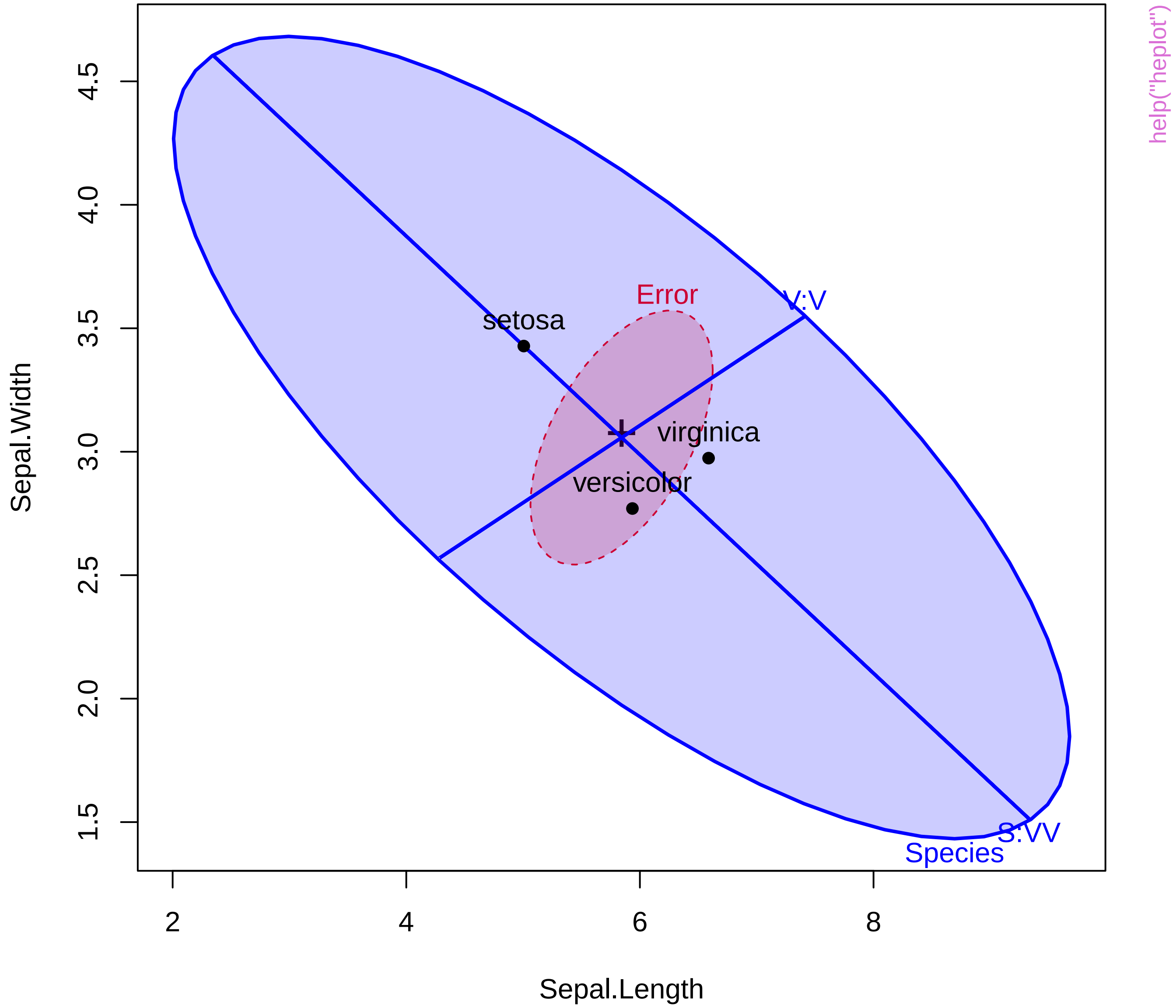


Error
na
ss
ns
s
n
90
80
70
60
PPVT
0
20
40
60
80
SAT
help("gsorth")

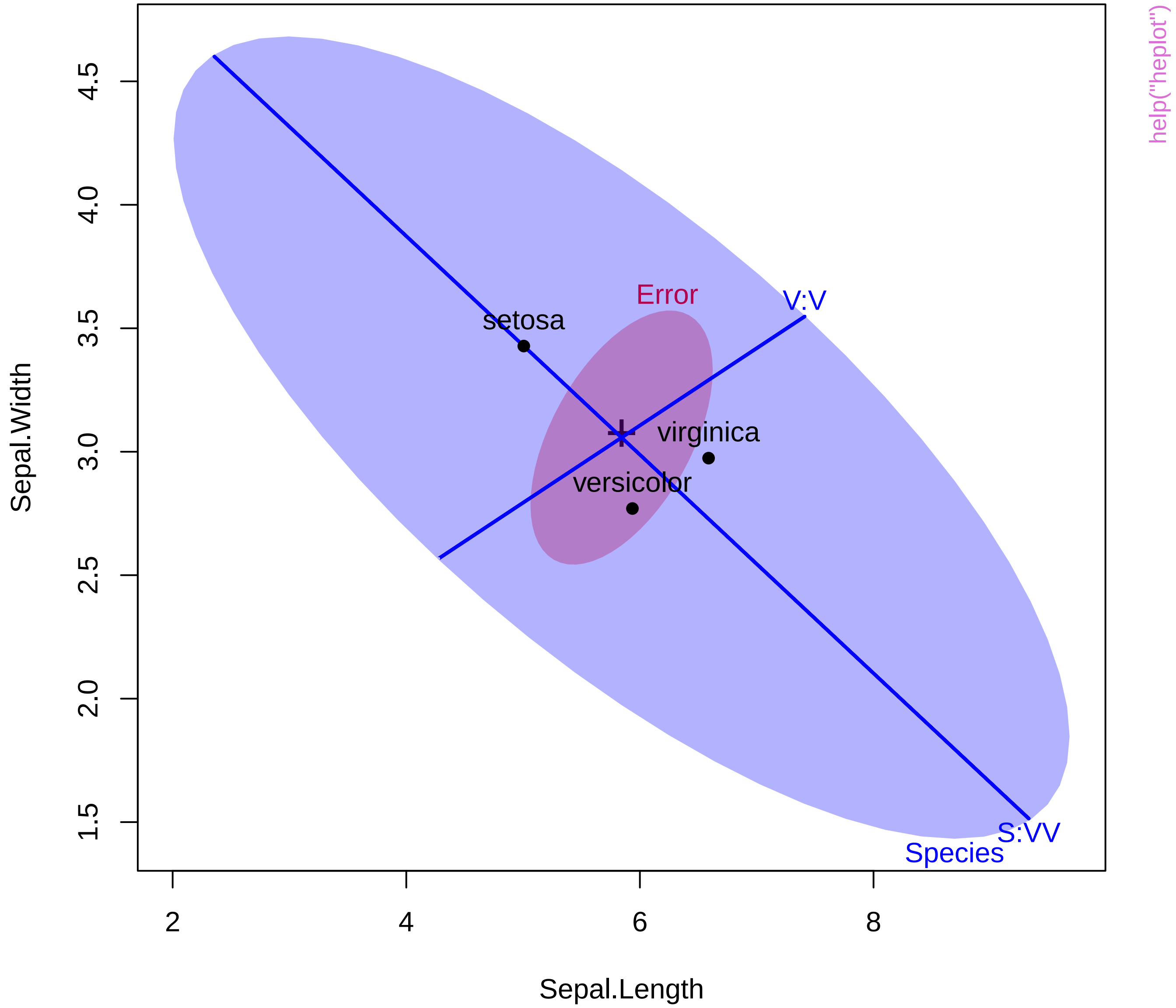


100
90
80
70
60
50
PPVT
0
20
40
60
80
SAT
Error
na
n
ss
s
ns
help("gsorth")

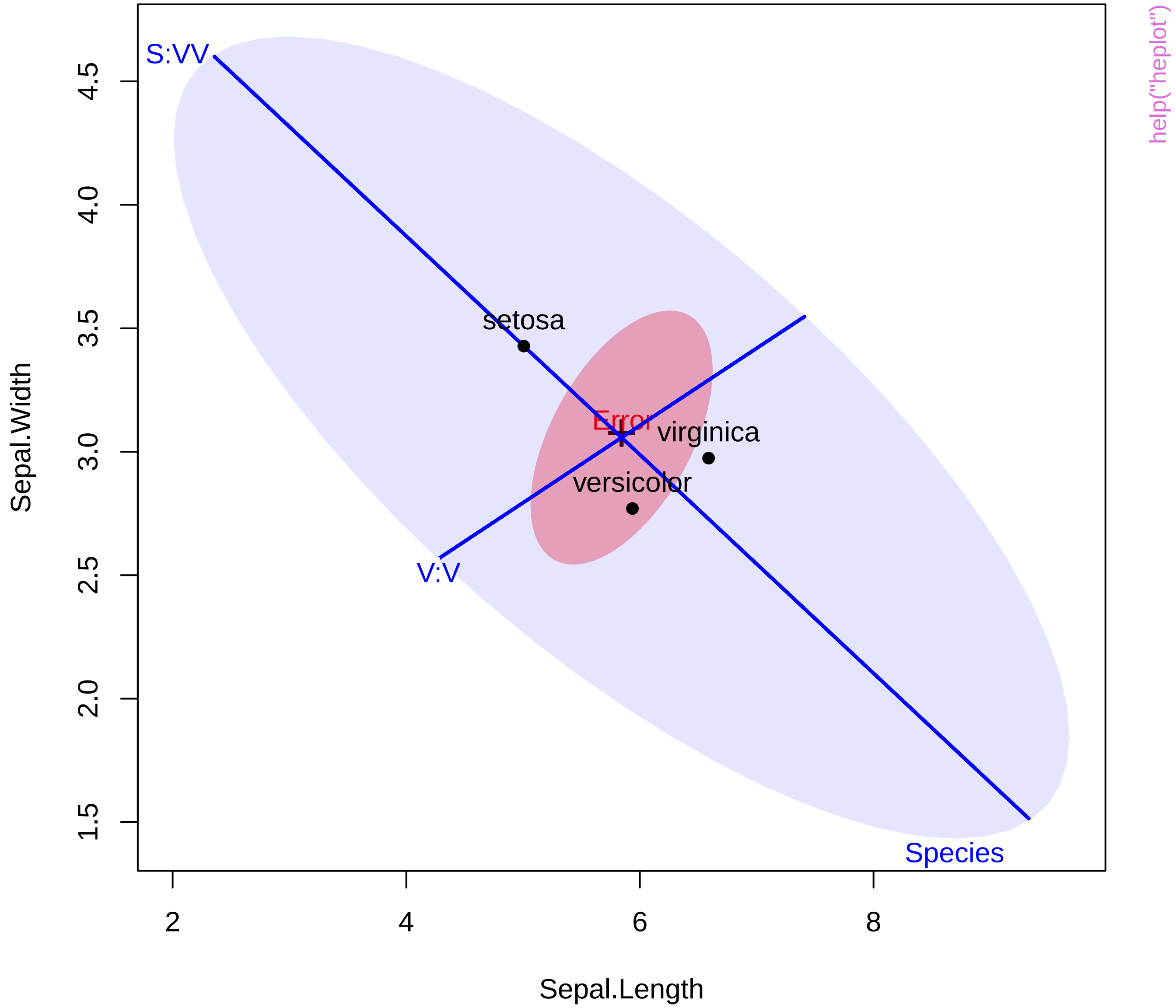

Error
V:V
setosa
V:V
virginica
versicolor
S:VV
Species
S:VV
Species
help("heplot")
Sepal.Width
Sepal.Length
4.5
4.0
3.5
3.0
2.5
2.0
1.5
2
4
6
8

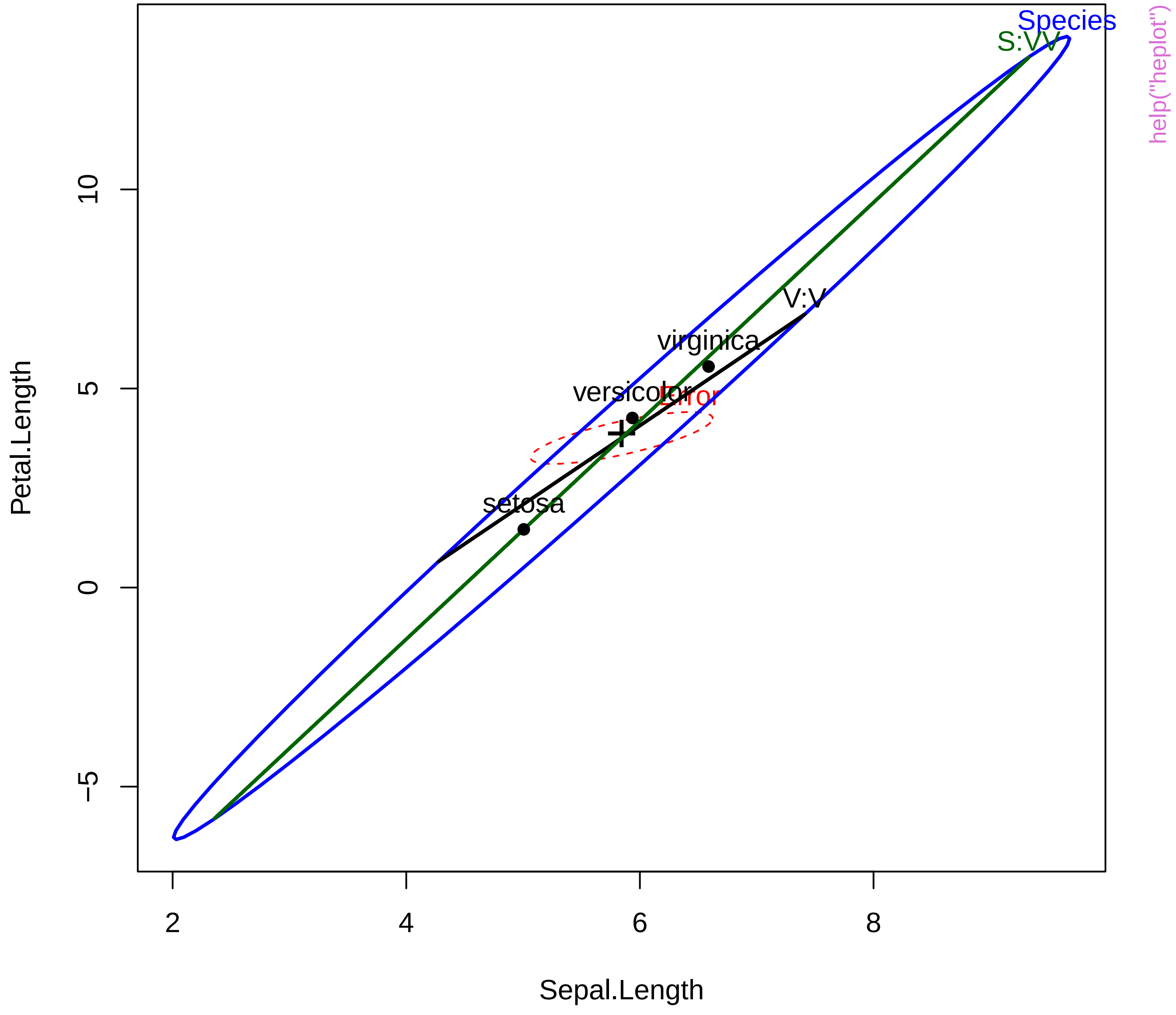


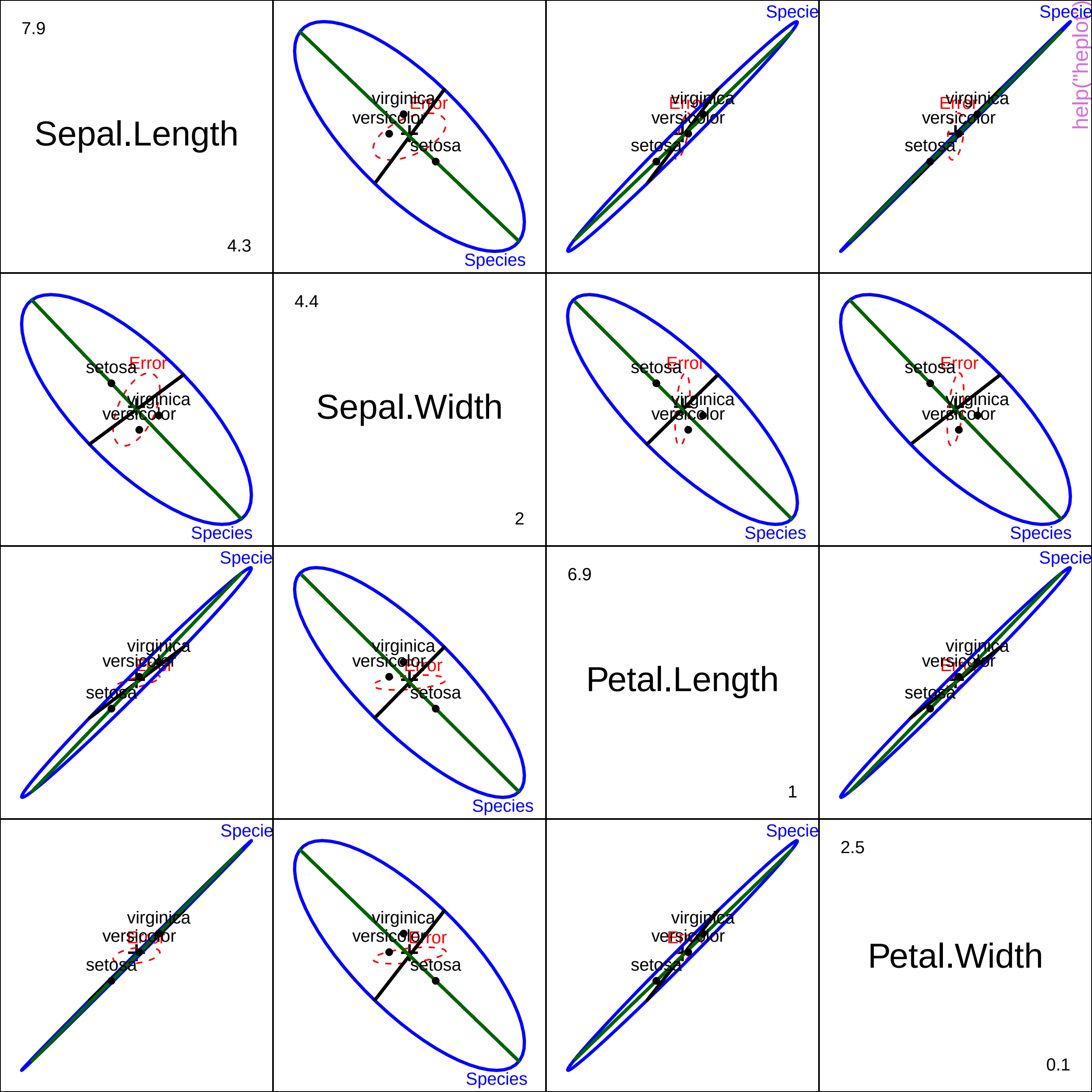
Sepal.Width
Sepal.Length
1.5
2.0
2.5
3.0
3.5
4.0
4.5
2
4
6
8
Error
V:V
setosa
virginica
versicolor
S:VV
Species
help("heplot")

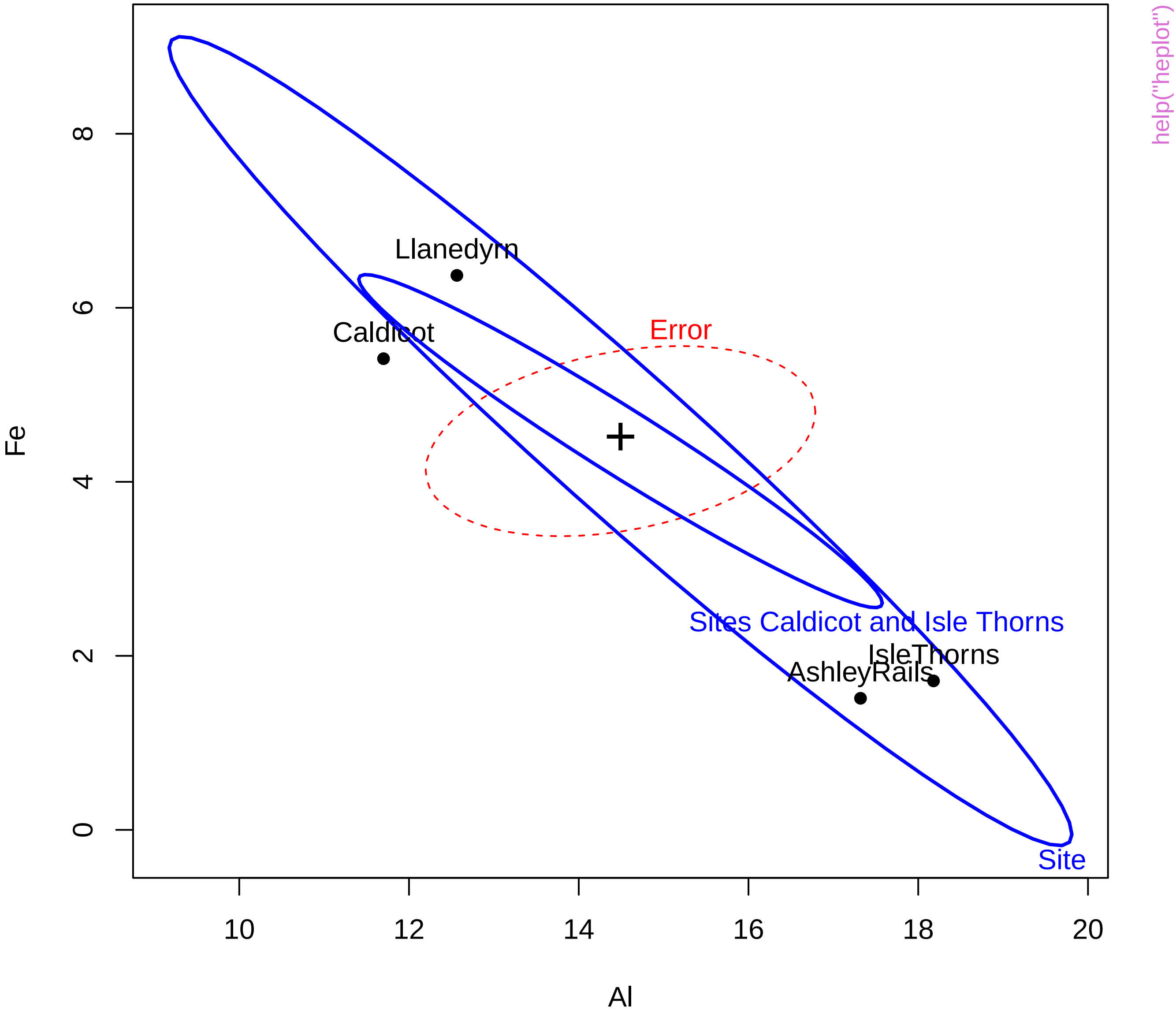


help("heplot")
Error
V:V
setosa
virginica
versicolor
S:VV
Species
Sepal.Width
Sepal.Length
4.5
4.0
3.5
3.0
2.5
2.0
1.5
2
4
6
8

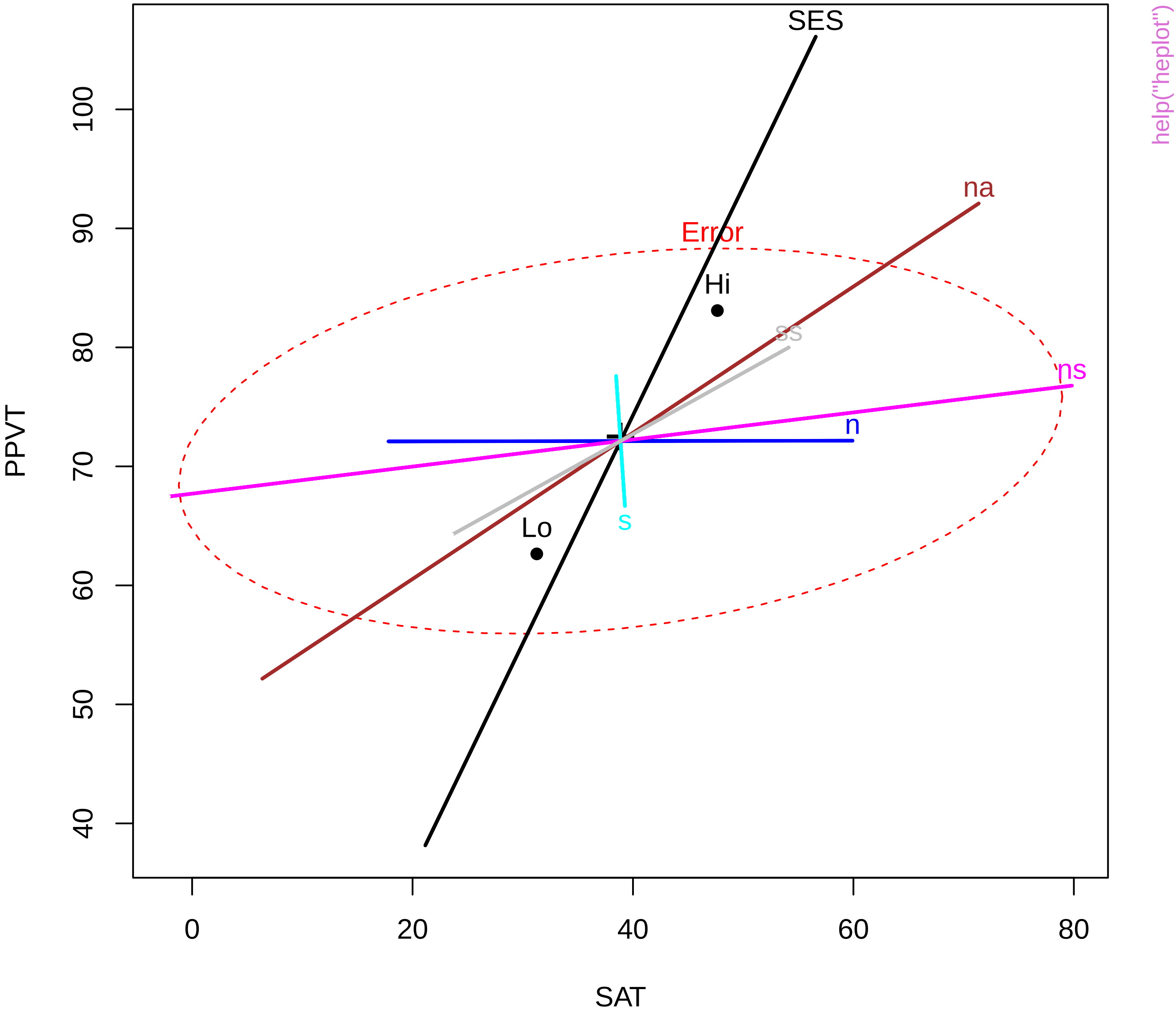

S:VV
V:V
Species
Error
setosa
virginica
versicolor
Sepal.Width
Sepal.Length
1.5
2.0
2.5
3.0
3.5
4.0
4.5
2
4
6
8
help("heplot")


Species
S:VV
help("heplot")
V:V
virginica
versicolor
Error
setosa
10
5
0
−5
Petal.Length
2
4
6
8
Sepal.Length

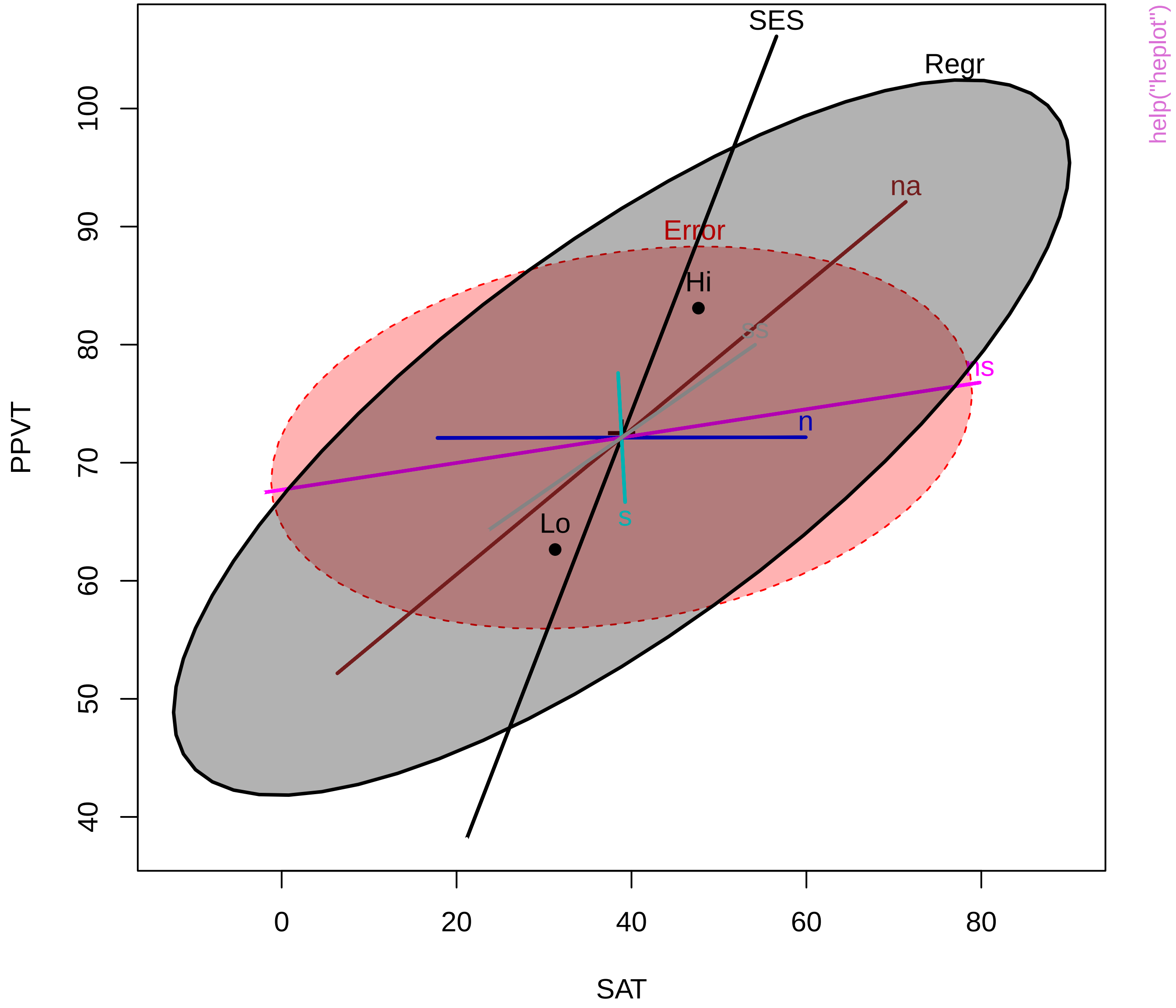

7.9
Sepal.Length
4.3
4.4
Sepal.Width
2
6.9
Petal.Length
1
2.5
Petal.Width
0.1
virginica
versicolor
setosa
Error
Species
help("heplot")

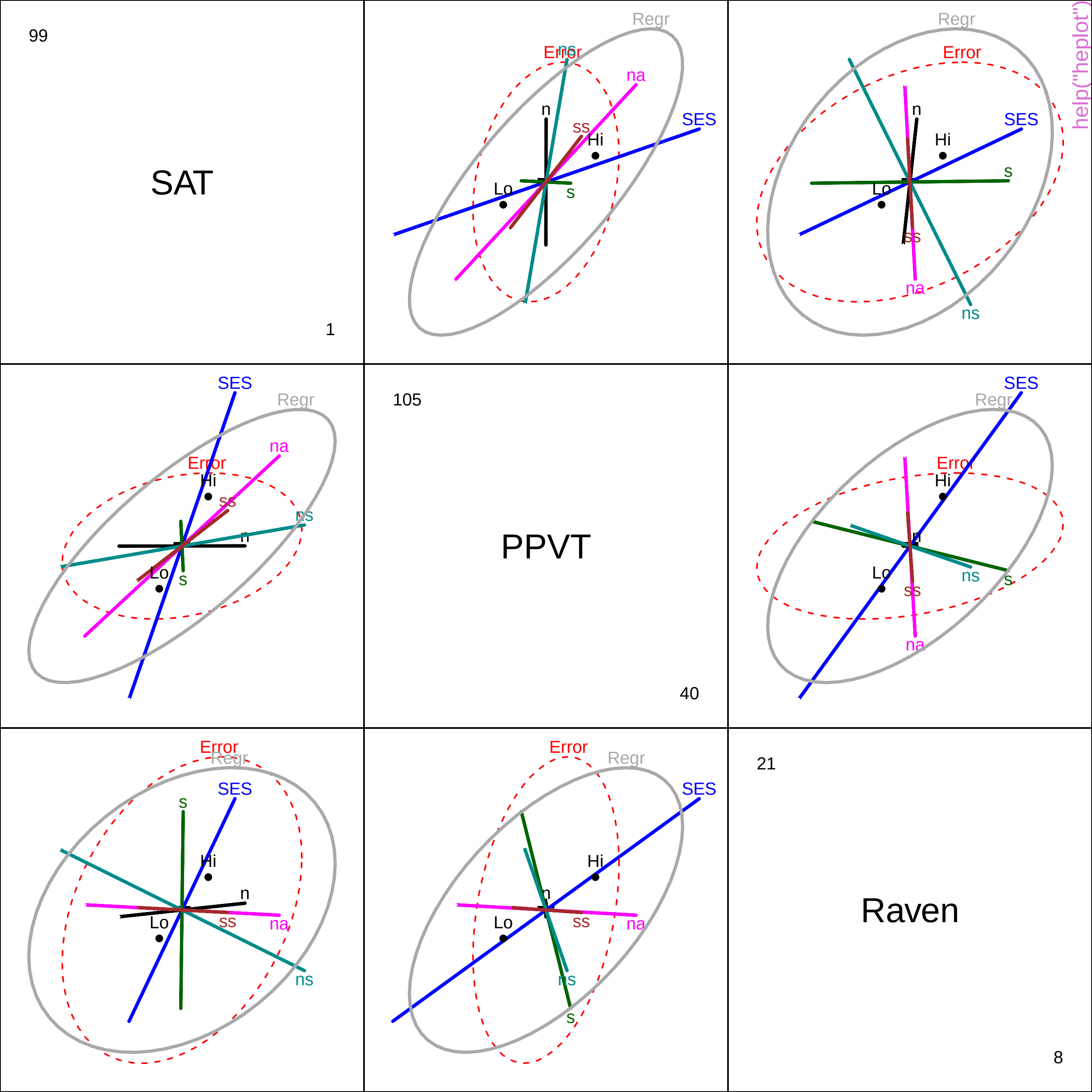


Llanedyrn
Caldicot
Error
Sites Caldicot and Isle Thorns
IsleThorns
AshleyRails
Site
Fe
Al
0
2
4
6
8
10
12
14
16
18
20
help("heplot")

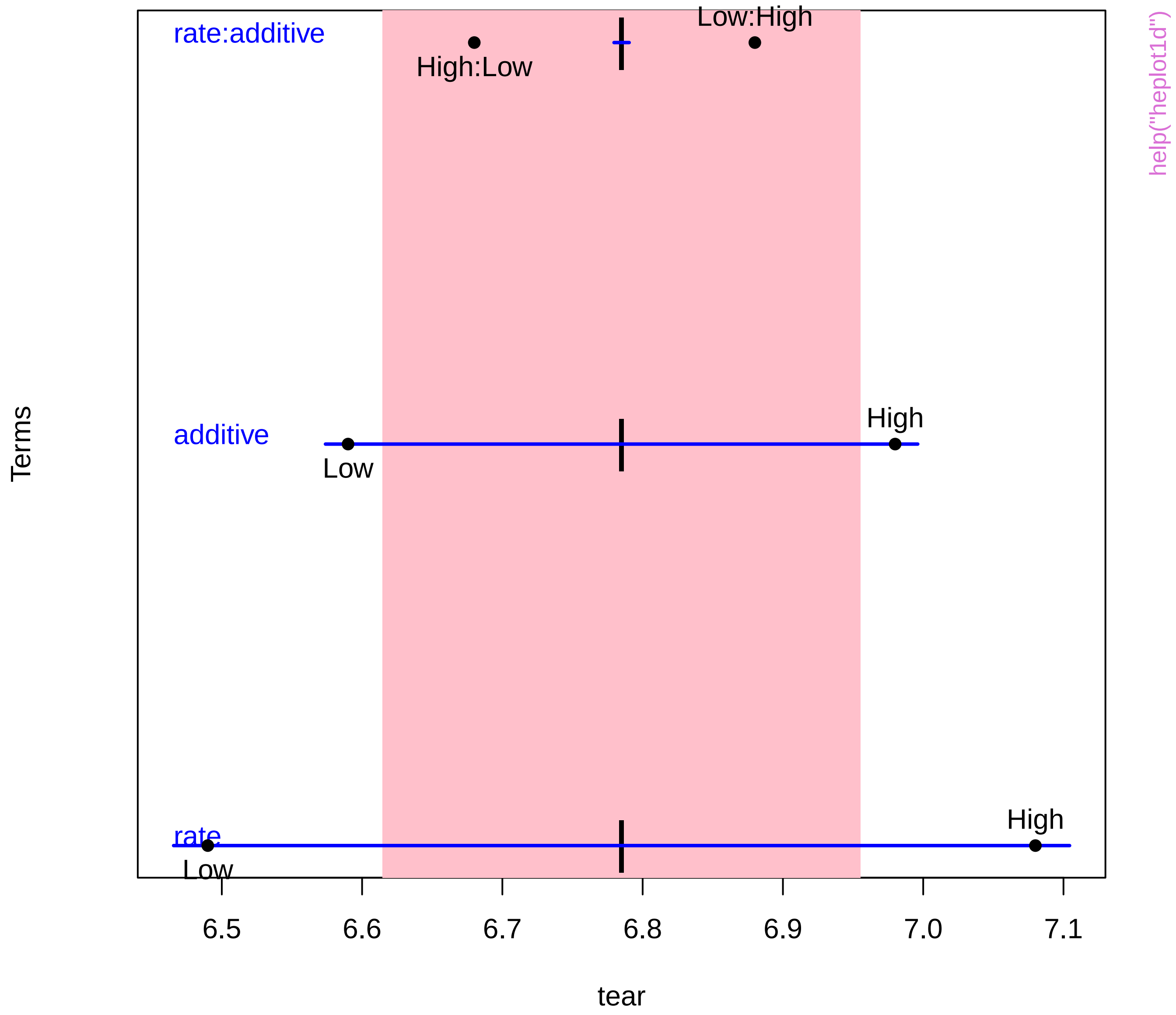


SES
na
Error
Hi
ss
ns
n
s
Lo
100
90
80
70
60
50
40
PPVT
0
20
40
60
80
SAT
help("heplot")

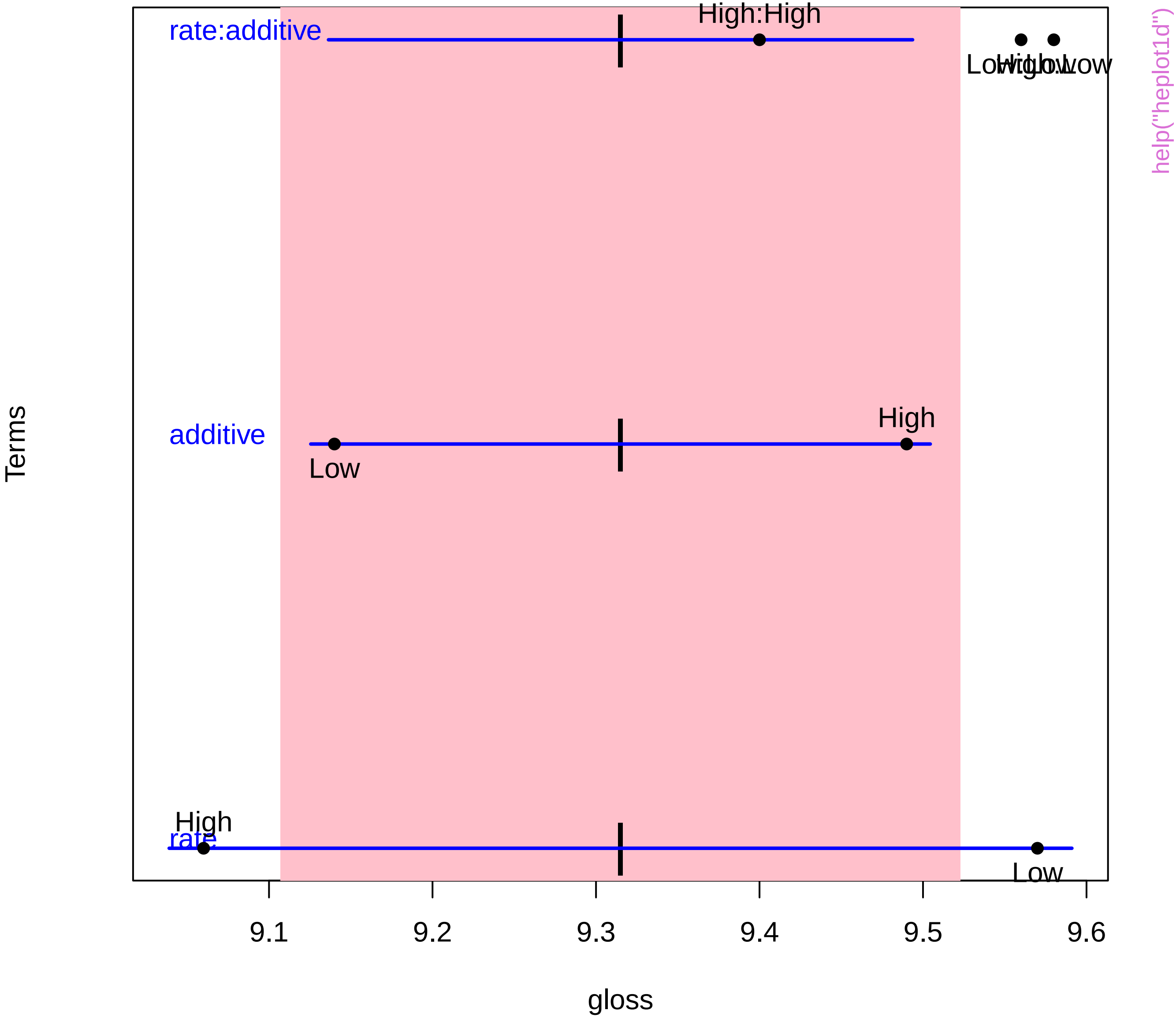

SES
Regr
na
Error
Hi
ss
ns
n
s
Lo
PPVT
SAT
0
20
40
60
80
40
50
60
70
80
90
100
help("heplot")



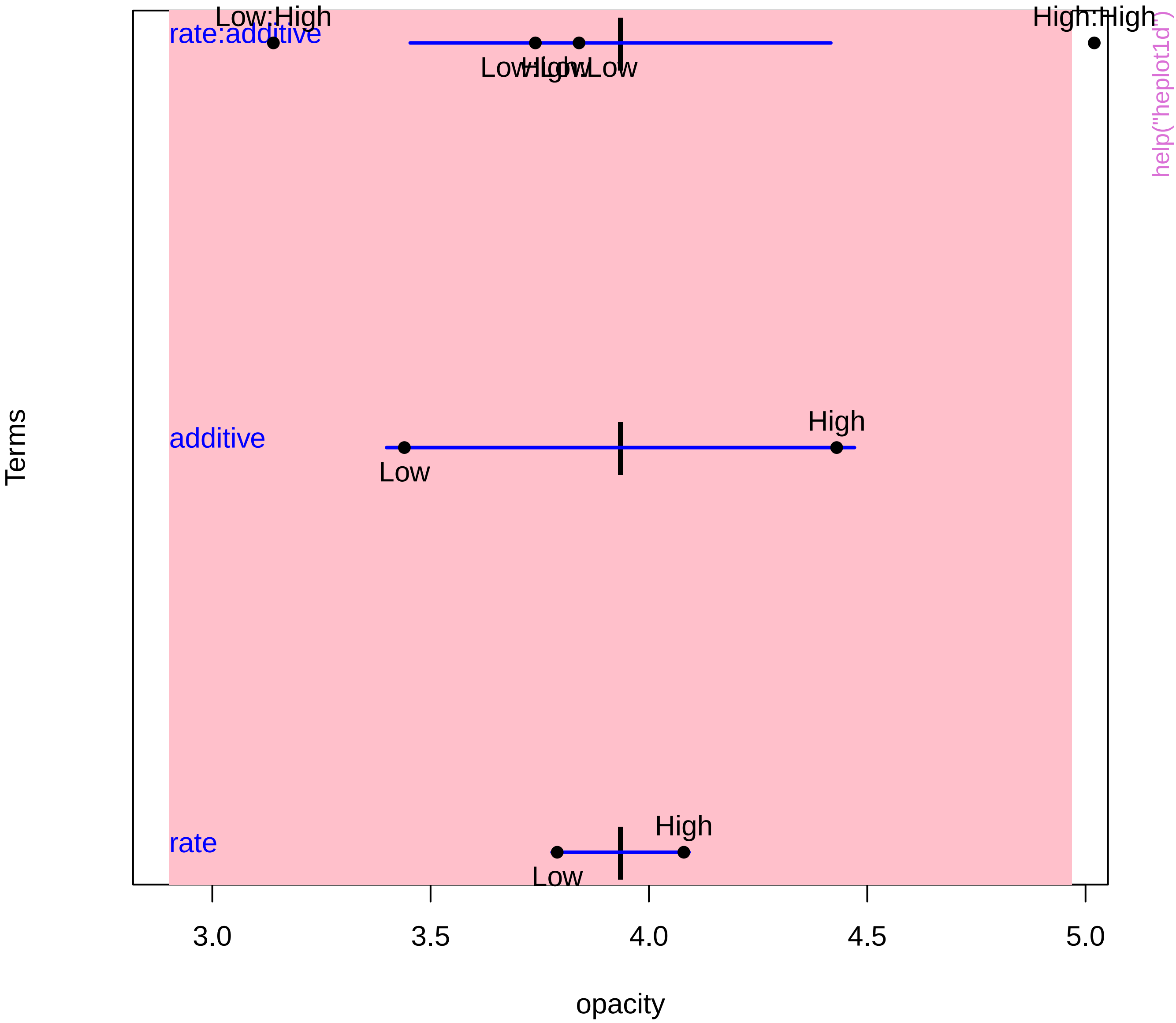
99
SAT
1
105
PPVT
40
21
Raven
8
Regr
Error
SES
ns
na
n
ss
s
Hi
Lo
help("heplot")



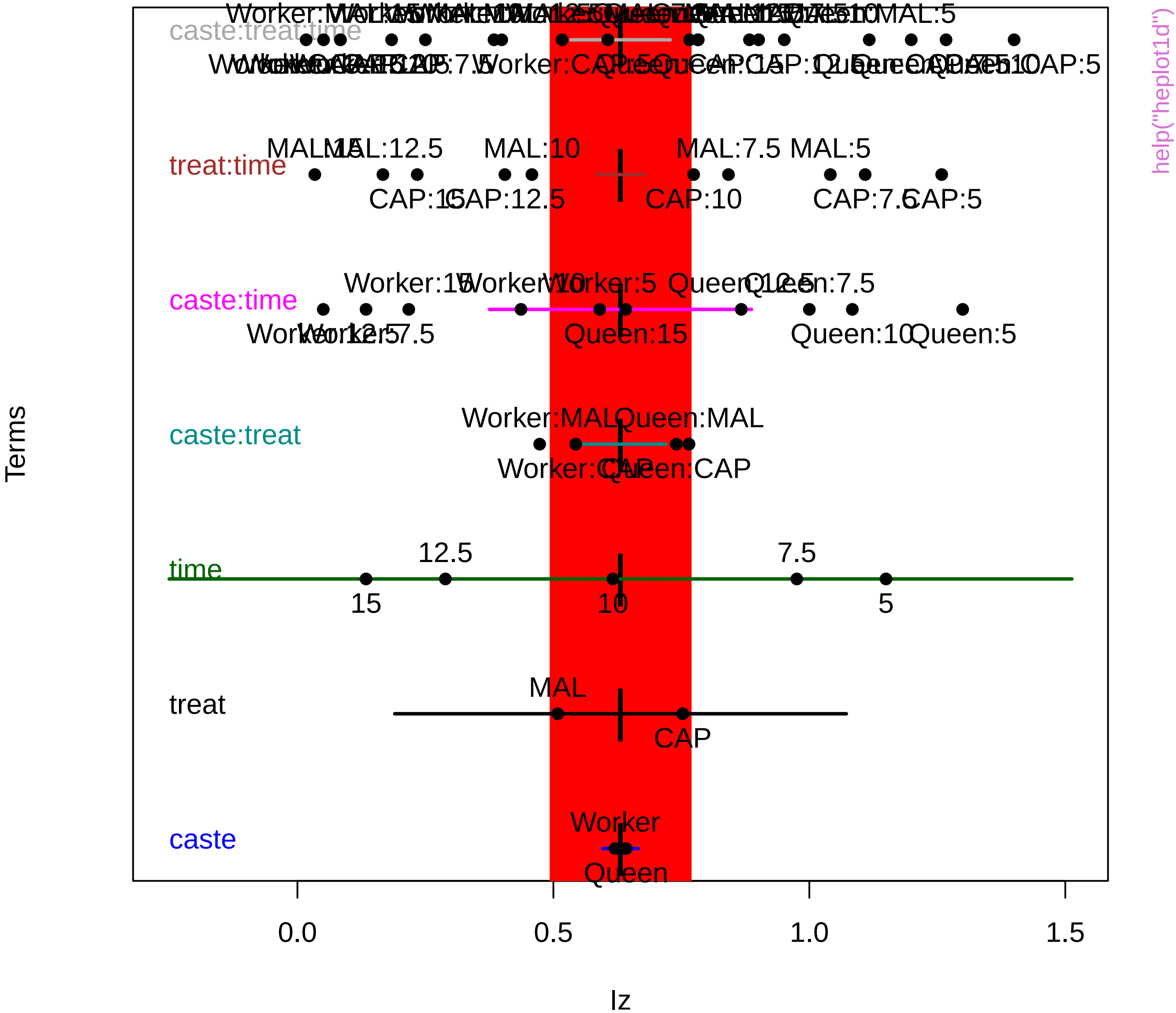
rate:additive
High:Low
Low:High
additive
Low
High
rate
Low
High
Terms
6.5
6.6
6.7
6.8
6.9
7.0
7.1
tear
help("heplot1d")



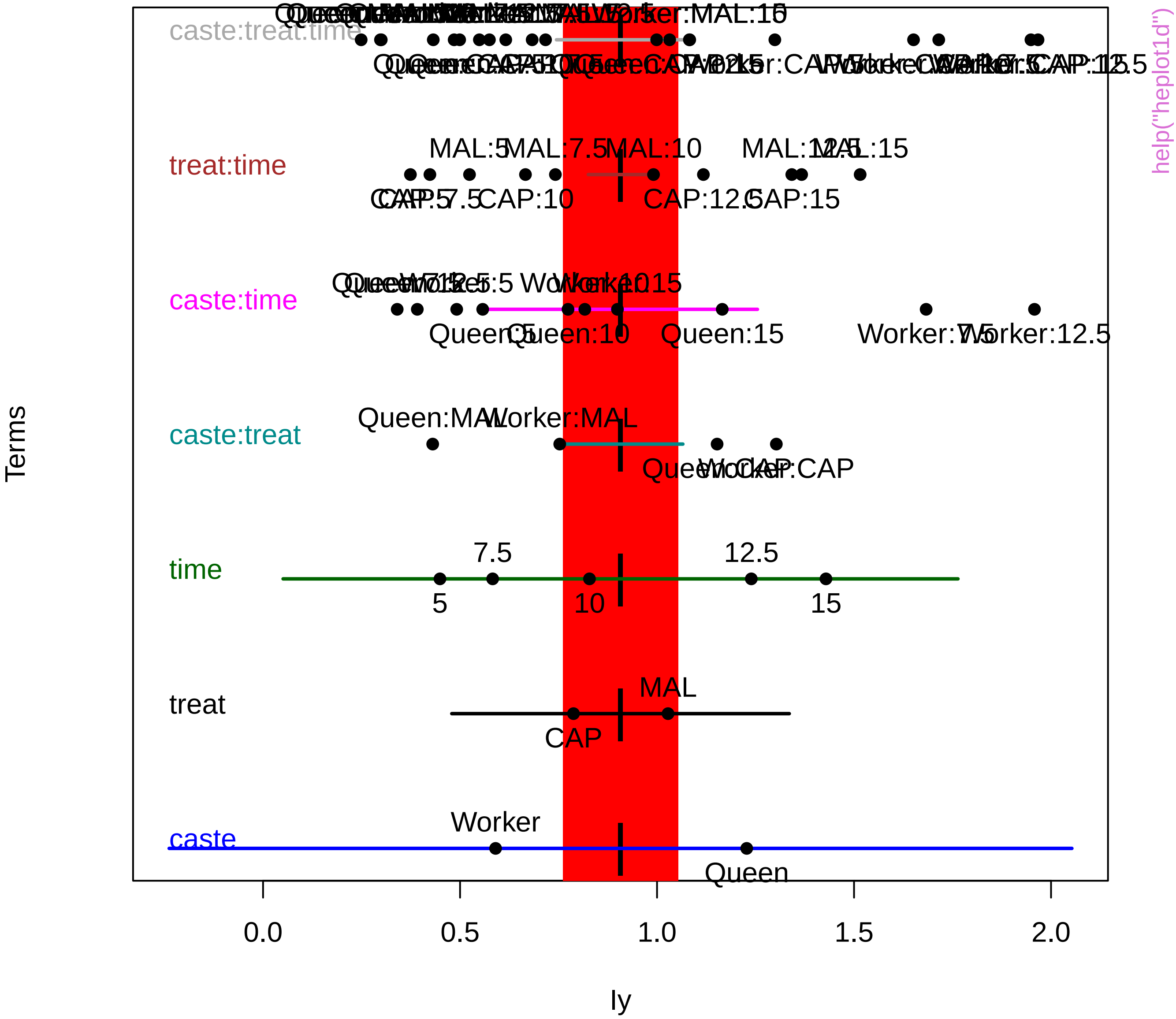
rate:additive
High:High
Low:High
High:Low
additive
High
Low
rate
High
Low
9.1
9.2
9.3
9.4
9.5
9.6
gloss
Terms
help("heplot1d")



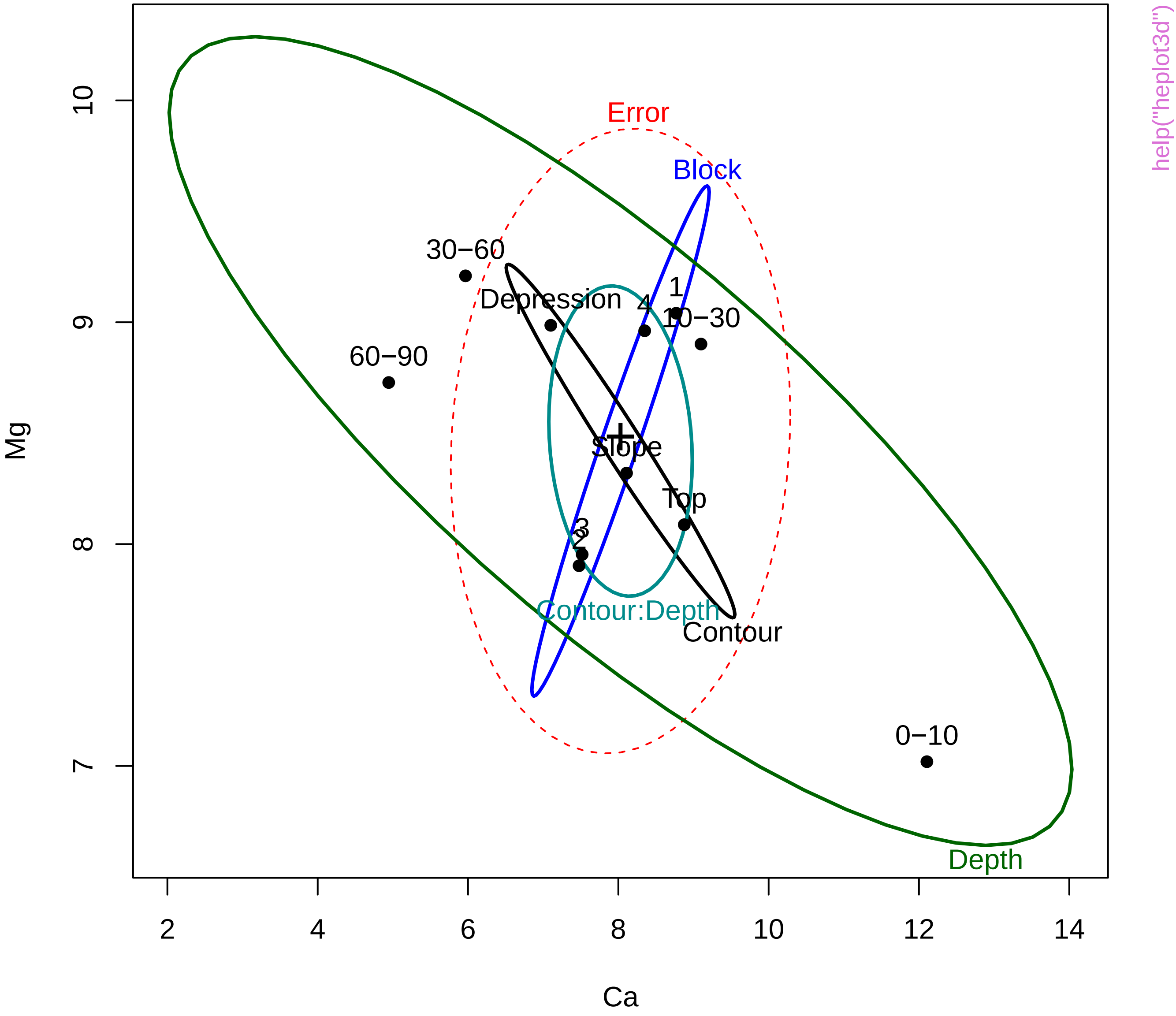
Low:High
High:High
rate:additive
Low:Low
High:Low
additive
High
Low
rate
High
Low
3.0
3.5
4.0
4.5
5.0
opacity
Terms
help("heplot1d")



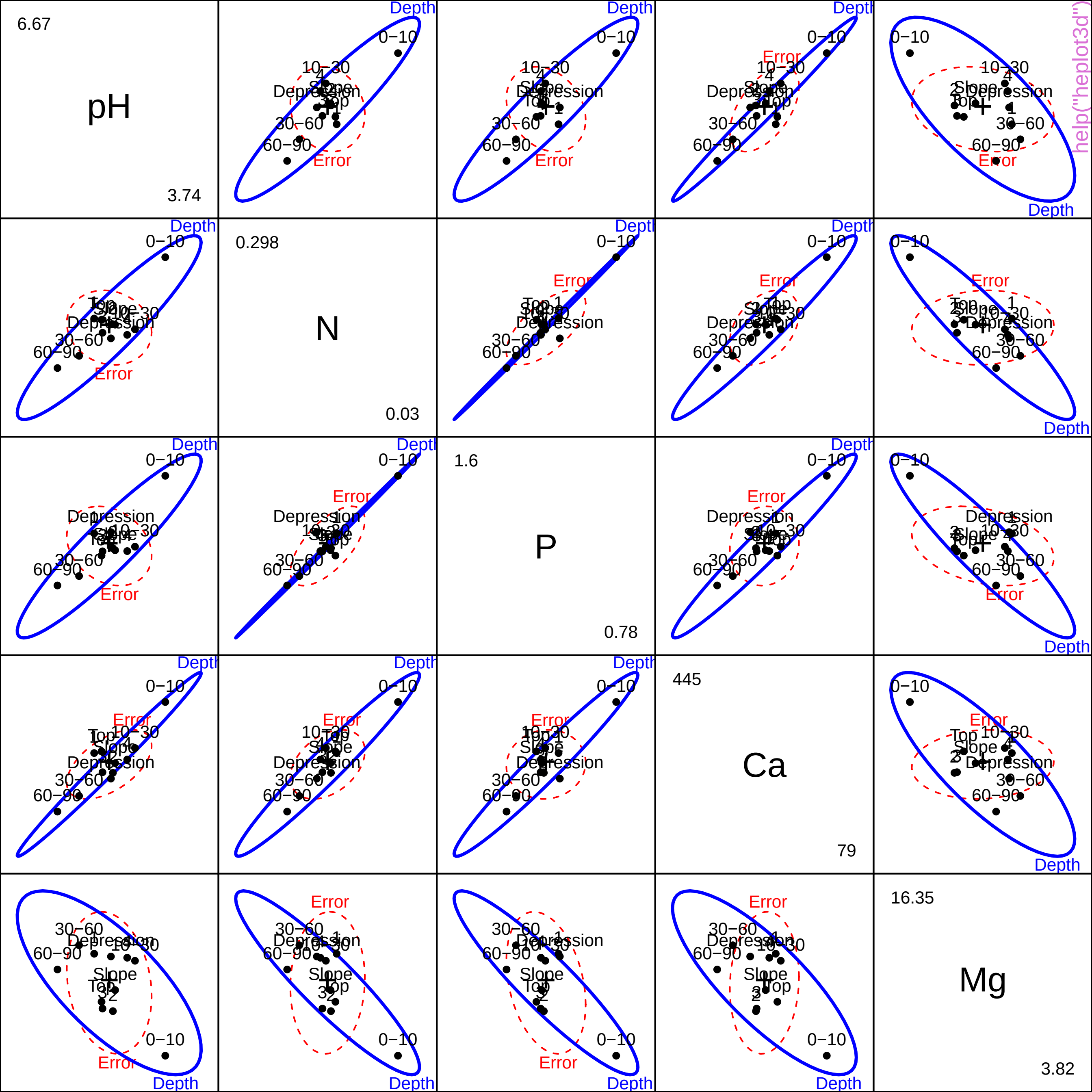
help("heplot1d")
caste:treat:time
treat:time
caste:time
caste:treat
time
treat
caste
Terms
lz
0.0
0.5
1.0
1.5
MAL:12.5
MAL:10
MAL:7.5
MAL:5
CAP:12.5
CAP:10
CAP:5
Queen:7.5
Queen:15
Queen:5
Worker:MAL
Queen:MAL
Queen:CAP
12.5
7.5
15
10
5
MAL
CAP
Worker
Queen



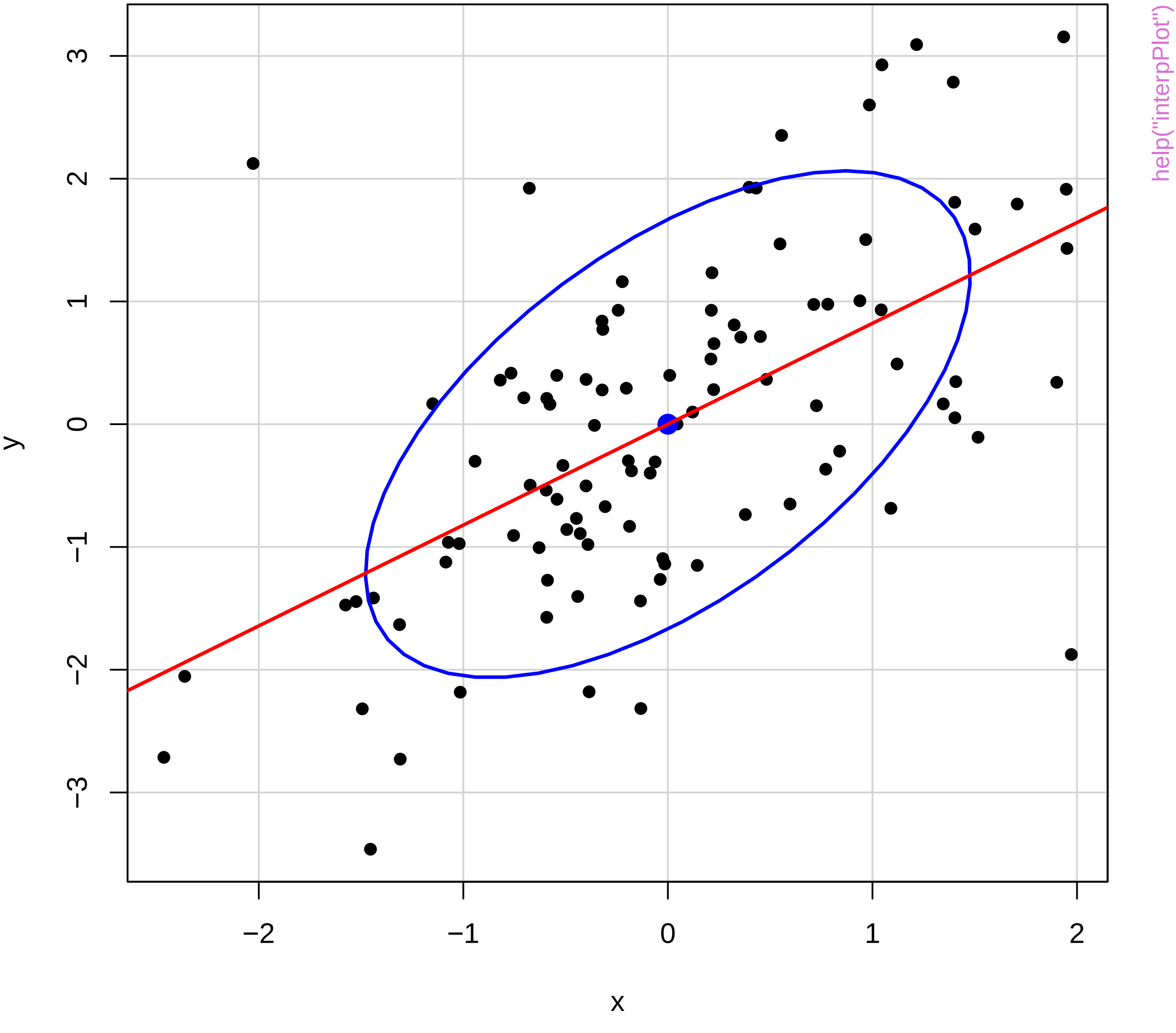
caste:treat:time
treat:time
caste:time
caste:treat
time
treat
caste
MAL:5
MAL:7.5
MAL:10
MAL:15
CAP:10
CAP:12.5
CAP:15
Queen:5
Queen:10
Queen:15
Worker:7.5
Worker:12.5
Queen:MAL
Worker:MAL
Worker:CAP
7.5
12.5
5
10
15
MAL
CAP
Worker
Queen
Worker:MAL:15
0.0
0.5
1.0
1.5
2.0
ly
Terms
help("heplot1d")

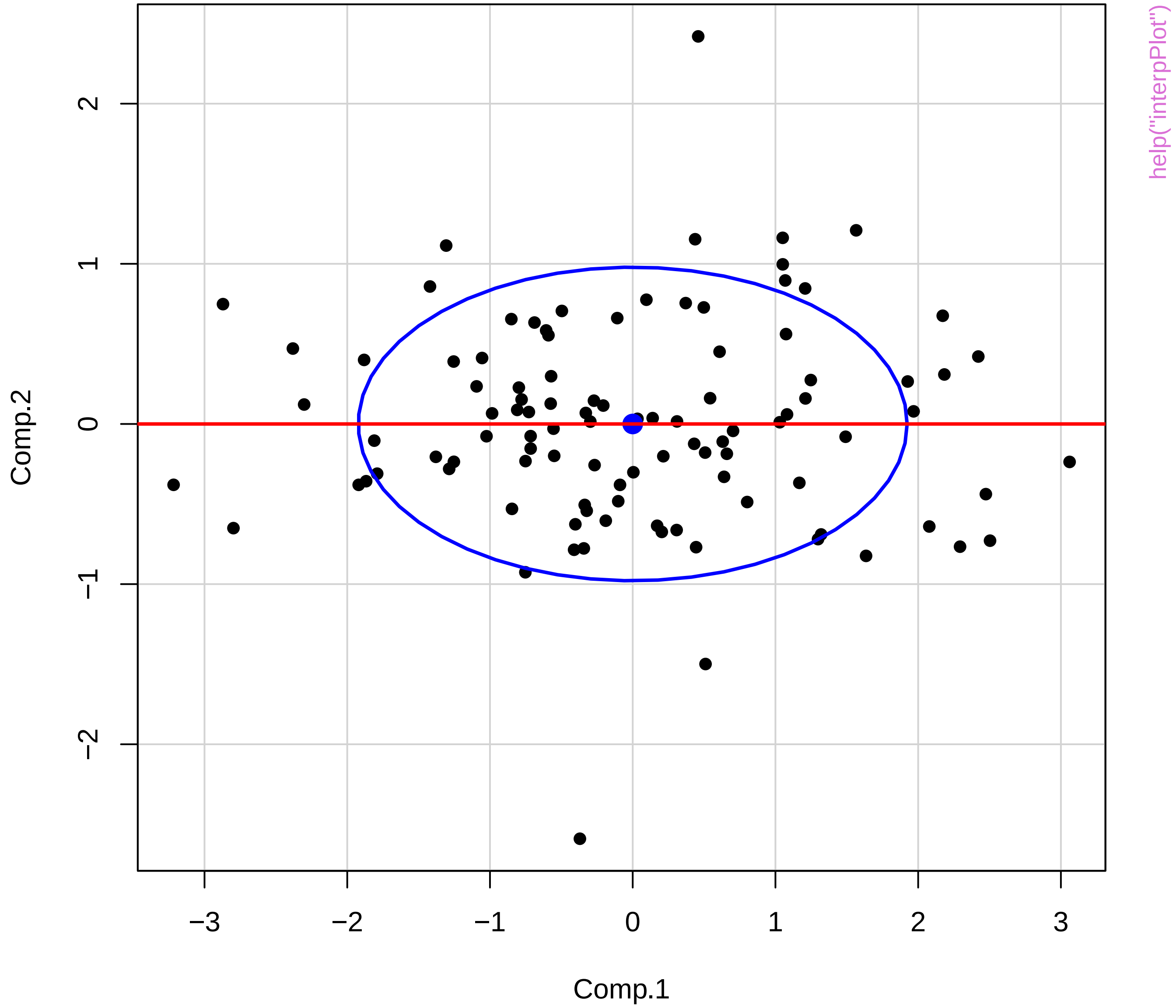


Error
Block
30−60
Depression
1
4
10−30
60−90
Slope
Top
3
2
Contour:Depth
Contour
0−10
Depth
10
9
8
7
Mg
2
4
6
8
10
12
14
Ca
help("heplot3d")

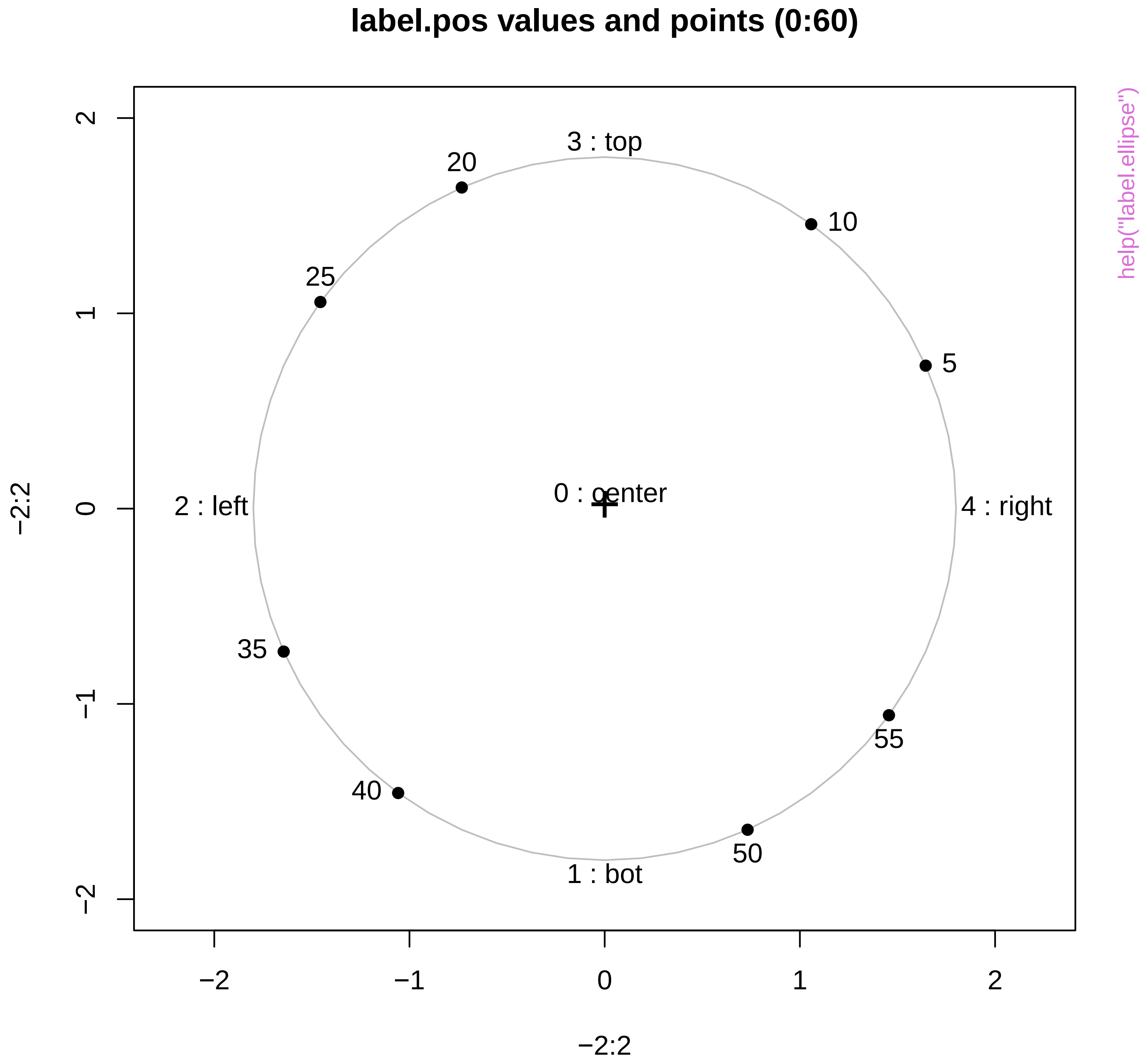


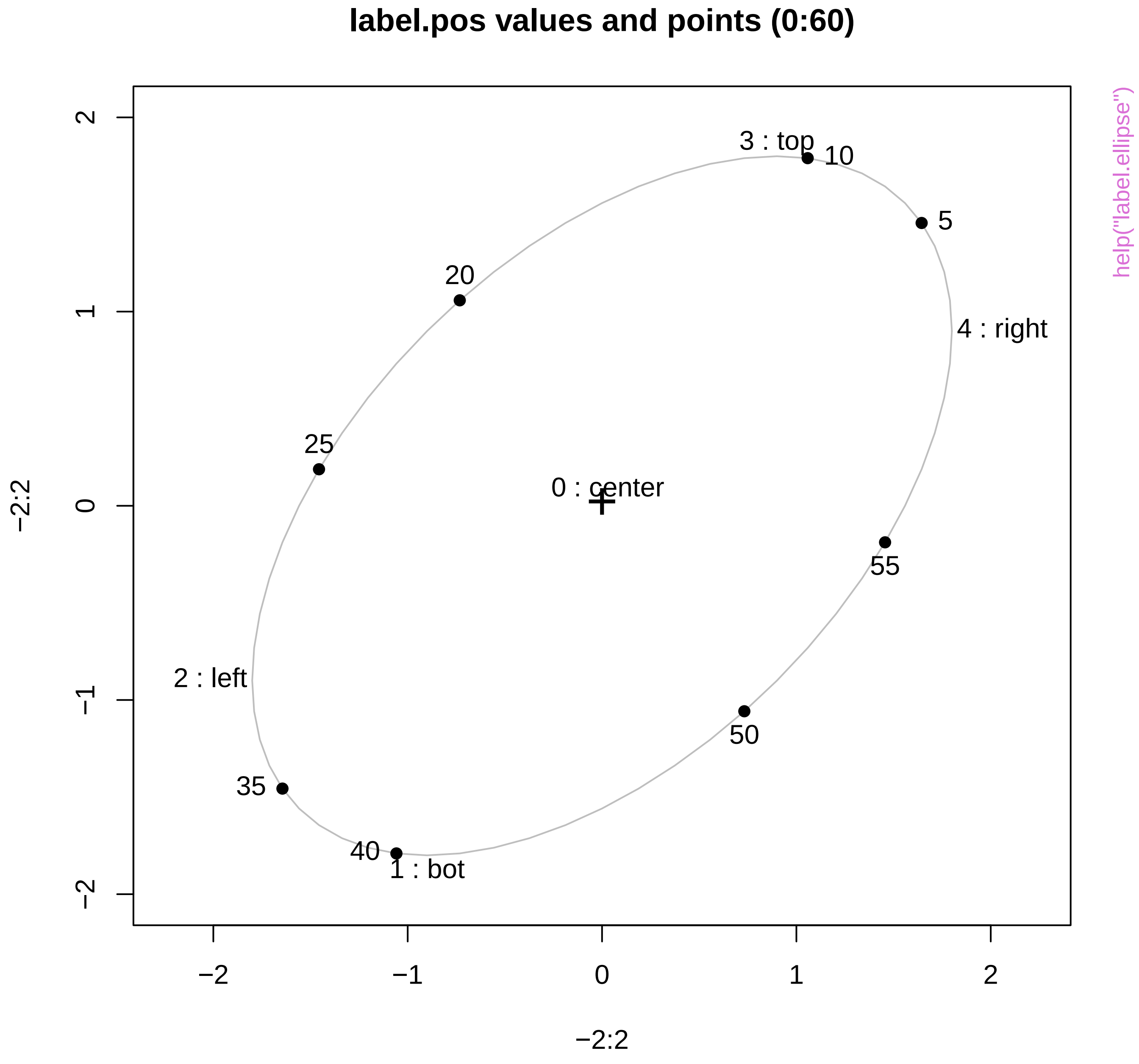
pH
6.67
3.74
N
0.298
0.03
P
1.6
0.78
Ca
445
79
Mg
16.35
3.82
Depth
Error
0−10
10−30
30−60
60−90
Depression
Slope
Top
help("heplot3d")

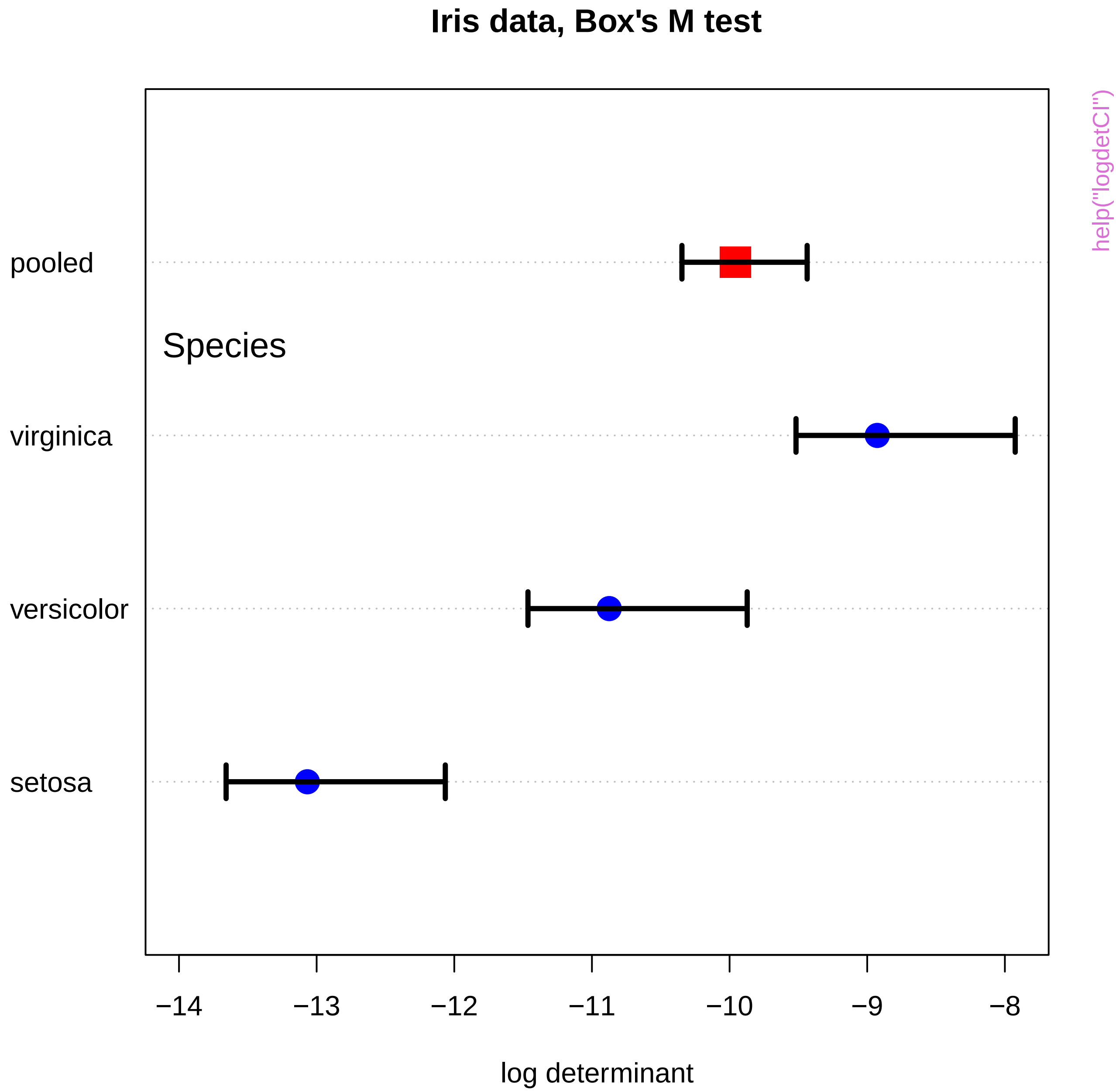

−2
−1
0
1
2
x
−3
−2
−1
0
1
2
3
y
help("interpPlot")

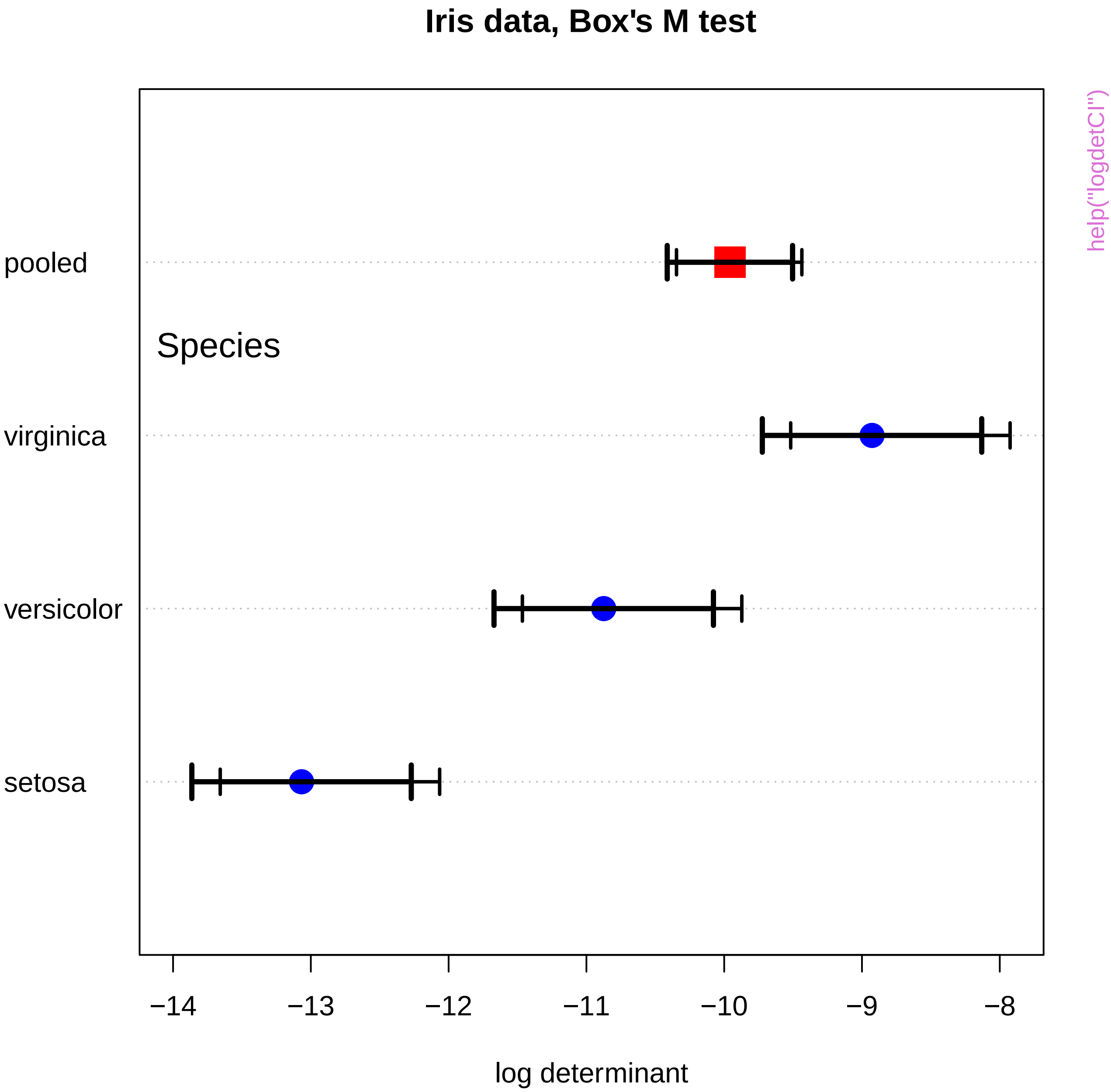


help("interpPlot")
Comp.2
2
1
0
−1
−2
Comp.1
−3
−2
−1
0
1
2
3

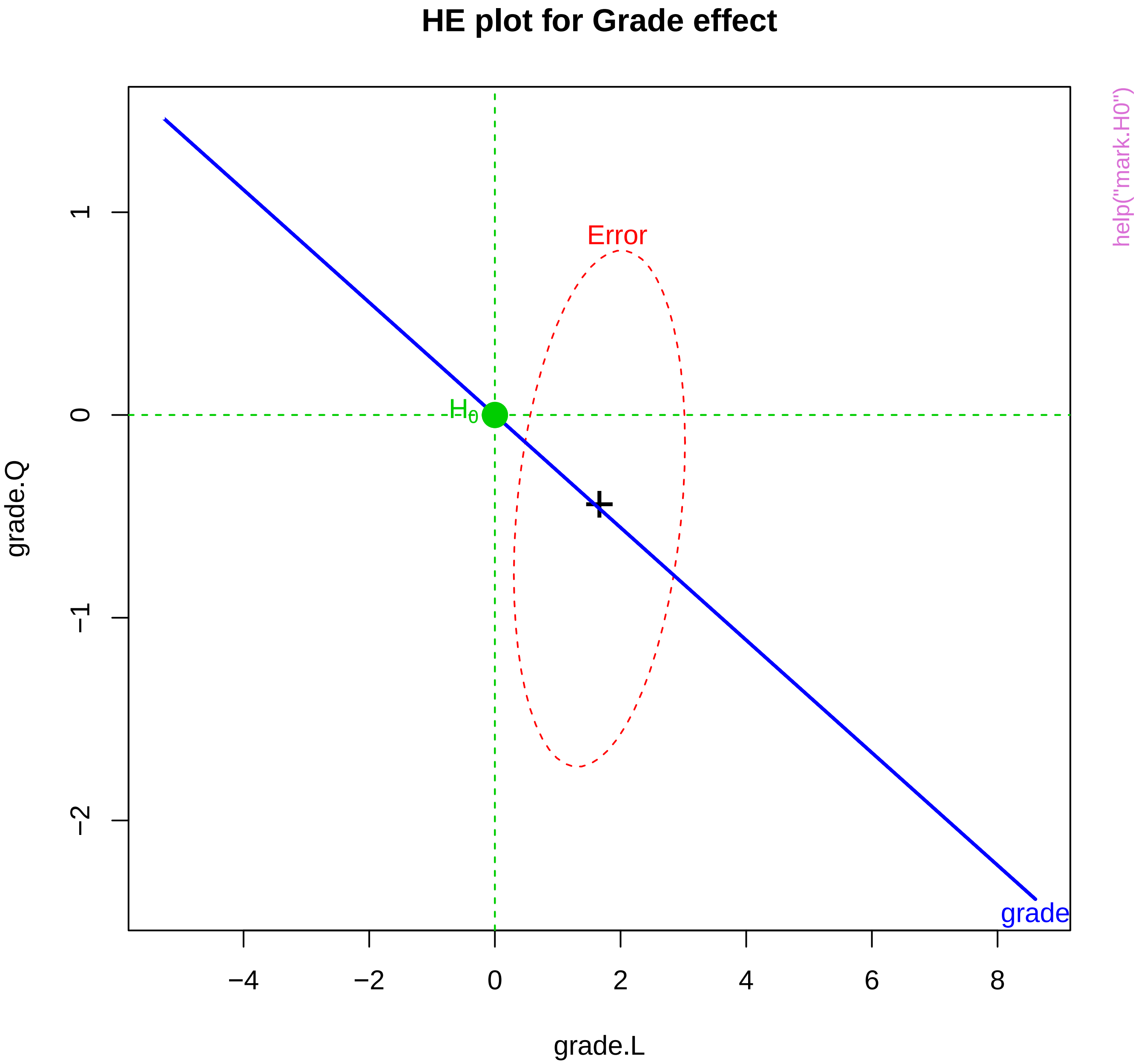

label.pos values and points (0:60)
3 : top
20
10
25
5
2 : left
0 : center
4 : right
35
55
40
50
1 : bot
−2
−1
0
1
2
−2:2
−2:2
help("label.ellipse")


label.pos values and points (0:60)
3 : top
10
5
20
4 : right
25
0 : center
55
2 : left
50
35
40
1 : bot
2
1
0
−1
−2
−2
−1
0
1
2
−2:2
−2:2
help("label.ellipse")

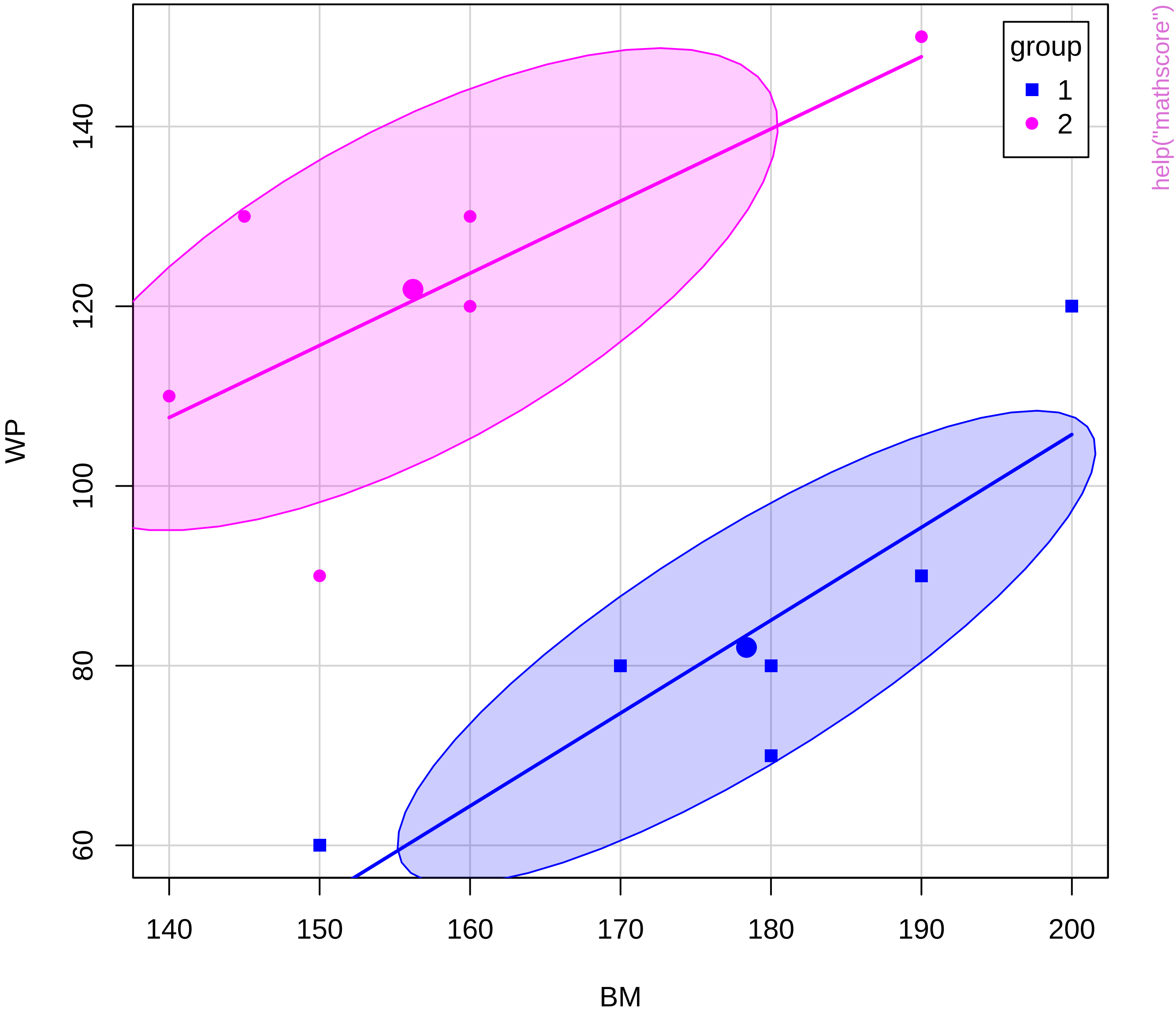

Iris data, Box's M test
pooled
Species
virginica
versicolor
setosa
−14
−13
−12
−11
−10
−9
−8
log determinant
help("logdetCI")



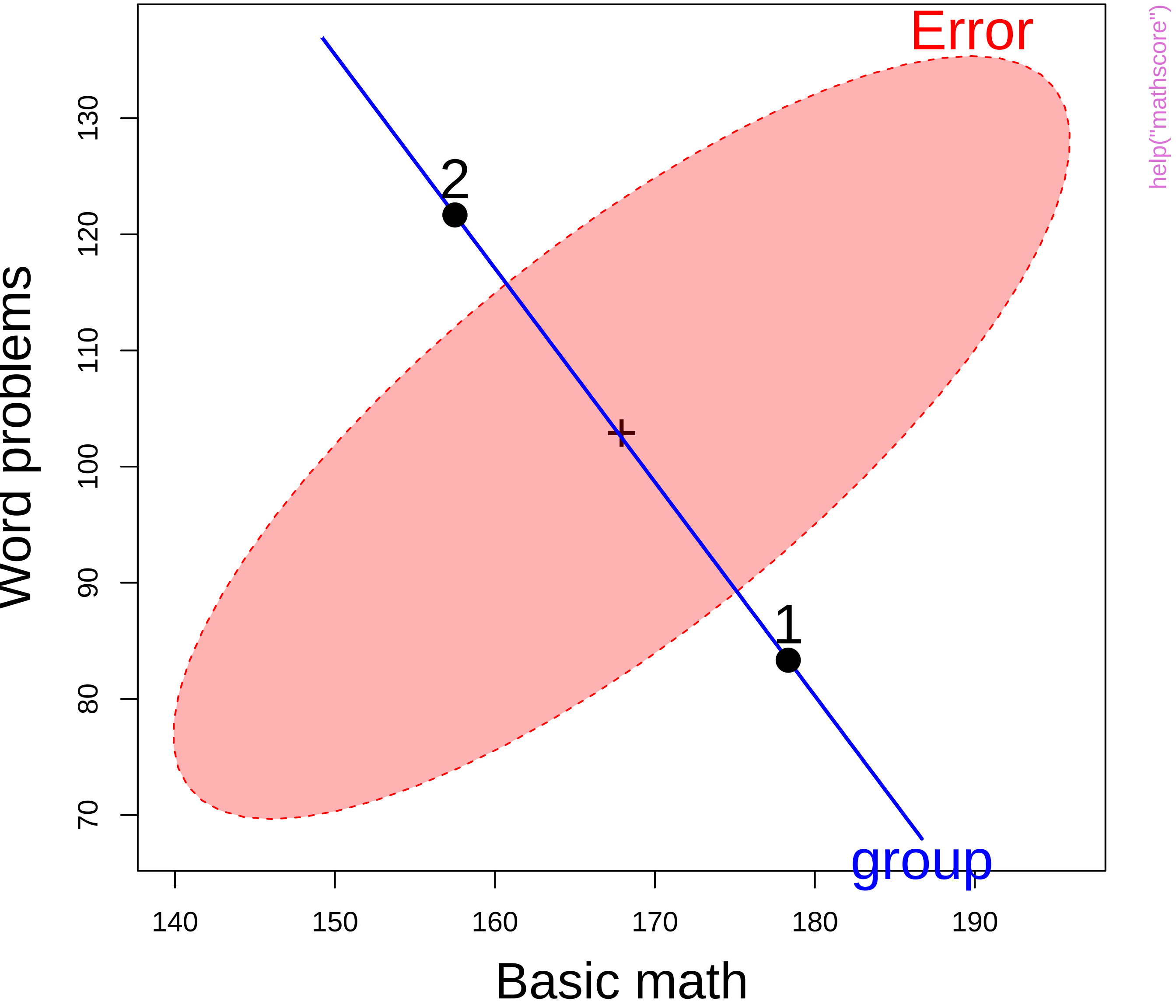
Iris data, Box's M test
pooled
Species
virginica
versicolor
setosa
−14
−13
−12
−11
−10
−9
−8
log determinant
help("logdetCI")



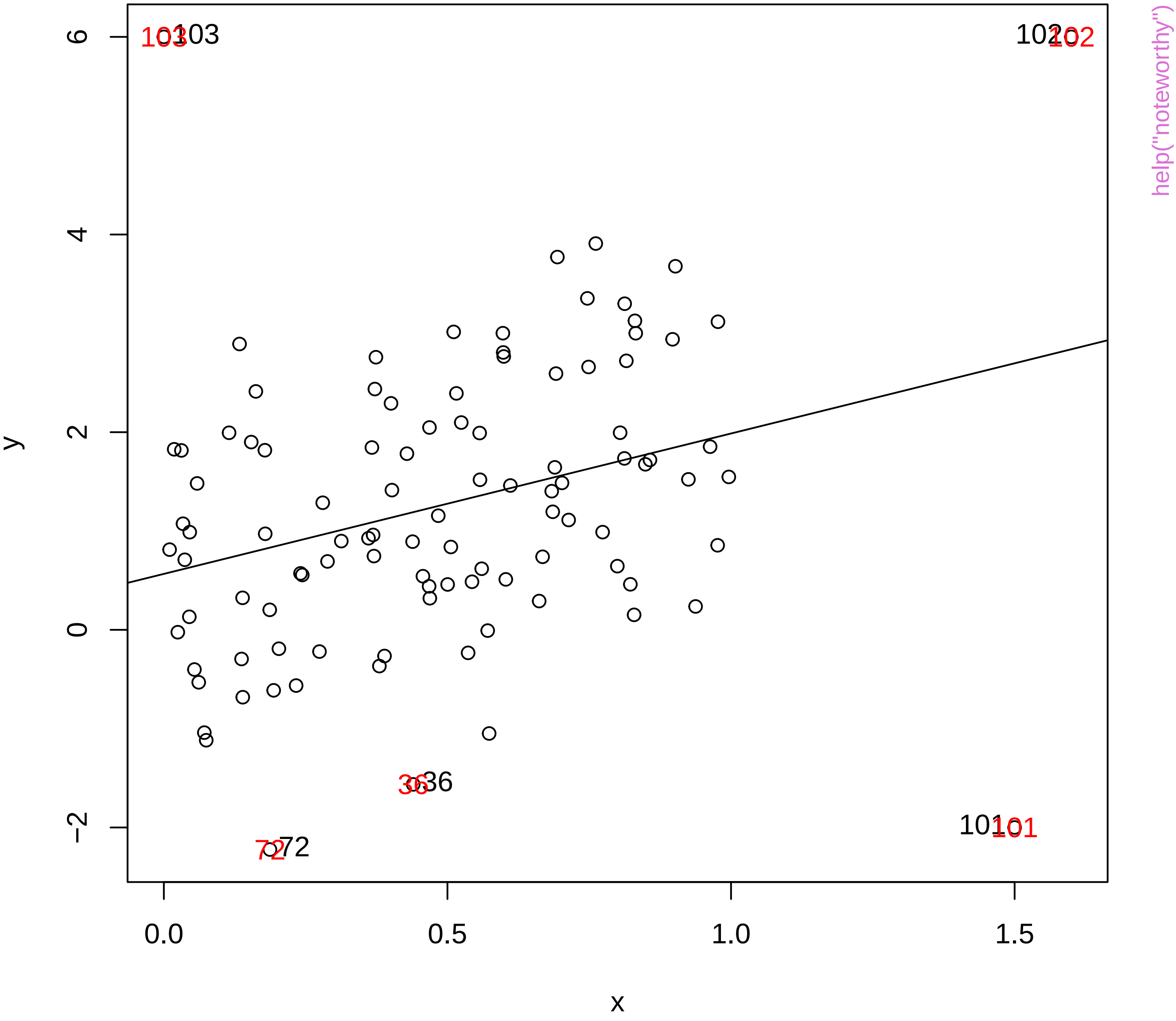
HE plot for Grade effect
Error
H_0
grade
grade.L
grade.Q
−4
−2
0
2
4
6
8
1
0
−1
−2
help("mark.H0")



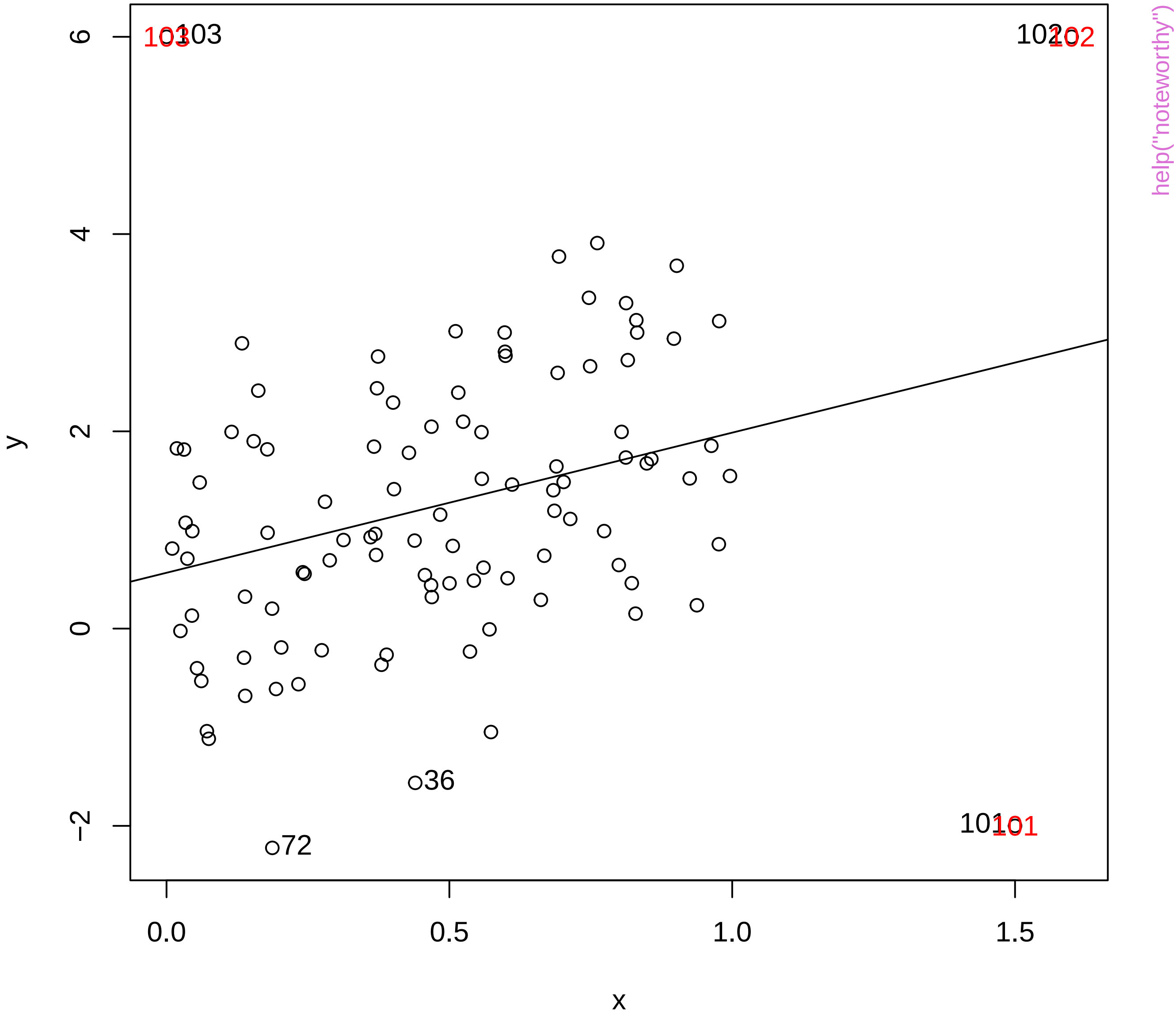
group
1
2
WP
BM
140
120
100
80
60
140
150
160
170
180
190
200
help("mathscore")



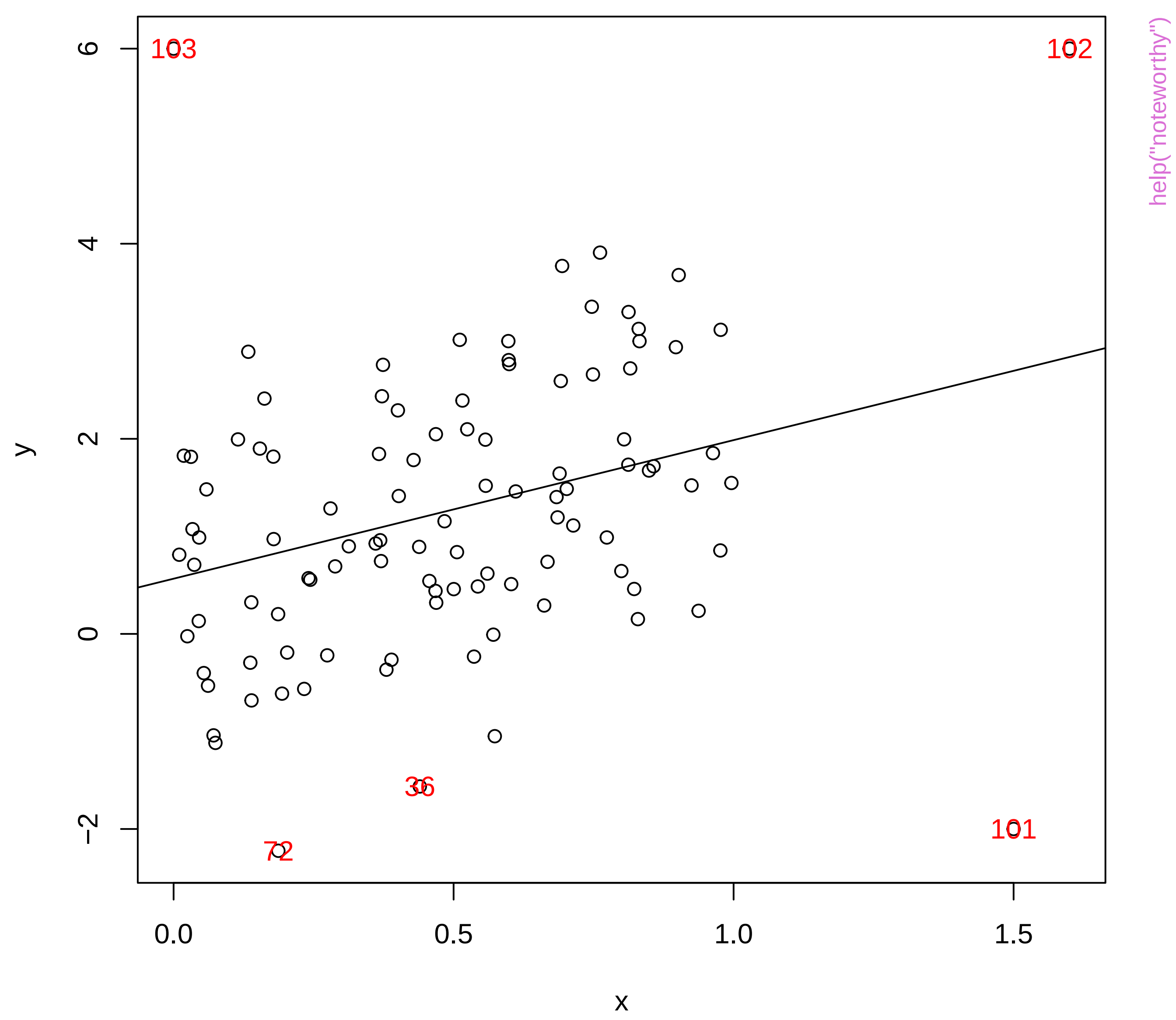
Error
group
2
1
help("mathscore")
Word problems
Basic math
70
80
90
100
110
120
130
140
150
160
170
180
190



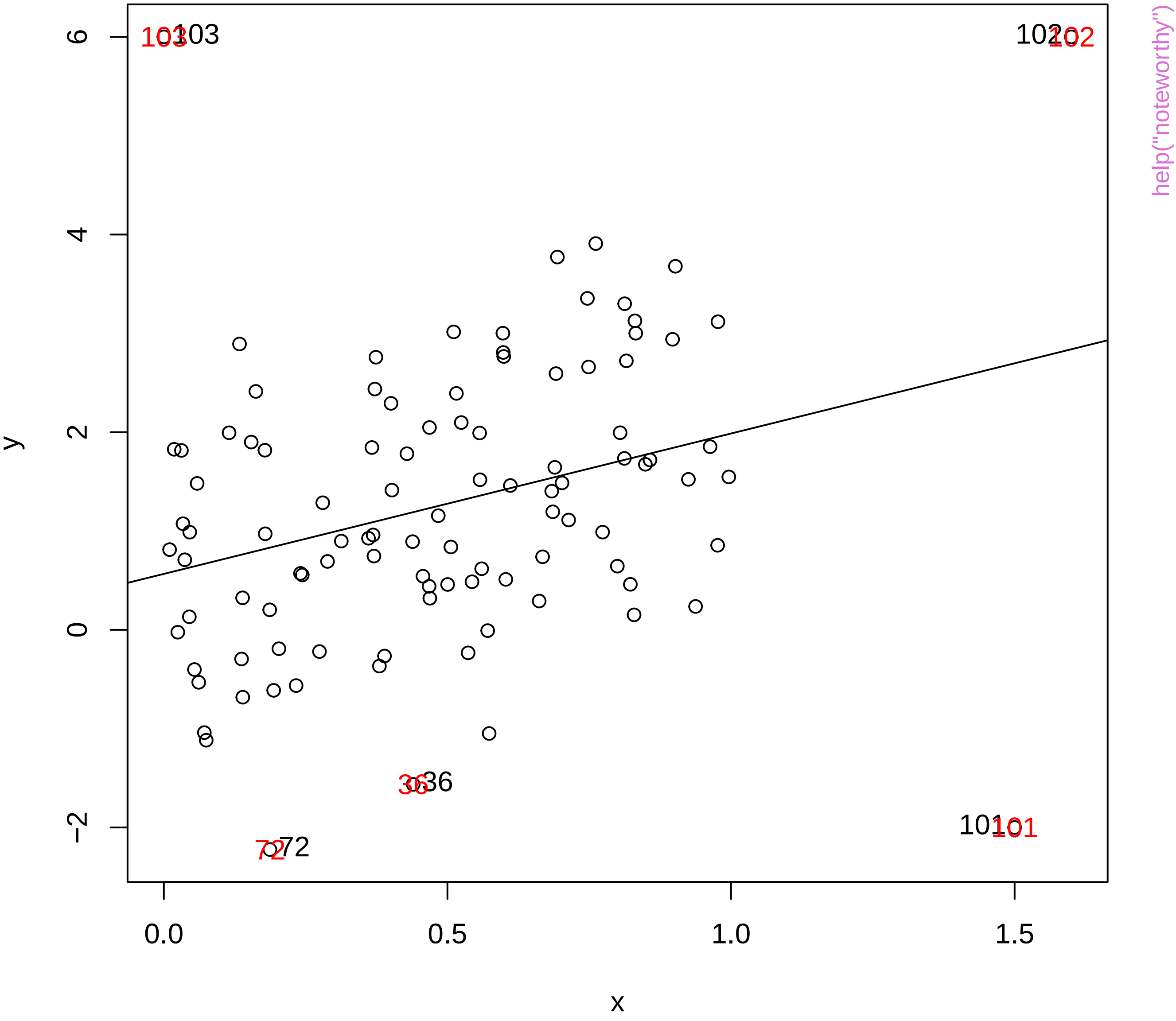
103
103
102
102
36
36
101
101
72
72
6
4
2
0
−2
y
0.0
0.5
1.0
1.5
x
help("noteworthy")



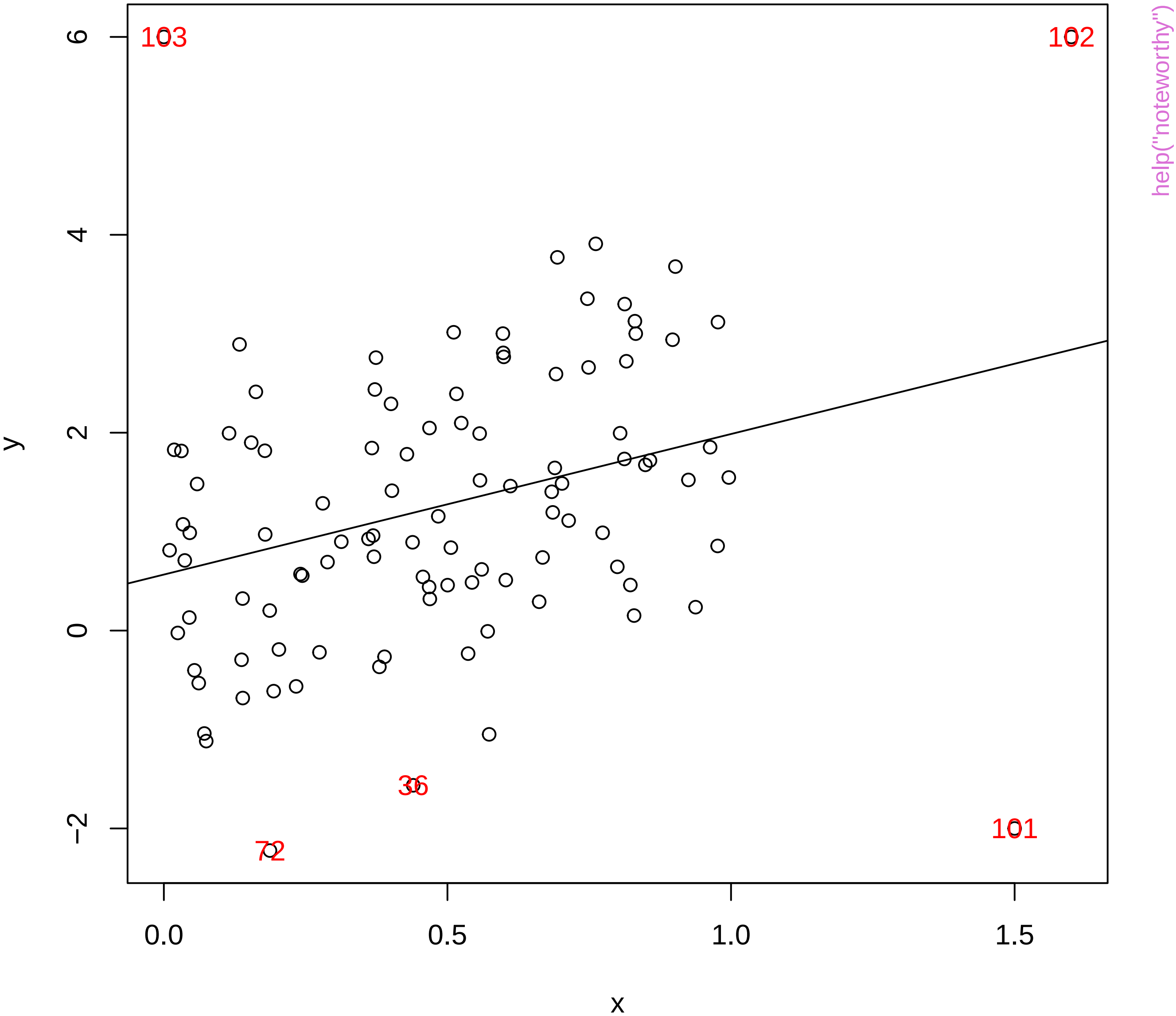
103
103
102
102
101
101
36
72
6
4
2
0
−2
y
0.0
0.5
1.0
1.5
x
help("noteworthy")



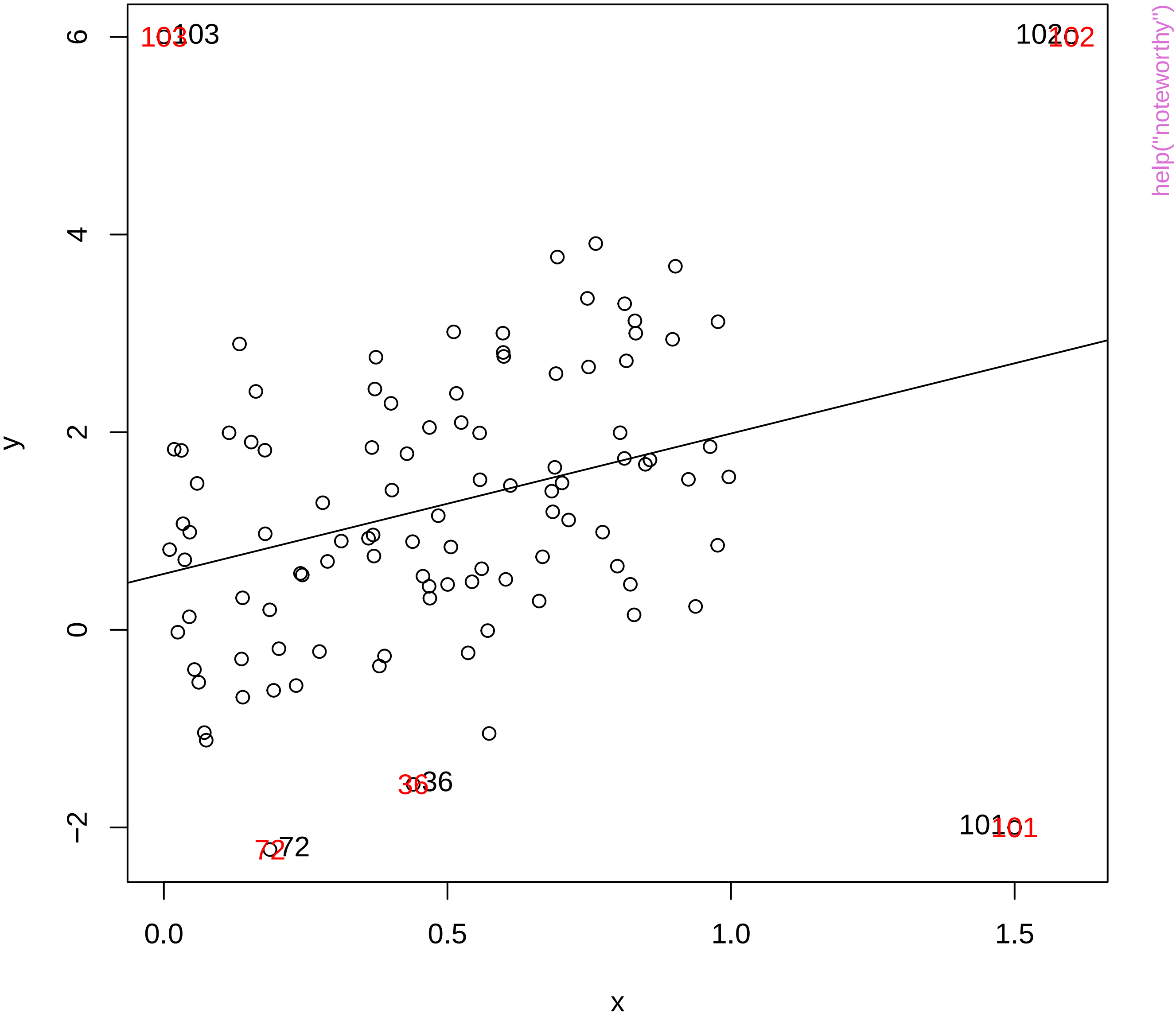
103
102
36
72
101
y
x
6
4
2
0
−2
0.0
0.5
1.0
1.5
help("noteworthy")



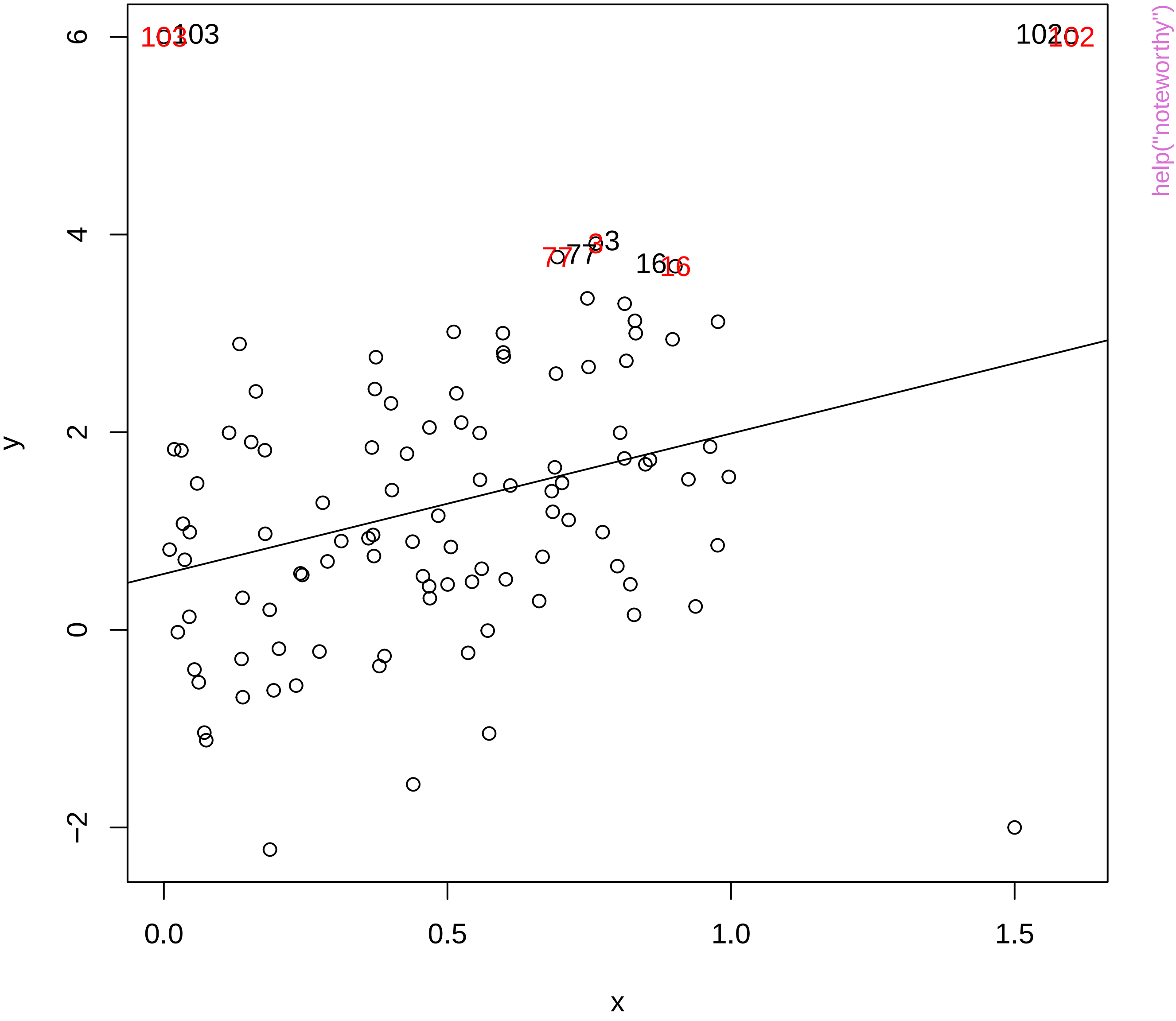
103
103
102
102
36
36
101
101
72
72
6
4
2
0
−2
y
0.0
0.5
1.0
1.5
x
help("noteworthy")



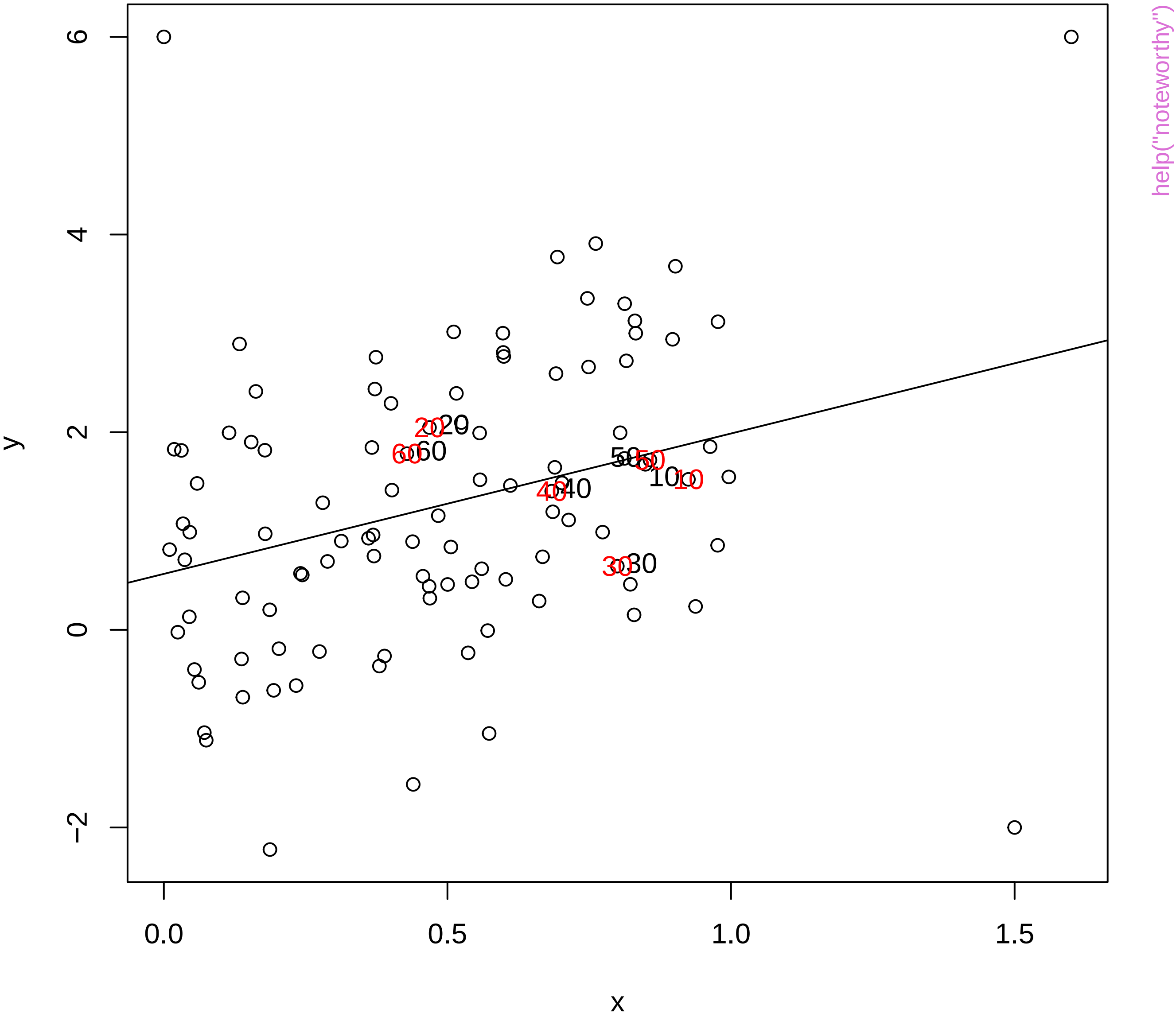
y
x
6
4
2
0
−2
0.0
0.5
1.0
1.5
103
102
36
72
101
help("noteworthy")



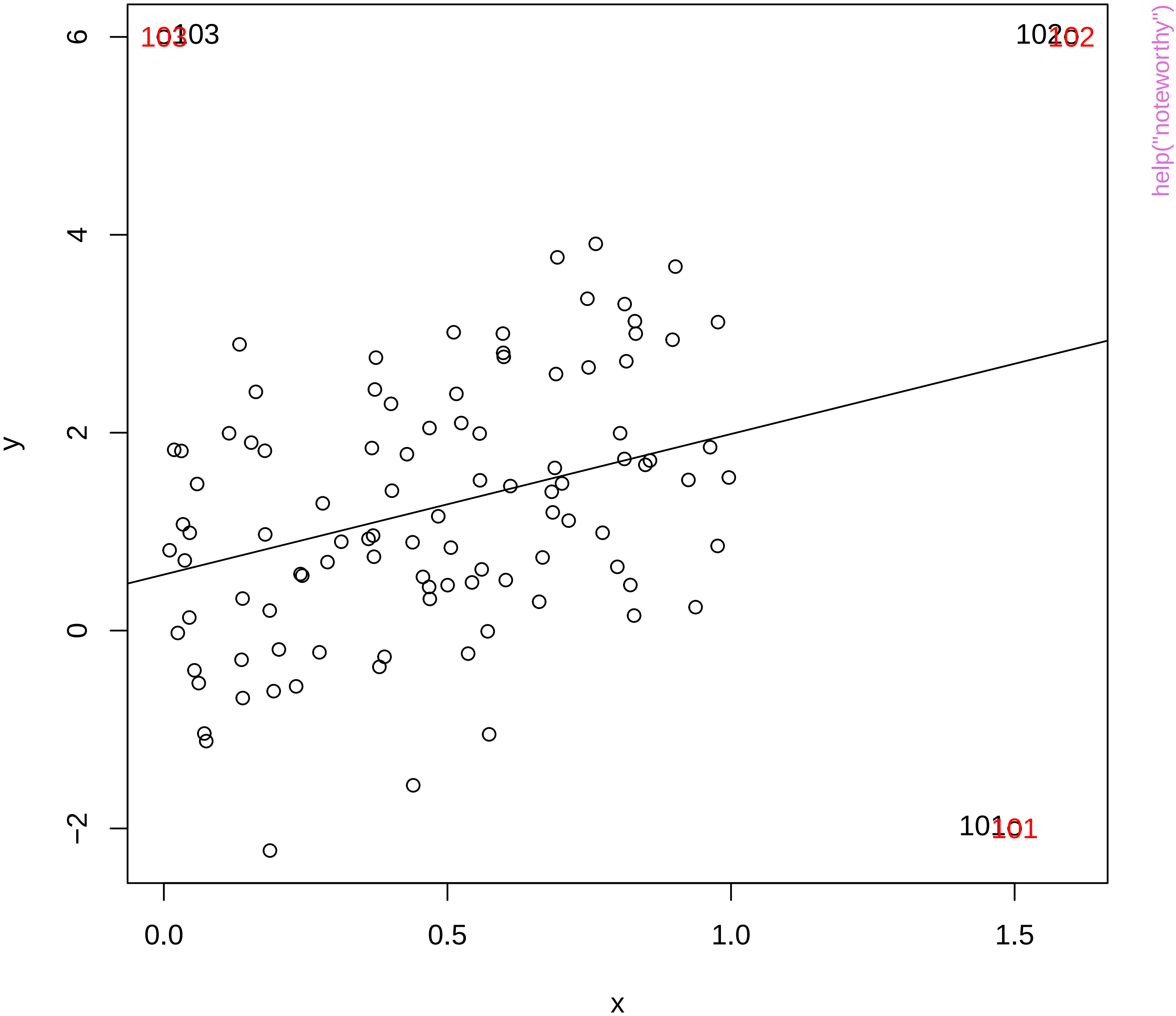
103
103
102
102
36
36
72
72
101
101
6
4
2
0
−2
y
0.0
0.5
1.0
1.5
x
help("noteworthy")



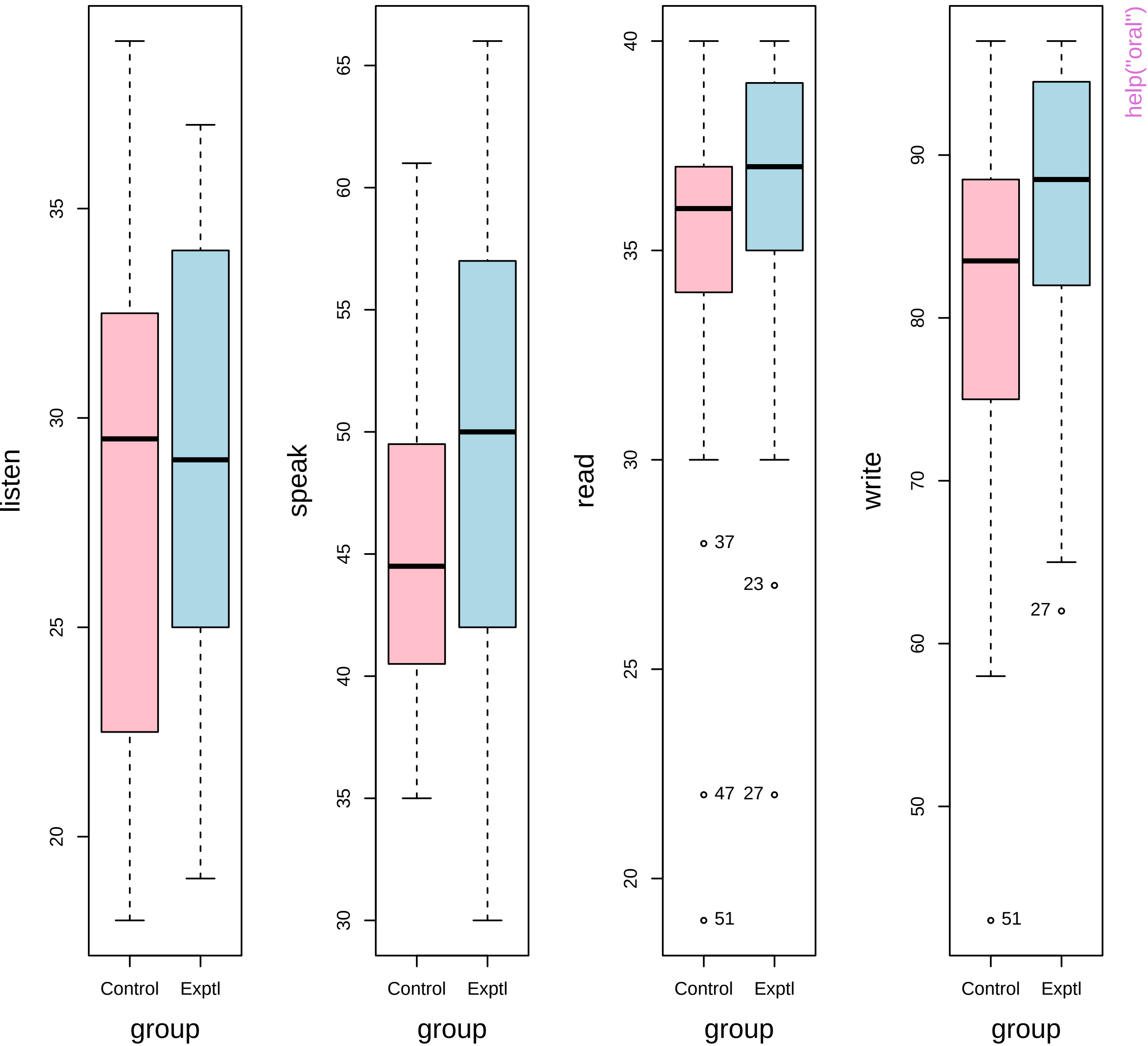
103
103
102
102
77
77
3
3
16
16
6
4
2
0
−2
y
0.0
0.5
1.0
1.5
x
help("noteworthy")



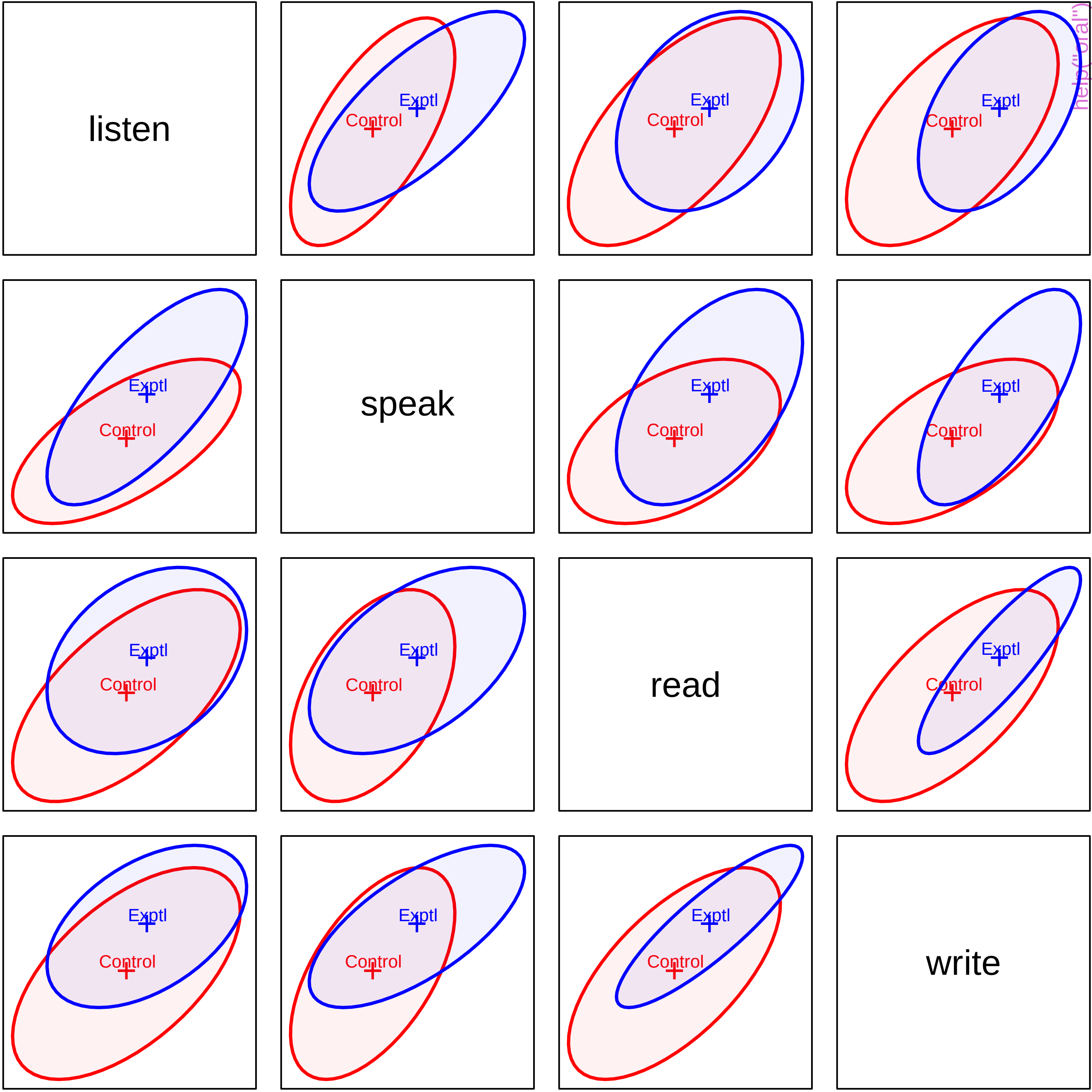
help("noteworthy")
y
x
6
4
2
0
−2
0.0
0.5
1.0
1.5
20
20
60
60
50
50
10
10
40
40
30
30

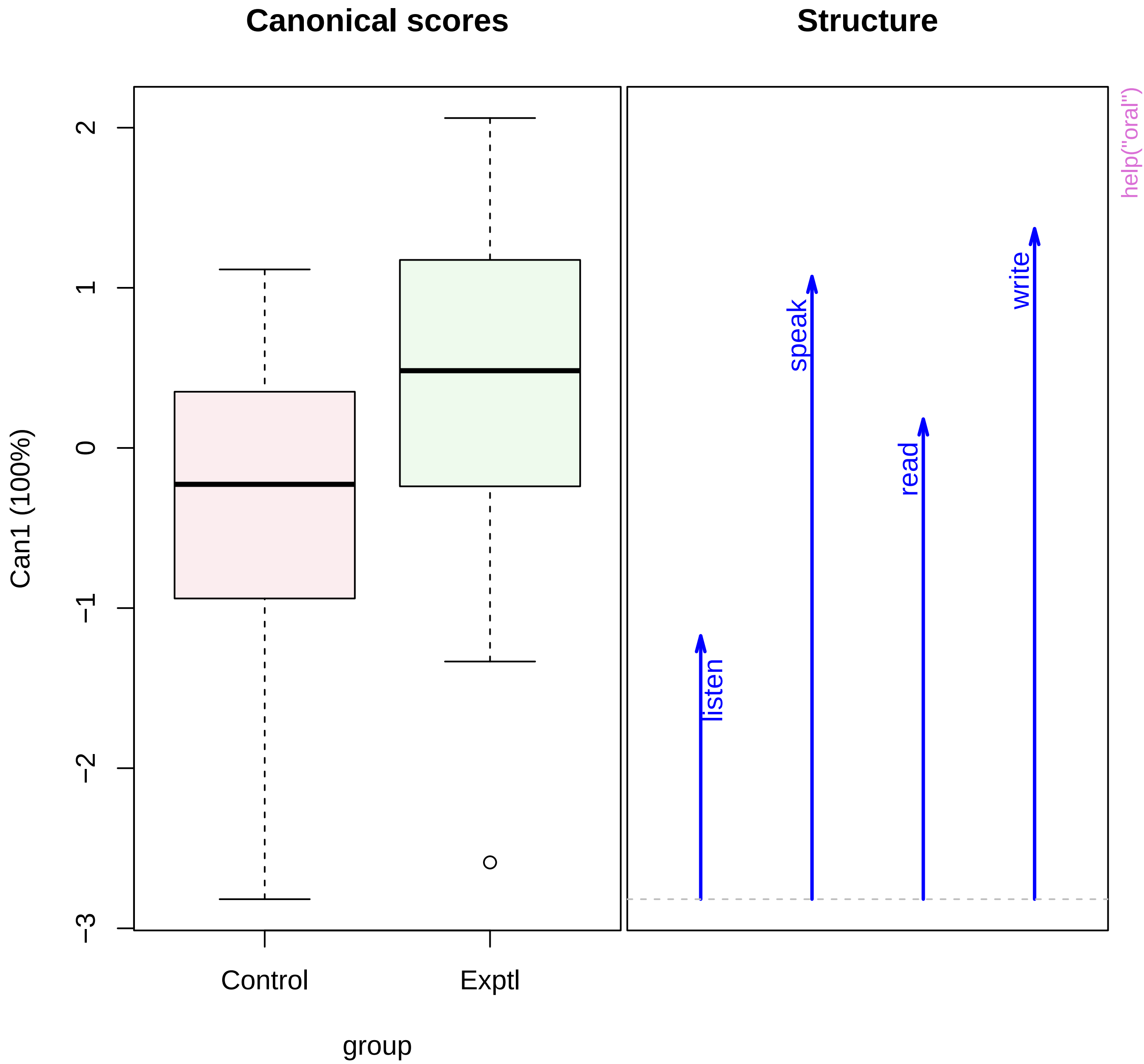


103
103
102
102
101
101
6
4
2
0
−2
y
0.0
0.5
1.0
1.5
x
help("noteworthy")

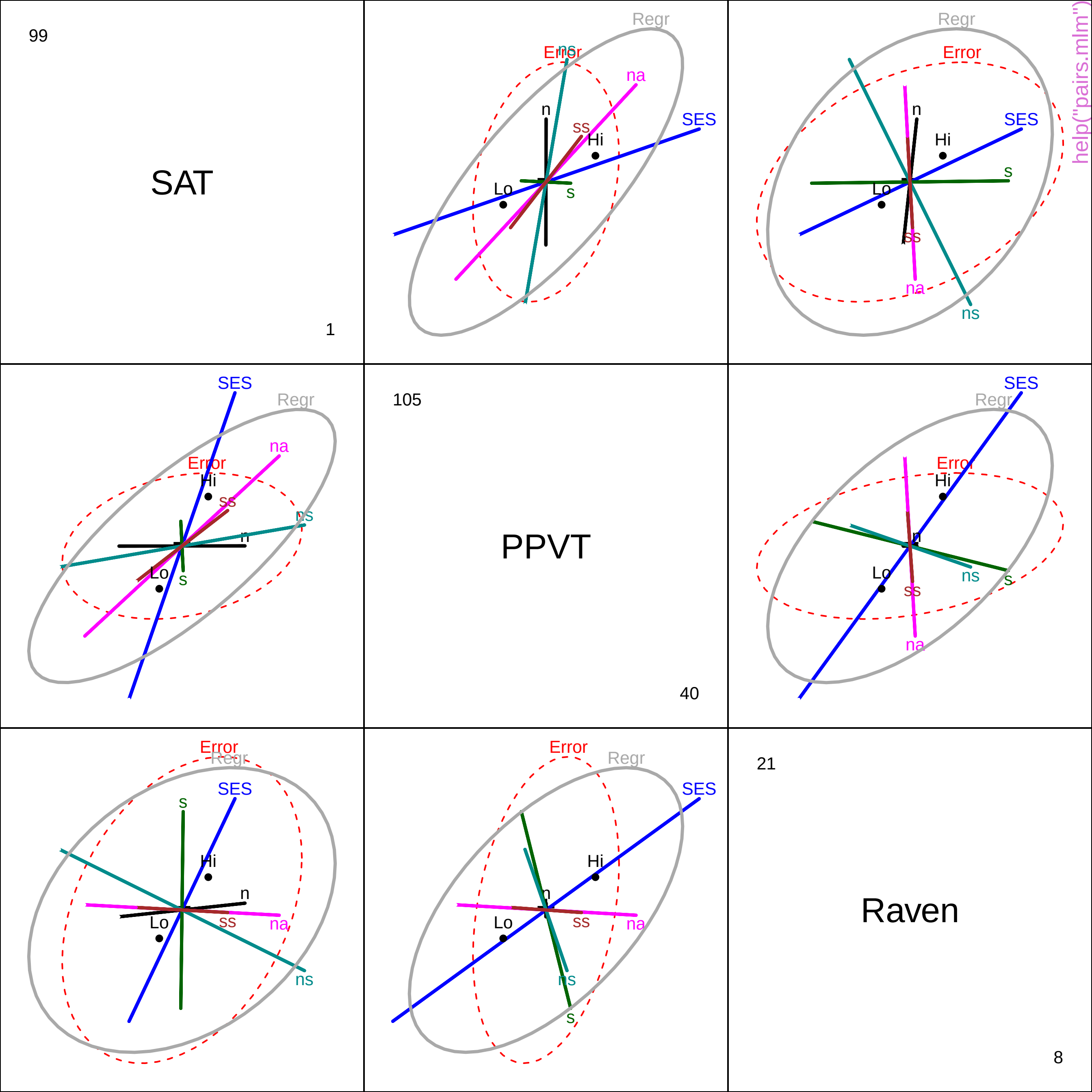


listen
speak
read
write
group
Control
Exptl
37
23
47
27
51
27
51
help("oral")

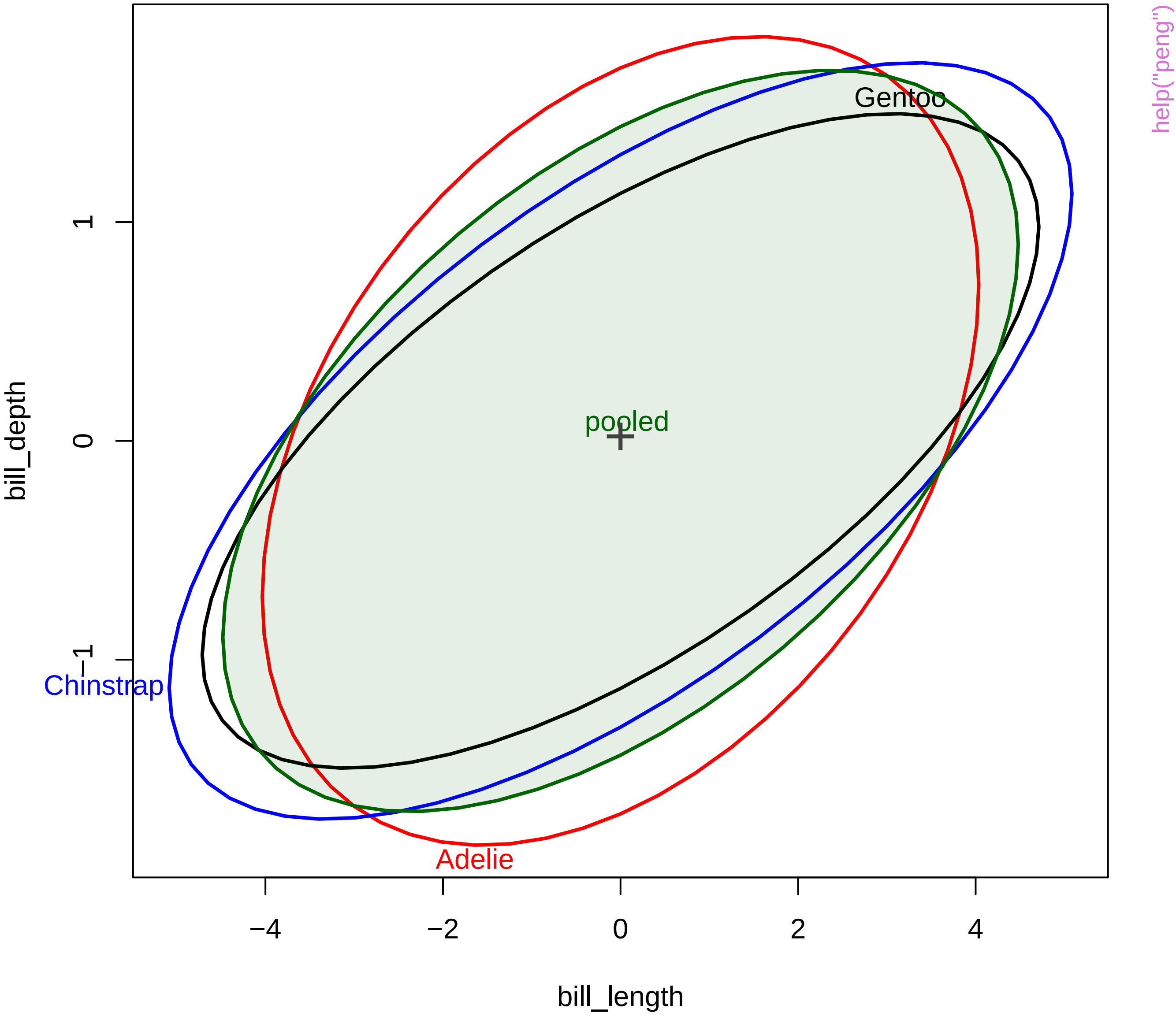

listen
speak
read
write
Exptl
Control
help("oral")

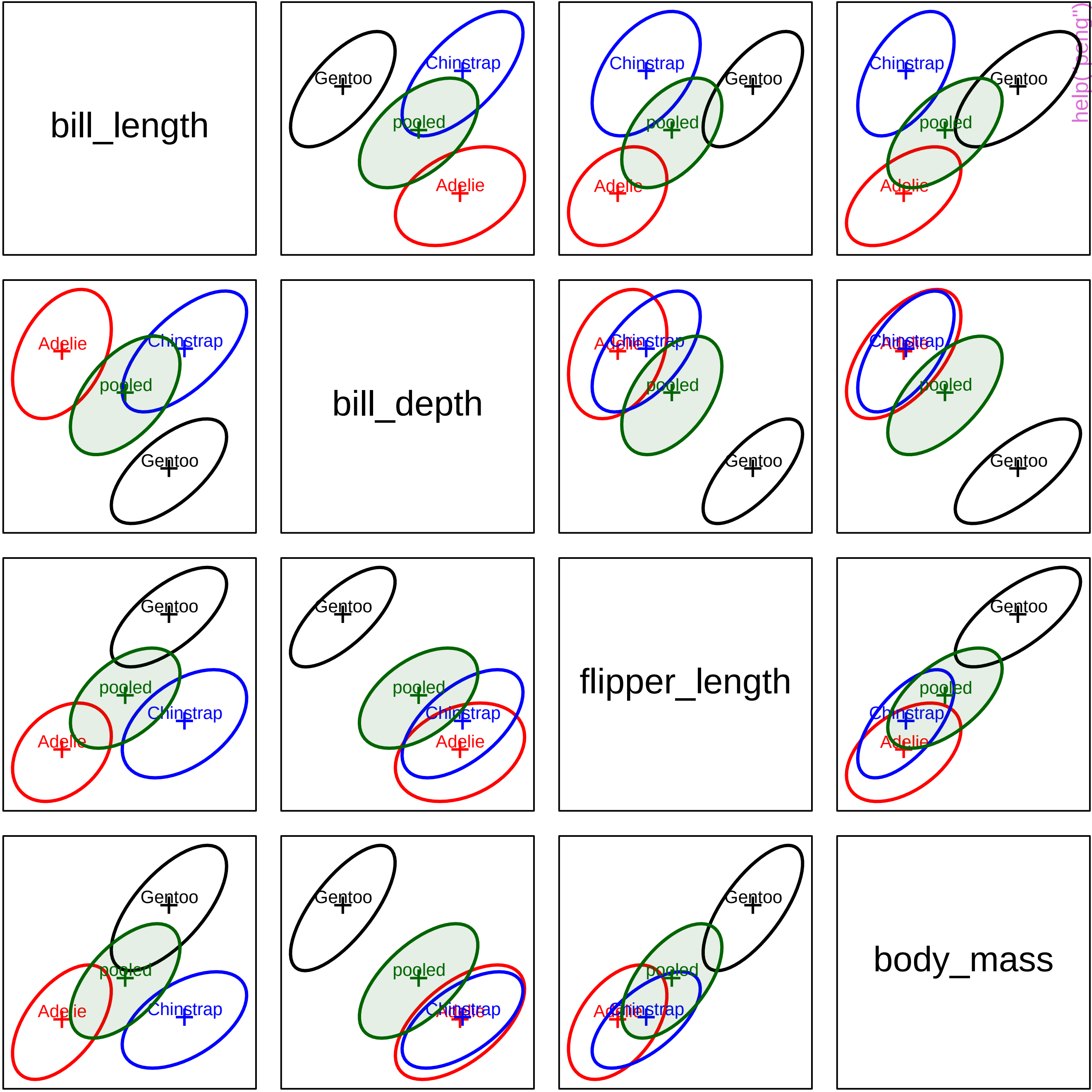


Canonical scores
Structure
Can1 (100%)
2
1
0
−1
−2
−3
Control
Exptl
group
listen
speak
read
write
help("oral")

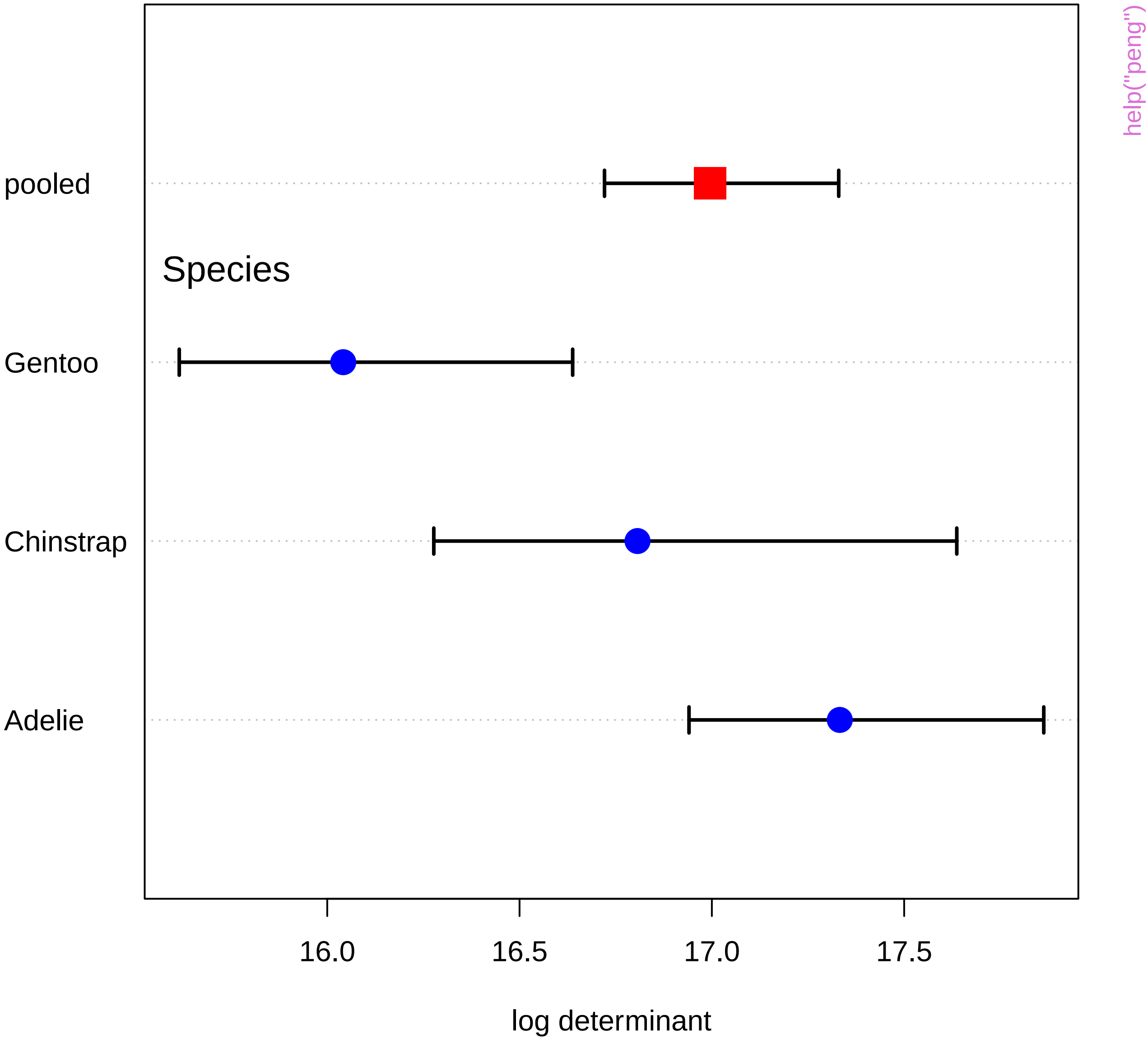


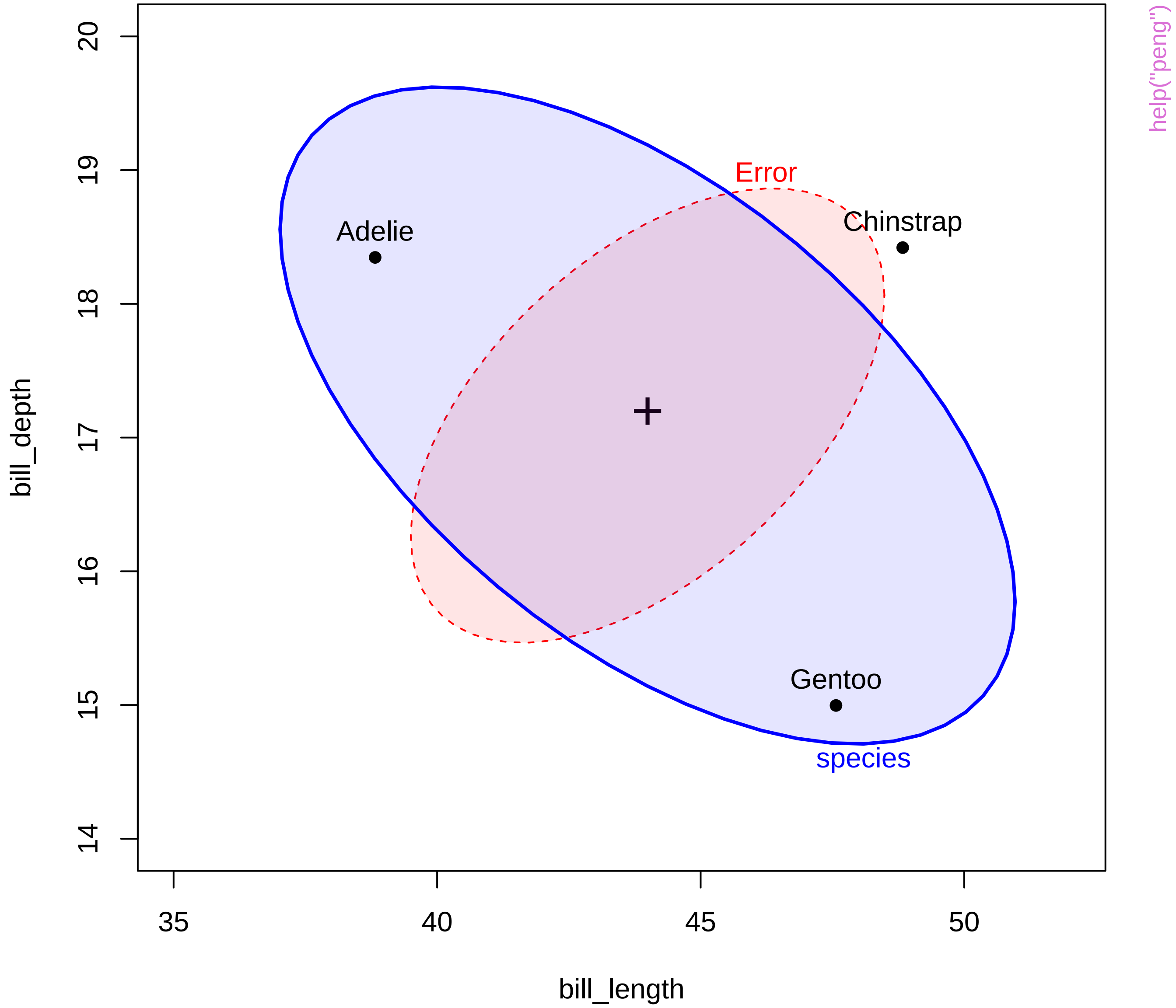
99
SAT
1
105
PPVT
40
21
Raven
8
Regr
Error
SES
na
ns
n
ss
s
Hi
Lo
help("pairs.mlm")


Gentoo
pooled
Chinstrap
Adelie
help("peng")
bill_depth
bill_length
−4
−2
0
2
4
−1
0
1

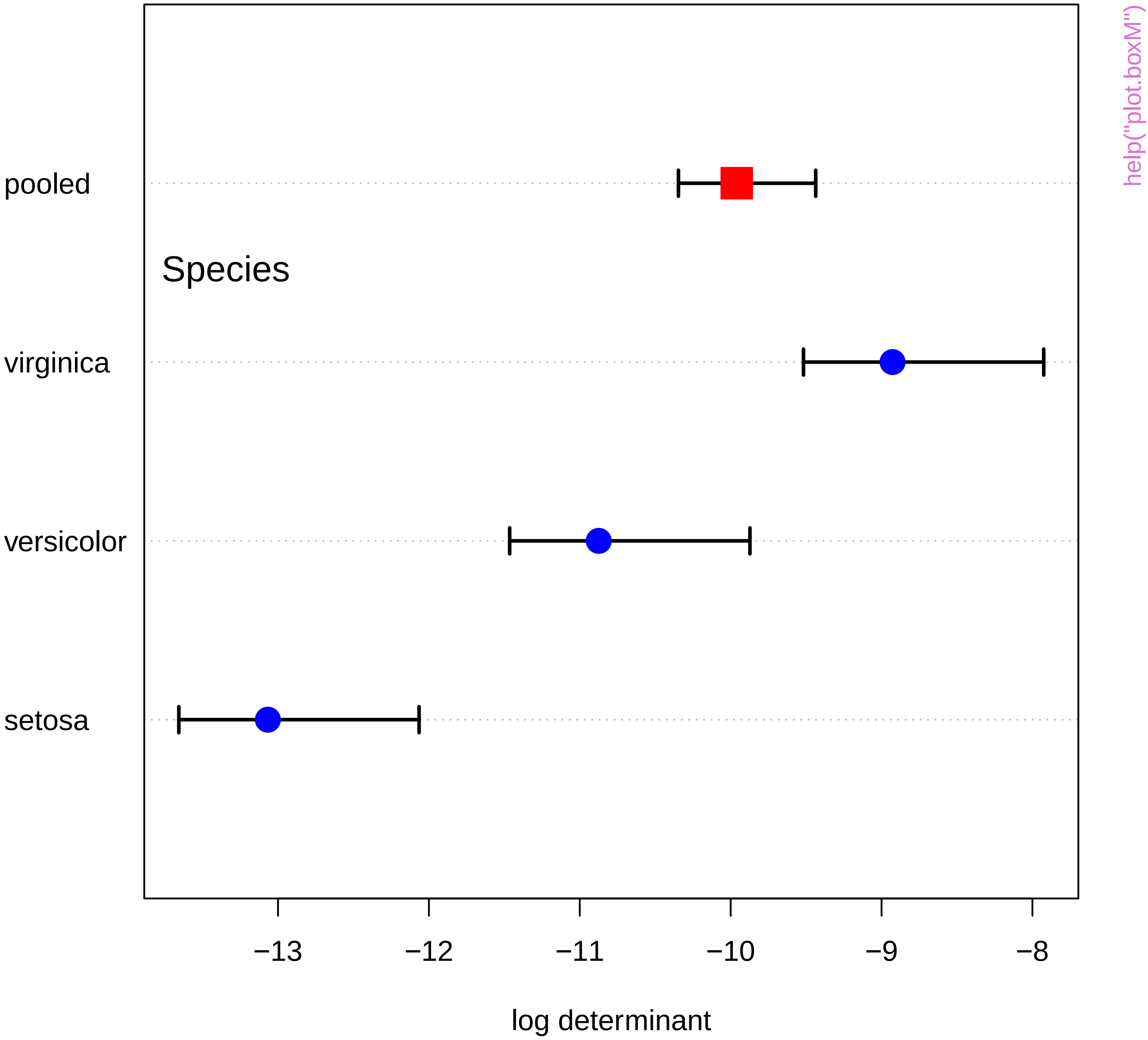


bill_length
Gentoo
Chinstrap
pooled
Adelie
bill_depth
flipper_length
body_mass
help("peng")

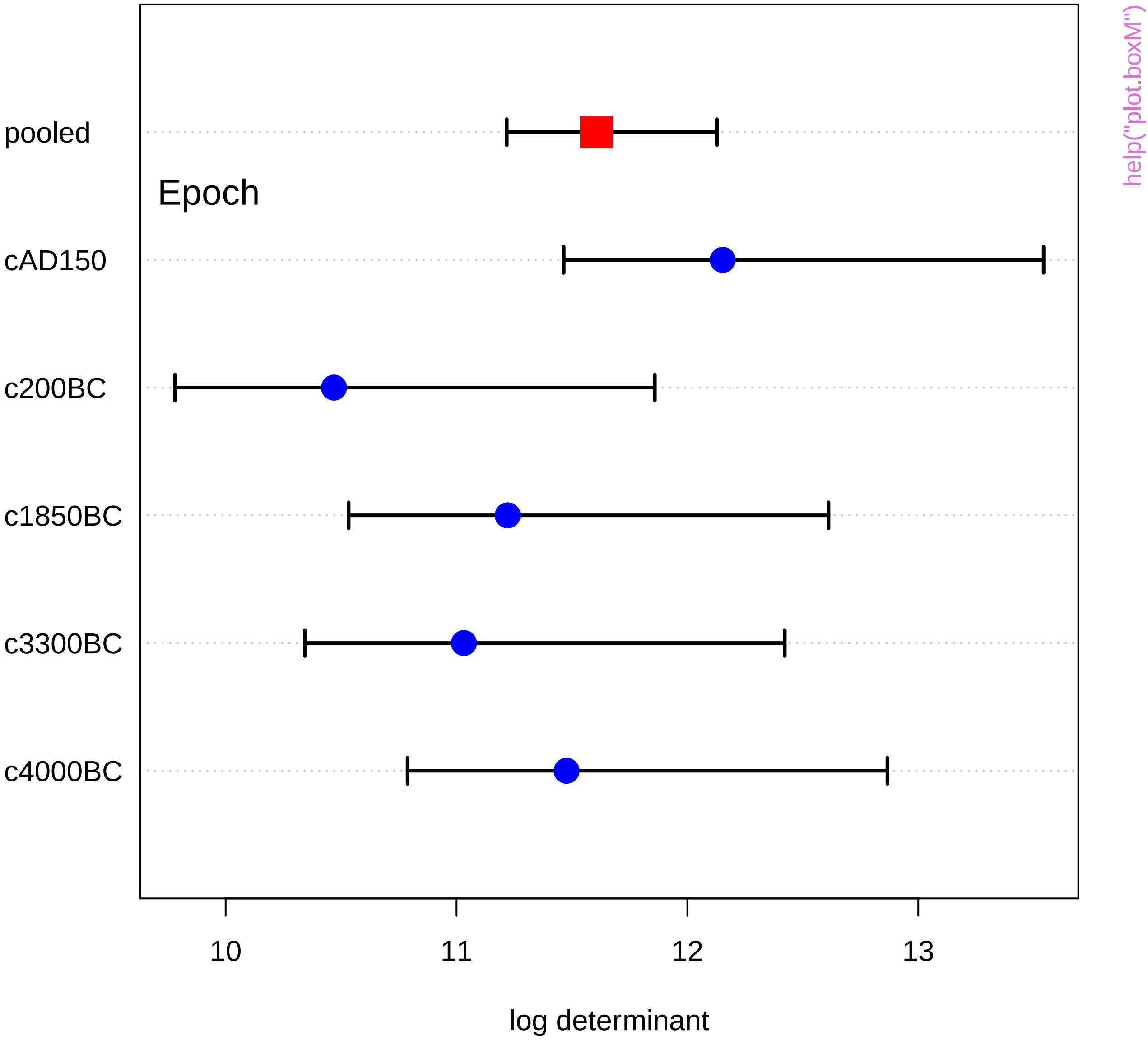

pooled
Species
Gentoo
Chinstrap
Adelie
16.0
16.5
17.0
17.5
log determinant
help("peng")

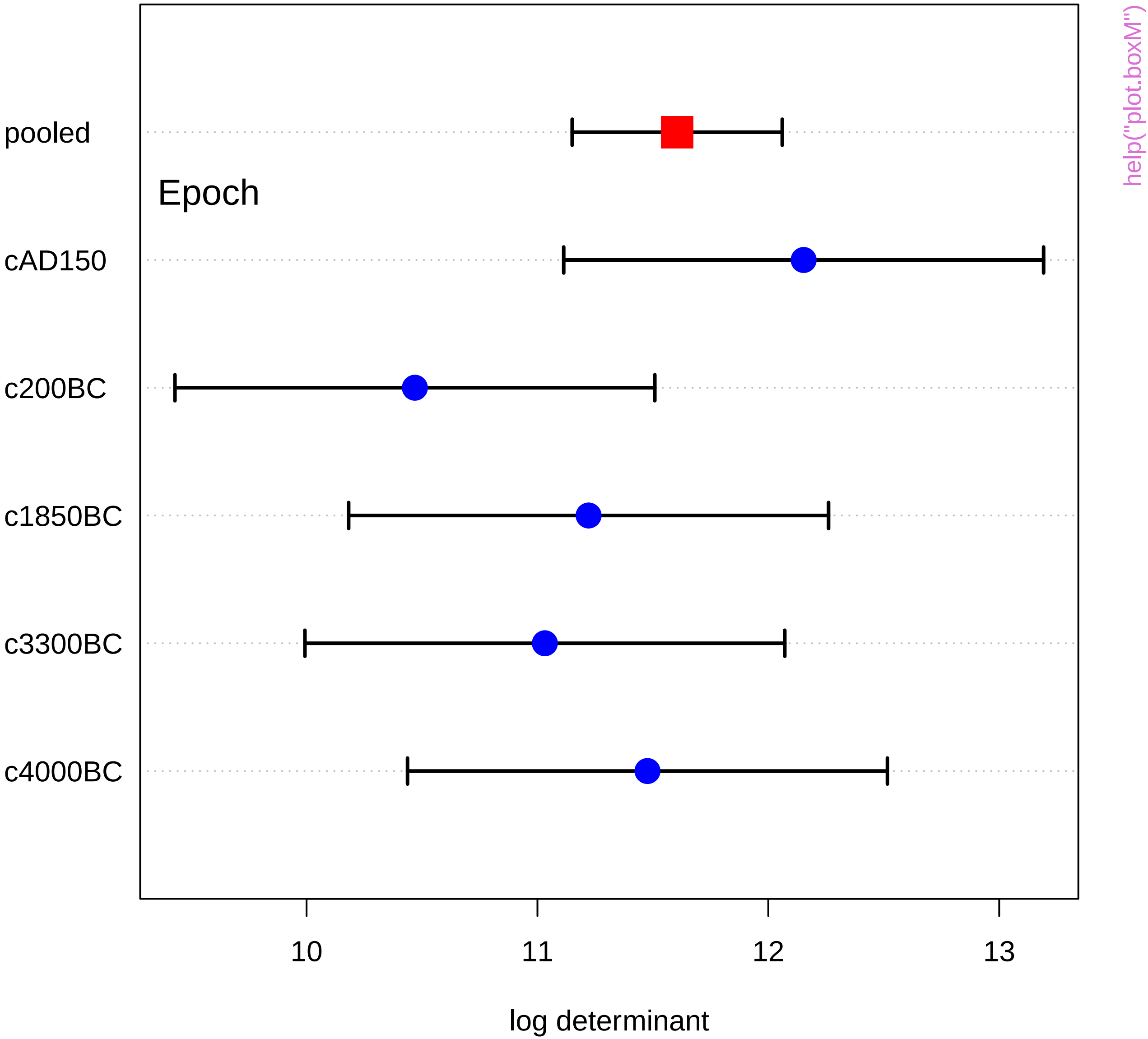


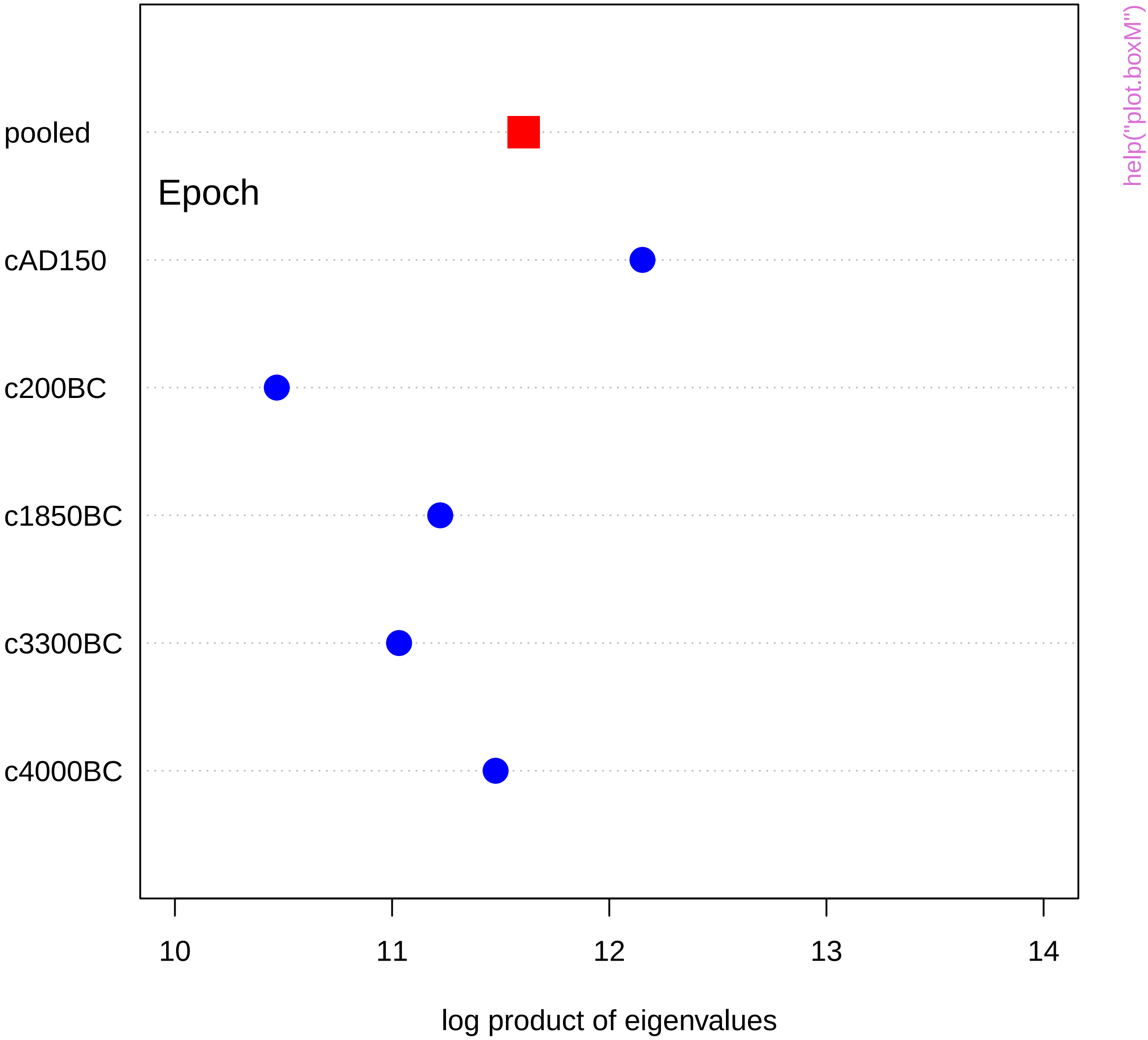
20
19
18
17
16
15
14
bill_depth
35
40
45
50
bill_length
Adelie
Chinstrap
Gentoo
Error
species
help("peng")



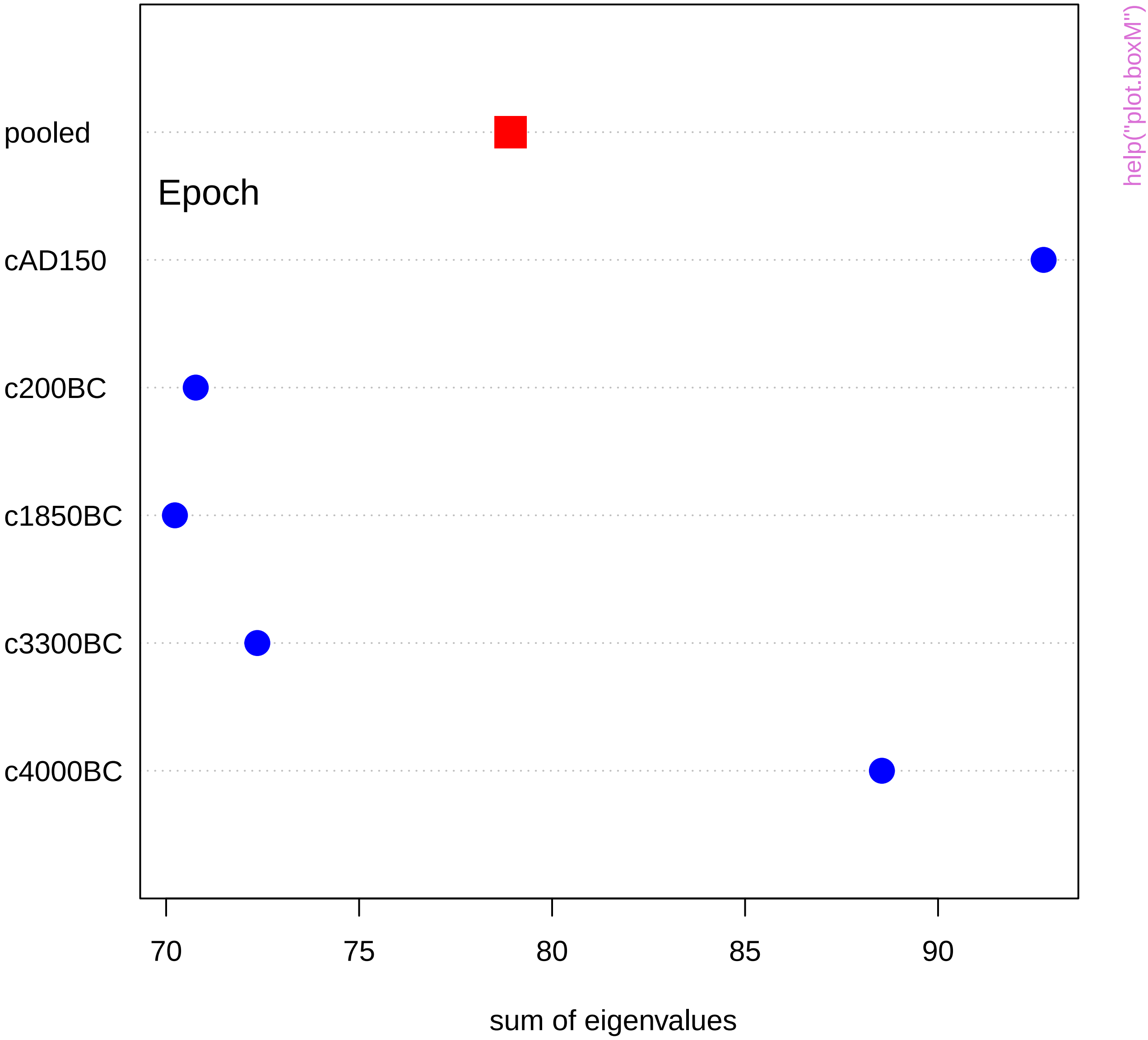
pooled
Species
virginica
versicolor
setosa
−13
−12
−11
−10
−9
−8
log determinant
help("plot.boxM")

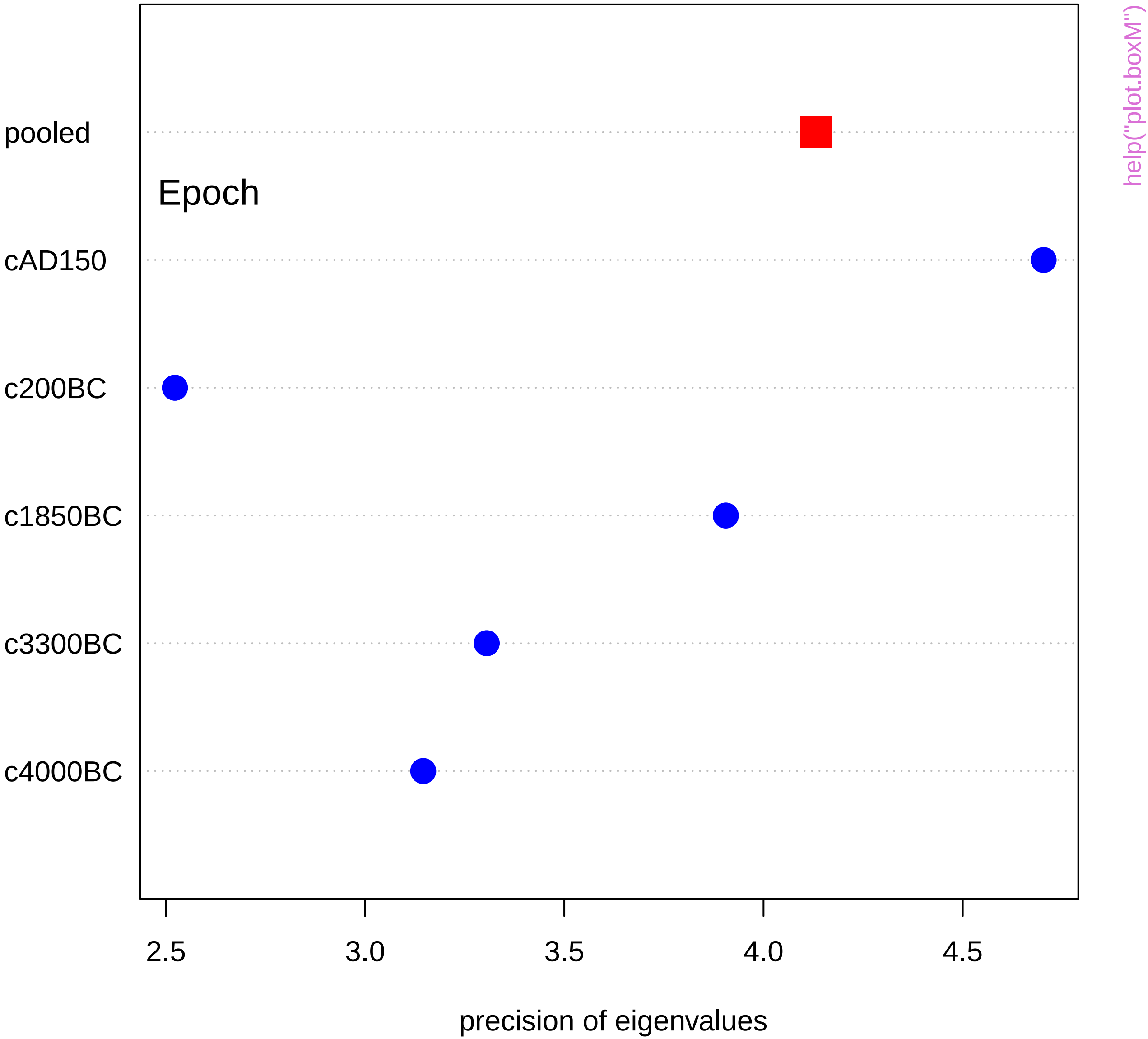

pooled
Epoch
cAD150
c200BC
c1850BC
c3300BC
c4000BC
10
11
12
13
log determinant
help("plot.boxM")

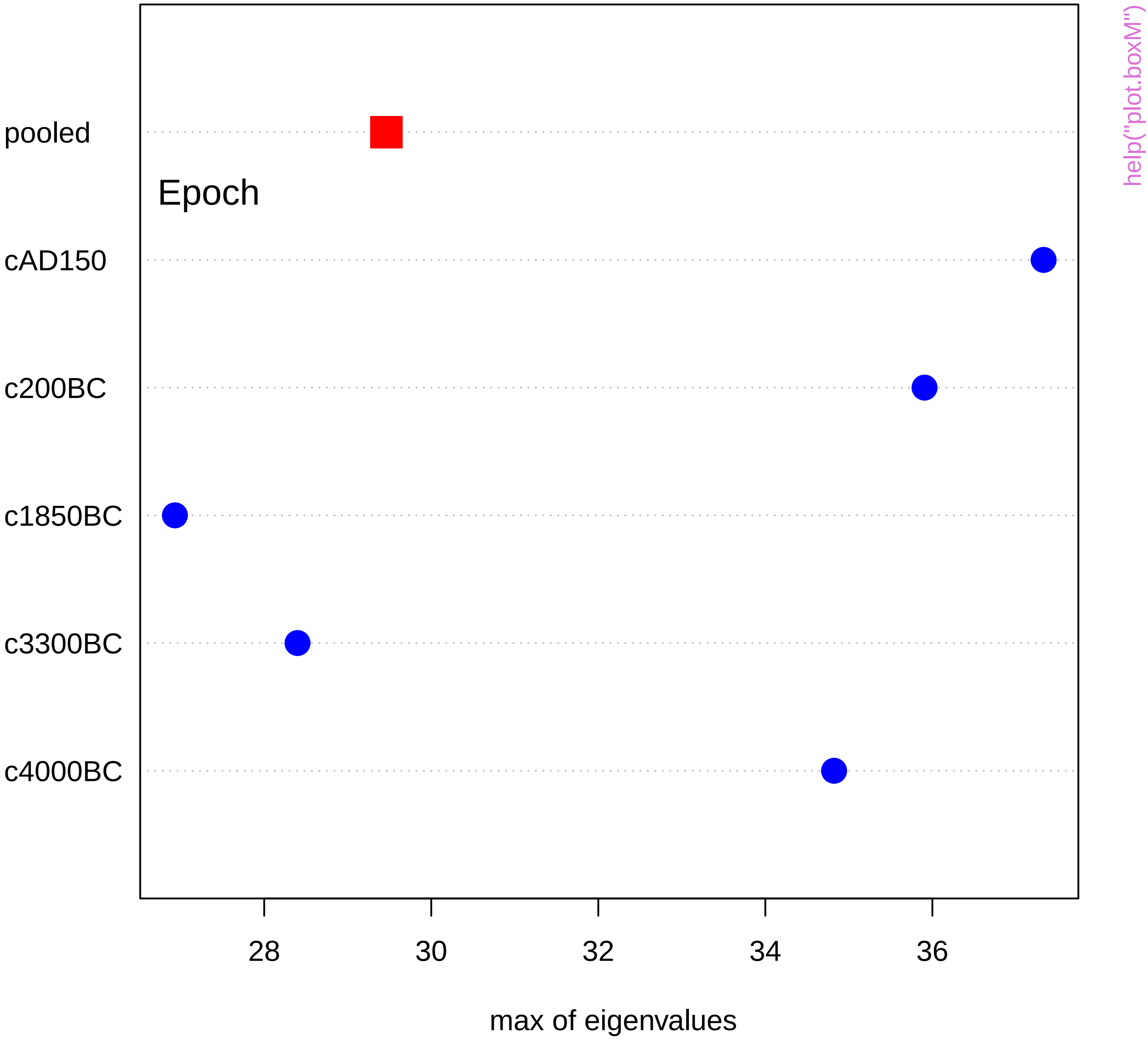

pooled
Epoch
cAD150
c200BC
c1850BC
c3300BC
c4000BC
10
11
12
13
log determinant
help("plot.boxM")

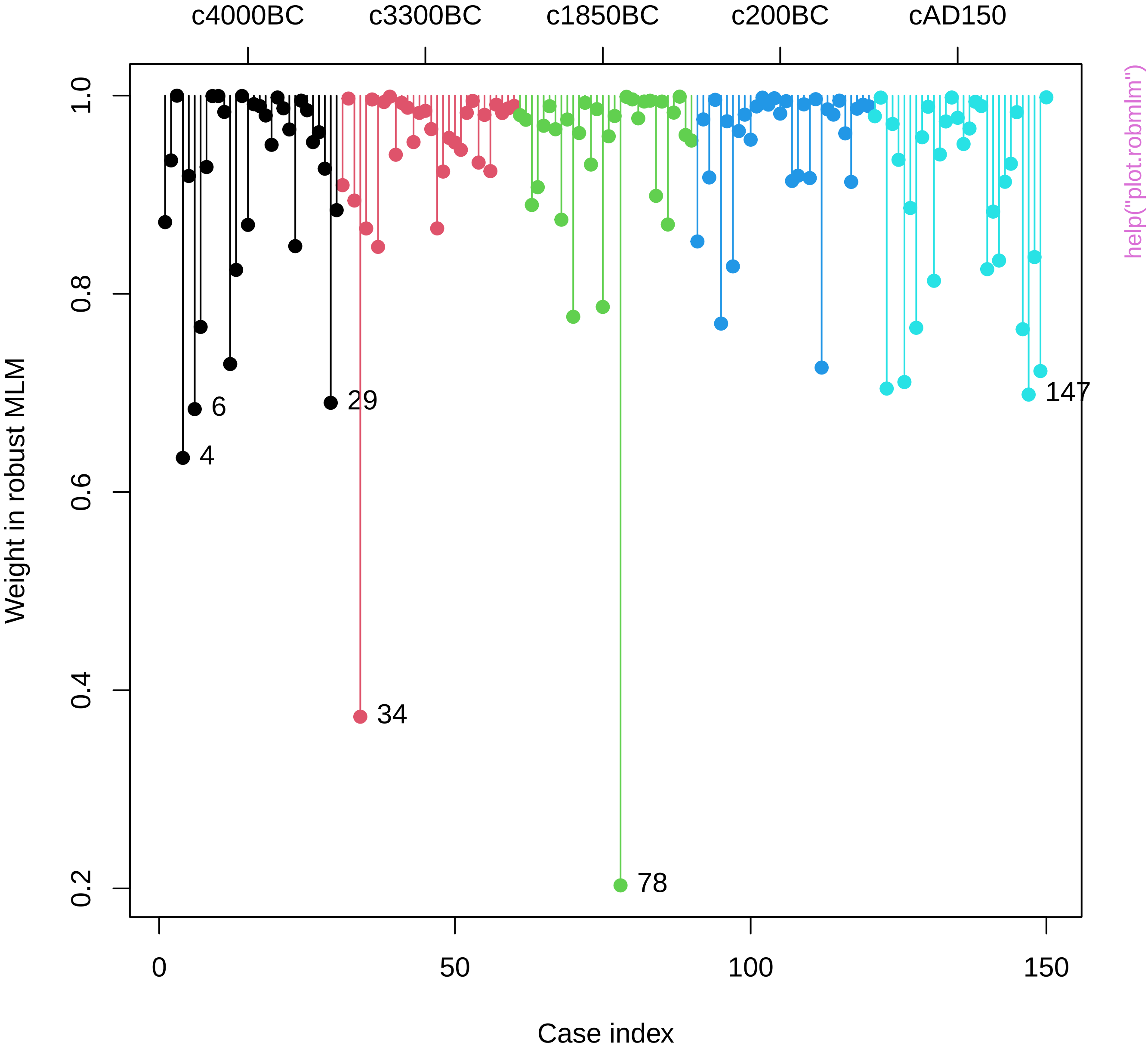


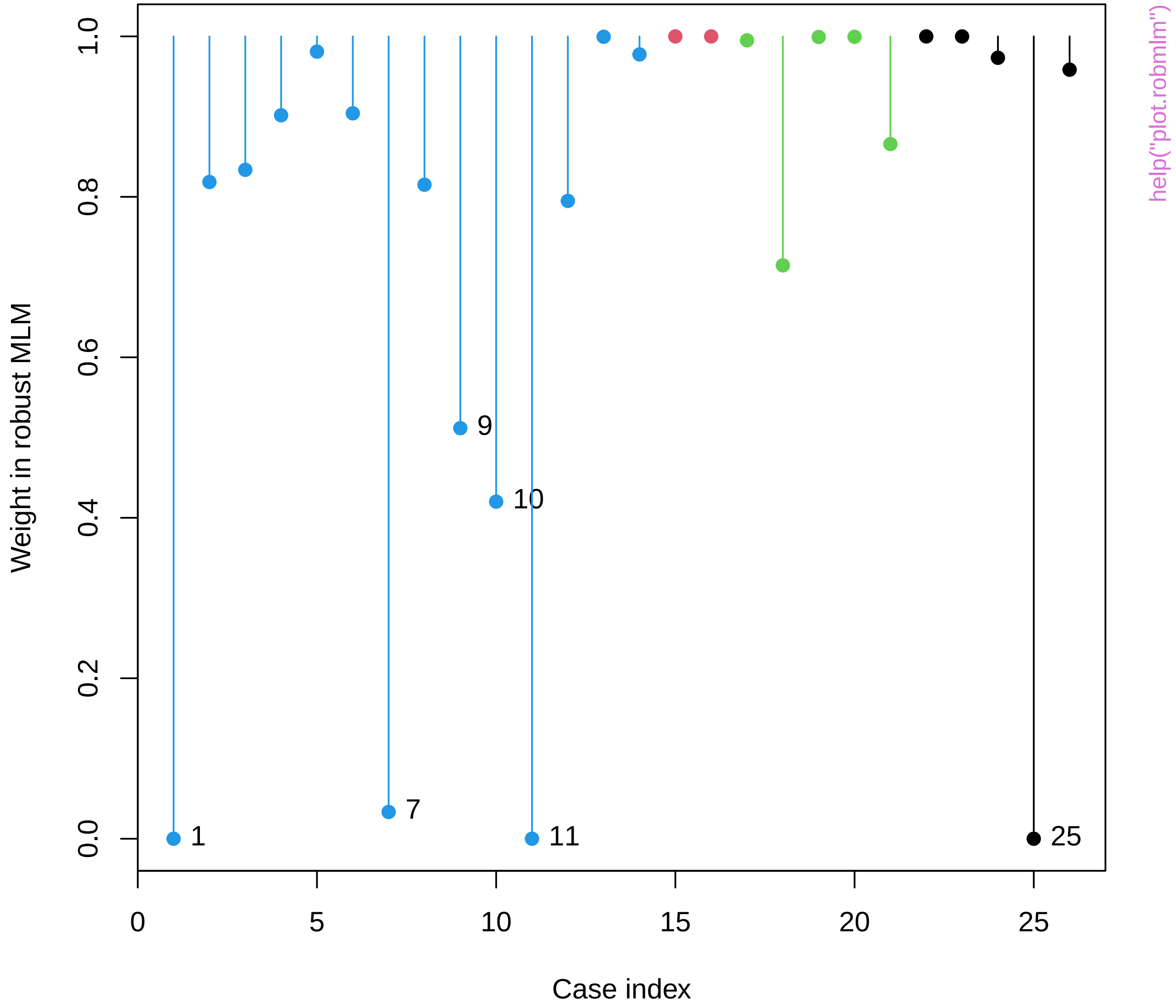
pooled
Epoch
cAD150
c200BC
c1850BC
c3300BC
c4000BC
10
11
12
13
14
log product of eigenvalues
help("plot.boxM")


pooled
Epoch
cAD150
c200BC
c1850BC
c3300BC
c4000BC
70
75
80
85
90
sum of eigenvalues
help("plot.boxM")

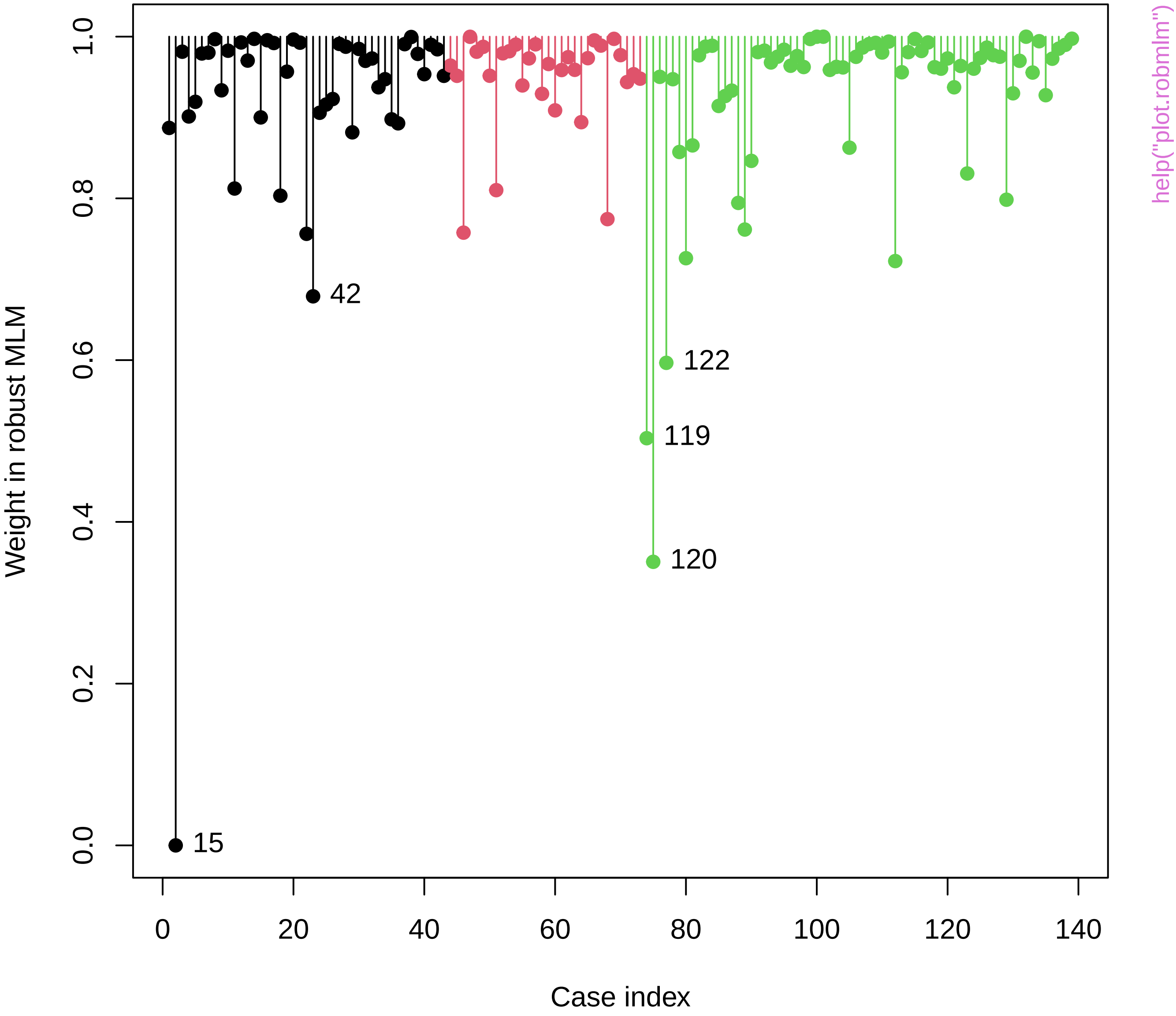

pooled
Epoch
cAD150
c200BC
c1850BC
c3300BC
c4000BC
2.5
3.0
3.5
4.0
4.5
precision of eigenvalues
help("plot.boxM")


help("plot.boxM")
pooled
Epoch
cAD150
c200BC
c1850BC
c3300BC
c4000BC
28
30
32
34
36
max of eigenvalues

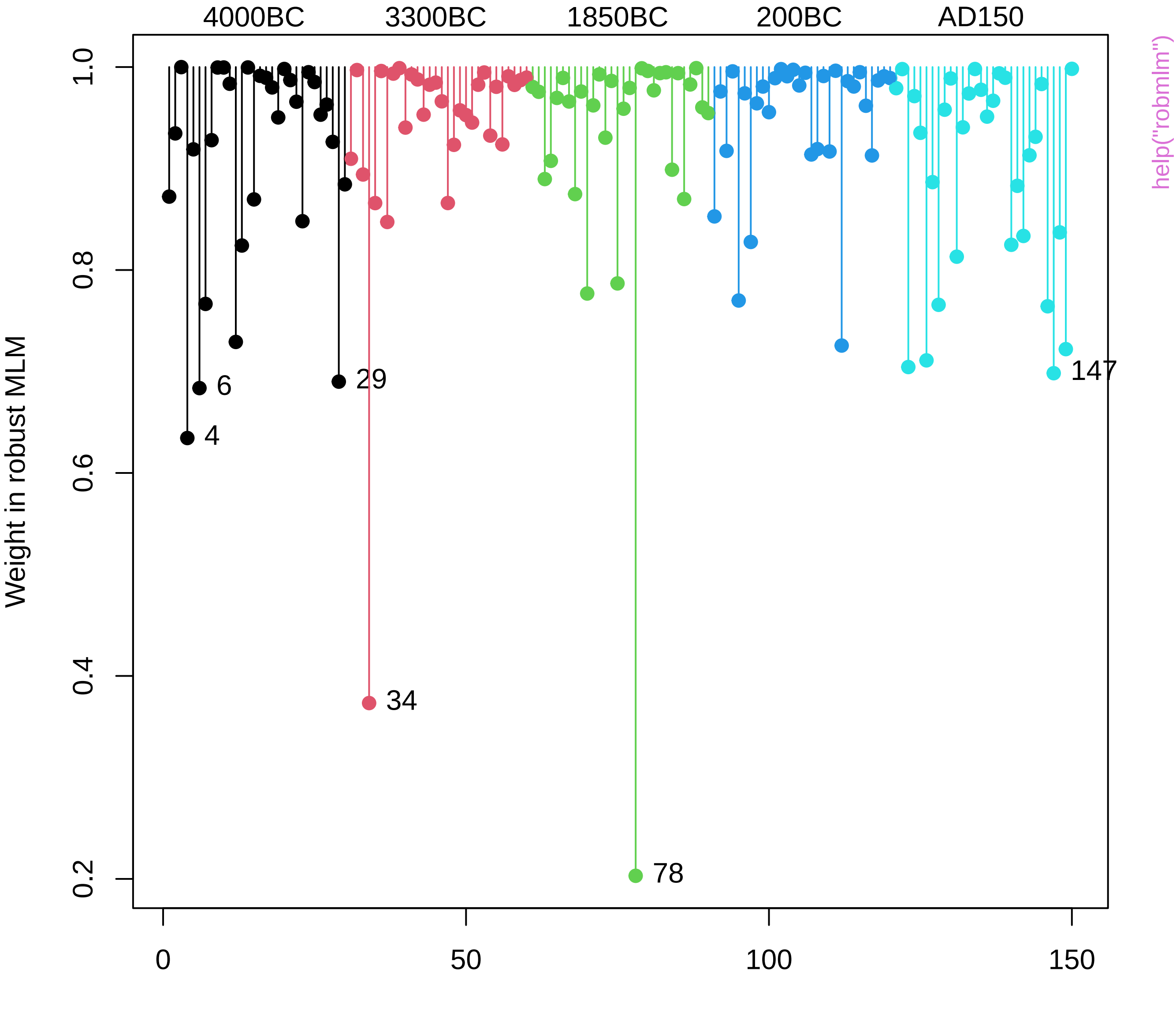

c4000BC
c3300BC
c1850BC
c200BC
cAD150
help("plot.robmlm")
Weight in robust MLM
1.0
0.8
0.6
0.4
0.2
6
4
29
34
78
147
0
50
100
150
Case index

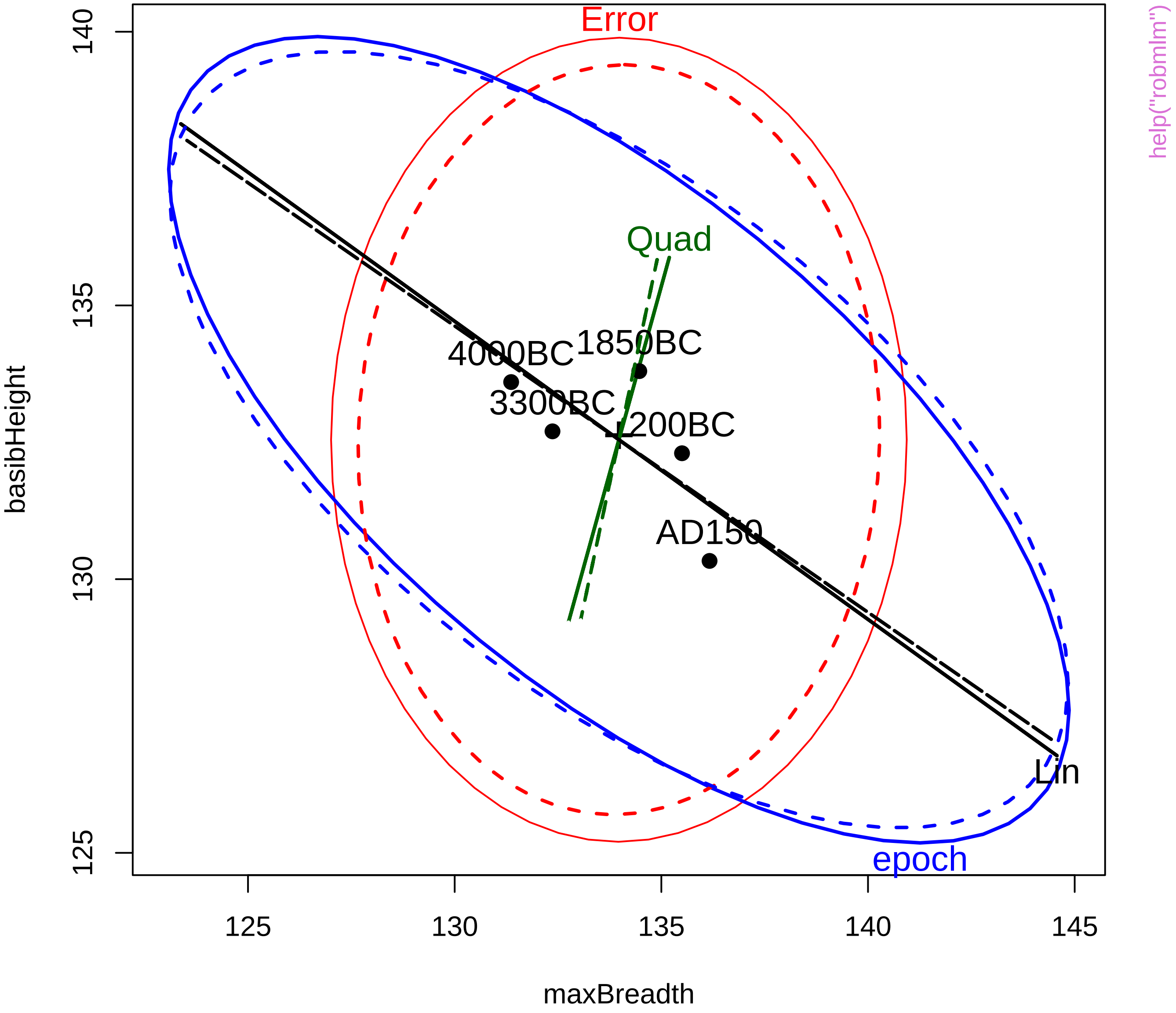


Weight in robust MLM
1.0
0.8
0.6
0.4
0.2
0.0
1
7
9
10
11
25
0
5
10
15
20
25
Case index
help("plot.robmlm")

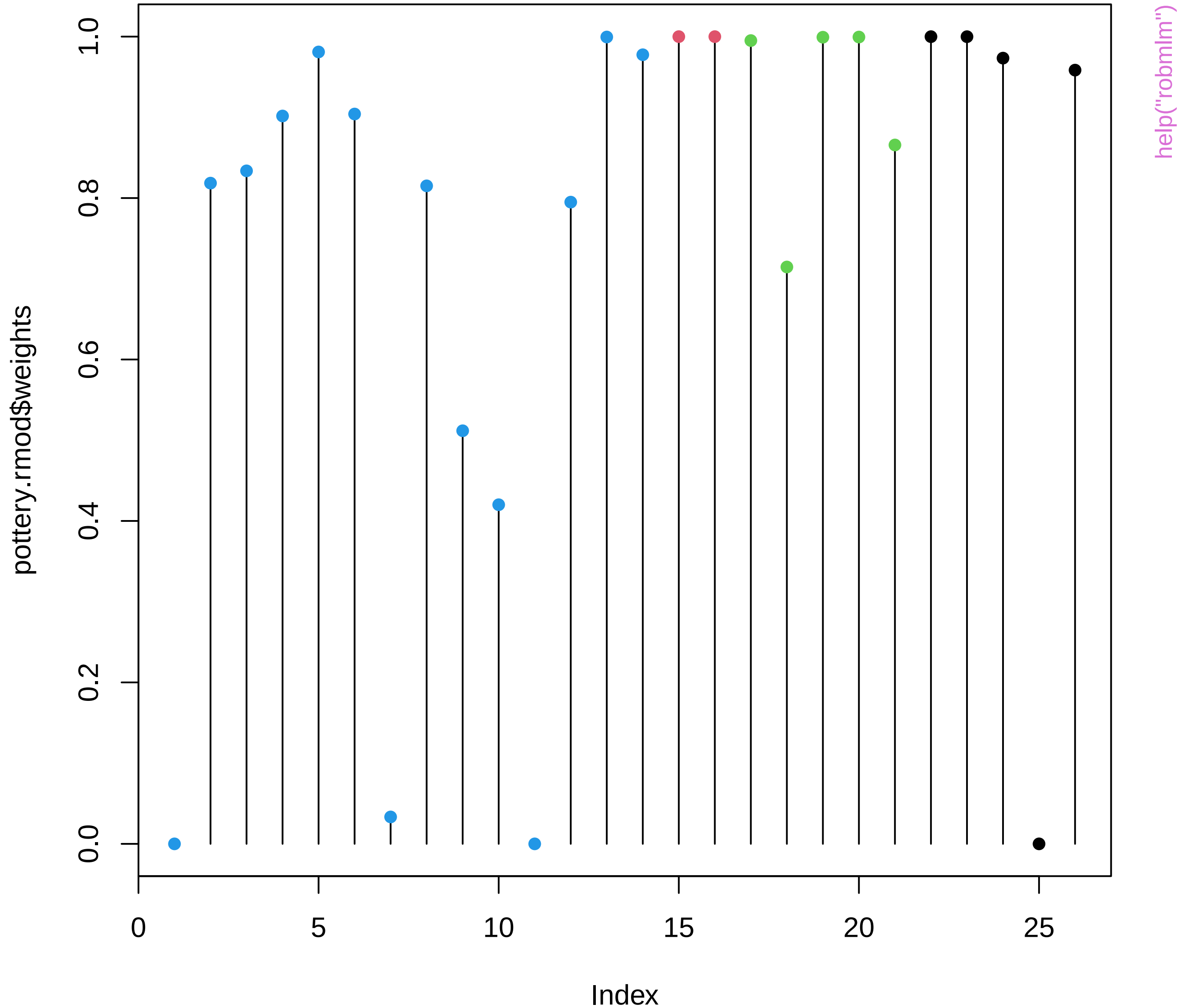


Weight in robust MLM
Case index
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
120
140
15
42
122
119
120
help("plot.robmlm")

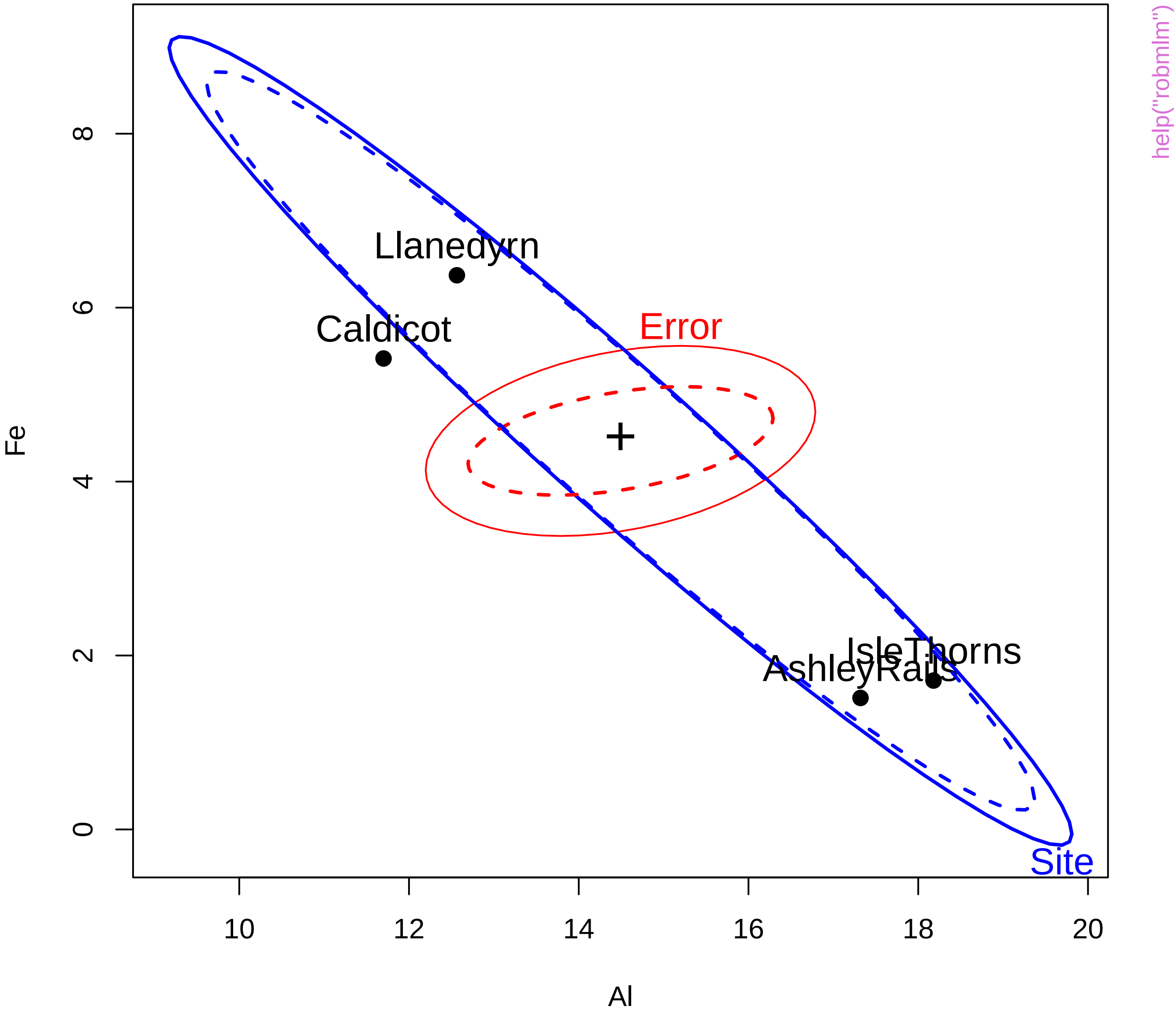

4000BC
3300BC
1850BC
200BC
AD150
help("robmlm")
Weight in robust MLM
1.0
0.8
0.6
0.4
0.2
Case index
0
50
100
150
4
6
29
34
78
147



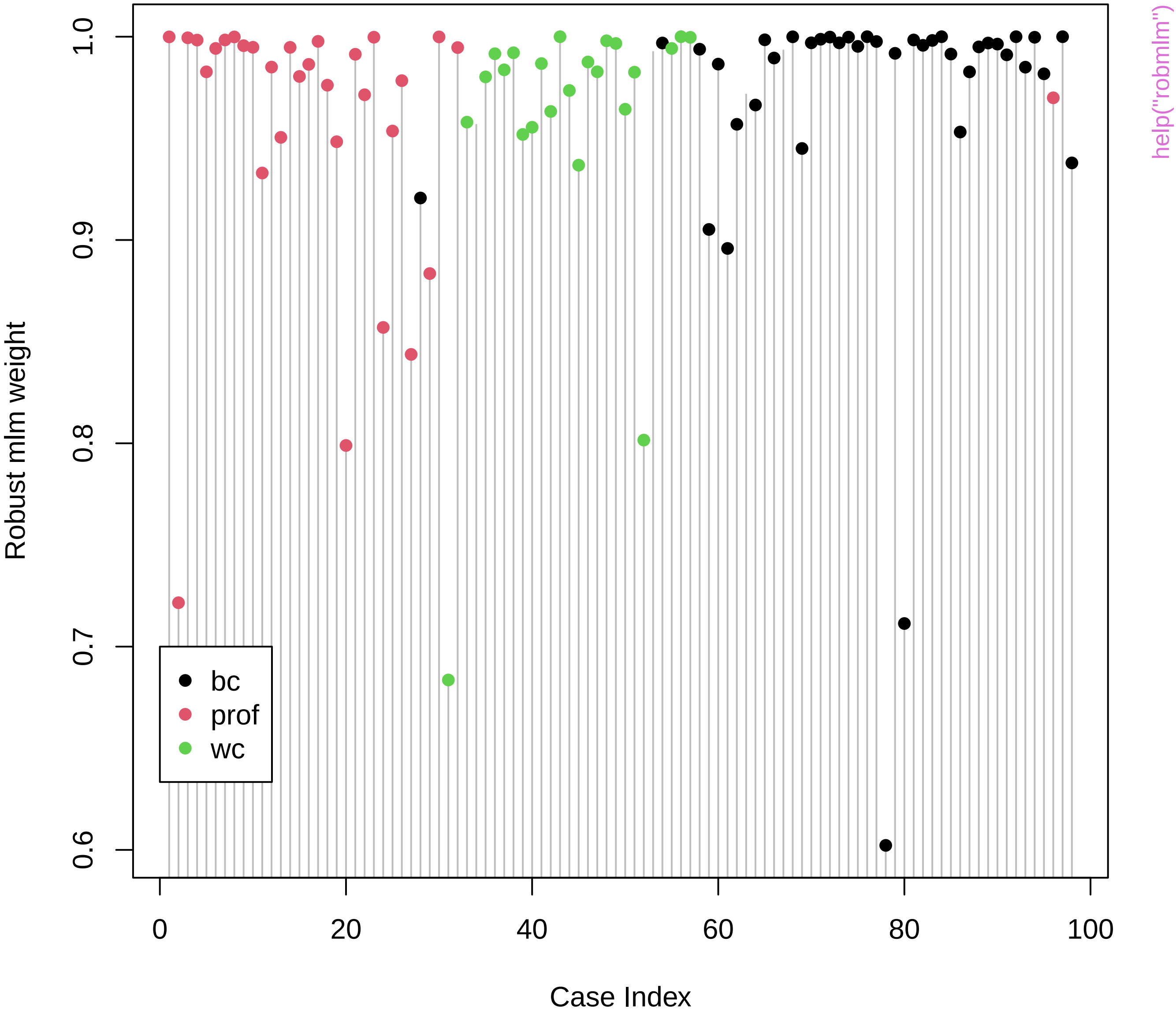
Error
Quad
4000BC
1850BC
3300BC
1200BC
AD150
Lin
epoch
140
135
130
125
125
130
135
140
145
basibHeight
maxBreadth
help("robmlm")



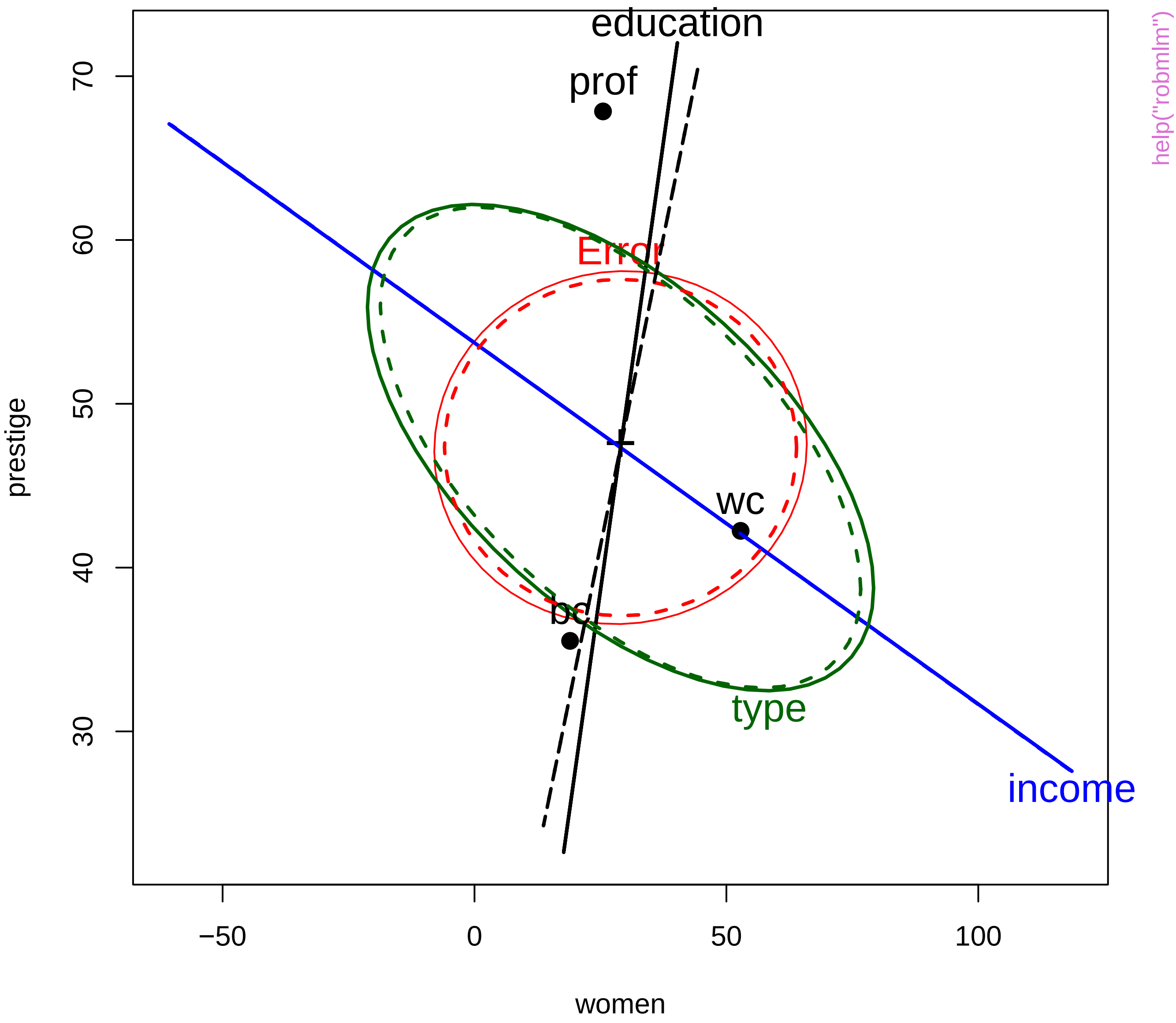
pottery.rmod$weights
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
Index
help("robmlm")



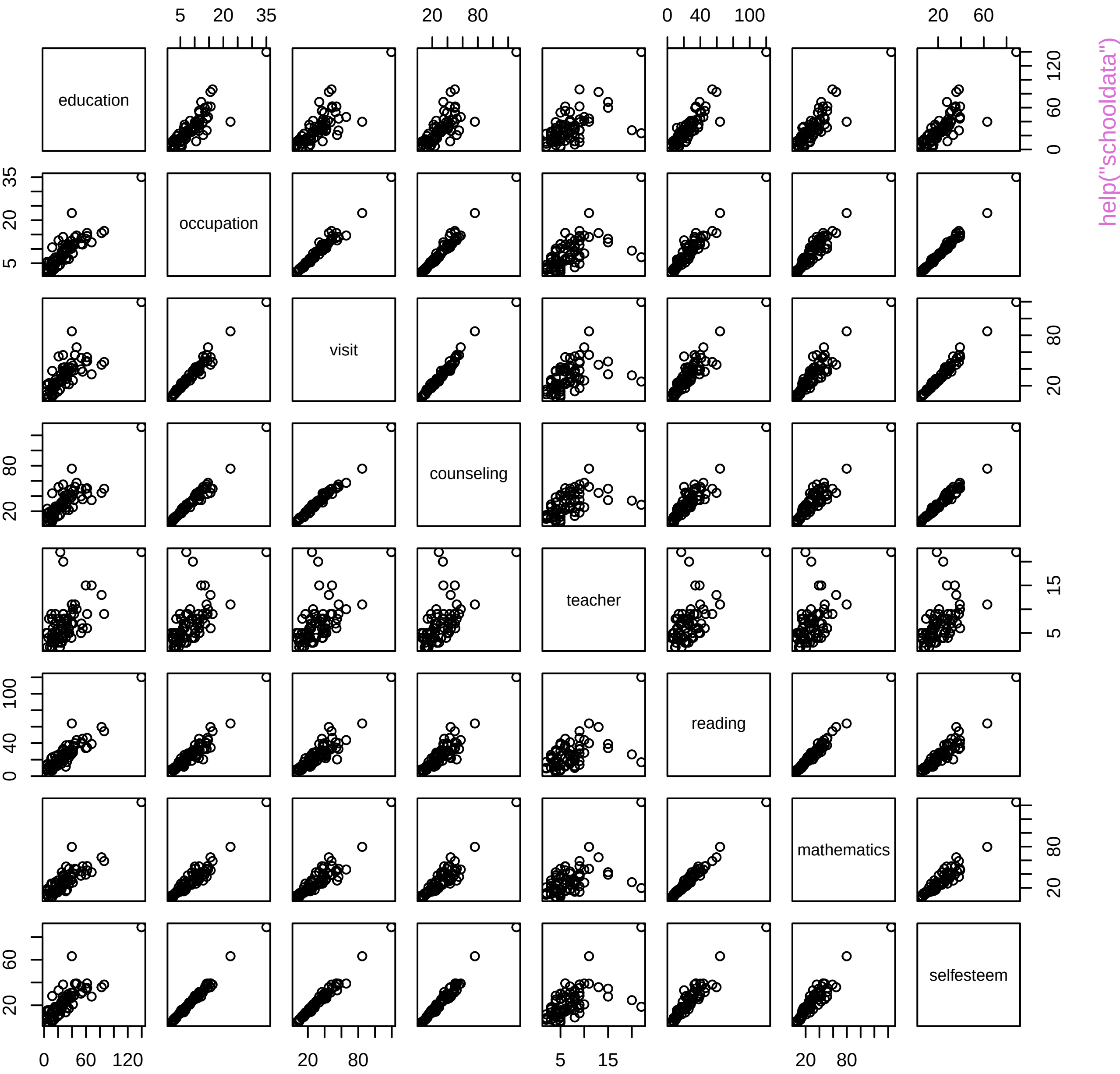
help("robmlm")
Fe
Al
Llanedyrn
Caldicot
Error
IsleThorns
AshleyRails
Site
0
2
4
6
8
10
12
14
16
18
20



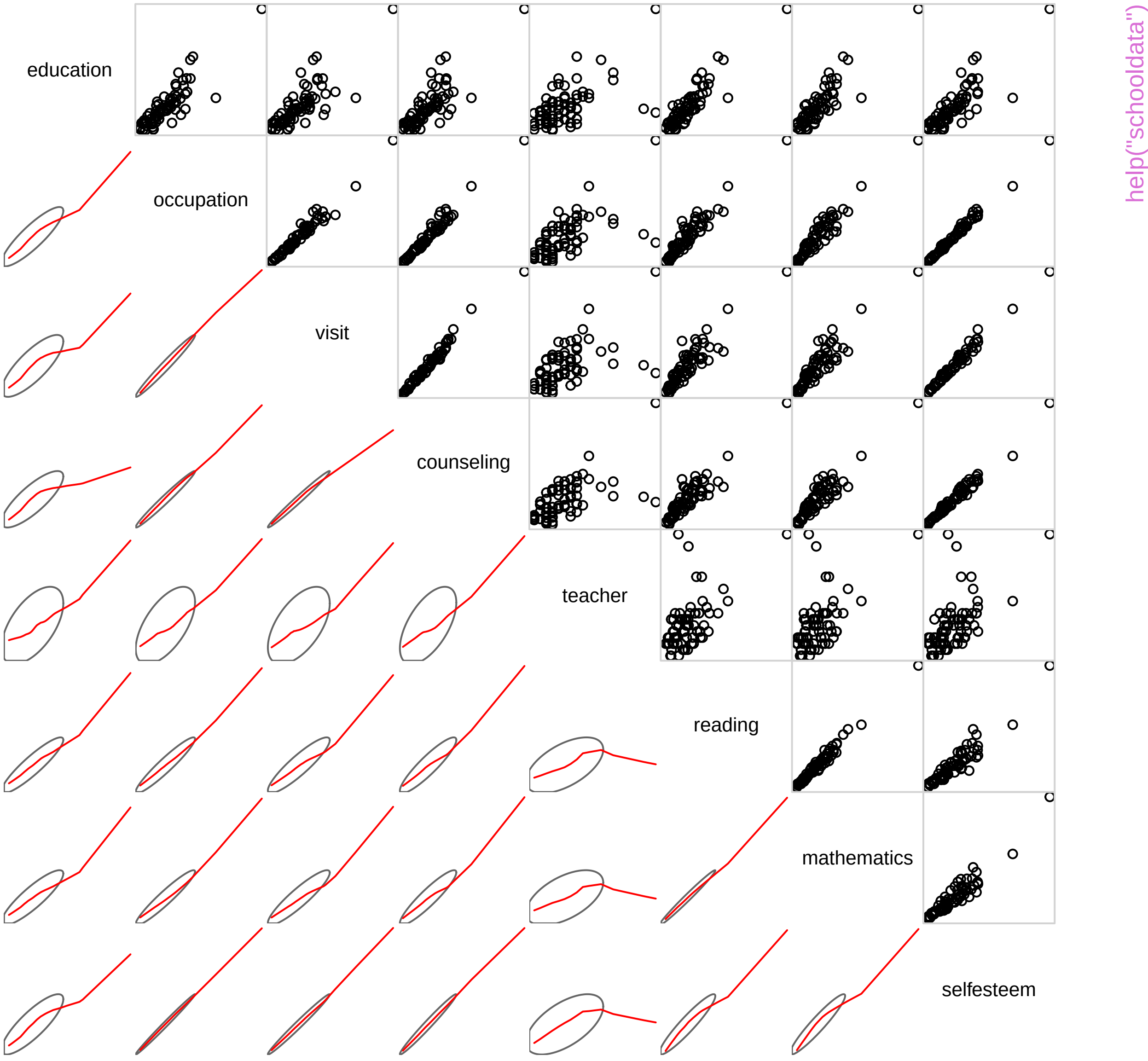
Robust mlm weight
Case Index
0.6
0.7
0.8
0.9
1.0
0
20
40
60
80
100
bc
prof
wc
help("robmlm")

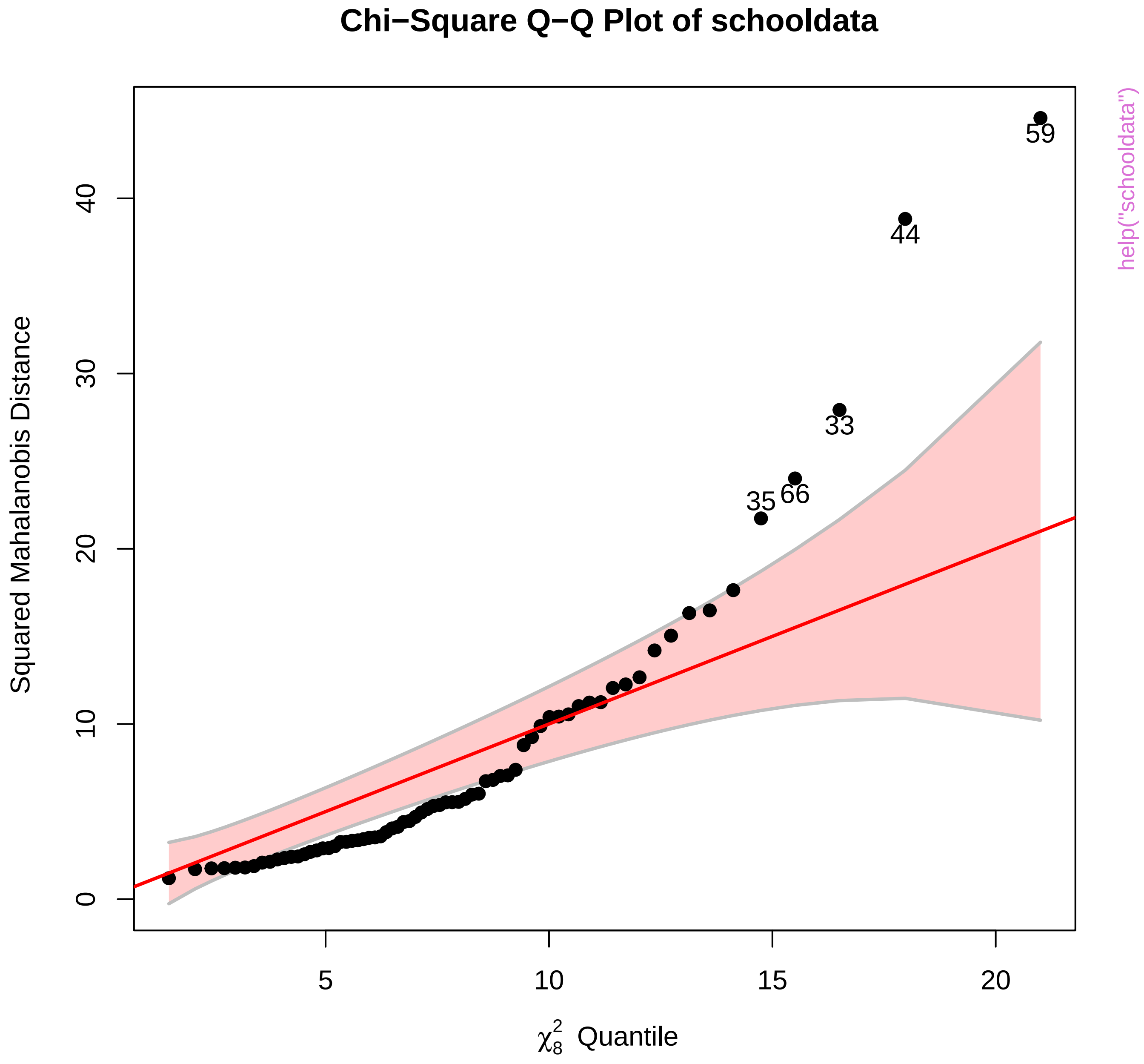


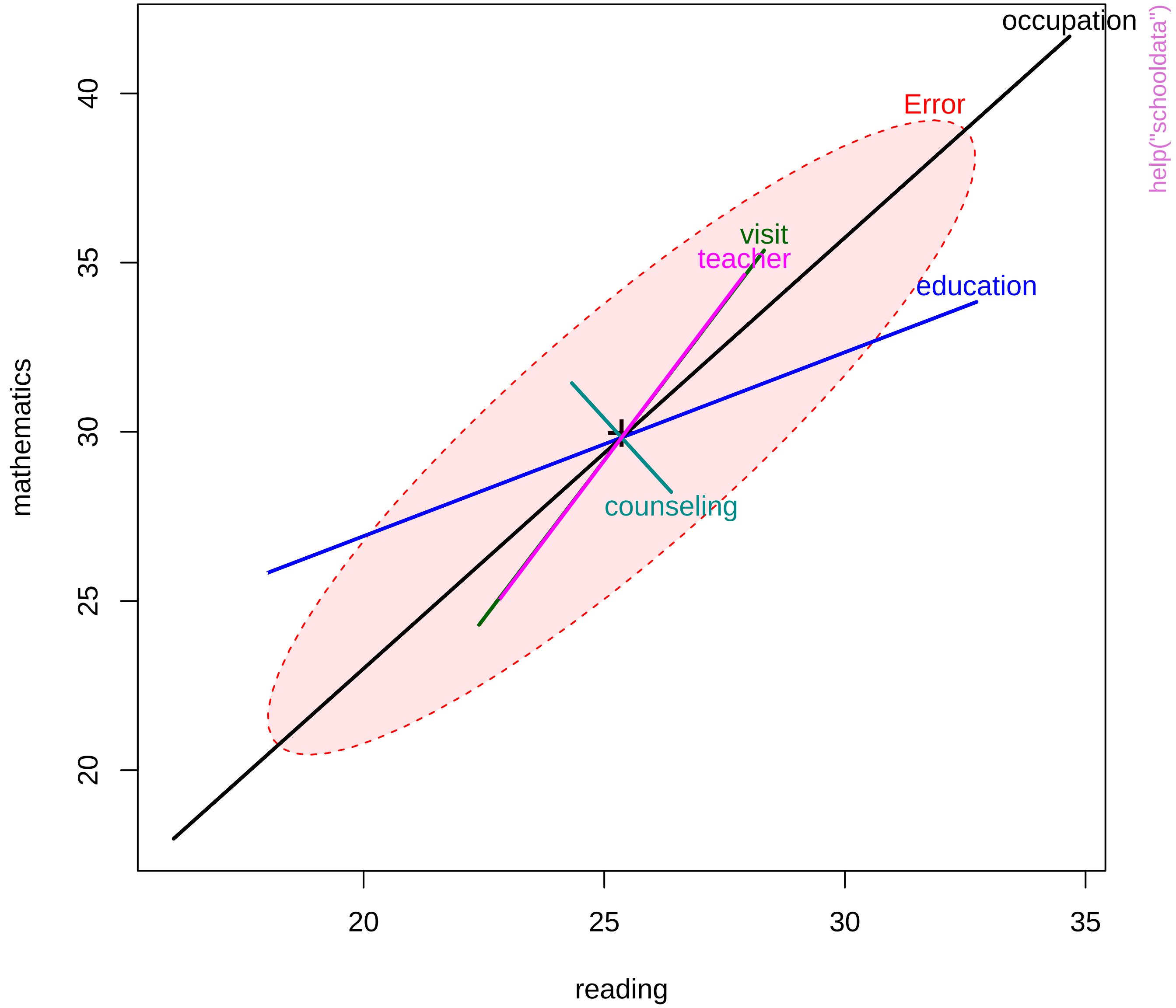
education
prof
Error
wc
bc
type
income
prestige
women
70
60
50
40
30
−50
0
50
100
help("robmlm")

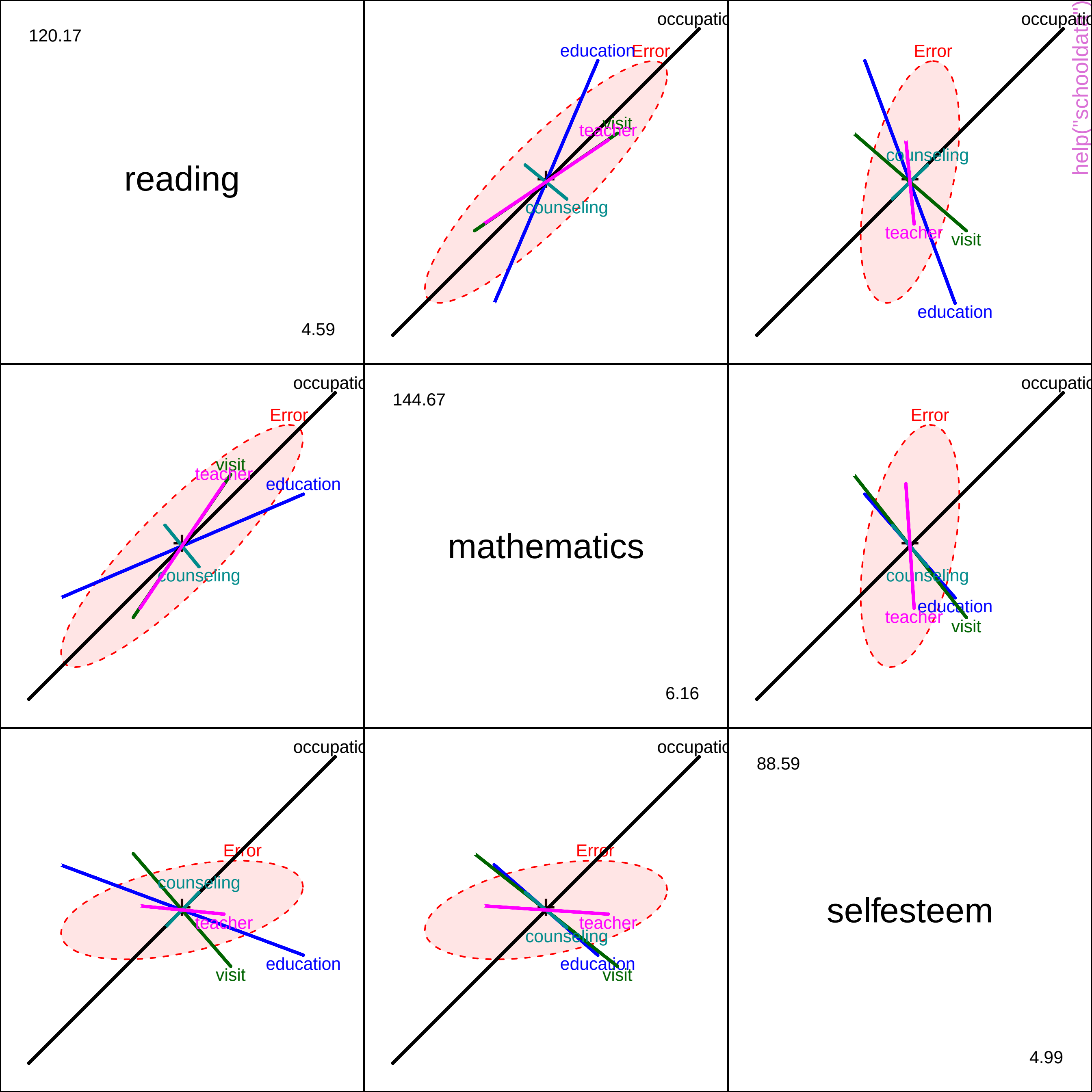


education
occupation
visit
counseling
teacher
reading
mathematics
selfesteem
help("schooldata")

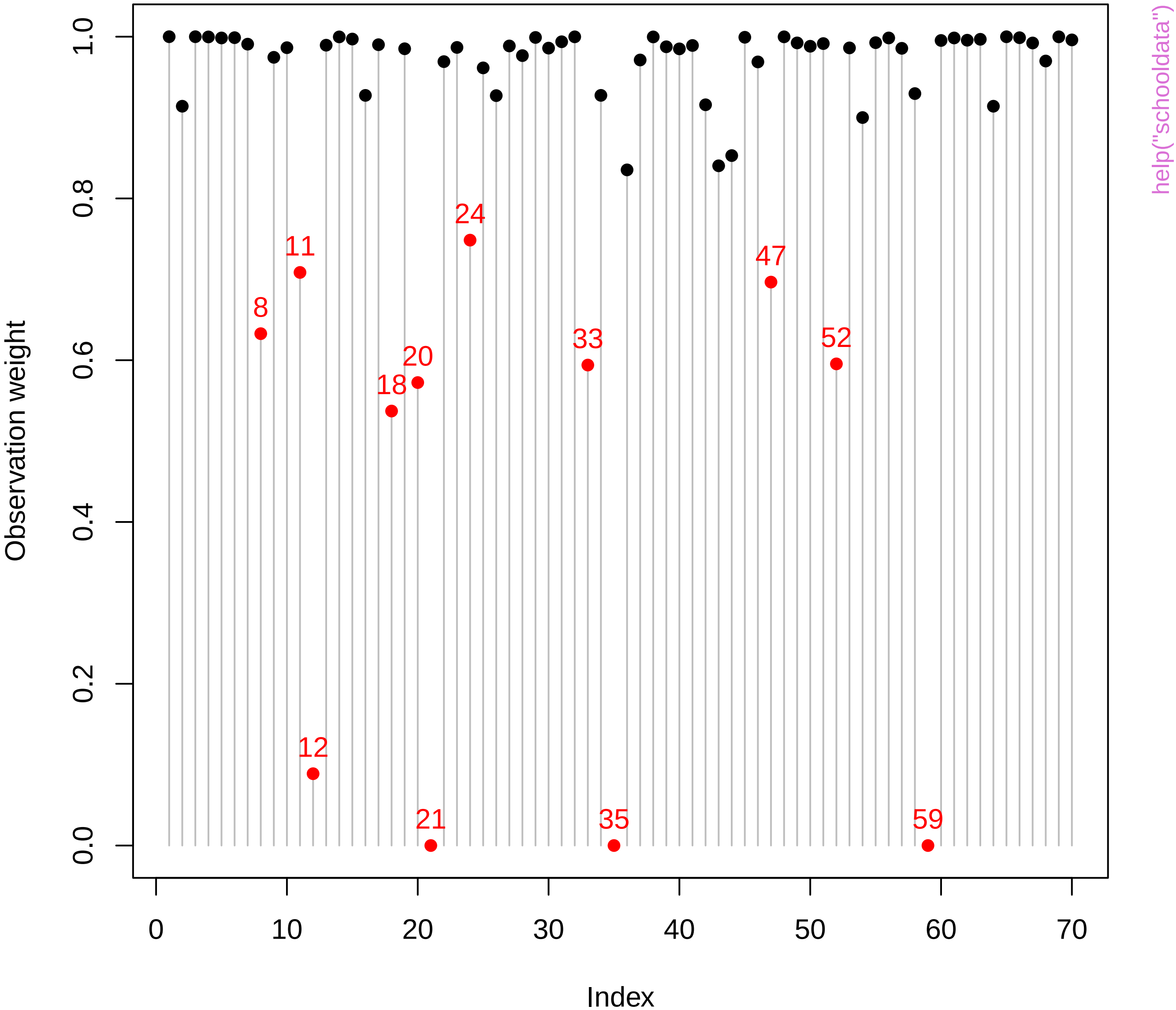

education
occupation
visit
counseling
teacher
reading
mathematics
selfesteem
help("schooldata")


Chi−Square Q−Q Plot of schooldata
Squared Mahalanobis Distance
0
10
20
30
40
5
10
15
20
χ^2_8 Quantile
59
44
33
66
35
help("schooldata")

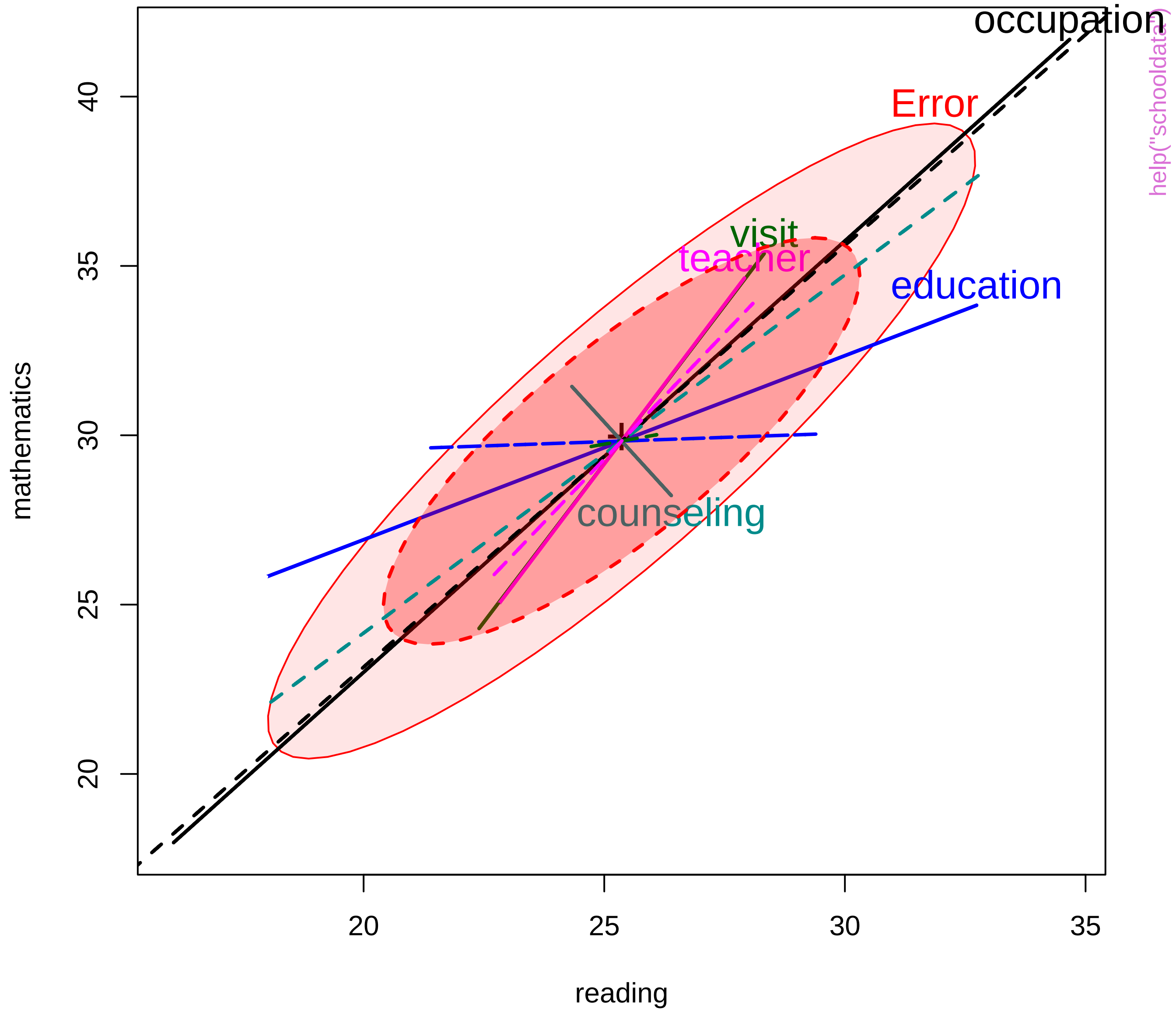

occupation
Error
visit
teacher
education
counseling
help("schooldata")
mathematics
reading
20
25
30
35
40



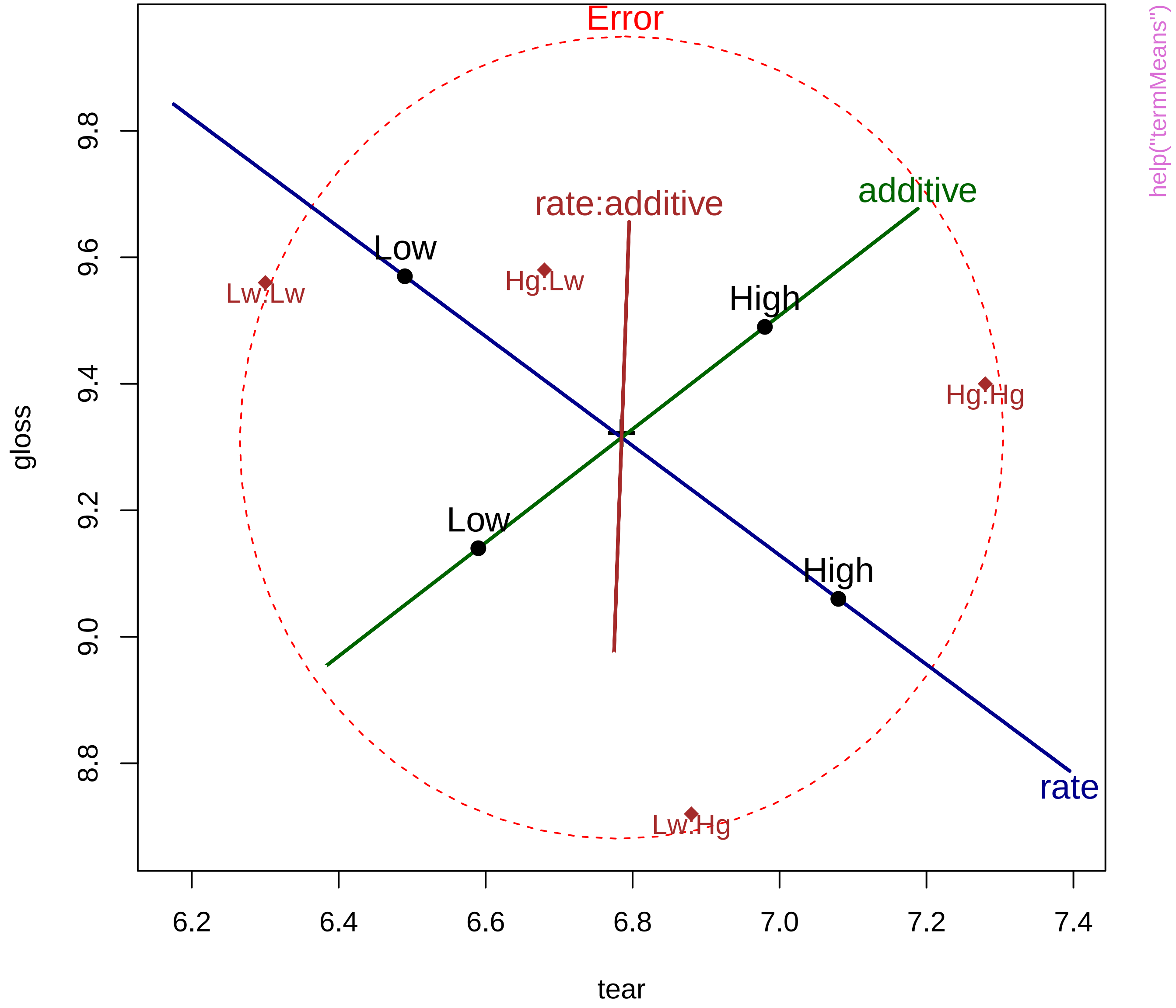
120.17
reading
4.59
occupatic
education
Error
visit
teacher
counseling
occupatic
Error
counseling
teacher
visit
education
occupatic
Error
visit
teacher
education
counseling
144.67
mathematics
6.16
occupatic
Error
counseling
education
teacher
visit
occupatic
Error
counseling
teacher
visit
education
occupatic
Error
teacher
counseling
education
visit
88.59
selfesteem
4.99



help("schooldata")
Observation weight
Index
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
60
70
8
11
12
18
20
21
24
33
35
47
52
59



occupation
help("schooldata")
Error
visit
teacher
education
counseling
mathematics
reading
20
25
30
35
40



Error
rate:additive
additive
Low
Hg:Lw
Lw:Lw
High
Hg:Hg
Low
High
Lw:Hg
rate
9.8
9.6
9.4
9.2
9.0
8.8
gloss
6.2
6.4
6.6
6.8
7.0
7.2
7.4
tear
help("termMeans")